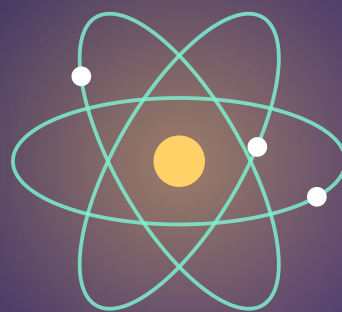
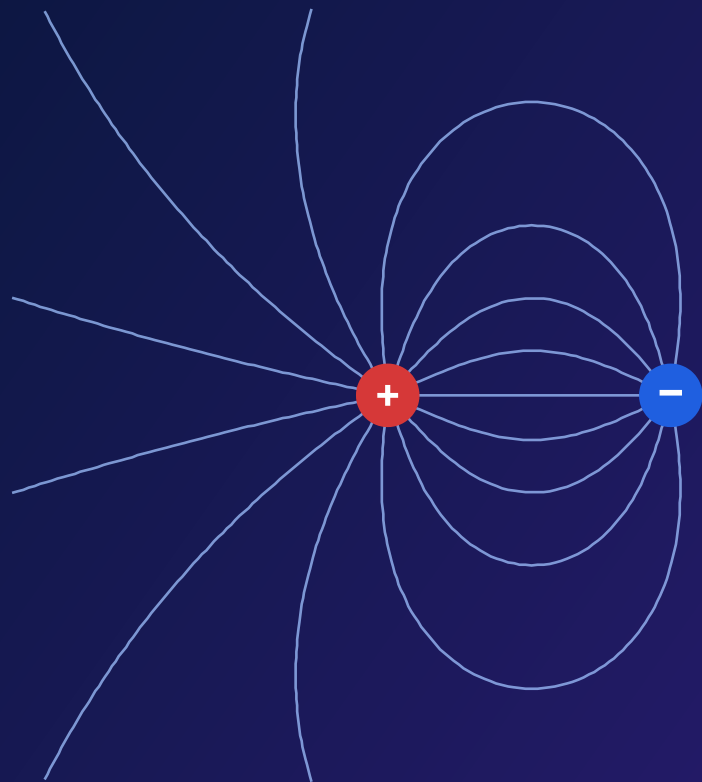




CLASS 12 · 2026-27

CBSE PHYSICS SCORE GUIDE

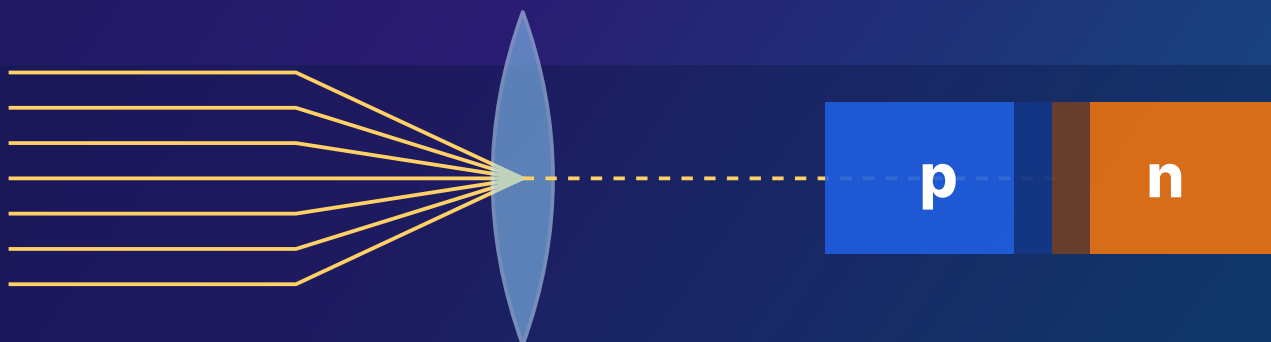
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Syllabus, paper pattern and marking scheme can change. Always confirm against the **official 2026–27 syllabus, sample question paper and marking scheme** published by the board before the examination. Where this book states a pattern, it is the recent pattern and is marked *provisional*.

HOW QUESTIONS ARE LABELLED (PREVIOUS-YEAR QUESTIONS POLICY)

VERIFIED PYQ

Copied from an official past paper and checked against it, with year and source in the Source Log.

ADAPTED PYQ

Reworded or re-numbered from a verified past question.

BOARD-STYLE PRACTICE

New question written for this book in board style. **Not** a past question.

Every question in this edition is BOARD-STYLE PRACTICE. No question is claimed to be a past board question, and no year is attached to any question.

ACCURACY AND CORRECTIONS

Worked numericals were computed in code and checked for arithmetic consistency while writing. Content has not been through an independent subject-expert review, so please report errors so they can be fixed in the next edition. Constants used: $c = 3 \times 10^8 \text{ m s}^{-1}$, $e = 1.6 \times 10^{-19} \text{ C}$, $h = 6.63 \times 10^{-34} \text{ J s}$, $\epsilon_0 = 8.85 \times 10^{-12}$, $\mu_0 = 4\pi \times 10^{-7} \text{ SI}$, unless a question states otherwise.

About the author



VS Sir (Vijay Kumar Sharma)

AI-Enabled Educator · 20+ years teaching NEET and IIT-JEE Physics, Kota

VS Sir has taught Physics to NEET and IIT-JEE aspirants of Class 11, Class 12 and droppers in Kota for more than 20 years, in English and Hindi medium. He is the founder of VCA Kota and the author of the *NEET + JEE Physics 30-Day Rapid Revision* books, *Electrostatics* and *Practical Guide to AI for Educators*. His method is simple: understand the idea, draw the diagram, then practise until the steps become automatic.

Doubts and feedback: WhatsApp +91 96024 05311

A message to the student

DEAR STUDENT,

Physics is not a subject to be memorised; it is a subject to be **seen**. Every chapter in this book is built on one habit: *draw it, name it, write the formula, then solve*. Most marks in the board paper are won by clear diagrams, correct formulas with units, and answers written in the steps an examiner expects.

Use this book with a pencil. Cover the green answer strips, try each question first, then compare. Mark what you got wrong in the revision page of each chapter and return to it before the exam. A little, every day, beats a lot, once.

— VS Sir

How to use this book

1. LEARN

Read the chapter opener (outcomes, prerequisites, exam note), then the concept pages. Look at every labelled diagram.

2. DERIVE & SOLVE

Practise each derivation (principle, steps, result). Solve numericals N1–N6 before reading the working.

3. TEST

Attempt MCQs and subjective questions with the answer strip covered. Note the question ID of anything you miss.

Colour code used throughout

● Blue	Core concept and definition
● Violet	Advanced / extension idea
● Teal	Correct method, checkpoint
● Orange	Formula, derivation, exam tip
● Red	Common mistake — avoid
● Gold	Revision and summary

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Syllabus coverage checklist

Tick each chapter when you have finished concepts, derivations, numericals, MCQs and the revision page. Chapter names follow the usual Class 12 NCERT Physics sequence; **confirm the exact 2026–27 syllabus and any deleted or added topics on the official board website.**

CHAPTER	UNIT	MAIN TOPICS	PAGE
☐ ● 1. Electric Charges and Fields	Electrostatics	Coulomb's law, field, dipole, Gauss's law	8
☐ ● 2. Electrostatic Potential and Capacitance	Electrostatics	Potential, dipole, conductors, capacitors, dielectrics	17
☐ ● 3. Current Electricity	Current Electricity	Drift, Ohm, cells, Kirchhoff, Wheatstone bridge, meter bridge, potentiometer	26
☐ ● 4. Moving Charges and Magnetism	Magnetic Effects of Current and Magnetism	Biot–Savart, Ampere, force, cyclotron, torque, galvanometer	35
☐ ● 5. Magnetism and Matter	Magnetic Effects of Current and Magnetism	Bar magnet, Earth's magnetism, magnetic materials	44
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★ TOPICS MARKED “CHECK”

Polarisation (Chapter 10), radioactive decay (Chapter 13) and special-purpose diodes (Chapter 14) are included as clearly marked **extension** boxes because their examinable status for 2026–27 could not be verified. If the official syllabus excludes them, skip those boxes.

Syllabus at a glance

1 Electric Charges and Fields	2 Electrostatic Potential and Capacitance	3 Current Electricity	4 Moving Charges and Magnetism	5 Magnetism and Matter	6 Electromagnetic Induction	7 Alternating Current
8 Electromagnetic Waves	9 Ray Optics and Optical Instruments	10 Wave Optics	11 Dual Nature of Radiation and Matter	12 Atoms	13 Nuclei	14 Semiconductor Electronics

Exam roadmap and revision strategy

★ PAPER PATTERN SNAPSHOT — PROVISIONAL

Recent CBSE Class 12 Physics theory papers are **70 marks, 3 hours**, with objective items (including assertion–reason and case-based questions), short-answer and long-answer questions. The exact section-wise count and internal choices for 2026–27 **must be confirmed from the official sample paper**; this book could not verify them and does not rely on them. The 3 practice papers at the end follow the recent pattern and are original, not past papers.

A 12-week roadmap

Weeks 1–3

Ch 1–3. Concepts + derivations + numericals

Weeks 4–6

Ch 4–7. Magnetism, EMI, AC

Weeks 7–9

Ch 8–11. EM waves, optics, dual nature

Weeks 10–11

Ch 12–14 + mixed MCQs

Week 12

Three full papers, formula sheets, diagrams

F FORMULA REVISION

- Maintain one A4 sheet per chapter (use the revision pages).
- Say each formula aloud with units and conditions.
- Test: write 10 formulas from memory each day; check, correct, repeat.
- Never memorise a formula without the situation it applies to.

I DIAGRAM REVISION

- Use the “diagrams to practise” list on each revision page.
- Draw each in 2 minutes with labels and arrows.
- Mark ray directions, field directions and units.
- Redraw missed ones next day.

★ LAST 7 DAYS

- Day 1–3: formula sheets and revision maps (back section).
- Day 4–6: one full paper per day under timed conditions.
- Day 7: light revision of mistakes list; sleep well.

! IN THE EXAM HALL

- Read all questions; start with those you are surest of.
- Show formula, substitution, answer with unit.
- Label diagrams and draw with a pencil.
- Do not leave case-based questions blank — read the passage twice.

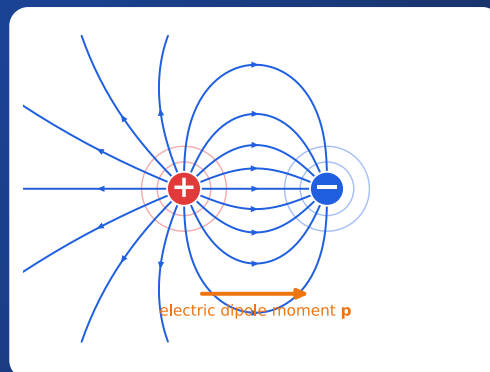
~ TIME PLAN FOR A 3-HOUR PAPER (GUIDE ONLY)

Reading: 10 min · objective: 25 min · short answers: 55 min · long answers and case-based: 70 min · revision: 20 min. Adjust once you know the official section pattern.

UNIT 1 · ELECTROSTATICS

1 Electric Charges and Fields

Everything in electrostatics starts here: how charges push and pull, how a field carries that push through space, and how symmetry lets Gauss's theorem find fields in one line.



Chapter roadmap

Charge & Coulomb's law

Electric field & field lines

Electric dipole

Flux & Gauss's theorem

Applications of Gauss

✓ LEARNING OUTCOMES

- Explain quantisation and conservation of charge and apply Coulomb's law in vector form
- Use the superposition principle for discrete and continuous charge distributions
- Sketch and interpret electric field lines; state their properties
- Derive the field of an electric dipole (axial and equatorial) and the torque on it
- State Gauss's theorem and apply it to a long wire, an infinite sheet and a thin spherical shell

I PREREQUISITES TO REVISE

- Vectors: addition, resolution, unit vector, dot and cross product
- Newton's laws; force as a vector
- Basic integration: $\int x^n dx$
- Area of a sphere $4\pi r^2$, area of a cylinder $2\pi r l$

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: electric charges; conservation of charge; Coulomb's law — force between two point charges, forces between multiple charges; superposition principle and continuous charge distribution; electric field, field due to a point charge, field lines; electric dipole, field due to a dipole, torque on a dipole in a uniform field; electric flux; **statement** of Gauss's theorem and its applications to an infinitely long straight wire, a uniformly charged infinite plane sheet and a uniformly charged thin spherical shell (inside and outside).

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs on Coulomb's law and field lines; 2-mark short answers (properties of field lines, dipole moment); 3-mark derivations (dipole field, torque, a Gauss's-law application); 5-mark long answers (axial/equatorial field or two Gauss's-law results with figures). *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

How this chapter connects to the rest of the syllabus

Ch 1

Field of fixed charges

Ch 2

Potential, capacitors: energy of the same charges

Ch 3

Charges in motion: current

Ch 4

Moving charges: magnetic force

F CONSTANTS USED IN THIS CHAPTER

Constant	Symbol	Value
Elementary charge	e	$1.602 \times 10^{-19} \text{ C}$
Coulomb constant	$k = \frac{1}{4\pi\epsilon_0}$	$9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Permittivity of free space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67 \times 10^{-27} \text{ kg}$

I THREE IDEAS TO CARRY FORWARD

- **Vector thinking:** forces and fields add as vectors, so direction is half the answer.
- **Symmetry:** choose the Gaussian surface (or components) so that the integral becomes one multiplication.
- **Fields are real:** the field stores and transmits the interaction — the idea behind capacitors, waves and light.

1.1 Charge and Coulomb's law

I FOUR PROPERTIES OF ELECTRIC CHARGE

- **Two kinds:** positive and negative. Like charges repel, unlike charges attract.
- **Additive:** total charge is the *algebraic* sum (charge is a scalar).
- **Conserved:** in an isolated system the net charge never changes; charge is only transferred.
- **Quantised:** $q = \pm ne$, $n = 1, 2, 3, \dots$, with $e = 1.602 \times 10^{-19}$ C. Because $1 \mu\text{C} \approx 6.24 \times 10^{12} e$, quantisation is invisible at the macroscopic scale.
- **Everyday link:** a rubbed comb attracts paper scraps because it polarises them — the nearer (opposite) charge is pulled harder than the farther (like) charge.

F COULOMB'S LAW

The force between two **point charges at rest** is directly proportional to the product of the charges and inversely proportional to the square of the distance; it acts along the line joining them.

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} = k \frac{|q_1 q_2|}{r^2}$$

$$k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}, \quad \epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}. \text{ In a medium: } F = \frac{F_{\text{vacuum}}}{\epsilon_r}.$$

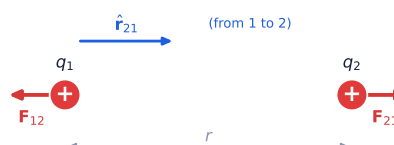


Fig. 1.1 Vector form: force on charge 2 due to charge 1 is $\mathbf{F}_{21} = k \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}_{21}$. For like charges it points along $\hat{\mathbf{r}}_{21}$ (repulsion). $\mathbf{F}_{12} = -\mathbf{F}_{21}$ (Newton's third law).

1.2 Superposition and continuous charge

I PRINCIPLE OF SUPERPOSITION

The force on a charge due to several others is the **vector sum** of the forces due to each one separately; the other charges do not alter the pair-wise force.

$$\mathbf{F}_1 = \mathbf{F}_{12} + \mathbf{F}_{13} + \dots + \mathbf{F}_{1n} = \frac{q_1}{4\pi\epsilon_0} \sum_{j=2}^n \frac{q_j}{r_{1j}^2} \hat{\mathbf{r}}_{j1}$$

Method: (i) draw each force with its true direction, (ii) resolve into x, y components, (iii) add components, (iv) find magnitude and angle.

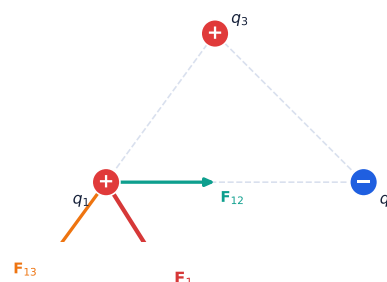


Fig. 1.2 Forces on q_1 : q_2 is negative (attracts q_1), q_3 is positive (repels q_1). The resultant $\mathbf{F}_1$ is the vector sum, not the sum of magnitudes.

~ CONTINUOUS CHARGE DISTRIBUTIONS (PRESCRIBED)

Treat the body as many tiny point charges dq ; add (integrate) their contributions vectorially. Use symmetry first — components that cancel by symmetry need not be integrated.

Distribution	Density	Charge element	Field element
Line (wire, ring)	$\lambda = \frac{dq}{dl} \text{ (C m}^{-1}\text{)}$	$dq = \lambda dl$	$d\mathbf{E} = \frac{dq}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}$
Surface (sheet, disc)	$\sigma = \frac{dq}{dA} \text{ (C m}^{-2}\text{)}$	$dq = \sigma dA$	integrate over the surface
Volume (solid sphere)	$\rho = \frac{dq}{dV} \text{ (C m}^{-3}\text{)}$	$dq = \rho dV$	integrate over the volume

★ HOW STRONG IS ELECTRICITY?

For an electron and proton:

$$\frac{F_{\text{electric}}}{F_{\text{gravity}}} = \frac{ke^2}{Gm_e m_p} \approx 2.3 \times 10^{39}. \text{ That is why gravity is negligible between charged particles but dominates for planets (which are neutral).}$$

! COMMON MISTAKES

- Using r in cm in $F = kq_1 q_2 / r^2$ — always convert to metres.
- Adding force *magnitudes* instead of vectors in superposition.
- Substituting signs of charges into the *magnitude* formula and then getting a negative 'magnitude'. Use signs only to decide direction.
- Applying Coulomb's law to extended bodies without treating them as point charges.

1.3 Electric field and field lines

F ELECTRIC FIELD

Field at a point = force per unit *test charge* (taken vanishingly small so that it does not disturb the source):

$$\mathbf{E} = \lim_{q_0 \rightarrow 0} \frac{\mathbf{F}}{q_0} \quad \mathbf{E}_{\text{point}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

Unit: N C^{-1} ($= \text{V m}^{-1}$). Force on charge q in a field: $\mathbf{F} = q\mathbf{E}$ — along $\mathbf{E}$ for $q > 0$, opposite for $q < 0$. For many charges:

$\mathbf{E} = \sum \mathbf{E}_i$ (vector sum).

I PROPERTIES OF ELECTRIC FIELD LINES

1. Start on positive charges, end on negative charges (or at infinity).
2. The **tangent** at any point gives the direction of $\mathbf{E}$ there.
3. Closer lines mean stronger field: density $\propto |\mathbf{E}|$.
4. Two field lines never cross — $\mathbf{E}$ would then have two directions at one point.
5. They do not form closed loops in electrostatics.
6. They meet a conductor's surface at right angles.
7. They are a visual aid; the field exists everywhere, not only on the lines.

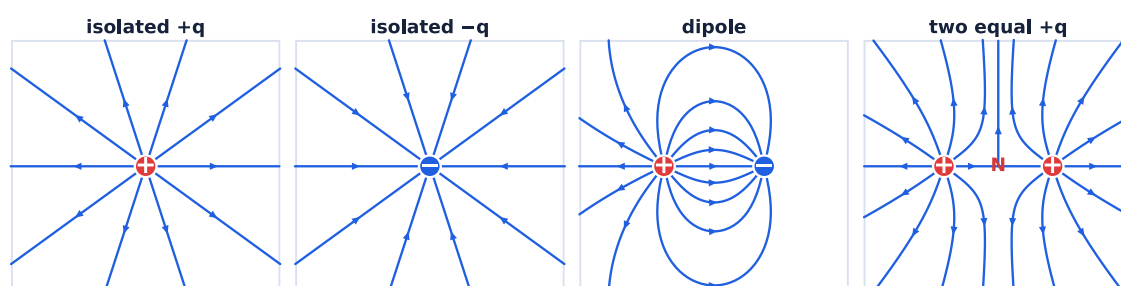


Fig. 1.3 Field-line patterns. For a lone $+q$ the lines point radially outward, for a lone $-q$ radially inward. The dipole pattern curves from $+q$ to $-q$. Between two equal $+q$ charges there is a **neutral point N** where the field is zero (lines repel and do not enter this region).

1.4 The electric dipole

I DEFINITION

Two equal and opposite charges $\pm q$ separated by a small distance $2a$. The **dipole moment** is a vector:

$$\mathbf{p} = q (2\mathbf{a}) \quad p = 2aq$$

Direction: from $-q$ to $+q$. SI unit: C m . Example: a water molecule has $p \approx 6.2 \times 10^{-30} \text{ C m}$. A *point (ideal) dipole* is the limit $a \rightarrow 0$, $q \rightarrow \infty$ with p fixed.

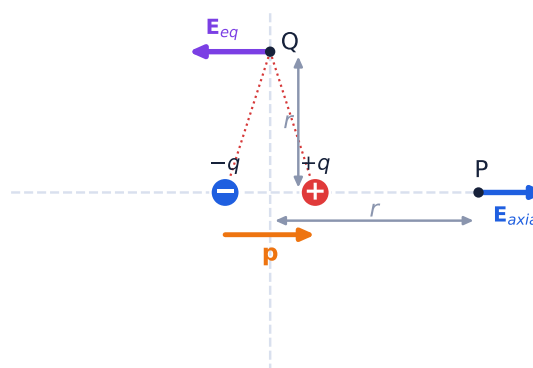


Fig. 1.4 Geometry used for the dipole field. **P** lies on the axial line, **Q** on the equatorial (perpendicular bisector) line, both at distance r from the centre. $\mathbf{E}_{\text{axial}}$ is along $\mathbf{p}$; $\mathbf{E}_{\text{eq}}$ is opposite to $\mathbf{p}$.

★ RESULTS TO REMEMBER (DERIVED ON THE NEXT PAGE)

Position	Exact	Short dipole ($r \gg a$)	Direction
Axial	$\frac{2pr}{4\pi\epsilon_0 (r^2 - a^2)^2}$	$\frac{2p}{4\pi\epsilon_0 r^3}$	along $\mathbf{p}$
Equatorial	$\frac{p}{4\pi\epsilon_0 (r^2 + a^2)^{3/2}}$	$\frac{p}{4\pi\epsilon_0 r^3}$	opposite to $\mathbf{p}$

Dipole field falls as $1/r^3$ (faster than a single charge's $1/r^2$) because the two fields partly cancel. Axial field is **twice** the equatorial field at the same r .

✓ CONCEPT CHECK (ANSWERS IN BRACKETS)

1. A positive test charge is placed at a point and feels a force $\mathbf{F}$. What is the force on an electron placed at the same point? [$-\mathbf{eE}$: opposite to $\mathbf{F}$ in direction]
2. Why can the electric field of a dipole at large distance not be $\propto 1/r^2$? [the two fields nearly cancel, leaving $1/r^3$]
3. A dipole is neutral. Does it feel a net force in a *uniform* field? [No — only a torque]

1.5 Derivations: dipole field and torque

Field of a dipole on the axial line

DERIVATION

Principle Superposition of the fields of $+q$ and $-q$.

Assumptions Short dipole; point P on the axis at distance r from the centre, $r > a$ (see Fig. 1.4).

- Distance of P from $+q$ is $(r - a)$ and from $-q$ is $(r + a)$.
- Field of $+q$ (away from $+q$, i.e. along **p**): $E_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{(r - a)^2}$.
- Field of $-q$ (toward $-q$, i.e. opposite to **p**): $E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{(r + a)^2}$.

Since $E_+ > E_-$, the resultant is along **p**.

- $E = E_+ - E_- = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(r - a)^2} - \frac{1}{(r + a)^2} \right] = \frac{q}{4\pi\epsilon_0} \frac{4ar}{(r^2 - a^2)^2}$
- Put $2aq = p$, so $4aqr = 2pr$: $E = \frac{1}{4\pi\epsilon_0} \frac{2pr}{(r^2 - a^2)^2}$. For $r \gg a$,
 $(r^2 - a^2)^2 \approx r^4$.

FINAL RESULT

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3} \quad (\text{along } \mathbf{p})$$

Limits: valid for a short dipole ($r \gg a$); exact form above is valid for any $r > a$ on the axis.

How to write it in the exam: Draw the dipole with P marked, show E_+ and E_- with their directions, write the four steps above and box the final result. (Typically: diagram 1 + working 1.5–2 + result 0.5–1 mark.)

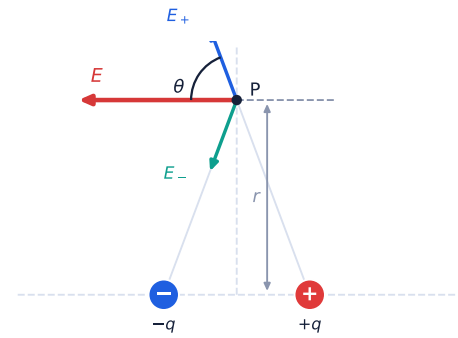


Fig. 1.5 Equatorial geometry: θ is the angle between E_+ and the axis, so the net field is $2E_+ \cos \theta$ opposite to **p**.

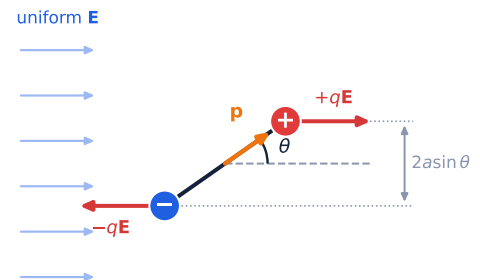


Fig. 1.6 Torque on a dipole in a uniform field. The two forces $\pm qE$ form a couple; their lines of action are $2a \sin \theta$ apart.

! SIGN CHECK

On the axial line the resultant is along **p**; on the equatorial line it is **opposite** to **p**. Both statements are asked often.

Field of a dipole on the equatorial line

DERIVATION

Principle Superposition; components perpendicular to the axis cancel.

Assumptions P on the perpendicular bisector at distance r from the centre; $d = \sqrt{r^2 + a^2}$ from each charge.

- Magnitudes are equal: $E_+ = E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2 + a^2}$ (E_+ away from $+q$, E_- toward $-q$).
- Components **perpendicular** to **p** (along the bisector) are equal and opposite $\rightarrow$ cancel.
- Components **parallel** to **p** both point opposite to **p** and add: $E = 2E_+ \cos \theta$ with $\cos \theta = \frac{a}{\sqrt{r^2 + a^2}}$.
- $$E = \frac{2}{4\pi\epsilon_0} \frac{q}{r^2 + a^2} \cdot \frac{a}{(r^2 + a^2)^{1/2}} = \frac{1}{4\pi\epsilon_0} \frac{2qa}{(r^2 + a^2)^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + a^2)^{3/2}}$$

FINAL RESULT

$$E_{\text{eq}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \quad (\text{opposite to } \mathbf{p}, r \gg a)$$

Limits: short dipole for the simplified form.

How to write it in the exam: State clearly that the perpendicular components cancel; many students lose marks by not saying why only the parallel components add.

Torque on a dipole in a uniform electric field

3 MARKS

Principle A pair of equal, opposite forces (a couple) produces rotation but no translation.

Assumptions Uniform field **E**; dipole makes angle θ with **E**.

- Force on $+q$: qE along **E**; on $-q$: qE opposite. Net force = 0.
- Perpendicular distance between the two forces: $2a \sin \theta$.
- Torque
 $\tau = (qE) (2a \sin \theta) = (2aq) E \sin \theta = pE \sin \theta$.

FINAL RESULT

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}, \quad \tau = pE \sin \theta$$

Limits: uniform field only. In a non-uniform field there is also a net force. τ is maximum (pE) at $\theta = 90^\circ$ and zero at 0° (stable equilibrium) and 180° (unstable equilibrium).

How to write it in the exam: Mention 'net force is zero, only torque acts' — it is a standard marking point.

1.6 Electric flux and Gauss's theorem

F FLUX AND GAUSS'S THEOREM

Electric flux through an area: $\Phi = \mathbf{E} \cdot \mathbf{A} = EA \cos \theta$ ($\text{N m}^2 \text{C}^{-1}$). The area vector $\mathbf{A}$ is normal to the surface (outward for a closed surface).

Gauss's theorem (statement): the net flux through any closed surface equals $1/\epsilon_0$ times the net charge enclosed.

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$\mathbf{E}$ is the total field of all charges. Flux depends only on enclosed charge, not on the surface shape. Always true; **handy for finding $\mathbf{E}$ only with symmetry.**

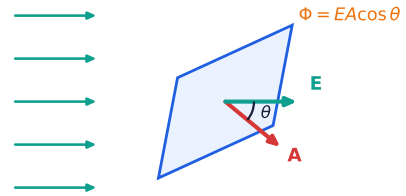


Fig. 1.7 Flux through a tilted area: only the component of $\mathbf{E}$ along $\mathbf{A}$ counts, $\Phi = EA \cos \theta$.

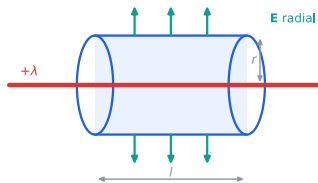


Fig. 1.8 Cylindrical Gaussian surface around a long wire; flux only through the curved part.

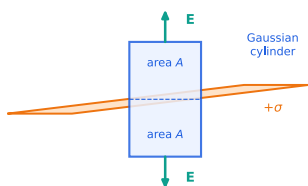


Fig. 1.9 Pillbox across an infinite sheet; flux through both end faces.

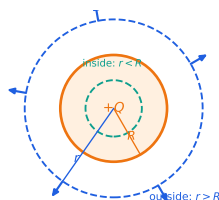


Fig. 1.10 Concentric spheres: $r > R$ encloses Q ; $r < R$ encloses nothing.

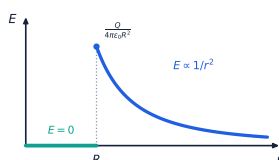


Fig. 1.11 E versus r for a thin charged shell.

A. Infinitely long straight charged wire

GAUSS

Principle Gauss's theorem with a coaxial cylinder.

Assumptions Uniform linear density λ ; wire very long, so by symmetry $\mathbf{E}$ is radial.

- Take a coaxial cylinder of radius r , length l . Flux through each flat end is zero ($\mathbf{E} \perp d\mathbf{A}$).
- Through the curved surface $\mathbf{E} \parallel d\mathbf{A}$ and E is constant:
 $\Phi = E (2\pi r l)$.
- Enclosed charge = λl . So $E (2\pi r l) = \lambda l / \epsilon_0$.

FINAL RESULT

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{radially outward for } \lambda > 0)$$

Limits: far from the ends ($r \ll \text{length}$). $E \propto 1/r$.

How to write it in the exam: Draw the cylinder, state the symmetry argument, flux only through the curved part.

B. Infinite uniformly charged plane sheet

GAUSS

Principle Gauss's theorem with a small cylindrical 'pillbox'.

Assumptions Surface density σ ; by symmetry $\mathbf{E}$ is perpendicular to the sheet and the same distance on both sides.

- Pillbox of end-area A placed symmetrically; flux through the curved side is zero.
- Flux through the two ends: $EA + EA = 2EA$.
- Enclosed charge = σA , so $2EA = \sigma A / \epsilon_0$.

FINAL RESULT

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{independent of distance from the sheet})$$

Limits: ideal infinite sheet. Two opposite sheets: $E = \sigma / \epsilon_0$ between, 0 outside.

How to write it in the exam: Two faces give the factor 2.

C. Uniformly charged thin spherical shell

GAUSS

Principle Gauss's theorem with concentric spherical surfaces.

Assumptions Total charge Q spread uniformly on a shell of radius R ; by symmetry $\mathbf{E}$ is radial and depends only on r .

- Outside ($r > R$):** $E (4\pi r^2) = Q / \epsilon_0 \Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2}$ — the shell acts like a point charge Q at its centre.
- On the surface ($r = R$):** $E = \frac{Q}{4\pi\epsilon_0 R^2} = \frac{\sigma}{\epsilon_0}$.
- Inside ($r < R$):** the Gaussian sphere encloses no charge, so $E (4\pi r^2) = 0 \Rightarrow E = 0$.

FINAL RESULT

$$E = 0 \quad (r < R), \quad E = \frac{Q}{4\pi\epsilon_0 r^2} \quad (r \geq R)$$

Limits: thin shell; E jumps by σ / ϵ_0 across the surface.

How to write it in the exam: Give both regions and sketch the E - r graph.

1.7 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$F = k \frac{ q_1 q_2 }{r^2}$	F (N), q (C), r (m); $k = 9 \times 10^9$	point charges at rest; direction: repulsive for like, attractive for unlike	$q_1 = q_2 = 1 \mu\text{C}$, $r = 0.1$ m: $F = 0.9$ N
$\mathbf{E} = \frac{kq}{r^2} \hat{\mathbf{r}}$	E (N C ⁻¹)	outward for $q > 0$, inward for $q < 0$	$q = 5$ nC, $r = 0.3$ m: $E = 500$ N/C
$\mathbf{F} = q\mathbf{E}$	F (N)	same direction as $\mathbf{E}$ for $q > 0$	e in 10^4 N/C: $F = 1.6 \times 10^{-15}$ N
$p = 2aq$	p (C m)	from $-q$ to $+q$	$2a = 1$ cm, $q = 10$ nC: $p = 10^{-10}$ C m
$E_{ax} = \frac{2kp}{r^3}$	$r \gg a$	along $\mathbf{p}$	$p = 10^{-10}$, $r = 0.1$: 1800 N/C
$E_{eq} = \frac{kp}{r^3}$	$r \gg a$	opposite to $\mathbf{p}$	half the axial value
$\tau = pE \sin \theta$	τ (N m)	$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$; uniform field	$\theta = 30^\circ$: $\tau = pE/2$
$\Phi = EA \cos \theta$	Φ (N m ² C ⁻¹)	θ between $\mathbf{E}$ and $\mathbf{A}$	$\mathbf{E} \parallel \mathbf{A}$: $\Phi = EA$
$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\epsilon_0}$	$\epsilon_0 = 8.854 \times 10^{-12}$	any closed surface	cube, charge q at centre: $q/6\epsilon_0$ per face
$E = \frac{\lambda}{2\pi\epsilon_0 r}; \frac{\sigma}{2\epsilon_0}$	wire ; sheet	infinitely long / infinite	sheet field is distance-independent

N1 C01-N01 Ch 1 Easy Numerical BOARD-STYLE PRACTICE 2 marks

Two point charges $4 \mu\text{C}$ and $6 \mu\text{C}$ are 30 cm apart in air. Find the force between them. What happens if they are placed in a medium of dielectric constant 4?

GIVEN $q_1 = 4 \times 10^{-6}$ C, $q_2 = 6 \times 10^{-6}$ C, $r = 30$ cm = 0.30 m

FIND Force F (air) and F' (medium)

FORMULA $F = \frac{kq_1 q_2}{r^2}$, $F' = F/\epsilon_r$

WORKING $F = \frac{9 \times 10^9 \times 4 \times 10^{-6} \times 6 \times 10^{-6}}{(0.30)^2} = \frac{0.216}{0.09} = 2.4$ N $\rightarrow F' = \frac{2.4}{4} = 0.6$ N

Answer: $F = 2.4$ N (repulsive); $F' = 0.6$ N

▲ Convert cm to m before squaring — a slip here changes the answer by 10^4 .

N2 C01-N02 Ch 1 Medium Numerical BOARD-STYLE PRACTICE 3 marks

Charges $+9$ nC and -9 nC are 20 cm apart. Find the electric field at the mid-point of the line joining them.

GIVEN $q = 9 \times 10^{-9}$ C; each charge is 10 cm = 0.10 m from the mid-point

FIND $\mathbf{E}$ at the mid-point

FORMULA $E = \frac{kq}{r^2}$ for each; fields add (same direction)

WORKING $E_+ = E_- = \frac{9 \times 10^9 \times 9 \times 10^{-9}}{(0.10)^2} = 8100$ N/C. Away from $+q$ and toward $-q$ are the same direction, so
 $E = 8100 + 8100 = 16200$ N/C.

Answer: 1.62×10^4 N C⁻¹ directed from $+9$ nC toward -9 nC

▲ Do not subtract: at the mid-point of opposite charges the two fields add.

N3 C01-N03 Ch 1 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A dipole has charges ± 5 nC separated by 2 cm. Find its dipole moment and the field at an axial point 10 cm from its centre.

GIVEN $q = 5 \times 10^{-9}$ C, $2a = 0.02$ m, $r = 0.10$ m

FIND p and E_{axial}

FORMULA $p = q(2a)$, $E = \frac{2kp}{r^3}$ (since $r \gg a$)

WORKING $p = 5 \times 10^{-9} \times 0.02 = 1.0 \times 10^{-10}$ C m. $E = \frac{2 \times 9 \times 10^9 \times 10^{-10}}{(0.1)^3} = \frac{1.8}{10^{-3}} = 1.8 \times 10^3$ N/C.

Answer: $p = 1.0 \times 10^{-10}$ C m; $E \approx 1.8 \times 10^3$ N C⁻¹ along $\mathbf{p}$

▲ Check: the exact formula gives 1.84×10^3 N/C — the $r \gg a$ approximation is good to about 2%.

1.8 Solved numericals (continued) and MCQ practice

N5 C01-N04 Ch 1 Hard Numerical BOARD-STYLE PRACTICE 3 marks

An infinite sheet carries uniform surface charge density $2 \mu\text{C m}^{-2}$. A thin spherical shell of radius 10 cm carries $8 \mu\text{C}$. Find (a) the field near the sheet, (b) the field of the shell at 20 cm and 5 cm from its centre.

GIVEN $\sigma = 2 \times 10^{-6} \text{ C m}^{-2}$; $Q = 8 \times 10^{-6} \text{ C}$, $R = 0.10 \text{ m}$

FIND E for the sheet; E (0.20) and E (0.05) for the shell

FORMULA sheet: $E = \frac{\sigma}{2\epsilon_0}$; shell: $E = \frac{kQ}{r^2}$ outside, 0 inside

WORKING (a) $E = \frac{2 \times 10^{-6}}{2 \times 8.854 \times 10^{-12}} = 1.13 \times 10^5 \text{ N/C}$. (b) at $r = 0.20 \text{ m}$: $E = \frac{9 \times 10^9 \times 8 \times 10^{-6}}{(0.20)^2} = 1.8 \times 10^6 \text{ N/C}$; at $r = 0.05 \text{ m}$ (inside): $E = 0$.

Answer: (a) $1.13 \times 10^5 \text{ N/C}$ (b) $1.8 \times 10^6 \text{ N/C}$ outside, 0 inside

▲ Inside a charged thin shell the field is zero for any point — no calculation needed.

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C01-M01 Ch 1 Easy Conceptual BOARD-STYLE PRACTICE

1. A glass rod rubbed with silk becomes positively charged because

- (a) electrons are transferred from silk to the rod
- (b) protons are transferred from the rod to silk
- (c) electrons are transferred from the rod to silk
- (d) protons are transferred from silk to the rod

Ans: (c) Only electrons (the loosely bound charge carriers) move; the rod loses electrons and becomes positive.

C01-M02 Ch 1 Easy Numerical BOARD-STYLE PRACTICE

2. Each of two point charges is doubled and the distance between them is also doubled. The force between them becomes

- (a) $F/4$
- (b) $F/2$
- (c) F
- (d) $2F$

Ans: (c) $F \propto q_1 q_2 / r^2 \rightarrow (2 \cdot 2) / (2^2) = 1$, so the force is unchanged.

C01-M03 Ch 1 Medium Conceptual

BOARD-STYLE PRACTICE

3. Two electric field lines cannot cross because

- (a) field lines are straight
- (b) the electric field at a point would have two directions
- (c) charges would be neutralised
- (d) the lines represent forces on neutral particles

Ans: (b) At the crossing point the tangent — hence $\mathbf{E}$ — would have two directions, which is impossible.

C01-M04 Ch 1 Medium Application

BOARD-STYLE PRACTICE

4. For a short dipole, the ratio of the field at an axial point to that at an equatorial point at the same distance r from the centre is

- (a) 1:2
- (b) 1:1
- (c) 2:1
- (d) 4:1

Ans: (c) $E_{ax} = 2kp/r^3$ and $E_{eq} = kp/r^3$.

C01-M05 Ch 1 Medium Numerical BOARD-STYLE PRACTICE

5. A dipole of moment $2 \times 10^{-8} \text{ C m}$ lies at 30° to a uniform field $5 \times 10^3 \text{ N/C}$. The torque is

- (a) $5 \times 10^{-5} \text{ N m}$
- (b) $1 \times 10^{-4} \text{ N m}$
- (c) $2.5 \times 10^{-5} \text{ N m}$
- (d) $8.7 \times 10^{-5} \text{ N m}$

Ans: (a) $\tau = pE \sin 30^\circ = 2 \times 10^{-8} \times 5 \times 10^3 \times 0.5 = 5 \times 10^{-5} \text{ N m}$.

C01-M06 Ch 1 Medium Application

BOARD-STYLE PRACTICE

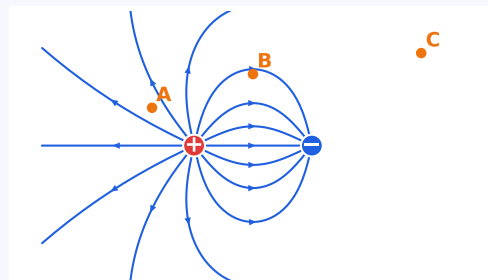
6. A point charge q is placed at the centre of a cube. The flux through one face is

- (a) q/ϵ_0
- (b) $q/4\epsilon_0$
- (c) $q/6\epsilon_0$
- (d) $6q/\epsilon_0$

Ans: (c) Total flux q/ϵ_0 is shared equally by six identical faces.

C01-M07 Ch 1 Medium Diagram BOARD-STYLE PRACTICE

7. In the dipole field-line figure, the field is strongest at point



- (a) A
- (b) B
- (c) C
- (d) equal at all three

Ans: (a) Lines are closest together near the $+q$ charge, and line density is proportional to $|\mathbf{E}|$.

C01-AR01 Ch 1 Medium Assertion-Reason

BOARD-STYLE PRACTICE

8. Assertion (A): Gauss's law holds only for charge distributions with spherical symmetry.

Reason (R): Gauss's law is used to find $\mathbf{E}$ only when symmetry makes E constant over the Gaussian surface.

(a) A and R true, R explains A; (b) A and R true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a
- (b) b
- (c) c
- (d) d

Ans: (d) Gauss's law is valid for every closed surface and every distribution; symmetry is needed only to use it for calculating E .

1.9 Board answer-writing guide and subjective practice

Mark type	What to write (Chapter 1 examples)	Typical step marking*
1 mark	Direct answer with unit — e.g. 'SI unit of dipole moment: C m'. Choose the option letter <i>and</i> value in MCQs.	1 for the correct answer
2 marks	Definition/statement + formula, <i>or</i> two properties — e.g. 'State Coulomb's law; write its vector form.'	1 statement + 1 formula/vector form
3 marks	Short derivation or reasoned numerical — e.g. torque on a dipole; Gauss's law for a sheet.	½ diagram, 1½ working, 1 result
5 marks	Full derivation with figure — e.g. dipole field (axial <i>and</i> equatorial) or two Gauss applications.	1 diagram, 3 derivation, 1 result and conditions

*Illustrative split. Check the official 2026–27 marking scheme for the exact allocation in each question.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C01-S01 Ch 1 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. State Coulomb's law. Express it in vector form for two like charges.

Model answer: The force between two stationary point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them, acting along the line joining them. $\mathbf{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}_{21}$, where $\hat{\mathbf{r}}_{21}$ points from q_1 to q_2 ; for like charges $\mathbf{F}_{21}$ is along $\hat{\mathbf{r}}_{21}$ (repulsion).

C01-S02 Ch 1 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. Two point charges $+q$ and $+4q$ are 0.6 m apart. Locate the point on the line joining them where the net field is zero.

Model answer: The neutral point lies between the charges (fields oppose there), nearer the smaller charge. Let it be x from $+q$:

$$\frac{kq}{x^2} = \frac{k(4q)}{(0.6-x)^2} \Rightarrow \frac{0.6-x}{x} = 2 \Rightarrow x = 0.2 \text{ m from } +q \text{ (0.4 m from } +4q).$$

C01-L01 Ch 1 Medium Long answer (5 marks) BOARD-STYLE PRACTICE

L1. Using Gauss's theorem, derive the electric field of a uniformly charged thin spherical shell at points (i) outside and (ii) inside the shell. Draw the E - r graph.

Plan (see derivation C): 1 mark Gaussian sphere figure · 1 mark symmetry and flux $E \cdot 4\pi r^2$ · 1 mark outside result $\frac{Q}{4\pi\epsilon_0 r^2}$ · 1 mark inside $q_{enc} = 0 \Rightarrow E = 0$ · 1 mark graph (flat zero up to R , jump, then $1/r^2$ fall).

C01-D01 Ch 1 Hard Derivation BOARD-STYLE PRACTICE

D1. Show that the net force on a dipole in a uniform field is zero, and that the torque is maximum when $\mathbf{p} \perp \mathbf{E}$. What is the position of stable equilibrium?

Hint: equal and opposite forces $\pm q\mathbf{E}$; $\tau = pE \sin \theta$ is maximum at $\theta = 90^\circ$; stable equilibrium at $\theta = 0^\circ$ (restoring torque), unstable at 180° .

C01-CB01 Ch 1 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. A long straight wire in an experiment carries a uniform linear charge density $\lambda = 2 \mu\text{C m}^{-1}$. A student uses Gauss's theorem with a coaxial cylindrical surface to study the field near it. (i) Find the field at 0.50 m from the wire. (ii) How does it change if the distance is doubled? (iii) What is the flux through a coaxial cylinder of length 20 cm? (iv) Why do the flat ends contribute no flux?

Answers: (i) $E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2 \times 10^{-6}}{2\pi \times 8.854 \times 10^{-12} \times 0.5} \approx 7.2 \times 10^4 \text{ N/C}$, radially outward. (ii) $E \propto 1/r$, so it halves. (iii) $\Phi = \lambda l / \epsilon_0 = \frac{2 \times 10^{-6} \times 0.2}{8.854 \times 10^{-12}} \approx 4.5 \times 10^4 \text{ N m}^2 \text{ C}^{-1}$. (iv) $\mathbf{E}$ is radial, so it is parallel to the end faces and perpendicular to their area vectors.

★ ABOUT PREVIOUS-YEAR QUESTIONS IN THIS BOOK

No past-paper question is reproduced word-for-word in this edition because the official papers could not be re-verified line by line. Every question here is labelled **BOARD-STYLE PRACTICE** — written for this book in the style of recent board papers. See the Source Log at the end of the book.

★ EXAMINER-STYLE REMINDERS

- Always draw the figure first — it carries marks even if the algebra slips.
- Write the formula before substituting values, and finish with the unit.
- For a derivation, state the *symmetry argument* in one line; it is a marking point.
- Use $\mathbf{p}$ direction (from $-q$ to $+q$) to state directions of dipole fields.

R Chapter 1 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

- Field lines: $+q$, $-q$, dipole, two equal $+q$ (mark the neutral point)
- Dipole with P on the axial line and Q on the equatorial line, with $\mathbf{E}$ arrows
- Dipole in a uniform field showing the two forces and the arm $2a \sin \theta$
- Cylindrical Gaussian surface (wire) and pillbox (sheet)
- Concentric Gaussian spheres for a shell, and the E - r graph

★ 7-DAY REVISION PLAN

Day 1	Coulomb's law, superposition numericals
Day 2	Field, field lines, dipole concept
Day 3	Dipole derivations (axial, equatorial, torque)
Day 4	Gauss's theorem — three applications
Day 5	Numericals N1–N5 without notes
Day 6	MCQs + assertion–reason
Day 7	Redraw all diagrams; self-test

Chapter 1 at a glance — Electric Charges and Fields

Charge

Quantised $q = ne$; conserved; additive (scalar).

Coulomb's law

$F = kq_1q_2/r^2$; vector form with $\hat{r}$; superposition for many charges.

Field and lines

$\mathbf{E} = \mathbf{F}/q_0$; lines start +, end –; never cross; density $\propto E$.

Dipole

$p = 2aq$ (– to +); $E_{ax} = 2kp/r^3$ along $\mathbf{p}$; $E_{eq} = kp/r^3$ against $\mathbf{p}$.

Torque

$\tau = pE \sin \theta$; net force zero in uniform E ; stable at 0° .

Gauss

$\oint \mathbf{E} \cdot d\mathbf{A} = q_{enc}/\epsilon_0$; wire $\lambda/2\pi\epsilon_0 r$; sheet $\sigma/2\epsilon_0$; shell 0 inside, kQ/r^2 outside.

Formula map

$$F = \frac{kq_1q_2}{r^2}$$

Coulomb

$$E = \frac{kq}{r^2}$$

point charge

$$\mathbf{F} = q\mathbf{E}$$

force on charge

$$p = 2aq$$

dipole moment

$$\frac{2kp}{r^3}$$

axial

$$\frac{kp}{r^3}$$

equatorial

$$\tau = pE \sin \theta$$

torque

$$\frac{\lambda}{2\pi\epsilon_0 r}$$

wire

$$\frac{\sigma}{2\epsilon_0}$$

sheet

! COMMON MISTAKES

- Forgetting that E_{eq} is **opposite** to $\mathbf{p}$.
- Using $2\epsilon_0$ for a shell (it is ϵ_0 in Q/ϵ_0).
- Treating field lines as particle paths.
- Not converting nC, μC , cm to SI.
- Quoting $E = \sigma/2\epsilon_0$ for a conductor surface (it is σ/ϵ_0).

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can state and use Coulomb's law in vector form
- ☐ I can draw field lines for 4 configurations
- ☐ I can derive axial and equatorial dipole fields
- ☐ I can derive torque on a dipole
- ☐ I can apply Gauss's law to wire, sheet and shell
- ☐ I solved N1–N5 without looking
- ☐ I attempted all 10 MCQs

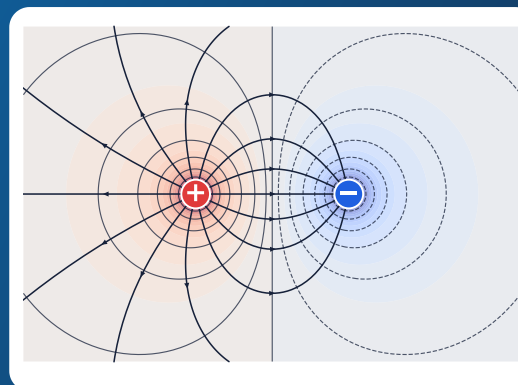
✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

- Charge on 5 electrons? | **Ans:** $-8.0 \times 10^{-19} \text{ C}$
- What is E inside a charged thin shell? | **Ans:** Zero
- $E_{ax} : E_{eq}$ for a short dipole? | **Ans:** 2:1
- SI unit of flux? | **Ans:** $\text{N m}^2 \text{ C}^{-1}$
- Flux through a closed surface enclosing no net charge? | **Ans:** Zero

UNIT 1 · ELECTROSTATICS

2 Electrostatic Potential and Capacitance

Chapter 1 described forces. Chapter 2 describes energy: how much work it takes to assemble charges, and how a capacitor stores that work as an electric field.



Chapter roadmap

Potential & potential energy

Equipotentials

Conductors & dielectrics

Capacitors

Combinations & energy

✓ LEARNING OUTCOMES

- Define electric potential and potential difference; compute the potential of a point charge, a dipole and a system of charges
- Draw and interpret equipotential surfaces; relate E and V
- Find the potential energy of a system of two charges and of a dipole in an electric field
- Explain free and bound charges in conductors, and polarisation of dielectrics
- Derive $C = \epsilon_0 A/d$ (with and without a dielectric) and combine capacitors in series and parallel; use $U = \frac{1}{2} CV^2$

I PREREQUISITES TO REVISE

- Work done by a force: $W = \mathbf{F} \cdot \mathbf{d}$; work–energy idea
- Coulomb's law and electric field (Chapter 1)
- Electric dipole and its moment
- Algebra of reciprocals for series combinations

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: electric potential, potential difference, potential due to a point charge, a dipole and a system of charges; equipotential surfaces; electrical potential energy of a system of two point charges and of an electric dipole in an electrostatic field; conductors and insulators, free and bound charges inside a conductor; dielectrics and electric polarisation; capacitors and capacitance; series and parallel combinations; capacitance of a parallel-plate capacitor with and without a dielectric; energy stored in a capacitor (**no derivation, formulae only**).

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (potential of dipole on equatorial line, capacitors in series, effect of dielectric); 2-mark properties of equipotentials / definition of capacitance; 3-mark derivation of parallel-plate capacitance or numericals on combinations; case-based questions on capacitors; 5-mark derivation (dipole potential or capacitance with dielectric). *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Three connected ideas

Force → energy

$V = \frac{W}{q_0}$: potential is work per unit charge, a **scalar**, so adding is easy.

Field from potential

$E = -\frac{dV}{dr}$: the field points toward falling potential, always **perpendicular** to equipotentials.

Storing charge

$C = \frac{Q}{V}$: a geometric property; dielectrics increase it by K .

★ UNITS TO BE FLUENT IN

$1 \text{ V} = 1 \text{ J C}^{-1}$; $1 \text{ F} = 1 \text{ C V}^{-1}$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$; $\mu\text{F} = 10^{-6} \text{ F}$, $\text{nF} = 10^{-9} \text{ F}$, $\text{pF} = 10^{-12} \text{ F}$. A field can be quoted as N C^{-1} or V m^{-1} — they are identical.

2.1 Electric potential and equipotential surfaces

F POTENTIAL AND POTENTIAL DIFFERENCE

Potential at a point = work done per unit positive test charge in bringing it from infinity to that point (slowly, without acceleration). **Potential difference** between A and B = work per unit charge in moving it from A to B.

$$V = \frac{W}{q_0}, \quad V_B - V_A = \frac{W_{A \rightarrow B}}{q_0} \quad (\text{unit: volt})$$

Point charge: $V = \frac{1}{4\pi\epsilon_0 r} q$ (sign of q included — positive near $+q$, negative near $-q$). **System:** $V = \sum V_i$ (scalar sum — no directions!).

Relation: $E = -\frac{dV}{dr}$; $\mathbf{E}$ points from higher to lower potential.

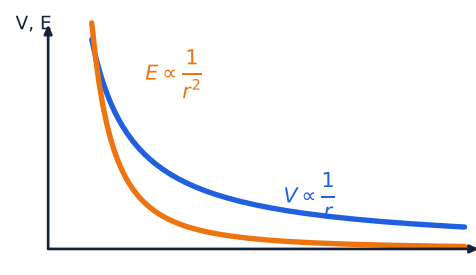


Fig. 2.1 For a point charge, $V \propto 1/r$ falls more slowly than $E \propto 1/r^2$. Both are measured from the same charge.

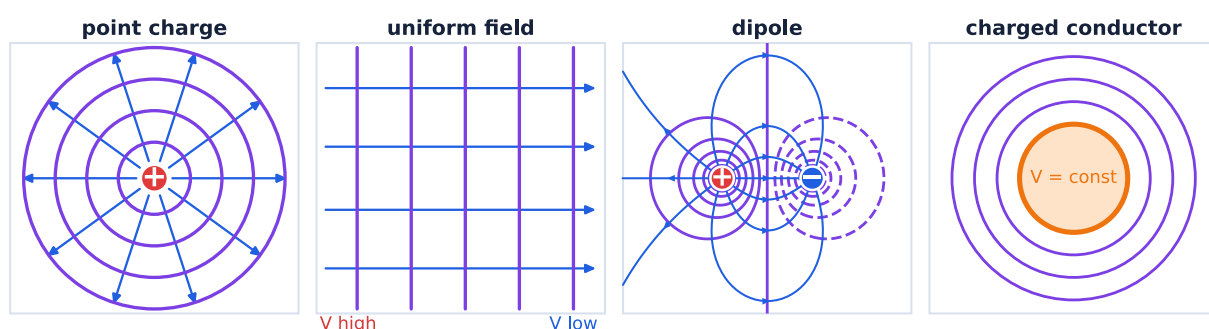


Fig. 2.2 Equipotential surfaces (violet) and field lines (blue) for a point charge, a uniform field, a dipole and a charged conductor. Every field line crosses every equipotential at right angles.

I EQUIPOTENTIAL SURFACES — PROPERTIES

1. Potential is the same at every point of the surface; **no work** is done in moving a charge along it.
2. $\mathbf{E}$ is always **perpendicular** to the surface, pointing toward lower potential.
3. Surfaces are closer together where the field is stronger.
4. Two equipotential surfaces never intersect.
5. Point charge: concentric spheres. Uniform field: parallel planes perpendicular to $\mathbf{E}$. Dipole: the equatorial plane is the surface $V = 0$.

! COMMON MISTAKES

- Adding potentials as vectors — potential is a **scalar**; only signs matter.
- Saying field is zero wherever potential is zero (at the dipole equatorial point $V = 0$ but $E \neq 0$) or vice-versa (inside a conductor $E = 0$ but $V \neq 0$).
- Forgetting that $V_\infty = 0$ is a *choice*; only differences are physical.
- Using $U = kq_1q_2/r$ without the signs of the charges.

F POTENTIAL ENERGY

Two point charges: $U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$ — work needed to bring them from infinity to separation r_{12} . For three charges add all three pairs. **Charge in an external potential:** $U = qV$. Work done by the field $= -\Delta U = q(V_A - V_B)$.

✓ CONCEPT CHECK (ANSWERS IN BRACKETS)

1. A charge moves from A to B along two different paths in an electrostatic field. Is the work the same? [Yes — the field is conservative; $W = q(V_A - V_B)$]
2. Can two points have the same potential but different field strengths? [Yes — e.g. two points on one sphere around a non-uniform source, or equatorial points of a dipole at different r]
3. If $V = 0$ at a point, must $E = 0$ there? [No — the dipole equatorial point]

2.2 Derivations: dipole potential, dipole energy

Potential due to an electric dipole

5 MARKS

Principle Superposition of the potentials of $+q$ and $-q$ (scalars).

Assumptions Short dipole, $r \gg a$; P at distance r from the centre, making angle θ with $\mathbf{p}$.

- From the geometry (figure), for $r \gg a$: $r_1 \approx r - a \cos \theta$ (distance from $+q$) and $r_2 \approx r + a \cos \theta$ (from $-q$).
- $$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \frac{q}{4\pi\epsilon_0} \frac{r_2 - r_1}{r_1 r_2} = \frac{q}{4\pi\epsilon_0} \frac{2a \cos \theta}{r^2 - a^2 \cos^2 \theta}.$$
- Since $a \ll r$, neglect $a^2 \cos^2 \theta$; put $p = 2aq$.

FINAL RESULT

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

Limits: short dipole. Axial point ($\theta = 0$): $V = \frac{p}{4\pi\epsilon_0 r^2}$; equatorial ($\theta = 90^\circ$): $V = 0$. Falls as $1/r^2$ (point charge: $1/r$).

How to write it in the exam: Draw the dipole with P, r_1 , r_2 , θ ; state the approximation $r \gg a$ explicitly.

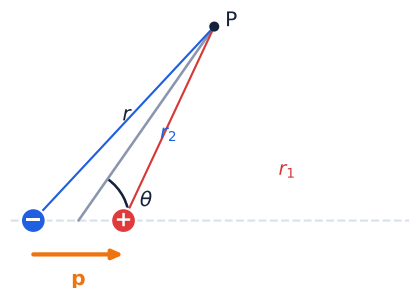


Fig. 2.3 Geometry for the dipole potential: r_1 and r_2 are the distances of P from $+q$ and $-q$.

Potential energy of a dipole in a uniform field

3 MARKS

Principle Work done by an external torque in rotating the dipole is stored as potential energy.

Assumptions Uniform $\mathbf{E}$; take $U = 0$ when $\theta = 90^\circ$ (a convenient reference).

- Torque due to the field: $\tau = pE \sin \theta$. The external torque needed to rotate slowly is equal and opposite.
- Work to turn the dipole from 90° to θ :
$$W = \int_{90^\circ}^{\theta} pE \sin \theta' d\theta' = -pE \cos \theta.$$
- This work is stored as $U(\theta)$.

FINAL RESULT

$$U = -pE \cos \theta = -\mathbf{p} \cdot \mathbf{E}$$

Limits: uniform field; reference $U(90^\circ) = 0$. Minimum $U = -pE$ at $\theta = 0$ (stable); maximum $+pE$ at 180° (unstable).

How to write it in the exam: Write the integral — many answers lose a mark by quoting the result only.

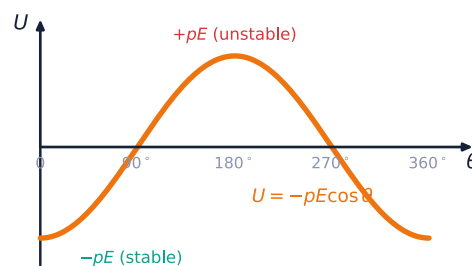


Fig. 2.4 $U(\theta)$ for a dipole: lowest at 0° (stable equilibrium), highest at 180° (unstable).

~ WORK NEEDED TO ROTATE A DIPOLE FROM θ_1 TO θ_2

$W = U(\theta_2) - U(\theta_1) = pE(\cos \theta_1 - \cos \theta_2)$. Example: from 0° to 180° , $W = 2pE$; from 0° to 90° , $W = pE$.

E1

C02-N01

Ch 2

Easy

Numerical

BOARD-STYLE PRACTICE

2 marks

A short dipole of moment 2×10^{-9} C m is at the origin. Find the potential at an axial point 20 cm from it, and at an equatorial point 20 cm away.

GIVEN $p = 2 \times 10^{-9}$ C m, $r = 0.20$ m

FIND V_{axial} , V_{eq}

FORMULA $V = \frac{kp \cos \theta}{r^2}$

WORKING Axial ($\theta = 0$): $V = \frac{9 \times 10^9 \times 2 \times 10^{-9}}{(0.20)^2} = 450$ V. Equatorial ($\theta = 90^\circ$): $\cos 90^\circ = 0 \Rightarrow V = 0$.

Answer: 450 V (axial); 0 (equatorial)

▲ Positive V on the $+q$ side of the dipole; negative on the $-q$ side.

2.3 Conductors, dielectrics and the parallel-plate capacitor

I CONDUCTORS IN ELECTROSTATICS

- **Free charges** (electrons in metals) move freely; **bound charges** stay attached to atoms.
- Inside a conductor $E = 0$; the surface is an equipotential, so E just outside is normal: $E = \sigma/\epsilon_0$.
- Net charge resides on the surface; a cavity inside a conductor is shielded from external fields (**electrostatic shielding**).

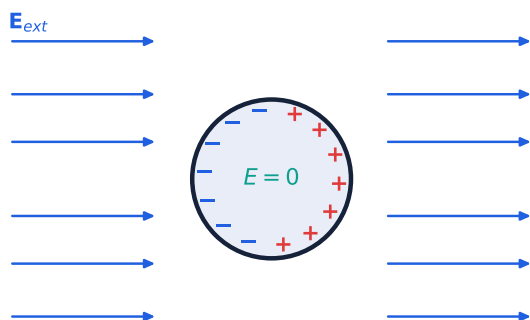


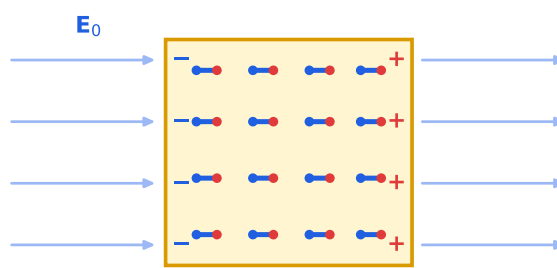
Fig. 2.5 A conductor in an external field: induced charges rearrange so that the net field inside is zero.

I DIELECTRICS AND POLARISATION

Insulators have no free charges. In an external field their molecules become (or align as) dipoles — **polarisation**. This leaves bound charges on the faces, producing a field E_p opposite to E_0 :

$$E = E_0 - E_p = \frac{E_0}{K}$$

K = dielectric constant ($K = \epsilon_r > 1$). **Non-polar** molecules get induced dipoles; **polar** molecules (e.g. water) are already dipoles and just align.



induced (bound) surface charges reduce the field inside: $E = E_0/K$

Fig. 2.6 Dielectric slab in a field: surface bound charges are induced and weaken the field inside.

Capacitance of a parallel-plate capacitor (with and without a dielectric)

3–5
MARKS

Principle Capacitance $C = Q/V$ with $V = Ed$ and E from Gauss's theorem.

Assumptions Plate area A , separation d (small compared with plate size so edge effects are neglected); plates carry $\pm Q$.

① Surface charge density $\sigma = Q/A$; field between the plates (vacuum) $E_0 = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$.

② Potential difference $V = E_0 d = \frac{Qd}{\epsilon_0 A}$, so $C_0 = \frac{Q}{V} = \frac{\epsilon_0 A}{d}$.

③ With a dielectric filling the gap: $E = E_0/K$, so $V = \frac{E_0 d}{K}$ for the same Q , and $C = \frac{Q}{V} = KC_0$.

FINAL RESULT

$$C_0 = \frac{\epsilon_0 A}{d}, \quad C = \frac{K\epsilon_0 A}{d} = KC_0$$

Limits: C depends only on geometry and medium (not on Q or V). Increases with A and K ; decreases as d increases.

How to write it in the exam: Show the capacitor figure with $\pm\sigma$, E , d , A ; then the three steps above.

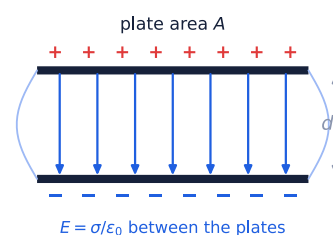


Fig. 2.7 Parallel-plate capacitor in vacuum.

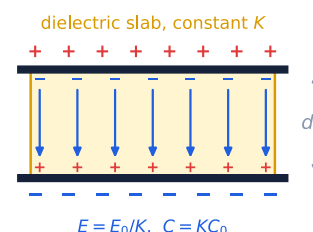


Fig. 2.8 With a dielectric the field (and V) drops by K at the same Q .

★ CAPACITORS IN EVERYDAY LIFE

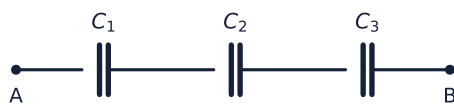
Camera flash and defibrillator (store energy, release in a burst), touch-screens (finger changes local capacitance), tuning circuits of radios, smoothing in power supplies. A good capacitor needs large A , small d and a high- K dielectric.

★ WHY IS A CAR SAFE IN LIGHTNING?

The metal body acts as a conducting shell: charge stays on the outside surface and the field inside is zero — electrostatic shielding in action.

2.4 Combinations of capacitors and stored energy

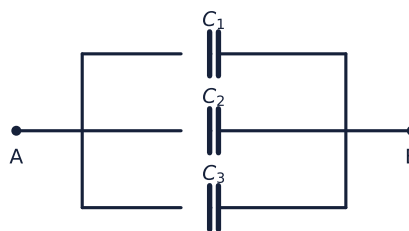
SERIES



same charge Q on each

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

PARALLEL



same voltage V across each: $C_p = C_1 + C_2 + C_3$

Fig. 2.9 Series: same charge Q , voltages add. Parallel: same voltage V , charges add.

Series and parallel combinations

3 MARKS

Principle Series: charge is the same; parallel: potential difference is the same.

Assumptions Ideal capacitors, initially uncharged.

① **Series:** $V = V_1 + V_2 + V_3$ and each has the same Q , so $\frac{Q}{C_s} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$.

② **Parallel:** $Q = Q_1 + Q_2 + Q_3$ with common V , so $C_p V = C_1 V + C_2 V + C_3 V$.

FINAL RESULT

$$\frac{1}{C_s} = \sum \frac{1}{C_i}, \quad C_p = \sum C_i$$

Limits: C_s is smaller than the smallest capacitor; C_p is larger than the largest.

How to write it in the exam: Draw both circuits; mention what is common (charge or voltage) before writing the sum.

F ENERGY STORED (FORMULAE ONLY — NO DERIVATION REQUIRED)

$$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$$

Energy equals the shaded area under the Q - V graph. U in joule.

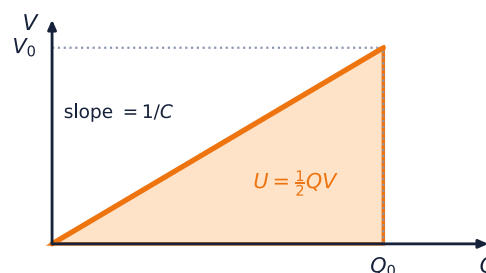


Fig. 2.10 Energy stored = area of the triangle $= \frac{1}{2}Q_0 V_0$.

★ INSERTING A DIELECTRIC SLAB (COMPLETELY FILLING THE GAP)

Quantity	Battery stays connected (V fixed)	Battery disconnected (Q fixed)
Capacitance C	$\rightarrow KC$	$\rightarrow KC$
Charge Q	$\rightarrow KQ$ (rises)	unchanged
Voltage V	unchanged	$\rightarrow V/K$
Field E	unchanged (V/d)	$\rightarrow E/K$
Energy U	$\rightarrow KU$	$\rightarrow U/K$

★ QUICK METHOD

- Series problems: find C_s , then $Q = C_s V$; the same Q gives each $V_i = Q/C_i$.
- Parallel problems: each capacitor has the full V ; total $Q = C_p V$.
- Mixed networks: collapse step by step; redraw after every step.

✓ MIXED NETWORK EXAMPLE

A $4 \mu\text{F}$ capacitor is in series with a parallel pair of $2 \mu\text{F}$ and $6 \mu\text{F}$. **Step 1:** parallel pair $= 2 + 6 = 8 \mu\text{F}$. **Step 2:** series with $4 \mu\text{F}$: $C_{eq} = \frac{4 \times 8}{4 + 8} = \frac{8}{3} \mu\text{F} \approx 2.67 \mu\text{F}$. On 12 V : $Q = \frac{8}{3} \times 12 = 32 \mu\text{C}$; the $4 \mu\text{F}$ capacitor has 8 V and the parallel pair 4 V .

2.5 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$V = \frac{kq}{r}$	$V(V), q(C), r(m)$	scalar; sign of q kept	$q = 4 \mu\text{C}, r = 0.2 \text{ m}: 1.8 \times 10^5 \text{ V}$
$V = \sum V_i$		scalar sum	3 charges: add three numbers
$E = -\frac{dV}{dr}$	V m^{-1}	toward lower V	uniform: $E = V/d$
$U = \frac{kq_1q_2}{r}$	$U(\text{J})$	negative for unlike charges	bound system has $U < 0$
$V_{\text{dip}} = \frac{kp \cos \theta}{r^2}$	$p(\text{C m})$	$r \gg a; V_{eq} = 0$	axial: kp/r^2
$U = -pE \cos \theta$	$U(\text{J})$	uniform field, $U(90^\circ) = 0$	$W = pE(\cos \theta_1 - \cos \theta_2)$
$C = \frac{Q}{V}$	F	definition	$Q = CV$
$C = \frac{K\epsilon_0 A}{d}$	$A(\text{m}^2), d(\text{m})$	slab fills the gap	$K = 1$ for vacuum/air
$\frac{1}{C_s} = \sum \frac{1}{C_i}; C_p = \sum C_i$	F	series / parallel	2, 3, 6 μF : 1 μF , 11 μF
$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	J	formula only	10 μF at 100 V: 0.05 J

N1 C02-N02 Ch 2 Easy Numerical BOARD-STYLE PRACTICE 2 marks

Find the potential at 20 cm from a $4 \mu\text{C}$ point charge. How much work is needed to bring a $2 \mu\text{C}$ charge from infinity to that point?

GIVEN $q = 4 \times 10^{-6} \text{ C}, r = 0.20 \text{ m}, q_0 = 2 \times 10^{-6} \text{ C}$ FIND V and W FORMULA $V = \frac{kq}{r}, W = q_0 V$ WORKING $V = \frac{9 \times 10^9 \times 4 \times 10^{-6}}{0.20} = 1.8 \times 10^5 \text{ V}; W = 2 \times 10^{-6} \times 1.8 \times 10^5 = 0.36 \text{ J}.$ Answer: $V = 1.8 \times 10^5 \text{ V}; W = 0.36 \text{ J}$

▲ Work is positive because both charges are positive (repulsion must be overcome).

N2 C02-N03 Ch 2 Medium Numerical BOARD-STYLE PRACTICE 3 marks

Charges $2 \mu\text{C}$, $3 \mu\text{C}$ and $-4 \mu\text{C}$ are at the vertices of an equilateral triangle of side 30 cm. Find the potential energy of the system.

GIVEN $q_1 = 2, q_2 = 3, q_3 = -4 \mu\text{C}; s = 0.30 \text{ m}$ FIND U FORMULA $U = k \left(\frac{q_1 q_2}{s} + \frac{q_1 q_3}{s} + \frac{q_2 q_3}{s} \right)$ WORKING $q_1 q_2 + q_1 q_3 + q_2 q_3 = (6 - 8 - 12) \times 10^{-12} = -14 \times 10^{-12} \text{ C}^2; U = \frac{9 \times 10^9 \times (-14 \times 10^{-12})}{0.30} = -0.42 \text{ J}.$ Answer: $U = -0.42 \text{ J}$

▲ Count every pair once (3 pairs for 3 charges).

N3 C02-N04 Ch 2 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A parallel-plate capacitor has plates of area 100 cm^2 separated by 1 mm. Find its capacitance in air and with a dielectric of $K = 5$. What charge does it hold when connected to 100 V in the second case?

GIVEN $A = 100 \text{ cm}^2 = 1.0 \times 10^{-2} \text{ m}^2, d = 10^{-3} \text{ m}, K = 5, V = 100 \text{ V}$ FIND C_0, C, Q FORMULA $C_0 = \frac{\epsilon_0 A}{d}, C = KC_0, Q = CV$ WORKING $C_0 = \frac{8.854 \times 10^{-12} \times 10^{-2}}{10^{-3}} = 8.85 \times 10^{-11} \text{ F} = 88.5 \text{ pF}; C = 5 \times 88.5 = 443 \text{ pF}; Q = 4.43 \times 10^{-10} \times 100 = 4.43 \times 10^{-8} \text{ C}.$ Answer: $C_0 = 88.5 \text{ pF}; C = 443 \text{ pF}; Q = 4.43 \times 10^{-8} \text{ C}$ ▲ Convert cm^2 to m^2 : $1 \text{ cm}^2 = 10^{-4} \text{ m}^2$.

2.6 More numericals, MCQs and subjective practice

N4

C02-N05

Ch 2

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

Capacitors of 2, 3 and 6 μF are joined (a) in series and (b) in parallel. For the series combination connected to 12 V find the charge, the voltage across each and the energy stored.

GIVEN $C_1 = 2, C_2 = 3, C_3 = 6 \mu\text{F}; V = 12 \text{ V}$

FIND C_s, C_p, Q, V_i, U

FORMULA $\frac{1}{C_s} = \sum \frac{1}{C_i}, C_p = \sum C_i, Q = C_s V, V_i = Q/C_i, U = \frac{1}{2} C_s V^2$

WORKING $\frac{1}{C_s} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1 \Rightarrow C_s = 1 \mu\text{F}; C_p = 11 \mu\text{F}. Q = 1 \times 12 = 12 \mu\text{C}. V_1 = \frac{12}{2} = 6 \text{ V}, V_2 = 4 \text{ V}, V_3 = 2 \text{ V (sum} = 12 \text{ V)}. U = \frac{1}{2} (10^{-6}) (144) = 72 \mu\text{J}.$

Answer: $C_s = 1 \mu\text{F}, C_p = 11 \mu\text{F}, Q = 12 \mu\text{C}, (6, 4, 2) \text{ V}, U = 72 \mu\text{J}$

▲ Check: voltages across series capacitors always add up to the supply voltage.

N5

C02-N06

Ch 2

Hard

Numerical

BOARD-STYLE PRACTICE

3 marks

A dipole of moment $2 \times 10^{-8} \text{ C m}$ is at 60° to a uniform field of 10^4 N C^{-1} . Find its potential energy and the work needed to turn it to 180° .

GIVEN $p = 2 \times 10^{-8} \text{ C m}, E = 10^4 \text{ N/C}, \theta_1 = 60^\circ, \theta_2 = 180^\circ$

FIND $U (60^\circ)$ and W

FORMULA $U = -pE \cos \theta, W = U_2 - U_1$

WORKING $U_1 = - (2 \times 10^{-8}) (10^4) \cos 60^\circ = -1 \times 10^{-4} \text{ J}; U_2 = +pE = +2 \times 10^{-4} \text{ J};$
 $W = U_2 - U_1 = 3 \times 10^{-4} \text{ J}.$

Answer: $U = -1 \times 10^{-4} \text{ J}; W = 3 \times 10^{-4} \text{ J}$

▲ The work is positive: an external agent works against the field.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C02-S01

Ch 2

Easy

Short answer (2 marks)

BOARD-STYLE PRACTICE

S1. What is an equipotential surface? State two of its properties.

Model answer: A surface on which every point has the same potential. (i) No work is done in moving a charge over it. (ii) The electric field at every point is perpendicular to it (and points toward lower potential).

C02-L01

Ch 2

Medium

Long answer (5 marks)

BOARD-STYLE PRACTICE

L1. Derive an expression for the potential at a point due to an electric dipole. Hence find it on the axial and equatorial lines.

Model answer: Use derivation on page 3: geometry ($r_1 \approx r - a \cos \theta, r_2 \approx r + a \cos \theta$), algebra, result $V = \frac{p \cos \theta}{4\pi\epsilon_0 r^2}$;

axial $V = \frac{p}{4\pi\epsilon_0 r^2}$, equatorial $V = 0$. Marks: figure 1, steps 2, result 1, special cases 1.

C02-CB01

Ch 2

Medium

Case-based (4 marks)

BOARD-STYLE PRACTICE

Case. A parallel-plate capacitor ($C = 10 \mu\text{F}$) is charged to 100 V and then disconnected. A dielectric slab of $K = 4$ fills the gap. (i) Charge on the plates? (ii) New capacitance and voltage? (iii) Energy before and after? (iv) Where did the missing energy go?

Model answer: (i) $Q = CV = 10^{-3} \text{ C}$ (unchanged). (ii) $C' = 40 \mu\text{F}, V' = 25 \text{ V}$. (iii) $U = \frac{1}{2} CV^2 = 0.05 \text{ J}$; after: $Q^2/2C' = 0.0125 \text{ J}$. (iv) The slab is pulled in by the field, doing work 0.0375 J on it.

2.7 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C02-M01 Ch 2 Easy Conceptual BOARD-STYLE PRACTICE

1. The work done in moving a charge between two points of the same equipotential surface is

- (a) positive (b) negative
(c) zero (d) depends on the path

Ans: (c) $W = q(V_A - V_B) = 0$ for equal potentials, whatever the path.

C02-M02 Ch 2 Medium Conceptual BOARD-STYLE PRACTICE

2. The electric potential at a point on the equatorial line of a short dipole is

- (a) $\frac{p}{4\pi\epsilon_0 r^2}$ (b) $\frac{2p}{4\pi\epsilon_0 r^3}$
(c) zero (d) $\frac{p}{4\pi\epsilon_0 r^3}$

Ans: (c) $V \propto \cos 90^\circ = 0$ (the field there is non-zero).

C02-M03 Ch 2 Easy Numerical BOARD-STYLE PRACTICE

3. Two $4\ \mu\text{F}$ capacitors are joined in series. The equivalent capacitance is

- (a) $8\ \mu\text{F}$ (b) $4\ \mu\text{F}$
(c) $2\ \mu\text{F}$ (d) $1\ \mu\text{F}$

Ans: (c) $1/C_s = 1/4 + 1/4 = 1/2$, so $C_s = 2\ \mu\text{F}$.

C02-M04 Ch 2 Medium Numerical BOARD-STYLE PRACTICE

4. The energy stored in a $2\ \mu\text{F}$ capacitor charged to $10\ \text{V}$ is

- (a) $10\ \mu\text{J}$ (b) $100\ \mu\text{J}$
(c) $200\ \mu\text{J}$ (d) $20\ \mu\text{J}$

Ans: (b) $U = \frac{1}{2}CV^2 = \frac{1}{2}(2 \times 10^{-6})(100) = 10^{-4}\ \text{J} = 100\ \mu\text{J}$.

C02-M05 Ch 2 Medium Application BOARD-STYLE PRACTICE

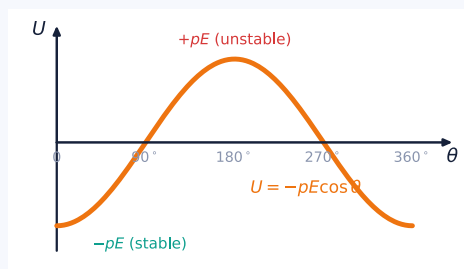
5. A charged capacitor is disconnected from the battery and a dielectric slab ($K = 4$) fills the gap. The stored energy becomes

- (a) four times (b) the same
(c) one-fourth (d) half

Ans: (c) Q fixed, $C \rightarrow 4C$, so $U = Q^2/2C \rightarrow U/4$.

C02-M06 Ch 2 Medium Graph BOARD-STYLE PRACTICE

6. From the $U-\theta$ graph of a dipole in a uniform field shown, stable equilibrium occurs at



- (a) $\theta = 0^\circ$ (b) $\theta = 90^\circ$
(c) $\theta = 180^\circ$ (d) $\theta = 270^\circ$

Ans: (a) U is minimum ($-pE$) at $\theta = 0^\circ$.

C02-AR01 Ch 2 Medium Assertion-Reason

BOARD-STYLE PRACTICE

7. A: The electric field inside a charged conductor is zero in electrostatics.

R: Free charges redistribute until the net internal field vanishes.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Charges move until $E_{\text{net}} = 0$ inside.

C02-AR02 Ch 2 Medium Assertion-Reason

BOARD-STYLE PRACTICE

8. A: Capacitance of a capacitor increases when a dielectric slab is inserted.

R: The dielectric reduces the potential difference for the same charge.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) $C = Q/V$ and $V \rightarrow V/K$ at fixed Q .

C02-M07 Ch 2 Hard Numerical BOARD-STYLE PRACTICE

9. A capacitor C charged to V is connected in parallel to an identical uncharged capacitor. The final stored energy compared with the initial is

- (a) the same (b) one-half
(c) one-fourth (d) double

Ans: (b) Charge CV is shared: $U_f = \frac{(CV)^2}{2(2C)} = \frac{1}{2}U_i$ (the rest is lost as heat/radiation).

C02-M08 Ch 2 Easy Numerical BOARD-STYLE PRACTICE

10. The potential falls by $50\ \text{V}$ over $10\ \text{cm}$ along a uniform field. The field strength is

- (a) $5\ \text{V/m}$ (b) $50\ \text{V/m}$
(c) $500\ \text{V/m}$ (d) $5000\ \text{V/m}$

Ans: (c) $E = \Delta V/d = 50/0.10 = 500\ \text{V/m}$.

★ EXAMINER-STYLE REMINDERS

- Series $\rightarrow$ same charge; parallel $\rightarrow$ same voltage.
- Potential is a scalar, field is a vector — say so.
- State the condition (battery connected / removed) for any dielectric question.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are BOARD-STYLE PRACTICE; none is claimed to be a past board question. See the Source Log at the end.

R Chapter 2 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

- Equipotentials with field lines: point charge, uniform field, dipole
- Dipole with r_1, r_2, θ
- Parallel-plate capacitor with and without dielectric (show $E, \pm\sigma$)
- Series and parallel circuits
- U - θ graph and Q - V graph

★ 7-DAY REVISION PLAN

Day 1	Potential, equipotentials
Day 2	Dipole potential and energy
Day 3	Conductors, dielectrics
Day 4	Parallel-plate capacitance derivation
Day 5	Combinations and energy
Day 6	Numericals and MCQs
Day 7	Redraw diagrams; self-test

Chapter 2 at a glance — Electrostatic Potential and Capacitance

Potential

$V = W/q_0 = kq/r$; scalar; $V_\infty = 0$.

Equipotentials

$E \perp$ surface; no work along it; closer = stronger.

Energy

$U = kq_1q_2/r$; $U_{dip} = -pE \cos \theta$.

Dipole

$V = kp \cos \theta/r^2$; $V_{eq} = 0$.

Conductor / dielectric

$E_{in} = 0$; $E = E_0/K$ in a dielectric.

Capacitor

$C = Q/V = K\epsilon_0 A/d$; series $1/C$; parallel $+$; $U = \frac{1}{2}CV^2$.

Formula map

$$V = \frac{kq}{r}$$

point charge

$$U = \frac{kq_1q_2}{r}$$

two charges

$$E = -\frac{dV}{dr}$$

field

$$V = \frac{kp \cos \theta}{r^2}$$

dipole

$$U = -pE \cos \theta$$

dipole energy

$$C = \frac{K\epsilon_0 A}{d}$$

parallel plate

$$\frac{1}{C_s} = \sum \frac{1}{C_i}; \sum C_i$$

$$U = \frac{1}{2}CV^2$$

energy

$$C = \frac{Q}{V}$$

definition

! COMMON MISTAKES

- Adding potentials as vectors.
- Using C_p rule for series (or vice-versa).
- Forgetting to convert μF , pF , cm^2 , mm .
- Mixing 'battery connected' and 'battery removed' cases for dielectric insertion.
- Writing $V = 0$ at a point and concluding $E = 0$.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can compute V and U for point-charge systems
- ☐ I can draw equipotentials for 3 configurations
- ☐ I can derive dipole potential and $U = -pE \cos \theta$
- ☐ I can derive $C = K\epsilon_0 A/d$
- ☐ I can reduce series/parallel networks
- ☐ I handled dielectric insertion in both cases
- ☐ I solved N1–N5

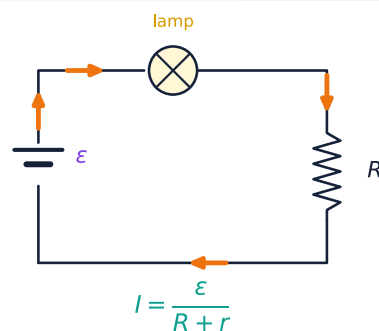
✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

- Unit of capacitance in base SI units? | **Ans:** $\text{kg}^{-1} \text{m}^{-2} \text{s}^4 \text{A}^2$
- V at the mid-point of $+q$ and $-q$? | **Ans:** Zero
- Effect of doubling plate separation on C ? | **Ans:** C halves
- Energy density concept: U for 3 μF at 10 V? | **Ans:** 150 μJ
- K for a conductor (idea)? | **Ans:** Infinite (field inside is zero)

UNIT II · CURRENT ELECTRICITY

3 Current Electricity

Charges that sat still in Chapter 1 now flow. Everything in a circuit follows from two conservation laws: charge (Kirchhoff's junction rule) and energy (the loop rule).



electrons drift opposite to the conventional current

Chapter roadmap

Current & drift

Ohm's law & resistivity

Energy & power

Cells & emf

Kirchhoff & bridge

✓ LEARNING OUTCOMES

- Define current, drift velocity, mobility, resistivity and conductivity; derive $I = neAv_d$
- State Ohm's law, interpret linear and non-linear V - I graphs and explain the temperature dependence of resistance
- Use $P = VI$, $I^2 R$, V^2/R for energy and power
- Explain emf, terminal potential difference and internal resistance; combine cells in series and parallel
- Apply Kirchhoff's rules and derive the Wheatstone-bridge balance condition

I PREREQUISITES TO REVISE

- Electric potential and potential difference (Chapter 2)
- Force on a charge in a field: $\mathbf{F} = q\mathbf{E}$
- Series and parallel arithmetic
- Simple linear equations in two unknowns

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: electric current; flow of charges in a metallic conductor; drift velocity, mobility and their relation with current; Ohm's law; V - I characteristics (linear and non-linear); electrical energy and power; electrical resistivity and conductivity; temperature dependence of resistance; internal resistance of a cell; potential difference and emf of a cell; combination of cells in series and parallel; Kirchhoff's rules; Wheatstone bridge.

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (drift velocity, resistivity vs T , cells); 2-mark derived relations ($I = neAv_d$, mobility); 3-mark numericals on internal resistance, Kirchhoff or Wheatstone; case-based circuits; 5-mark derivation of $\rho = m/ne^2\tau$, or Wheatstone balance. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Combinations at a glance

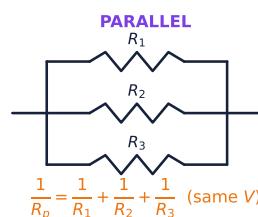
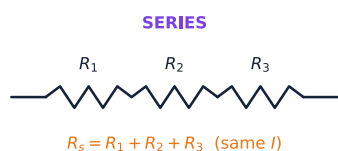


Fig. 3.1 Resistors in series share the current; in parallel they share the voltage.

F HANDY VALUES

$e = 1.6 \times 10^{-19}$ C; copper: $n \approx 8.5 \times 10^{28} \text{ m}^{-3}$;
 $\rho_{Cu} \approx 1.7 \times 10^{-8} \Omega \text{ m}$; nichrome $\rho \approx 1.1 \times 10^{-6} \Omega \text{ m}$; 1 kWh = 3.6×10^6 J.

★ CIRCUIT-SOLVING STRATEGY

1) Mark currents with arrows. 2) Junction rule at each node. 3) Loop rule on independent loops. 4) Solve; a negative current means the real direction is opposite to the arrow.

3.1 Current, Ohm's law and resistivity

1 CURRENT AND DRIFT

Electric current $I = \frac{dq}{dt}$ (ampere, scalar; direction = conventional, i.e. of positive charge). In a metal the carriers are electrons: free electrons move randomly at $\sim 10^6 \text{ m s}^{-1}$ but acquire a very small net **drift velocity** v_d (opposite to $\mathbf{E}$), typically $\sim 10^{-4} \text{ m s}^{-1}$.

$$I = neAv_d, \quad \mu = \frac{v_d}{E} \left(\text{m}^2 \text{V}^{-1} \text{s}^{-1} \right)$$

n = free electrons per m^3 , A = cross-section. **Mobility** μ = drift speed per unit field. Current density $J = I/A = nev_d$.

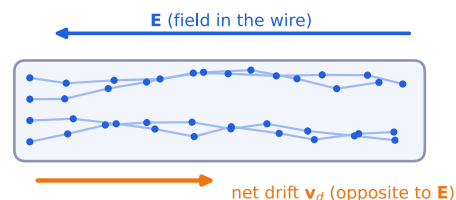


Fig. 3.2 Electrons collide with ions (zig-zag paths) yet drift slowly against $\mathbf{E}$. The signal that sets them all moving travels at nearly the speed of light, so a bulb lights instantly.

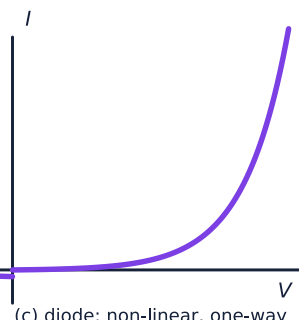
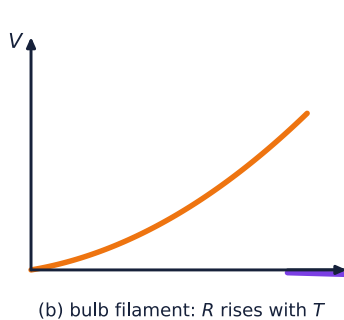
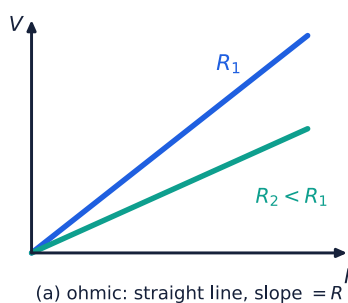


Fig. 3.3 V - I characteristics: (a) ohmic (straight line, slope = R); (b) non-ohmic filament, whose resistance rises with temperature; (c) non-ohmic diode.

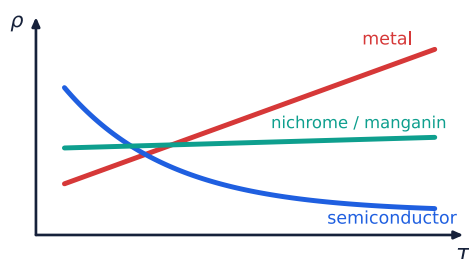


Fig. 3.4 Resistivity against temperature for a metal, a semiconductor and an alloy.

1 TEMPERATURE DEPENDENCE

Metals: $\rho = \rho_0 [1 + \alpha (T - T_0)]$, $\alpha > 0$ — more lattice vibrations, so shorter relaxation time τ . **Semiconductors:** ρ falls with T (more carriers). **Alloys** (nichrome, manganin): large ρ , almost no change with T — used for heating elements and standard resistors.

! COMMON MISTAKES

- Saying electrons travel across the wire at the speed of light — the *field* does.
- R depends on l and A ; ρ does not.
- Thinking $V = IR$ defines resistance for all conductors — the law ($V \propto I$) holds only for ohmic ones.

3.2 Derivations: drift, mobility and resistivity

Drift velocity, current and resistivity of a metal

5 MARKS

Principle Newton's second law for an electron between collisions, averaged over relaxation time τ .

Assumptions Electrons collide with ions on average every τ (relaxation time); after a collision the electron's velocity is random, so the average initial velocity is zero.

- Acceleration of an electron in field E : $a = \frac{eE}{m}$ (force eE opposite to $\mathbf{E}$).
- Average velocity gained between collisions (drift): $v_d = a\tau = \frac{eE\tau}{m}$. So **mobility** $\mu = \frac{v_d}{E} = \frac{e\tau}{m}$.
- Current: $I = neAv_d = \frac{ne^2\tau A}{m}E$. With $E = V/l$: $I = \frac{ne^2\tau A}{ml}V$, i.e. $V = IR$ with $R = \frac{ml}{ne^2\tau A}$.
- Comparing with $R = \rho l/A$ gives the resistivity; also $\sigma = ne\mu$.

FINAL RESULT

$$v_d = \frac{eE\tau}{m}, \quad \rho = \frac{m}{ne^2\tau}, \quad \sigma = ne\mu$$

Limits: free-electron model; τ and n depend on the material and temperature. ρ is independent of l and A .

How to write it in the exam: State the model (average over τ), show $v_d = a\tau$ and the substitution $E = V/l$; box ρ .

$$I = neAv_d$$

2 MARKS

Principle Charge crossing a section in time Δt .

Assumptions Uniform n , A , drift speed v_d .

- In time Δt all electrons within a length $v_d\Delta t$ cross the section: number $= nAv_d\Delta t$.
- Charge $\Delta q = (nAv_d\Delta t)e$, so
 $I = \Delta q/\Delta t = neAv_d$.

FINAL RESULT

$$I = neAv_d$$

Limits: uniform conductor. Same current at every section of a series wire even if A varies ($v_d \propto 1/A$).

How to write it in the exam: Draw the cylinder of length $v_d\Delta t$.

Temperature dependence of resistivity (metals)

2 MARKS

Principle $\rho = m/ne^2\tau$; heating increases ion vibrations.

Assumptions n roughly constant for a metal.

- As T rises, collisions are more frequent, so τ decreases.
- Since $\rho \propto 1/\tau$, ρ increases:
 $\rho_T = \rho_0 [1 + \alpha (T - T_0)]$.
- For a semiconductor n increases rapidly with T and outweighs the fall of τ , so ρ decreases.

FINAL RESULT

$$\alpha = \frac{\rho_T - \rho_0}{\rho_0 (T - T_0)} \quad (\text{per } ^\circ\text{C})$$

Limits: valid over moderate temperature ranges.

How to write it in the exam: Explain via τ , then via n for semiconductors.

ELECTRICAL ENERGY AND POWER

$$P = VI = I^2R = \frac{V^2}{R}, \quad W = Pt \quad (1 \text{ kWh} = 3.6 \times 10^6 \text{ J})$$

Heating effect (Joule): $H = I^2Rt$. At fixed voltage a *lower* R gives more power; at fixed current a *higher* R does.

3.3 Cells, emf and internal resistance

I EMF AND TERMINAL POTENTIAL DIFFERENCE

emf ε = potential difference between the terminals when no current is drawn (work done by the source per unit charge). The cell's electrolyte offers **internal resistance** r , so when current I flows:

$$V = \varepsilon - Ir \text{ (discharging)}, \quad V = \varepsilon + Ir \text{ (charging)}, \quad I =$$

Hence $r = \left(\frac{\varepsilon}{V} - 1 \right) R$. In open circuit $V = \varepsilon$; in short circuit $V = 0$ and $I = \varepsilon/r$.

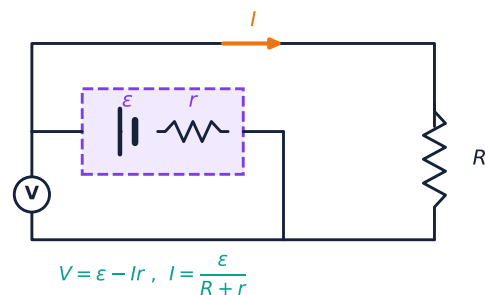


Fig. 3.5 Cell with internal resistance r driving current through external resistance R ; the voltmeter reads terminal voltage V .

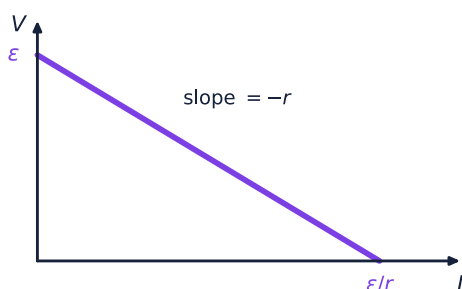
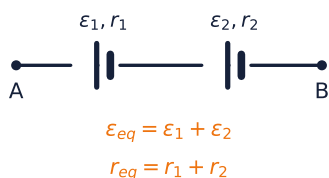


Fig. 3.6 V against I for a cell: intercept ε , slope $-r$.

SERIES



PARALLEL

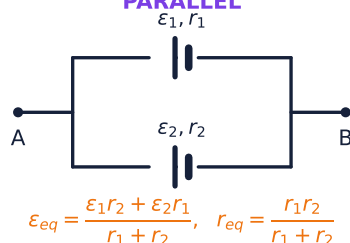


Fig. 3.7 Cells in series and in parallel.

Combination of cells

3 MARKS

Principle Apply $V = \varepsilon - Ir$ to the combination and to a single equivalent cell.

Assumptions Cells joined with like terminals together (for parallel) or end-to-end in the same sense (for series).

① **Series:** the current I is the same; voltages add:

$$V_{AB} = (\varepsilon_1 - Ir_1) + (\varepsilon_2 - Ir_2) = (\varepsilon_1 + \varepsilon_2) - I(r_1 + r_2).$$

② **Parallel:** $I = I_1 + I_2$ with $I_1 = \frac{\varepsilon_1 - V}{r_1}$, $I_2 = \frac{\varepsilon_2 - V}{r_2}$. Solving for V : $V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I \frac{r_1 r_2}{r_1 + r_2}$.

FINAL RESULT

Series: $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$, $r_{eq} = r_1 + r_2$; Parallel: $\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$, $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$

Limits: for opposite polarity use $\varepsilon_1 - \varepsilon_2$. n identical cells in series: $n\varepsilon$, nr ; in parallel: ε , r/n .

How to write it in the exam: Compare with $V = \varepsilon_{eq} - Ir_{eq}$ and read off the coefficients.

3.4 Kirchhoff's rules and the Wheatstone bridge

F KIRCHHOFF'S RULES

Junction rule (conservation of charge): $\sum I_{in} = \sum I_{out}$. **Loop rule** (conservation of energy): $\sum \Delta V = 0$ around any closed loop.

Sign convention while going round a loop: through a resistor *along* the current $-IR$, *against* it $+IR$; through a cell from $-$ to $+$: $+\varepsilon$, from $+$ to $-$: $-\varepsilon$ (internal resistance counts as a resistor).

★ METHOD

1. Label currents (any direction).
2. Junction equations: one fewer than the number of junctions.
3. Loop equations for the remaining unknowns.
4. Solve; negative current = opposite direction.

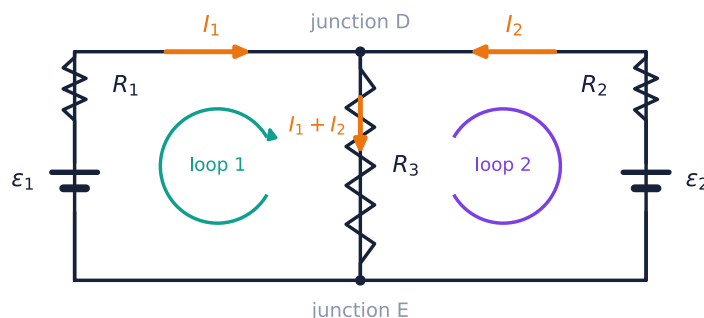


Fig. 3.8 Two-loop network: junction D gives $I_3 = I_1 + I_2$; each loop gives one equation.

Wheatstone bridge: balance condition

3 MARKS

Principle Kirchhoff's rules applied to the bridge with zero galvanometer current.

Assumptions Bridge arms P, Q, R, S and galvanometer between B and D ; cell connected between A and C .

- ① At balance no current flows through G , so $I_G = 0$ and the same current I_1 flows in P and Q ; I_2 in R and S .
- ② Loop $ABDA$ with $I_G = 0$: $I_1 P - I_2 R = 0$. Loop $BCDB$: $I_1 Q - I_2 S = 0$.
- ③ Divide: $\frac{P}{Q} = \frac{R}{S}$.

FINAL RESULT

$$\frac{P}{Q} = \frac{R}{S} \quad (\text{balanced bridge: points } B \text{ and } D \text{ at the same potential})$$

Limits: applies only when $I_G = 0$; the cell and galvanometer may be interchanged. Basis of the metre bridge.

How to write it in the exam: Draw the bridge with I_1, I_2 marked; say 'no current through G '.

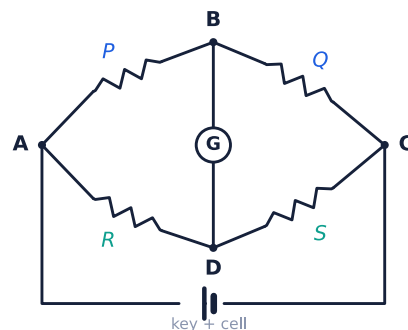


Fig. 3.9 Wheatstone bridge. When balanced, B and D are at equal potential and the galvanometer shows no deflection.

~ WORKED LOOP-RULE CHECK (USED IN N4)

For the network above with $\varepsilon_1 = 12 \text{ V}$, $\varepsilon_2 = 6 \text{ V}$, $R_1 = 2 \Omega$, $R_2 = 3 \Omega$, $R_3 = 6 \Omega$: loop 1:

$12 - 2I_1 - 6(I_1 + I_2) = 0$; loop 2: $6 - 3I_2 - 6(I_1 + I_2) = 0$. A negative I_2 simply means cell 2 is being charged.

3.5 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$I = \frac{dq}{dt}$	A	direction of positive charge	$q = 6 \text{ C}$ in 3 s: 2 A
$I = neAv_d$	$n \text{ (m}^{-3}\text{)}, A \text{ (m}^2\text{)}, v_d \text{ (m s}^{-1}\text{)}$	uniform conductor	Cu, 2 A, 1 mm ² : 0.15 mm s ⁻¹
$v_d = \frac{eE\tau}{m}; \mu = \frac{e\tau}{m}$	$\tau \text{ (s)}$	free-electron model	μ in m ² V ⁻¹ s ⁻¹
$V = IR; R = \frac{\rho l}{A}$	V, A, Ω	ohmic, constant T	stretching: $R \rightarrow n^2 R$
$\rho = \frac{m}{ne^2\tau}; \sigma = ne\mu$	$\Omega \text{ m}; \text{S/m}$	metal	ρ rises with T
$R_T = R_0 [1 + \alpha (T - T_0)]$	$\alpha \text{ (}^\circ\text{C}^{-1}\text{)}$	metals ($\alpha > 0$)	10 Ω , $\alpha = 0.0039$, +80°C: 13.1 Ω
$P = VI = I^2 R = \frac{V^2}{R}$	W	1 kWh = 3.6 × 10 ⁶ J	2 kW heater, 5 h: 10 kWh
$V = \varepsilon - Ir; I = \frac{\varepsilon}{R + r}$	V	discharging	$r = (\varepsilon/V - 1) R$
$\frac{P}{Q} = \frac{R}{S}$	Ω	balanced bridge	$P = 100, Q = 150, R = 40$: $S = 60 \Omega$

N1

C03-N01

Ch 3

Easy

Numerical

BOARD-STYLE PRACTICE

2 marks

A copper wire of cross-section 1 mm² carries 2 A. Find the drift speed of electrons ($n = 8.5 \times 10^{28} \text{ m}^{-3}$).

GIVEN $I = 2 \text{ A}, A = 10^{-6} \text{ m}^2, n = 8.5 \times 10^{28} \text{ m}^{-3}, e = 1.6 \times 10^{-19} \text{ C}$

FIND v_d

FORMULA $v_d = \frac{I}{neA}$

WORKING $v_d = \frac{2}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} = \frac{2}{1.36 \times 10^4} = 1.47 \times 10^{-4} \text{ m s}^{-1}$.

Answer: $v_d \approx 1.5 \times 10^{-4} \text{ m s}^{-1}$ (0.15 mm/s)

▲ Convert mm² to m² (10⁻⁶). Tiny v_d does not delay the current.

N2

C03-N02

Ch 3

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

A nichrome wire ($\rho = 1.1 \times 10^{-6} \Omega \text{ m}$) of length 2 m and diameter 0.4 mm is stretched uniformly to double its length. Find its resistance before and after.

GIVEN $\rho = 1.1 \times 10^{-6} \Omega \text{ m}, l = 2 \text{ m}, d = 0.4 \text{ mm} \Rightarrow r = 0.2 \times 10^{-3} \text{ m}$

FIND R and R'

FORMULA $R = \frac{\rho l}{A}, A = \pi r^2$; stretching: $R' = 4R$ (volume constant)

WORKING $A = \pi (0.2 \times 10^{-3})^2 = 1.257 \times 10^{-7} \text{ m}^2; R = \frac{1.1 \times 10^{-6} \times 2}{1.257 \times 10^{-7}} = 17.5 \Omega$. Doubling l halves A , so $R' = 4R = 70 \Omega$.

Answer: $R = 17.5 \Omega; R' = 70 \Omega$

▲ Do not simply double R : area shrinks as length grows.

N3

C03-N03

Ch 3

Medium

Numerical

BOARD-STYLE PRACTICE

2 marks

A cell of emf 1.5 V sends 0.3 A through an external resistance of 4 Ω . Find its internal resistance and terminal voltage.

GIVEN $\varepsilon = 1.5 \text{ V}, I = 0.3 \text{ A}, R = 4 \Omega$

FIND r and V

FORMULA $I = \frac{\varepsilon}{R + r}, V = IR$

WORKING $R + r = \frac{1.5}{0.3} = 5 \Omega \Rightarrow r = 1 \Omega; V = IR = 0.3 \times 4 = 1.2 \text{ V}$ (check: $\varepsilon - Ir = 1.5 - 0.3 = 1.2 \text{ V}$).

Answer: $r = 1 \Omega; V = 1.2 \text{ V}$

▲ Terminal voltage is always less than emf when current is drawn.

3.6 More numericals and subjective practice

N4 C03-N04 Ch 3 **Hard** Numerical **BOARD-STYLE PRACTICE** 3 marks

For the network of Fig. 3.7 take $\varepsilon_1 = 12\text{ V}$, $\varepsilon_2 = 6\text{ V}$, $R_1 = 2\ \Omega$, $R_2 = 3\ \Omega$, $R_3 = 6\ \Omega$ (cells ideal). Find I_1 , I_2 and I_3 .

GIVEN as stated; take I_1, I_2 towards D and $I_3 = I_1 + I_2$ through R_3

FIND I_1, I_2, I_3

FORMULA junction rule at D; loop rule on both loops

WORKING Loop 1: $12 - 2I_1 - 6(I_1 + I_2) = 0 \Rightarrow 8I_1 + 6I_2 = 12$. Loop 2: $6 - 3I_2 - 6(I_1 + I_2) = 0 \Rightarrow 6I_1 + 9I_2 = 6$.
Solving: $I_1 = 2\text{ A}$, $I_2 = -\frac{2}{3}\text{ A}$. Then $I_3 = I_1 + I_2 = \frac{4}{3}\text{ A}$.

Answer: $I_1 = 2\text{ A}$; $I_2 = -0.67\text{ A}$ (cell 2 is being charged); $I_3 = 1.33\text{ A}$

▲ Check loop 2: $6 - 3(-\frac{2}{3}) - 6(\frac{4}{3}) = 6 + 2 - 8 = 0$ ✓.

N5 C03-N05 Ch 3 **Medium** Numerical **BOARD-STYLE PRACTICE** 3 marks

An electric heater is rated 2 kW , 220 V . Find its resistance and current. What is the cost of using it for 5 hours at ₹7 per unit?

GIVEN $P = 2000\text{ W}$, $V = 220\text{ V}$, $t = 5\text{ h}$

FIND R , I , cost

FORMULA $R = \frac{V^2}{P}$, $I = \frac{P}{V}$, energy = Pt (kWh)

WORKING $R = \frac{220^2}{2000} = 24.2\ \Omega$; $I = \frac{2000}{220} = 9.09\text{ A}$; energy = $2\text{ kW} \times 5\text{ h} = 10\text{ kWh}$; cost = $10 \times 7 = ₹70$.

Answer: $24.2\ \Omega$; 9.09 A ; ₹70

▲ The 'unit' in electricity bills is 1 kWh , not 1 kW .

N6 C03-N06 Ch 3 **Medium** Numerical **BOARD-STYLE PRACTICE** 2 marks

A bridge has $P = 100\ \Omega$, $Q = 150\ \Omega$ and $R = 40\ \Omega$. Find S for balance. A copper coil has $10\ \Omega$ at 20°C ($\alpha = 0.0039^\circ\text{C}^{-1}$); find its resistance at 100°C .

GIVEN as stated

FIND S and R_{100}

FORMULA $\frac{P}{Q} = \frac{R}{S}$; $R_T = R_0 [1 + \alpha\Delta T]$

WORKING $S = \frac{QR}{P} = \frac{150 \times 40}{100} = 60\ \Omega$. $R_{100} = 10 [1 + 0.0039 \times 80] = 13.12\ \Omega$.

Answer: $S = 60\ \Omega$; $R_{100} = 13.1\ \Omega$

▲ Use $\Delta T = 100 - 20 = 80$, not 100.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C03-S01 Ch 3 **Easy** Short answer (2 marks) **BOARD-STYLE PRACTICE**

S1. Define drift velocity and mobility. Write the SI unit of each.

Model answer: **Drift velocity:** average velocity with which free electrons move in a conductor under an electric field (unit m s^{-1}). **Mobility** $\mu = v_d/E$: drift velocity per unit field (unit $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$).

C03-S02 Ch 3 **Medium** Short answer (3 marks) **BOARD-STYLE PRACTICE**

S2. A 12 V battery of internal resistance $1\ \Omega$ is connected to $5\ \Omega$ and $10\ \Omega$ resistors in series. Find the current and the terminal voltage.

Model answer: $R_{\text{tot}} = 5 + 10 + 1 = 16\ \Omega$; $I = 12/16 = 0.75\text{ A}$; $V = \varepsilon - Ir = 12 - 0.75 = 11.25\text{ V}$.

C03-CB01 Ch 3 **Medium** Case-based (4 marks) **BOARD-STYLE PRACTICE**

Case. A student measures V across a cell for different currents and plots V against I : a straight line cutting the V -axis at 1.5 V and the I -axis at 3.0 A . (i) What is the emf? (ii) Find r . (iii) What does the I -intercept represent? (iv) Terminal voltage at 0.5 A ?

Model answer: (i) $\varepsilon = 1.5\text{ V}$. (ii) $r = \varepsilon/I_{\text{sc}} = 1.5/3 = 0.5\ \Omega$. (iii) short-circuit current. (iv) $V = 1.5 - 0.5 \times 0.5 = 1.25\text{ V}$.

3.7 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C03-M01 Ch 3 Easy Conceptual BOARD-STYLE PRACTICE

1. The drift velocity of free electrons in a copper wire is of the order of

- (a) 10^8 m/s (b) 10^6 m/s
(c) 1 m/s (d) 10^{-4} m/s

Ans: (d) Typically ~ 0.1 mm/s for ordinary currents.

C03-M02 Ch 3 Medium Numerical BOARD-STYLE PRACTICE

2. A wire of resistance R is stretched to twice its length (volume constant). Its new resistance is

- (a) R (b) $2R$
(c) $4R$ (d) $R/2$

Ans: (c) $l \rightarrow 2l$, $A \rightarrow A/2$, so $R \rightarrow 4R$.

C03-M03 Ch 3 Medium Conceptual BOARD-STYLE PRACTICE

3. With rise in temperature, the resistivity of a semiconductor

- (a) increases (b) decreases
(c) stays constant (d) first increases then vanishes

Ans: (b) More carriers are released; this outweighs the fall in τ .

C03-M04 Ch 3 Medium Numerical BOARD-STYLE PRACTICE

4. A cell of emf 2 V and internal resistance 0.5Ω is shorted by a wire of negligible resistance. The current is

- (a) 1 A (b) 2 A
(c) 4 A (d) 0.25 A

Ans: (c) $I = \varepsilon/r = 2/0.5 = 4$ A.

C03-M05 Ch 3 Medium Application BOARD-STYLE PRACTICE

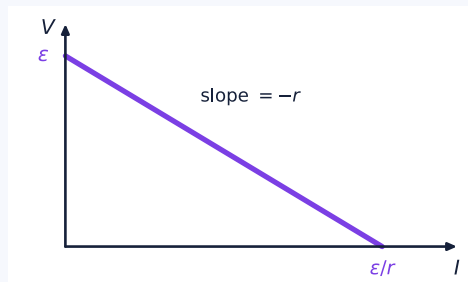
5. Three 6Ω resistors are connected in parallel. The equivalent resistance is

- (a) 18Ω (b) 6Ω
(c) 3Ω (d) 2Ω

Ans: (d) $1/R_p = 3/6 \Rightarrow R_p = 2 \Omega$.

C03-M06 Ch 3 Medium Graph BOARD-STYLE PRACTICE

6. The graph shows V against I for a cell. The internal resistance is the



- (a) intercept on the V -axis (b) magnitude of the slope
(c) intercept on the I -axis (d) area under the line

Ans: (b) $V = \varepsilon - Ir$; slope = $-r$.

C03-M07 Ch 3 Hard Application BOARD-STYLE PRACTICE

7. In a balanced Wheatstone bridge, interchanging the cell and the galvanometer

- (a) unbalances the bridge (b) does not affect the balance
(c) reverses the current in P (d) doubles S

Ans: (b) The balance condition $P/Q = R/S$ is symmetric.

C03-M08 Ch 3 Medium Numerical BOARD-STYLE PRACTICE

8. A 100 W, 220 V bulb is used on a 110 V supply. The power consumed is

- (a) 100 W (b) 50 W
(c) 25 W (d) 12.5 W

Ans: (c) $P = V^2/R$; halving V gives $P/4 = 25$ W.

C03-AR01 Ch 3 Medium Assertion-Reason BOARD-STYLE PRACTICE

9. A: The drift speed is very small, yet a bulb lights instantly when the switch is closed.

R: The electric field sets up almost instantly throughout the wire and moves all the free electrons together.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Field propagation, not electron travel, causes the instant response.

C03-AR02 Ch 3 Easy Assertion-Reason BOARD-STYLE PRACTICE

10. A: The terminal voltage of a cell is less than its emf when it supplies current.

R: There is a potential drop across the internal resistance.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) $V = \varepsilon - Ir$.

★ EXAMINER-STYLE REMINDERS

- Draw the circuit and mark the currents before writing equations.
- In Kirchhoff problems state the sign convention you use.
- In a derivation of ρ , mention the relaxation time and the averaging.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are BOARD-STYLE PRACTICE; none is claimed to be a past board question. See the Source Log at the end.

R Chapter 3 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Drift of electrons in a wire with $\mathbf{E}$ and $\mathbf{v}_d$
2. $V-I$ graphs: ohmic, bulb, diode
3. $\rho-T$ curves for metal and semiconductor
4. Cell with r and the $V-I$ graph
5. Two-loop Kirchhoff network and Wheatstone bridge

★ 7-DAY REVISION PLAN

Day 1	Current, drift, Ohm
Day 2	Resistivity and temperature
Day 3	Power and energy
Day 4	Cells and internal resistance
Day 5	Kirchhoff and bridge
Day 6	Numericals N1–N6
Day 7	MCQs; redraw circuits

Chapter 3 at a glance — Current Electricity

Current

$$I = dq/dt = neAv_d; v_d \sim 10^{-4} \text{ m/s.}$$

Drift

$$v_d = eE\tau/m; \mu = e\tau/m.$$

Ohm

$$V = IR; R = \rho l/A; \rho = m/ne^2\tau.$$

T -dependence

metal $\rho \uparrow$; semiconductor $\rho \downarrow$; alloy flat.

Cells

$$V = \varepsilon - Ir; \text{series } \sum \varepsilon, \sum r.$$

Kirchhoff / bridge

$$\sum I = 0; \sum \Delta V = 0; P/Q = R/S.$$

Formula map

$$I = neAv_d$$

current

$$v_d = \frac{eE\tau}{m}$$

drift

$$R = \frac{\rho l}{A}$$

resistance

$$\rho = \frac{m}{ne^2\tau}$$

resistivity

$$P = I^2 R$$

power

$$V = \varepsilon - Ir$$

terminal V

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

parallel cells

$$\frac{P}{Q} = \frac{R}{S}$$

bridge

$$R_T = R_0 (1 + \alpha \Delta T)$$

temperature

! COMMON MISTAKES

- Treating drift speed as the speed of signal.
- Forgetting $R \rightarrow n^2 R$ on stretching.
- Wrong sign of IR in a loop equation.
- Using $r_{eq} = r/n$ for unequal cells.
- Using $\Delta T = T$ instead of $T - T_0$.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can derive v_d , μ and ρ
- ☐ I can read $V-I$ graphs
- ☐ I can use $V = \varepsilon - Ir$ and combine cells
- ☐ I can apply Kirchhoff's rules to two loops
- ☐ I can derive the bridge balance condition
- ☐ I solved N1–N6
- ☐ I attempted all MCQs

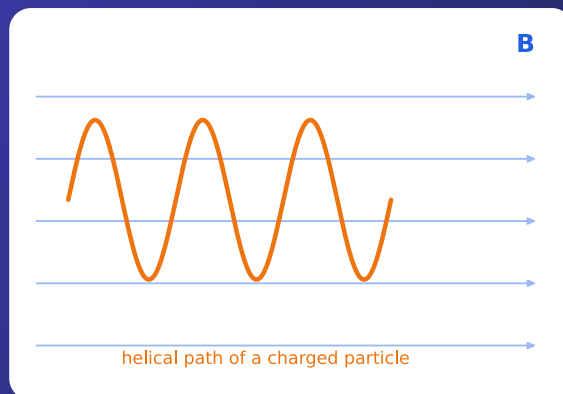
✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

1. SI unit of resistivity? | **Ans:** $\Omega \text{ m}$
2. V of a cell on open circuit? | **Ans:** Its emf
3. Resistors R and R in parallel? | **Ans:** $R/2$
4. Effect of T on nichrome ρ ? | **Ans:** Almost none
5. Wheatstone balance condition? | **Ans:** $P/Q = R/S$

UNIT III · MAGNETIC EFFECTS OF CURRENT AND MAGNETISM

4 Moving Charges and Magnetism

A magnetic force never changes a particle's speed; it only turns it. That one fact explains circular orbits, cyclotrons and why the magnetic force does no work.



Chapter roadmap

Biot–Savart

Ampere's law

Force on charges

Force on wires

Galvanometer

✓ LEARNING OUTCOMES

- State Oersted's observation and the Biot–Savart law; find B on the axis of a circular loop
- Apply Ampere's law to a long wire, a solenoid and a toroid
- Find the force on a moving charge ($q\mathbf{v} \times \mathbf{B}$) and describe circular and helical paths
- Find the force on a current element, the force between parallel wires and the torque on a loop
- Explain the moving-coil galvanometer and convert it to an ammeter or voltmeter

I PREREQUISITES TO REVISE

- Vector cross product and the right-hand rule
- Circular motion: $F = mv^2/r$
- Current and resistance (Chapter 3)
- Torque and moment of a couple

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: concept of magnetic field; Oersted's experiment; Biot–Savart law and its application to a current-carrying circular loop; Ampere's law and its applications to an infinitely long straight wire, straight and toroidal solenoids (qualitative treatment only for solenoids); force on a moving charge in uniform magnetic and electric fields; cyclotron; force on a current-carrying conductor in a uniform magnetic field; force between two parallel current-carrying conductors and definition of the ampere; torque on a current loop in a uniform magnetic field; moving-coil galvanometer, its current sensitivity and conversion to ammeter and voltmeter.

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (direction of force, radius of path); 2-mark relations (Biot–Savart, force between wires); 3-mark numericals (circular path, solenoid, galvanometer conversion); 5-mark derivations (B on the axis of a loop, force between parallel wires, torque on a loop). *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Three rules to remember

F FORCE ON A MOVING CHARGE

$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$; magnitude $qvB \sin \theta$. Zero if $\mathbf{v} \parallel \mathbf{B}$; maximum if perpendicular. Direction by the right-hand rule (reverse for a negative charge).

★ SOURCES OF B

Moving charges and currents. Long wire: $\mu_0 I / 2\pi r$. Loop centre: $\mu_0 NI / 2R$. Solenoid: $\mu_0 nI$.
 $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$.

4.1 Magnetic field and the Biot–Savart law

1 BIOT–SAVART LAW

Oersted showed that a current deflects a compass needle: currents produce magnetic fields. The field due to a current element $I d\mathbf{l}$ at displacement $\mathbf{r}$ is

$$d\mathbf{B} = \frac{\mu_0 I d\mathbf{l} \times \mathbf{r}}{4\pi r^3}$$

Magnitude $\frac{\mu_0 I d\mathbf{l} \sin \theta}{4\pi r^2}$; $d\mathbf{B} \perp$ both $d\mathbf{l}$ and $\mathbf{r}$; zero along the line of the element. Unit of B : tesla (T) = $\text{N A}^{-1} \text{m}^{-1}$.

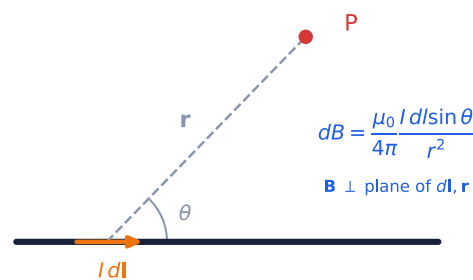


Fig. 4.1 Geometry of the Biot–Savart law.

Magnetic field on the axis of a circular current loop

5
MARKS

Principle Biot–Savart law; symmetry cancels the components perpendicular to the axis.

Assumptions Loop of radius R , current I , point P at distance x along the axis; $d\mathbf{l} \perp \mathbf{r}$ for every element, so $\sin \theta = 1$.

- ① Each element gives $dB = \frac{\mu_0 I dl}{4\pi r^2}$ with $r^2 = R^2 + x^2$.
- ② Components perpendicular to the axis cancel in pairs of diametrically opposite elements. Since $d\mathbf{B} \perp \mathbf{r}$, its axial component is $dB \cdot \frac{R}{r}$ (the angle between $\mathbf{r}$ and the loop plane has $\sin = R/r$).
- ③ Integrate over the loop ($\oint dl = 2\pi R$):

$$B = \frac{\mu_0 I R}{4\pi r^3} 2\pi R = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

FINAL RESULT

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}; \quad x = 0: B = \frac{\mu_0 I}{2R}; \quad x \gg R: B \approx \frac{\mu_0}{2}$$

Limits: for a coil of N turns multiply by N . Direction by the right-hand grip rule (fingers along I , thumb along $\mathbf{B}$).

How to write it in the exam: Draw the loop, $d\mathbf{B}$ components, state that perpendicular components cancel.

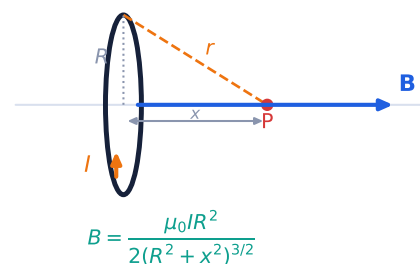


Fig. 4.2 Field on the axis of a circular loop.

! COMMON MISTAKES

- Using $B = \mu_0 I / 2R$ for points off the centre.
- Forgetting N for coils.
- Confusing μ_0 (free space) with permeability of a medium.

4.2 Ampere's circuital law and its applications

F AMPERE'S LAW

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$$

The line integral of $\mathbf{B}$ around any closed loop equals μ_0 times the net current through it. Useful when symmetry makes B constant along the loop.

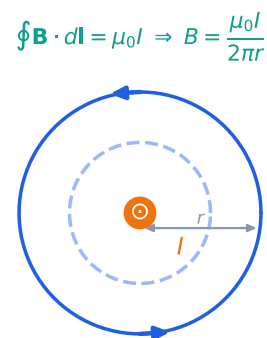


Fig. 4.3 Amperian loop around a long straight wire.

Field of a long straight wire

2 MARKS

Principle Ampere's law with a circular amperian loop.

Assumptions Infinitely long wire; circular loop of radius r centred on the wire in a plane perpendicular to it.

- By symmetry B is constant in magnitude and tangential on the loop, so $\oint \mathbf{B} \cdot d\mathbf{l} = B (2\pi r)$.
- Current enclosed $= I$. So $B (2\pi r) = \mu_0 I$.

FINAL RESULT

$$B = \frac{\mu_0 I}{2\pi r}$$

Limits: valid outside the wire; field lines are concentric circles (direction by the right-hand rule).

How to write it in the exam: Draw the amperian loop; write both sides.

I SOLENOID AND TOROID (QUALITATIVE TREATMENT)

A **solenoid** of n turns per unit length carrying I : field inside is nearly uniform and axial, outside is almost zero.

$$B_{\text{solenoid}} = \mu_0 n I, \quad B_{\text{toroid}} = \frac{\mu_0 N I}{2\pi r}$$

Ideal long solenoid: apply Ampere's law to a rectangle with one side inside: $Bl = \mu_0 (nl) I$. A toroid (circular solenoid with N turns) has field confined within its core; $B = 0$ outside and in the hollow.

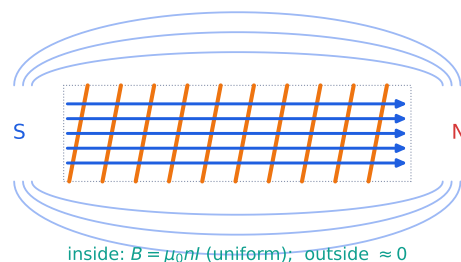


Fig. 4.4 Field of a solenoid: uniform inside, weak outside.

Force between two parallel currents; the ampere

3 MARKS

Principle B of one long wire acts on the current in the other ($F = BIl$).

Assumptions Parallel straight wires of length l , separation d , currents I_1, I_2 .

- Field of wire 1 at wire 2: $B_1 = \frac{\mu_0 I_1}{2\pi d}$, perpendicular to the plane of the wires.
- Force on length l of wire 2: $F = B_1 I_2 l = \frac{\mu_0 I_1 I_2 l}{2\pi d}$. By the right-hand rule it is attractive for parallel currents.

FINAL RESULT

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

Limits: Definition of ampere: 1 A is the current which, in two infinite parallel wires 1 m apart in vacuum, produces a force of 2×10^{-7} N per metre of length. Parallel currents attract; antiparallel repel.

How to write it in the exam: Write $F = BIl$ for wire 2 and state attractive/repulsive.

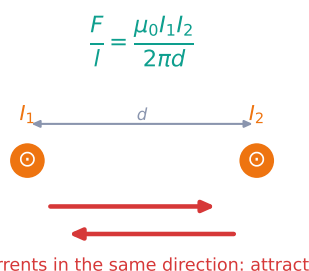


Fig. 4.5 Parallel currents attract.

4.3 Force on a moving charge; the cyclotron

F LORENTZ FORCE

$$\mathbf{F} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Magnetic part $q\mathbf{v} \times \mathbf{B}$ is always perpendicular to $\mathbf{v}$: it does **no work** and changes direction, not speed. Electric part $q\mathbf{E}$ can speed the particle up.

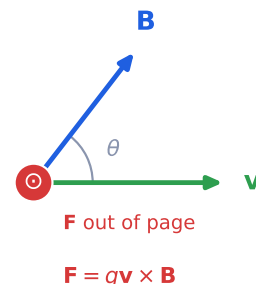


Fig. 4.6 Direction of $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$.

Circular motion in a uniform magnetic field

3 MARKS

Principle Magnetic force provides the centripetal force.

Assumptions $\mathbf{v} \perp \mathbf{B}$; uniform $\mathbf{B}$; no other force.

- ① $qvB = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB} = \frac{p}{qB}$.
- ② Period $T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$ — independent of v and r ;
frequency $f = \frac{qB}{2\pi m}$.
- ③ If $\mathbf{v}$ has a component $v_{\parallel}$ along $\mathbf{B}$ the path is a **helix** of radius $mv_{\perp}/qB$ and pitch $p = v_{\parallel}T$.

FINAL RESULT

$$r = \frac{mv}{qB}, \quad T = \frac{2\pi m}{qB}, \quad \text{pitch} = \frac{2\pi m v \cos \theta}{qB}$$

Limits: non-relativistic speeds.

How to write it in the exam: Show that T does not depend on speed (basis of the cyclotron).

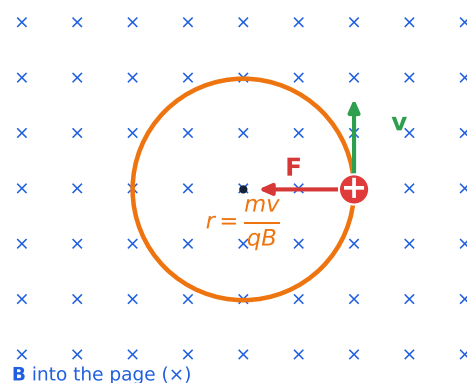


Fig. 4.7 Charge moving in $\mathbf{B}$ (into page).

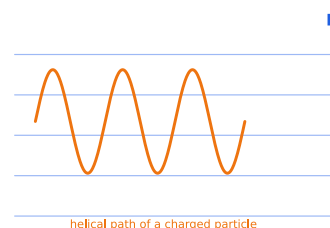


Fig. 4.8 Helical path.

~ CYCLOTRON

Two hollow D-shaped electrodes in a vertical $\mathbf{B}$; a high-frequency alternating voltage across the gap accelerates the ions each time they cross. Because the period $T = 2\pi m/qB$ is independent of speed, resonance requires $f_{osc} = \frac{qB}{2\pi m}$.

Final kinetic energy $K = \frac{q^2 B^2 R^2}{2m}$ (R = radius of the D). Limit: relativistic mass increase for light particles; unsuitable for electrons.

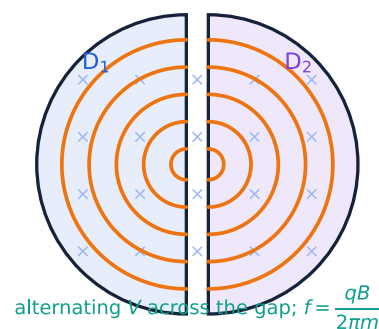


Fig. 4.9 Schematic cyclotron.

4.4 Force on conductors, torque and the galvanometer

F FORCE ON A CURRENT ELEMENT AND TORQUE ON A LOOP

$$\mathbf{F} = I\mathbf{l} \times \mathbf{B} \quad (F = BIl \sin \theta), \quad \boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}, \quad m = NIA,$$

θ for torque is the angle between $\mathbf{m}$ (normal to the loop, by the right-hand rule) and $\mathbf{B}$. Net force on a loop in a *uniform* field is zero; torque is maximum when the plane of the loop is parallel to $\mathbf{B}$ ($\theta = 90^\circ$).

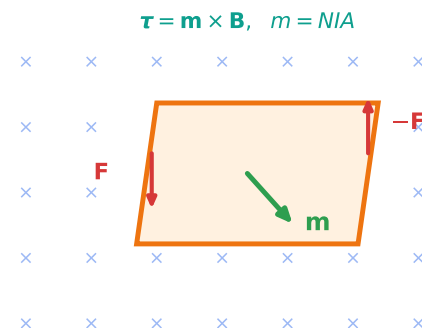


Fig. 4.10 Rectangular loop in a uniform magnetic field.

Torque on a rectangular current loop

5 MARKS

Principle $F = BIl$ on each side; forces on the two opposite sides form a couple.

Assumptions Rectangular loop $a \times b$ in uniform $\mathbf{B}$; normal makes angle θ with $\mathbf{B}$.

- ① Forces on the two sides of length b : $F = IbB$ each, equal and opposite, collinear (along the axis), so they cancel and give no torque.
- ② Forces on the two sides of length a : $F = IaB$ each, equal and opposite, forming a couple with arm $b \sin \theta$.
- ③ $\tau = (IaB)(b \sin \theta) = IAB \sin \theta$; for N turns $\tau = NIAB \sin \theta$.

FINAL RESULT

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}, \quad m = NIA$$

Limits: uniform field; net force zero. A loop behaves as a magnetic dipole of moment m .

How to write it in the exam: Draw the loop, the two forces and the arm; derive $\tau = NIAB \sin \theta$.

I MOVING-COIL GALVANOMETER

A coil on a soft-iron core in the radial field of a permanent magnet. Deflecting torque $NIAB$ is balanced by the spring's restoring torque $k\phi$:

$$\phi = \frac{NAB}{k}I \Rightarrow \text{current sensitivity } \frac{\phi}{I} = \frac{NAB}{k}$$

Radial field makes the torque independent of the coil's position, so the scale is linear. Voltage sensitivity

$$= \phi/V = \frac{NAB}{kR}. \text{ Increasing } N \text{ raises current sensitivity but also resistance, so voltage sensitivity may not rise.}$$

F CONVERSION

Ammeter: low shunt S in parallel: $S = \frac{I_g G}{I - I_g}$;

$R_A \approx S$ (very small). **Voltmeter:** high series resistance $R = \frac{V}{I_g} - G$; $R_V = G + R$ (very large).

I_g = full-scale current, G = galvanometer resistance.

4.5 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$d\mathbf{B} = \frac{\mu_0 I d\mathbf{l} \times \mathbf{r}}{4\pi r^3}$	T	$\mu_0 = 4\pi \times 10^{-7}$	$B \perp$ plane of $d\mathbf{l}$, $\mathbf{r}$
$B = \frac{\mu_0 NI}{2R}$	T	centre of circular coil	$N = 100$, $R = 0.1$ m, $I = 2$ A: 1.26 mT
$B = \frac{\mu_0 I}{2\pi r}$	T	long straight wire	at $r = 5$ cm, 10 A: $40 \mu\text{T}$
$B = \mu_0 nI$	T	long solenoid, inside	$n = 2000 \text{ m}^{-1}$, 2 A: 5.0 mT
$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$	N	no work done	$r = mv/qB$
$T = \frac{2\pi m}{qB}$	s	independent of v	cyclotron frequency
$\mathbf{F} = I\mathbf{l} \times \mathbf{B}$	N	straight conductor	$F = BIl \sin \theta$
$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$	N m ⁻¹	parallel wires	def. of ampere
$\tau = NIAB \sin \theta$	N m	uniform $\mathbf{B}$	$m = NIA$
$S = \frac{I_g G}{I - I_g}$; $R = \frac{V}{I_g} - G$	Ω	ammeter ; voltmeter	shunt ; series

N1 C04-N01 Ch 4 Easy Numerical BOARD-STYLE PRACTICE 3 marks

An electron moving at $2 \times 10^6 \text{ m s}^{-1}$ enters a uniform field $B = 1.0 \text{ mT}$ perpendicular to its velocity. Find the radius of its path and its period.

GIVEN $v = 2 \times 10^6 \text{ m/s}$, $B = 10^{-3} \text{ T}$, $m = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$

FIND r , T

FORMULA $r = \frac{mv}{eB}$, $T = \frac{2\pi m}{eB}$

WORKING $r = \frac{9.1 \times 10^{-31} \times 2 \times 10^6}{1.6 \times 10^{-19} \times 10^{-3}} = 1.14 \times 10^{-2} \text{ m}$; $T = \frac{2\pi \times 9.1 \times 10^{-31}}{1.6 \times 10^{-19} \times 10^{-3}} = 3.57 \times 10^{-8} \text{ s}$.

Answer: $r \approx 1.14 \text{ cm}$; $T \approx 3.6 \times 10^{-8} \text{ s}$

▲ T does not depend on v .

N2 C04-N02 Ch 4 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A proton is accelerated from rest through 500 V and enters a perpendicular field of 0.2 T. Find its speed and radius of path.

GIVEN $V = 500 \text{ V}$, $B = 0.2 \text{ T}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$

FIND v , r

FORMULA $qV = \frac{1}{2}mv^2$; $r = \frac{mv}{qB}$

WORKING $v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 500}{1.67 \times 10^{-27}}} = 3.1 \times 10^5 \text{ m/s}$; $r = \frac{1.67 \times 10^{-27} \times 3.1 \times 10^5}{1.6 \times 10^{-19} \times 0.2} = 1.6 \times 10^{-2} \text{ m}$.

Answer: $v \approx 3.1 \times 10^5 \text{ m/s}$; $r \approx 1.6 \text{ cm}$

▲ Energy gain qV first; magnetic field does no work.

N3 C04-N03 Ch 4 Easy Numerical BOARD-STYLE PRACTICE 2 marks

A circular coil of 100 turns and radius 10 cm carries 2 A. Find B at its centre.

GIVEN $N = 100$, $R = 0.1 \text{ m}$, $I = 2 \text{ A}$

FIND B

FORMULA $B = \frac{\mu_0 NI}{2R}$

WORKING $B = \frac{4\pi \times 10^{-7} \times 100 \times 2}{2 \times 0.1} = 1.26 \times 10^{-3} \text{ T}$.

Answer: $B \approx 1.26 \text{ mT}$

▲ Radius in metres.

N4 C04-N04 Ch 4 Medium Numerical BOARD-STYLE PRACTICE 2 marks

Two long parallel wires 10 cm apart carry 10 A and 5 A in the same direction. Find the force per metre and its nature.

GIVEN $d = 0.1 \text{ m}$, $I_1 = 10 \text{ A}$, $I_2 = 5 \text{ A}$

FIND F/l

FORMULA $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$

WORKING $\frac{F}{l} = \frac{2 \times 10^{-7} \times 10 \times 5}{0.1} = 1.0 \times 10^{-4} \text{ N/m}$.

Answer: $1.0 \times 10^{-4} \text{ N/m}$, attractive

▲ Same-direction currents attract.

4.6 More numericals and subjective practice

N6 C04-N05 Ch 4 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A solenoid of length 25 cm has 500 turns and carries 2 A. Find B inside. If a 2-cm² cross-section iron-free coil of 50 turns carrying 0.5 A is placed with its plane parallel to this field, find the torque on it.

GIVEN $l = 0.25$ m, $N = 500$, $I = 2$ A; coil: $N' = 50$, $A = 2 \times 10^{-4}$ m², $I' = 0.5$ A

FIND B , τ

FORMULA $B = \mu_0 nI$; $\tau = N' I' A B \sin \theta$

WORKING $n = 500/0.25 = 2000$ m⁻¹; $B = 4\pi \times 10^{-7} \times 2000 \times 2 = 5.03 \times 10^{-3}$ T. Plane parallel to $\mathbf{B}$ means $\mathbf{m} \perp \mathbf{B}$, $\sin \theta = 1$; $\tau = 50 \times 0.5 \times 2 \times 10^{-4} \times 5.03 \times 10^{-3} = 2.5 \times 10^{-5}$ N m.

Answer: $B = 5.03$ mT; $\tau = 2.5 \times 10^{-5}$ N m

▲ Torque uses the angle between the *normal* and $\mathbf{B}$.

N5 C04-N06 Ch 4 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A galvanometer has $G = 50 \Omega$ and full-scale current 5 mA. How can it be made (a) a 0–1 A ammeter, (b) a 0–10 V voltmeter?

GIVEN $G = 50 \Omega$, $I_g = 5 \times 10^{-3}$ A

FIND S ; R

FORMULA $S = \frac{I_g G}{I - I_g}$; $R = \frac{V}{I_g} - G$

WORKING (a) $S = \frac{0.005 \times 50}{1 - 0.005} = 0.251 \Omega$ in parallel. (b) $R = \frac{10}{0.005} - 50 = 1950 \Omega$ in series.

Answer: (a) 0.25 Ω shunt; (b) 1950 Ω series

▲ Shunt is parallel, multiplier is series.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C04-S01 Ch 4 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. Why does a magnetic field not change the kinetic energy of a moving charge?

Model answer: The force $q\mathbf{v} \times \mathbf{B}$ is perpendicular to $\mathbf{v}$, so the work $\mathbf{F} \cdot d\mathbf{s} = 0$; speed and kinetic energy remain constant while direction changes.

C04-S02 Ch 4 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. Why is a radial field used in a moving-coil galvanometer? Why is the coil wound on a soft-iron core?

Model answer: A radial field keeps the plane of the coil always parallel to $\mathbf{B}$, so torque $= NIAB$ for every deflection and the scale is linear. The soft-iron core strengthens the field and makes it radial, increasing sensitivity.

C04-S03 Ch 4 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S3. A charged particle enters a uniform $\mathbf{B}$ at 60° to the field. Describe its path.

Model answer: A helix: the perpendicular component $v \sin 60^\circ$ gives circular motion of radius $mv \sin 60^\circ / qB$ while the parallel component $v \cos 60^\circ$ moves it along $\mathbf{B}$ uniformly, pitch $= v \cos 60^\circ \times 2\pi m / qB$.

C04-CB01 Ch 4 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. A cyclotron has D-radius 0.5 m and $B = 1.2$ T and accelerates protons. (i) Find the oscillator frequency. (ii) Find the maximum kinetic energy (in MeV). (iii) Does the frequency depend on speed? (iv) Why can't it accelerate electrons effectively?

Model answer: (i) $f = \frac{qB}{2\pi m} = \frac{1.6 \times 10^{-19} \times 1.2}{2\pi \times 1.67 \times 10^{-27}} = 1.83 \times 10^7$ Hz. (ii)

$K = \frac{q^2 B^2 R^2}{2m} = \frac{(1.6 \times 10^{-19} \times 1.2 \times 0.5)^2}{2 \times 1.67 \times 10^{-27}} = 2.76 \times 10^{-12}$ J = 17.2 MeV. (iii) No (non-relativistic). (iv) Electrons quickly become relativistic, changing m and losing resonance.

4.7 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C04-M01 Ch 4 Easy Conceptual

BOARD-STYLE PRACTICE

1. A charge moving parallel to a uniform magnetic field experiences a force of

- (a) qvB (b) $qvB/2$
(c) zero (d) qB/v

Ans: (c) $\sin 0^\circ = 0$.

C04-M02 Ch 4 Medium Numerical

BOARD-STYLE PRACTICE

2. A proton and an alpha particle have the same kinetic energy and enter the same B perpendicularly. The ratio $r_p : r_\alpha$ is

- (a) 1:1 (b) 1:2
(c) 2:1 (d) 1:4

Ans: (a) $r = \sqrt{2mK}/qB$; $m_\alpha = 4m_p$, $q_\alpha = 2q_p$, so
 $r \propto \sqrt{m}/q$; $\sqrt{4}/2 = 1$.

C04-M03 Ch 4 Medium Conceptual

BOARD-STYLE PRACTICE

3. Magnetic field due to a long straight wire varies with distance r as

- (a) r (b) r^2
(c) $1/r$ (d) $1/r^2$

Ans: (c) $B = \mu_0 I/2\pi r$.

C04-M04 Ch 4 Medium Numerical

BOARD-STYLE PRACTICE

4. A solenoid with 1000 turns/m carries 5 A. The field inside is nearly ($\mu_0 = 4\pi \times 10^{-7}$)

- (a) 6.3×10^{-3} T (b) 6.3×10^{-5} T
(c) 1.3×10^{-3} T (d) 0.63 T

Ans: (a) $B = \mu_0 nI = 4\pi \times 10^{-7} \times 5000 = 6.28 \times 10^{-3}$ T.

C04-M05 Ch 4 Medium Application

BOARD-STYLE PRACTICE

5. Two parallel wires carry currents in opposite directions. They

- (a) attract (b) repel
(c) experience no force (d) rotate

Ans: (b) Antiparallel currents repel.

C04-M06 Ch 4 Medium Graph

BOARD-STYLE PRACTICE

6. The graph shows period T of a charged particle in B against its speed v (non-relativistic). It is

- (a) a line through the origin (b) a hyperbola
(c) a horizontal line (d) a parabola

Ans: (c) $T = 2\pi m/qB$ is independent of v .

C04-M07 Ch 4 Medium Numerical

BOARD-STYLE PRACTICE

7. A galvanometer of $G = 100 \Omega$ and $I_g = 2$ mA is converted to a 10 V voltmeter. The series resistance is

- (a) 4900Ω (b) 5000Ω
(c) 5100Ω (d) 100Ω

Ans: (a) $R = 10/0.002 - 100 = 4900 \Omega$.

C04-M08 Ch 4 Hard Application

BOARD-STYLE PRACTICE

8. A rectangular loop in a uniform field experiences

- (a) a net force and torque (b) a torque but no net force
(c) a net force but no torque (d) neither, always torque

Ans: (b) Forces on opposite sides cancel in a uniform field; they form a couple unless $\mathbf{m} \parallel \mathbf{B}$.

C04-AR01 Ch 4 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: A magnetic force can accelerate a charge but cannot change its kinetic energy.

R: The magnetic force is always perpendicular to velocity.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Perpendicular force does no work.

C04-AR02 Ch 4 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: A cyclotron cannot accelerate electrons.

R: The cyclotron frequency is independent of the particle's mass.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (c) A is true (relativistic effects) but $f = qB/2\pi m$ depends on mass, so R is false.

★ EXAMINER-STYLE REMINDERS

- Show the direction (right-hand rule) with a labelled sketch.
- State 'perpendicular' when using $r = mv/qB$.
- In the galvanometer answer quote the radial field reason.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are **BOARD-STYLE PRACTICE**; none is claimed to be a past board question.

R Chapter 4 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Loop with $d\mathbf{B}$ components on the axis
2. Circular and helical path in $\mathbf{B}$
3. Rectangular loop with forces and arm
4. Moving-coil galvanometer (radial field)
5. Parallel currents with force arrows

★ 7-DAY REVISION PLAN

Day 1	Biot–Savart and loop
Day 2	Ampere's law
Day 3	Charge in $\mathbf{B}$
Day 4	Wires and torque
Day 5	Galvanometer
Day 6	N1–N6
Day 7	MCQs; redraw diagrams

Chapter 4 at a glance — Moving Charges and Magnetism

Biot–Savart

$$d\mathbf{B} \propto I d\mathbf{l} \times \mathbf{r}/r^3$$

Loop

$$B = \mu_0 NI/2R \text{ at centre.}$$

Ampere

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I; \text{ wire, solenoid, toroid.}$$

Charge in $\mathbf{B}$

$$r = mv/qB; T = 2\pi m/qB; \text{ helix.}$$

Wires

$$F = BIl; F/l = \mu_0 I_1 I_2 / 2\pi d.$$

Galvanometer

$$\tau = NIAB; \text{ shunt / series conversion.}$$

Formula map

$$\frac{\mu_0 NI}{2R}$$

loop centre

$$\frac{\mu_0 I}{2\pi r}$$

wire

$$\mu_0 nI$$

solenoid

$$\frac{mv}{qB}$$

radius

$$\frac{2\pi m}{qB}$$

period

$$BIl \sin \theta$$

wire force

$$\frac{\mu_0 I_1 I_2}{2\pi d}$$

F per length

$$NIAB \sin \theta$$

torque

$$\frac{I_g G}{I - I_g}$$

shunt

! COMMON MISTAKES

- Using $\sin \theta$ with the wrong angle (normal vs plane).
- Thinking $\mathbf{B}$ does work on charges.
- Reversing attraction/repulsion for parallel currents.
- Putting the ammeter shunt in series.
- Using r in cm in μ_0 formulas.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can derive B on the loop axis
- ☐ I can use Ampere's law for a wire
- ☐ I can derive r and T in $\mathbf{B}$
- ☐ I know the force between parallel wires
- ☐ I can derive $\tau = NIAB \sin \theta$
- ☐ I converted a galvanometer
- ☐ I solved N1–N6

✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

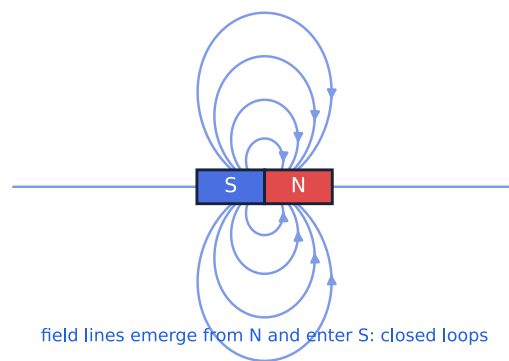
1. Unit of B ? | **Ans:** Tesla
2. Force on charge at rest in $\mathbf{B}$? | **Ans:** Zero
3. Does T depend on v ? | **Ans:** No
4. Solenoid field outside? | **Ans:** ≈ 0
5. Ammeter resistance? | **Ans:** Very small

UNIT III · MAGNETIC EFFECTS OF CURRENT AND MAGNETISM

5

Magnetism and Matter

Magnetic poles never appear alone — cut a magnet and each piece is a new magnet. Every magnet is, at heart, a tiny current loop.



Chapter roadmap

Bar magnet

Dipole fields

Torque & energy

Earth's field

Magnetic materials

✓ LEARNING OUTCOMES

- Describe a bar magnet as a magnetic dipole and as an equivalent solenoid
- Find the field of a dipole on its axis and on the equatorial line
- Use $\tau = mB \sin \theta$ and $U = -\mathbf{m} \cdot \mathbf{B}$; find the magnetic moment of a revolving electron
- Define the elements of the Earth's magnetic field (declination, dip, horizontal component)
- Classify dia-, para- and ferromagnetic materials; interpret the hysteresis loop and choose materials for magnets

I PREREQUISITES TO REVISE

- Biot–Savart law and the loop as a dipole (Chapter 4)
- Torque on a loop in a field
- Vector addition
- Bohr model of hydrogen

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: bar magnet; bar magnet as an equivalent solenoid; magnetic field lines; current loop as a magnetic dipole and its dipole moment; dipole moment of a revolving electron; magnetic field intensity due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis; torque on a magnetic dipole in a uniform field; Earth's magnetic field and magnetic elements; para-, dia- and ferromagnetic substances with examples; electromagnets and factors affecting their strengths; permanent magnets.

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (properties of dia/para/ferro, dip); 2-mark relations (dipole fields, Bohr magneton); 3-mark numericals (torque, Earth's field); 5-mark derivation of axial/equatorial field or torque; case-based on hysteresis. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Why it matters

★ BIG IDEAS

1) $\mathbf{B}$ lines are closed loops (no monopoles). 2) A loop of area A and current I has moment $\mathbf{m} = IA$. 3) A dipole in a field feels torque, not net force, in a uniform field.

F QUICK REFERENCE

$$B_{ax} = \frac{\mu_0 2m}{4\pi r^3}; B_{eq} = \frac{\mu_0 m}{4\pi r^3}; \tau = mB \sin \theta;$$

$$U = -mB \cos \theta; \mu_B = 9.27 \times 10^{-24} \text{ J T}^{-1}.$$

5.1 The bar magnet as a dipole

1 BAR MAGNET AND FIELD LINES

Properties: it points N–S when freely suspended; like poles repel, unlike attract; poles are inseparable. **Field lines** are continuous closed curves, leave N and enter S outside the magnet and continue from S to N inside; tangent gives the direction of $\mathbf{B}$; closer lines mean a stronger field; two lines never cross. Gauss's law for magnetism: net flux through any closed surface is zero.

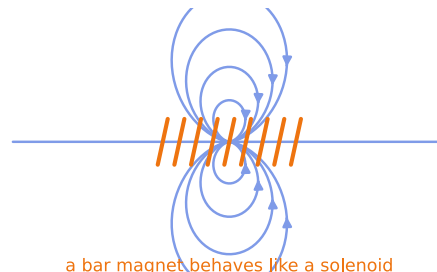


Fig. 5.1 A bar magnet and the equivalent solenoid have the same external field.

1 MAGNETIC MOMENT AND EQUIVALENT SOLENOID

A current loop is a dipole of moment $\mathbf{m} = NIA$ (unit A m^2 or J T^{-1}). A solenoid behaves as a bar magnet. For a magnet of pole strength q_m and length $2l$: $m = q_m 2l$ (directed S to N).

Revolving electron: orbital moment

$$\mu_l = \frac{e}{2m_e} L = \frac{evr}{2} \text{ (opposite to } \mathbf{L}\text{); for } L = \hbar,$$

$$\mu_B = \frac{e\hbar}{2m_e} = 9.27 \times 10^{-24} \text{ J T}^{-1} \text{ (Bohr magneton)}.$$

Moment of a revolving electron

3 MARKS

Principle Orbiting electron is a tiny current loop:
 $\mu = IA$.

Assumptions Circular orbit of radius r , speed v .

① Period $T = \frac{2\pi r}{v}$; current $I = \frac{e}{T} = \frac{ev}{2\pi r}$.

② Area $A = \pi r^2$, so $\mu_l = IA = \frac{evr}{2}$.

③ Angular momentum $L = m_e vr \Rightarrow \mu_l = \frac{e}{2m_e} L$;
direction opposite to $\mathbf{L}$ (negative charge).

FINAL RESULT

$$\mu_l = \frac{e}{2m_e} L, \quad \mu_B = \frac{e\hbar}{2m_e} \approx 9.27 \times 10^{-24} \text{ J T}^{-1}$$

Limits: Bohr model; the gyromagnetic ratio
 $e/2m_e = 8.8 \times 10^{10} \text{ C kg}^{-1}$.

How to write it in the exam: Write $I = e/T$, $A = \pi r^2$, then eliminate v with L .

! COMMON MISTAKES

- Believing a magnet can be cut into separate N and S poles.
- Taking field lines as paths of particles.
- Forgetting that the electron's moment is opposite to $\mathbf{L}$.

5.2 Field of a dipole, torque and energy

Field of a bar magnet on its axis and equatorial line

5 MARKS

Principle Treat the magnet as a short solenoid (loop analogy) of moment m , or add the fields of poles $\pm q_m$ at $\pm l$.

Assumptions Short dipole: $r \gg l$.

① **Axial point (distance r from centre):**

$$B = \frac{\mu_0 q_m}{4\pi} \left[\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2} \right] = \frac{\mu_0 q_m}{4\pi} \frac{4rl}{(r^2 - l^2)^2} \approx \frac{\mu_0 2m}{4\pi r^3}$$

, along $\mathbf{m}$.

② **Equatorial point:** each pole is at distance $\sqrt{r^2 + l^2}$; components along the axis add, perpendicular ones cancel:

$$B = 2 \frac{\mu_0 q_m}{4\pi} \frac{1}{r^2 + l^2} \cdot \frac{l}{\sqrt{r^2 + l^2}} \approx \frac{\mu_0 m}{4\pi r^3}, \text{ opposite to } \mathbf{m}.$$

FINAL RESULT

$$B_{ax} = \frac{\mu_0 2m}{4\pi r^3}, \quad B_{eq} = \frac{\mu_0 m}{4\pi r^3}, \quad B_{ax} = 2B_{eq}$$

Limits: valid for $r \gg l$ (short magnet). Direction: axial parallel to $\mathbf{m}$, equatorial antiparallel.

How to write it in the exam: Draw the two pole fields at the equatorial point and show cancellation of the perpendicular parts.

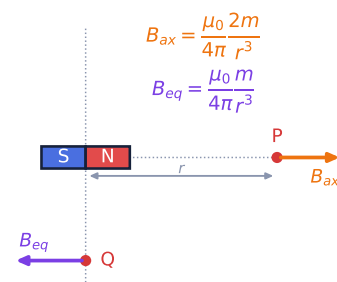


Fig. 5.2 Axial (P) and equatorial (Q) points of a short magnet.

Torque and potential energy of a dipole in a uniform field

3 MARKS

Principle Forces $\pm q_m B$ on the poles form a couple.

Assumptions Uniform $\mathbf{B}$; dipole of length $2l$ making angle θ with $\mathbf{B}$.

① Forces on the poles: $q_m B$ each, equal and opposite, so net force is zero.

② Couple arm $= 2l \sin \theta$: $\tau = q_m B \cdot 2l \sin \theta = mB \sin \theta$, i.e. $\tau = \mathbf{m} \times \mathbf{B}$.

③ Work to rotate from θ_0 to θ :

$$W = \int mB \sin \theta d\theta = mB (\cos \theta_0 - \cos \theta); \text{ with } U(90^\circ) = 0, U = -mB \cos \theta.$$

FINAL RESULT

$$\tau = mB \sin \theta, \quad U = -\mathbf{m} \cdot \mathbf{B} = -mB \cos \theta$$

Limits: stable equilibrium at $\theta = 0$ ($U = -mB$), unstable at $\theta = 180^\circ$ ($U = +mB$). Small oscillations: $T = 2\pi \sqrt{I/mB}$.

How to write it in the exam: Mention both τ and the zero-net-force statement.

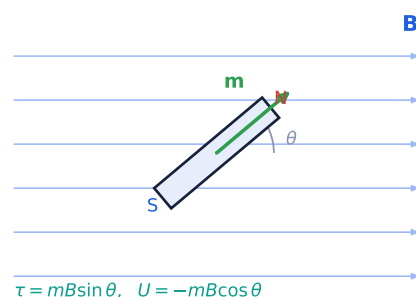


Fig. 5.3 A dipole in a uniform field.

5.3 Earth's magnetic field and magnetic materials

I EARTH'S MAGNETIC ELEMENTS

The Earth behaves like a dipole whose magnetic S pole lies near the geographic north. Three elements specify the field at a place: **declination** D (angle between the geographic and magnetic meridians), **dip** δ (angle of $\mathbf{B}_E$ with the horizontal) and **horizontal component** $H = B_E \cos \delta$. At the magnetic equator $\delta = 0$; at the poles $\delta = 90^\circ$.

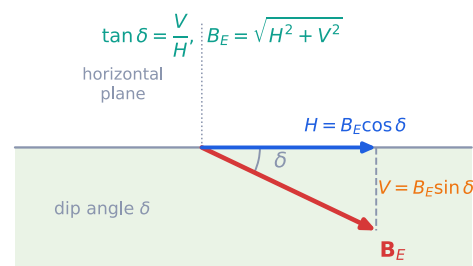


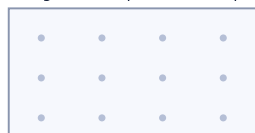
Fig. 5.4 Dip angle, horizontal and vertical components.

I MAGNETISATION AND CLASSIFICATION

Magnetic intensity $H = B_0/\mu_0$; magnetisation $M = m_{\text{net}}/V$; $\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$; $M = \chi H$; $\mu_r = 1 + \chi$.

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Susceptibility χ	small, negative	small, positive	large, positive
μ_r	slightly < 1	slightly > 1	$\gg 1$
In a field	weakly repelled	weakly attracted	strongly attracted
Dipoles	none permanent	random, align weakly	domains, aligned
T effect	independent	Curie law $\chi = C/T$	Curie temp: becomes para
Examples	Bi, Cu, Au, water	Al, Na, Pt, O_2	Fe, Co, Ni, alnico

Diamagnetic: no permanent dipoles



$\chi < 0$; weakly repelled

Paramagnetic: random dipoles



$\chi > 0$ small; weakly attracted

Ferromagnetic: aligned domains



$\chi \gg 1$; strongly attracted

Fig. 5.5 Atomic dipoles in the three classes of material.

★ CURIE TEMPERATURE AND CURIE'S LAW

Above its **Curie temperature** (about 1043 K for iron) a ferromagnet turns paramagnetic because thermal motion destroys domain alignment. For paramagnets $\chi = C/T$; for ferromagnets above T_c , $\chi = C/(T - T_c)$.

★ QUICK CONCEPTUAL Q&A

- Why do iron filings show field lines? — they become induced dipoles and align along $\mathbf{B}$.
- Where is the field strongest? — near the poles, where lines crowd.
- Why is the dip zero at the equator? — $\mathbf{B}_E$ is horizontal there.

5.4 Hysteresis, electromagnets and permanent magnets

I HYSTERESIS

Cycling H on a ferromagnet gives the loop B – H : **retentivity** = B left at $H = 0$; **coercivity** = reverse H needed to bring B to zero; the loop **area** is the energy dissipated per unit volume per cycle.

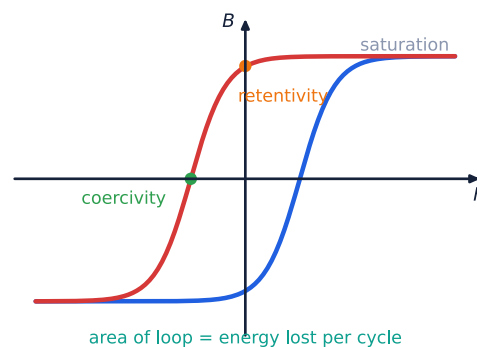


Fig. 5.6 Hysteresis loop of a ferromagnet.

Use	Needed properties	Material
Permanent magnet	high retentivity, high coercivity	steel, alnico, ticonal
Electromagnet / transformer core	low retentivity, low coercivity, high permeability, narrow loop	soft iron, mu-metal

~ ELECTROMAGNETS

A soft-iron core inside a current-carrying solenoid is strongly magnetised ($B = \mu_0 \mu_r nI$). The strength increases with **current**, **number of turns per unit length** and the **permeability of the core**. It can be switched on and off, so it is used in cranes, bells, relays and loudspeakers. A **permanent magnet** keeps its magnetism because the core has high retentivity and coercivity.

! COMMON MISTAKES

- Using steel for transformer cores (wide loop, high loss) — soft iron is needed.
- Thinking diamagnetism is shown only by some substances — it is present in all, but is masked by para/ferro effects.
- Mixing H (intensity) with B (flux density).

F ENERGY LOSS IN HYSTERESIS

Energy dissipated per unit volume per cycle = area of the B – H loop (J m^{-3}). Power loss in a transformer core $\propto f \times \text{loop area}$, so soft iron (narrow loop) is used.

★ COMPARE IN THE EXAM

Write materials in a 2-column table: *permanent* (high retentivity, high coercivity, wide loop) vs *core* (low retentivity, low coercivity, narrow loop, high μ_r).

5.5 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$m = NIA$	$A\ m^2$	loop / dipole	bar magnet $m = 2lq_m$
$B_{ax} = \frac{\mu_0 2m}{4\pi r^3}$	T	$r \gg l$	$m = 5\ A\ m^2, r = 0.2\ m$: $1.25 \times 10^{-4}\ T$
$B_{eq} = \frac{\mu_0 m}{4\pi r^3}$	T	$r \gg l$	half of axial
$\tau = mB \sin \theta$	N m	uniform B	$m = 2, B = 0.5, \theta = 30^\circ$: 0.5 N m
$U = -mB \cos \theta$	J	$U(90^\circ) = 0$	$W = mB (\cos \theta_0 - \cos \theta)$
$\mu_l = \frac{e}{2m_e} L$	$J\ T^{-1}$	Bohr orbit	$\mu_B = 9.27 \times 10^{-24}$
$H = B_E \cos \delta; V = B_E \sin \delta$	T	Earth	$\tan \delta = V/H$
$\mu_r = 1 + \chi$	—	linear media	$B = \mu_0 \mu_r H$
$\chi = C/T$	—	paramagnet (Curie law)	T doubles, χ halves

N1

C05-N01

Ch 5

Easy

Numerical

BOARD-STYLE PRACTICE

2 marks

A short bar magnet has moment $5\ A\ m^2$. Find B at $0.2\ m$ from its centre on (a) the axis, (b) the equatorial line.

GIVEN $m = 5\ A\ m^2, r = 0.2\ m, \mu_0/4\pi = 10^{-7}$

FIND B_{ax}, B_{eq}

FORMULA $B_{ax} = \frac{\mu_0 2m}{4\pi r^3}; B_{eq} = \frac{\mu_0 m}{4\pi r^3}$

WORKING $r^3 = 8 \times 10^{-3}. B_{ax} = \frac{10^{-7} \times 10}{8 \times 10^{-3}} = 1.25 \times 10^{-4}\ T; B_{eq} = \frac{10^{-7} \times 5}{8 \times 10^{-3}} = 6.25 \times 10^{-5}\ T.$

Answer: $1.25 \times 10^{-4}\ T$ (axis); $6.25 \times 10^{-5}\ T$ (equator)

▲ $B_{ax} = 2B_{eq}$ is a quick check.

N2

C05-N02

Ch 5

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

A magnet of moment $2\ A\ m^2$ lies at 30° to a uniform field of $0.5\ T$. Find the torque. How much work is needed to turn it from 0° to 60° ?

GIVEN $m = 2, B = 0.5\ T, \theta = 30^\circ$

FIND τ, W

FORMULA $\tau = mB \sin \theta; W = mB (\cos \theta_1 - \cos \theta_2)$

WORKING $\tau = 2 \times 0.5 \times \sin 30^\circ = 0.5\ N\ m. W = 2 \times 0.5 (\cos 0^\circ - \cos 60^\circ) = 1 \times 0.5 = 0.5\ J.$

Answer: $\tau = 0.5\ N\ m; W = 0.5\ J$

▲ Work equals the increase in U .

N3

C05-N03

Ch 5

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

In the Bohr model the electron orbits at $r = 0.53 \times 10^{-10}\ m$ with $v = 2.19 \times 10^6\ m\ s^{-1}$. Find the equivalent current and the magnetic moment; compare with the Bohr magneton.

GIVEN $r = 5.3 \times 10^{-11}\ m, v = 2.19 \times 10^6\ m/s, e = 1.6 \times 10^{-19}\ C$

FIND I, μ

FORMULA $I = \frac{ev}{2\pi r}, \mu = IA$

WORKING $I = \frac{1.6 \times 10^{-19} \times 2.19 \times 10^6}{2\pi \times 5.3 \times 10^{-11}} = 1.05 \times 10^{-3}\ A; A = \pi r^2 = 8.8 \times 10^{-21}\ m^2; \mu = IA = 9.3 \times 10^{-24}\ J\ T^{-1}.$

Answer: $I \approx 1.05\ mA; \mu \approx 9.3 \times 10^{-24}\ J\ T^{-1} = \mu_B$

▲ Matches $e\hbar/2m_e$.

5.6 More numericals and subjective practice

N4

C05-N04

Ch 5

Medium

Numerical

BOARD-STYLE PRACTICE

2 marks

At a place $H = 3 \times 10^{-5}$ T and the dip is 60° . Find B_E and the vertical component.

GIVEN $H = 3 \times 10^{-5}$ T, $\delta = 60^\circ$

FIND B_E, V

FORMULA $H = B_E \cos \delta; V = B_E \sin \delta = H \tan \delta$

WORKING $B_E = \frac{3 \times 10^{-5}}{\cos 60^\circ} = 6 \times 10^{-5}$ T; $V = 3 \times 10^{-5} \times \tan 60^\circ = 5.2 \times 10^{-5}$ T.

Answer: $B_E = 6 \times 10^{-5}$ T; $V = 5.2 \times 10^{-5}$ T

▲ $\tan 60^\circ = \sqrt{3}$.

N5

C05-N05

Ch 5

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

A paramagnetic sample has $\chi = 2.3 \times 10^{-5}$ at 300 K. Find χ at 600 K. A solenoid with 1000 turns/m carries 1 A and has an iron core of $\mu_r = 1000$; find B inside.

GIVEN $\chi_{300} = 2.3 \times 10^{-5}; n = 1000, I = 1$ A, $\mu_r = 1000$

FIND χ_{600}, B

FORMULA $\chi \propto 1/T; B = \mu_0 \mu_r nI$

WORKING $\chi_{600} = 2.3 \times 10^{-5} \times \frac{300}{600} = 1.15 \times 10^{-5}$. $B = 4\pi \times 10^{-7} \times 1000 \times 1000 \times 1 = 1.26$ T.

Answer: $\chi = 1.15 \times 10^{-5}; B \approx 1.26$ T

▲ Curie law applies only to paramagnets.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C05-S01

Ch 5

Easy

Short answer (2 marks)

BOARD-STYLE PRACTICE

S1. State two differences between a diamagnetic and a paramagnetic substance.

Model answer: Diamagnetic: χ small and negative, weakly repelled by a magnet, no permanent dipoles.
Paramagnetic: χ small and positive, weakly attracted, has permanent dipoles that align partially.

C05-S02

Ch 5

Medium

Short answer (3 marks)

BOARD-STYLE PRACTICE

S2. Why is soft iron preferred for an electromagnet core and steel for a permanent magnet?

Model answer: Soft iron has high permeability, low retentivity and low coercivity, so it is magnetised and demagnetised easily with small loss. Steel has high retentivity and coercivity, so it retains its magnetism.

C05-S03

Ch 5

Medium

Short answer (3 marks)

BOARD-STYLE PRACTICE

S3. A short bar magnet is placed with its axis at 30° to a uniform field of 0.25 T and experiences a torque of 4.5×10^{-2} N m. Find its moment and the work done in rotating it to the equilibrium position.

Model answer: $m = \frac{\tau}{B \sin \theta} = \frac{0.045}{0.25 \times 0.5} = 0.36$ A m².
 $W = mB (\cos 0 - \cos 30^\circ) = 0.36 \times 0.25 \times (1 - 0.866) = 0.012$ J.

C05-CB01

Ch 5

Medium

Case-based (4 marks)

BOARD-STYLE PRACTICE

Case. The hysteresis loop of a soft magnetic material has a very narrow area. (i) Is it suitable for a transformer core? (ii) What does the loop area represent? (iii) Name one material for permanent magnets. (iv) What happens above the Curie temperature?

Model answer: (i) Yes — a very narrow loop means little energy loss per cycle, which is what a transformer core needs. (ii) Energy lost per unit volume per cycle. (iii) Alnico or steel. (iv) It becomes paramagnetic.

5.7 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C05-M01 Ch 5 Easy Conceptual BOARD-STYLE PRACTICE

1. Magnetic field lines

- (a) start at N and end at S only
(b) are closed curves
(c) can intersect
(d) are always straight

Ans: (b) $\nabla \cdot \mathbf{B} = 0$: lines form closed loops.

C05-M02 Ch 5 Medium Numerical

BOARD-STYLE PRACTICE

2. For a short magnet the ratio of axial to equatorial field at the same distance is

- (a) 1:2
(b) 2:1
(c) 1:1
(d) 4:1

Ans: (b) $B_{ax} = 2B_{eq}$.

C05-M03 Ch 5 Medium Conceptual

BOARD-STYLE PRACTICE

3. The SI unit of magnetic dipole moment is

- (a) A m
(b) A m²
(c) T m
(d) N m²

Ans: (b) $m = IA$.

C05-M04 Ch 5 Medium Numerical

BOARD-STYLE PRACTICE

4. A magnet of moment m in field B has potential energy $-mB$ when the angle is

- (a) 0°
(b) 90°
(c) 180°
(d) 45°

Ans: (a) $U = -mB \cos \theta$.

C05-M05 Ch 5 Medium Application

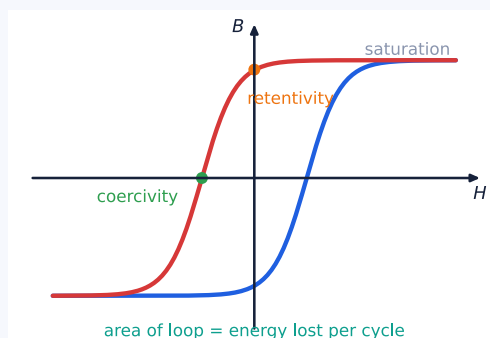
BOARD-STYLE PRACTICE

5. At the magnetic equator the dip angle is

- (a) 0°
(b) 45°
(c) 90°
(d) 180°

Ans: (a) B_E is horizontal at the equator.

C05-M06 Ch 5 Medium Graph BOARD-STYLE PRACTICE

6. In the hysteresis loop shown, the intercept on the B -axis at $H = 0$ is called

- (a) coercivity
(b) retentivity
(c) susceptibility
(d) permeability

Ans: (b) B remaining at $H = 0$ is retentivity.

C05-M07 Ch 5 Medium Application

BOARD-STYLE PRACTICE

7. Which material is best for the core of a transformer?

- (a) steel
(b) alnico
(c) soft iron
(d) copper

Ans: (c) Soft iron: low retentivity, narrow loop.

C05-M08 Ch 5 Hard Numerical BOARD-STYLE PRACTICE

8. A paramagnetic substance has $\chi = 4 \times 10^{-5}$ at 250 K. At 500 K its χ is

- (a) 8×10^{-5}
(b) 4×10^{-5}
(c) 2×10^{-5}
(d) 1×10^{-5}

Ans: (c) $\chi \propto 1/T$.

C05-AR01 Ch 5 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: A magnet cannot be broken into separate poles.
R: Magnetic field lines are closed loops.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a
(b) b
(c) c
(d) d

Ans: (a) No monopoles because $\mathbf{B}$ lines are closed.

C05-AR02 Ch 5 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: A bar magnet placed in a uniform field experiences no net force.

R: The forces on its two poles are equal and opposite.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a
(b) b
(c) c
(d) d

Ans: (a) Couple, not net force.

★ EXAMINER-STYLE REMINDERS

- Draw the field lines with arrows and a labelled magnet.
- State 'short magnet, $r \gg l$ ' before quoting dipole formulas.
- Name the property (retentivity/coercivity) when choosing materials.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are BOARD-STYLE PRACTICE; none is claimed to be a past board question.

R Chapter 5 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Bar-magnet field lines
2. Axial and equatorial points of a dipole
3. Dip, H and V components of the Earth's field
4. Hysteresis loop with retentivity and coercivity
5. Domain diagrams for dia/para/ferro

★ 7-DAY REVISION PLAN

Day 1	Dipole and field lines
Day 2	Axial / equatorial fields
Day 3	Torque and energy
Day 4	Earth's field
Day 5	Materials
Day 6	Hysteresis; N1–N5
Day 7	MCQs; redraw diagrams

Chapter 5 at a glance — Magnetism and Matter

Dipole

$m = NIA$; closed field lines; no monopoles.

Fields

$$B_{ax} = 2B_{eq} = \mu_0 2m/4\pi r^3.$$

Torque

$$\tau = mB \sin \theta; U = -mB \cos \theta.$$

Earth

$$D, \delta, H = B_E \cos \delta.$$

Materials

dia $\chi < 0$; para $\chi > 0$; ferro $\chi \gg 1$.

Hysteresis

retentivity, coercivity, loop area.

Formula map

$$\frac{m = NIA}{\text{moment}}$$

$$\frac{\mu_0 2m}{4\pi r^3}$$

axial

$$\frac{\mu_0 m}{4\pi r^3}$$

equatorial

$$\frac{mB \sin \theta}{\text{torque}}$$

$$\frac{-mB \cos \theta}{\text{energy}}$$

$$\frac{e\hbar}{2m_e}$$

Bohr magneton

$$\frac{B_E \cos \delta}{H}$$

$$\frac{1 + \chi}{\mu_r}$$

$$\frac{C/T}{\text{Curie law}}$$

! COMMON MISTAKES

- Cutting a magnet to isolate poles.
- Axial vs equatorial factor of 2.
- Wrong sign of U at $\theta = 180^\circ$.
- Using steel for a transformer core.
- Forgetting dia-magnetism is universal.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can explain the equivalent solenoid
- ☐ I can derive axial and equatorial fields
- ☐ I can derive τ and U of a dipole
- ☐ I know the Earth's magnetic elements
- ☐ I can compare dia/para/ferro
- ☐ I can read a hysteresis loop
- ☐ I solved N1–N5

✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

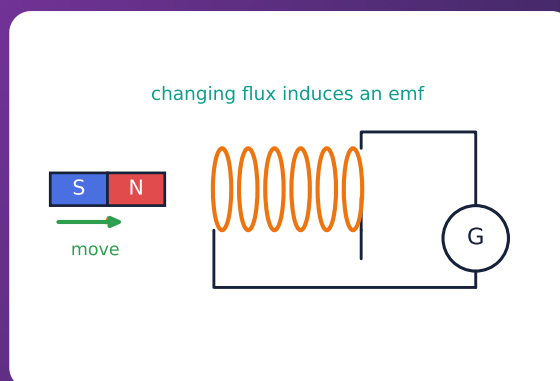
1. Unit of magnetic moment? | **Ans:** A m^2
2. B_{ax}/B_{eq} ? | **Ans:** 2
3. Dip at the poles? | **Ans:** 90°
4. Material for a permanent magnet? | **Ans:** Steel / alnico
5. Loop area means? | **Ans:** Energy loss per cycle

UNIT IV · ELECTROMAGNETIC INDUCTION AND ALTERNATING CURRENTS

6

Electromagnetic Induction

Chapters 4–5 showed that currents create fields. Faraday's discovery is the reverse: a *changing* magnetic field creates an electric field, an emf. This is how power stations, transformers and wireless chargers work.



Chapter roadmap

Flux

Faraday & Lenz

Motional emf

Inductance

Eddy currents

✓ LEARNING OUTCOMES

- State Faraday's laws and Lenz's law and apply them to find magnitude and direction of induced emf
- Derive motional emf $\varepsilon = Blv$ and relate the mechanical power to electrical power
- Define self and mutual inductance; derive L of a long solenoid and the energy $\frac{1}{2}LI^2$
- Explain eddy currents, their uses and ways of reducing them
- Solve numericals on changing flux, motional emf and inductance

I PREREQUISITES TO REVISE

- Magnetic force on a moving charge (Chapter 4)
- Magnetic flux and area vector
- Derivatives as rates of change
- Energy and power

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: Faraday's laws; induced emf and current; Lenz's law; eddy currents; self and mutual induction.

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (Lenz direction, units of L); 2-mark (motional emf, eddy currents); 3-mark numericals (induced emf, M , L); 5-mark derivation of motional emf or L of a solenoid; case-based on a moving rod on rails. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Master equations

F FARADAY + LENZ

$\varepsilon = -N \frac{d\Phi_B}{dt}$, with $\Phi_B = BA \cos \theta$. The minus sign is Lenz's law: the induced current opposes the *change* in flux.

★ THREE WAYS TO CHANGE FLUX

Change B (moving magnet, varying current); change A (moving rod, loop shrinking); change θ (rotating coil — leads to the AC generator in Chapter 7).

6.1 Faraday's laws and Lenz's law

I MAGNETIC FLUX AND FARADAY'S LAWS

Magnetic flux $\Phi_B = \mathbf{B} \cdot \mathbf{A} = BA \cos \theta$ (weber; $1 \text{ Wb} = 1 \text{ T m}^2$). **First law:** an emf is induced whenever the flux through a circuit changes. **Second law:** the magnitude of the induced emf equals the rate of change of flux (for N turns, of flux linkage):

$$\varepsilon = -N \frac{d\Phi_B}{dt}, \quad I = \frac{|\varepsilon|}{R}, \quad Q = \frac{N\Delta\Phi_B}{R}$$

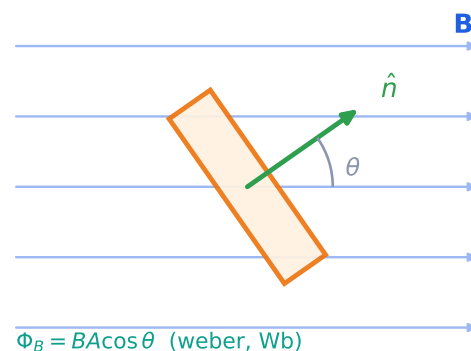
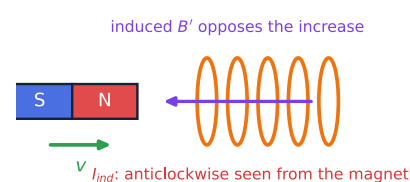


Fig. 6.1 Flux through a loop: $\Phi_B = BA \cos \theta$.

I LENZ'S LAW

The direction of the induced current is such that it **opposes** the change in flux that produced it. It is a statement of conservation of energy: if the induced current aided the change, we would get energy for nothing. Approaching N pole $\rightarrow$ the near face of the coil becomes N (repels); receding N pole $\rightarrow$ near face becomes S (attracts).



Lenz: induced current opposes the change that causes it

Fig. 6.2 Lenz's law for an approaching magnet.

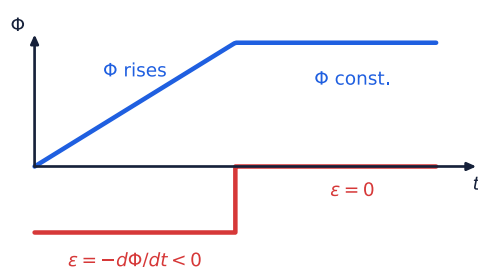


Fig. 6.3 Induced emf is non-zero only while the flux changes.

! COMMON MISTAKES

- Saying Lenz's law says the induced field is *opposite* to the original field — it opposes the *change*.
- Using θ between $\mathbf{B}$ and the plane instead of the normal.
- Forgetting N for coils.

6.2 Motional emf

Motional emf of a rod moving in a uniform field

5 MARKS

Principle Faraday's law applied to a loop whose area changes; equivalently the Lorentz force on free charges in the rod.

Assumptions Rod of length l slides with speed v on rails in a uniform $\mathbf{B}$ perpendicular to the plane.

- In time dt the rod moves $dx = v dt$, so the area of the loop grows by $dA = l v dt$.
- Flux $\Phi_B = B l x$, so $|\varepsilon| = \frac{d\Phi_B}{dt} = B l \frac{dx}{dt} = B l v$.
- Check by Lorentz force: on a free charge q in the rod, $F = qvB$; work per unit charge over length l is vBl . Same result.
- Current $I = \frac{Blv}{R}$; direction by Lenz's law (here the loop flux increases, so the induced current opposes it).

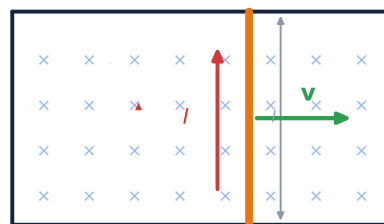
FINAL RESULT

$$\varepsilon = Blv, \quad I = \frac{Blv}{R}$$

Limits: uniform $\mathbf{B} \perp$ plane of motion; $\mathbf{v} \perp$ rod. General: $\varepsilon = \mathbf{l} \cdot (\mathbf{v} \times \mathbf{B})$.

How to write it in the exam: Draw the rails, mark $\mathbf{B}$, $\mathbf{v}$, I ; show both derivations briefly.

$$\varepsilon = Blv$$



$\mathbf{B}$ into the page

Fig. 6.4 Rod moving on rails in a field into the page.

F FORCE, POWER AND ENERGY CONSERVATION

$$F = BlI = \frac{B^2 l^2 v}{R}, \quad P_{\text{mech}} = Fv = \frac{B^2 l^2 v^2}{R} = I^2 R = P_{\text{elec}}$$

The magnetic force on the rod opposes its motion (Lenz); the external agent must supply F to keep v constant. The mechanical work done is exactly the heat produced — energy is conserved.

★ QUICK CHECKS

- If v doubles, ε doubles but P becomes four times.
- If the rod is released with no external force it slows down (induced force opposes motion).
- No emf if the rod moves *along* $\mathbf{B}$.

~ DIRECTION BY FLEMING'S RIGHT-HAND RULE

Stretch thumb, forefinger and middle finger mutually perpendicular: forefinger along $\mathbf{B}$, thumb along the motion $\mathbf{v}$, middle finger gives the direction of the induced current in the conductor. It is Lenz's law in a convenient form for a moving conductor.

✓ CHECKPOINT

A 1 m rod moves at 2 m/s along rails in a 0.5 T field. $\text{emf} = ?$ 1 V. If the loop resistance is 0.5Ω , power? $I = 2$ A, $P = I^2 R = 2$ W.

6.3 Self and mutual induction

I MUTUAL INDUCTANCE

Changing current I_1 in one coil changes the flux linked with a neighbouring coil and induces an emf there:

$$\Phi_2 = MI_1, \quad \varepsilon_2 = -M \frac{dI_1}{dt}$$

M (henry, H = Wb A⁻¹ = V s A⁻¹) depends on geometry, number of turns, separation, orientation and the core material. $M_{12} = M_{21}$.

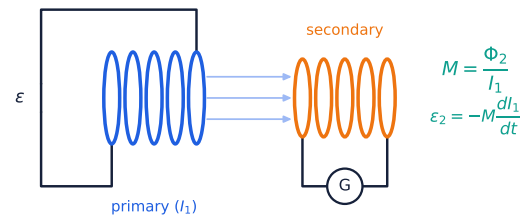


Fig. 6.5 Two coaxial coils.

Mutual inductance of two coaxial solenoids

3 MARKS

Principle Flux of the long inner solenoid linked with the outer coil.

Assumptions Inner solenoid S_1 : n_1 turns/m, radius r_1 ; outer S_2 : N_2 turns total over length l , wound over S_1 .

- ① Field inside S_1 : $B_1 = \mu_0 n_1 I_1$.
- ② Flux through each turn of S_2 is $B_1 \pi r_1^2$; for $N_2 = n_2 l$ turns: $\Phi_2 = \mu_0 n_1 I_1 \pi r_1^2 n_2 l$.
- ③ So $M = \frac{\Phi_2}{I_1} = \mu_0 n_1 n_2 \pi r_1^2 l$.

FINAL RESULT

$$M = \mu_0 n_1 n_2 \pi r_1^2 l$$

Limits: ideal long solenoids, tight coupling. With a core of permeability μ_r multiply by μ_r .

How to write it in the exam: State assumptions and the flux per turn.

I SELF-INDUCTANCE AND ENERGY

A changing current in a coil induces an emf in the same coil which opposes the change (back emf):

$$\Phi = LI, \quad \varepsilon = -L \frac{dI}{dt}. \quad L \text{ is the electrical inertia of a circuit.}$$

Self-inductance of a long solenoid and the stored energy

5 MARKS

Principle $\Phi = LI$; work done against back emf is stored as magnetic energy.

Assumptions Solenoid of length l , area A , N turns, $n = N/l$.

- ① $B = \mu_0 n I$; flux linkage
 $N\Phi = N(BA) = \mu_0 n^2 A l I$, so
 $L = \mu_0 n^2 A l = \frac{\mu_0 N^2 A}{l}$.
- ② Work to raise the current:
 $dW = |\varepsilon| dI = L I dI \Rightarrow W = \frac{1}{2} L I^2$.
- ③ Energy density $u = \frac{W}{A l} = \frac{B^2}{2\mu_0}$.

FINAL RESULT

$$L = \frac{\mu_0 N^2 A}{l}, \quad U = \frac{1}{2} L I^2, \quad u = \frac{B^2}{2\mu_0}$$

Limits: with a core: $\mu_0 \rightarrow \mu_0 \mu_r$.

How to write it in the exam: Show $dW = L I dI$ and integrate.

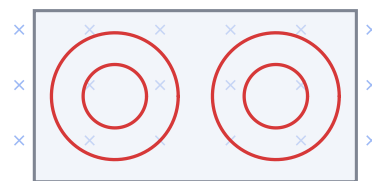
Mechanics	Electromagnetism
Mass m (inertia)	Self-inductance L
Velocity v	Current I
Kinetic energy $\frac{1}{2}mv^2$	Magnetic energy $\frac{1}{2}LI^2$
Force $F = m dv/dt$	emf $\varepsilon = L dI/dt$

6.4 Eddy currents and applications

I EDDY CURRENTS

When a bulk conductor (a metal plate) experiences a changing flux, circulating currents — **eddy currents** — are induced. By Lenz's law they oppose the change that produces them (a moving plate is braked) and they heat the metal ($I^2 R$).

B into page, plate pulled out



eddy currents in a moving plate: opposing, heating

Fig. 6.6 Eddy currents in a plate leaving a magnetic field.

Useful	Harmful
Magnetic braking in trains	Heating and energy loss in transformer cores and motors
Induction furnace (melting metals)	Reduced efficiency
Electromagnetic damping in galvanometers and balances	—
Induction cooktops	—

★ REDUCING EDDY CURRENTS

- Use laminated cores: thin insulated sheets raise the resistance of the eddy path.
- Cut slots in the plate.
- Use ferrites (high resistivity).

~ WHY A MAGNET FALLS SLOWLY THROUGH A COPPER TUBE

The magnet's motion changes flux through each ring of the tube; induced currents create a field opposing the motion (Lenz), so the magnet reaches a small terminal speed. Energy lost to heat comes from gravitational potential energy.

F ORDER-OF-MAGNITUDE FEEL

A neodymium magnet dropped through a 1-m copper pipe takes several seconds instead of 0.45 s (free fall). Induction cooktops run at 20–100 kHz so that eddy-current heating happens in the pan itself.

★ FARADAY, LENZ AND ENERGY — ONE CHAIN

Change in flux → emf (Faraday) → current opposing the change (Lenz) → force opposing the motion → mechanical work converted to heat. Each arrow is a statement of energy conservation.

✓ CHECKPOINT

Predict: a copper plate with slots swings through a magnet gap. Does it stop sooner or later than a solid plate? *Later — slots cut the eddy loops, so the braking force is smaller.*

6.5 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$\Phi_B = BA \cos \theta$	Wb	θ from the normal	$1 \text{ Wb} = 1 \text{ T m}^2$
$\varepsilon = -N \frac{d\Phi}{dt}$	V	Faraday + Lenz	$N = 200, A = 0.05, \Delta B / \Delta t = 6 : 60 \text{ V}$
$Q = \frac{N\Delta\Phi}{R}$	C	independent of time	$N = 10, \Delta\Phi = 0.04, R = 10 : 0.04 \text{ C}$
$\varepsilon = Blv$	V	$\mathbf{v} \perp \mathbf{B} \perp \mathbf{l}$	$l = 0.5, v = 4, B = 0.3 : 0.6 \text{ V}$
$P = \frac{B^2 l^2 v^2}{R}$	W	constant v	$= I^2 R$
$\varepsilon_2 = -M \frac{dI_1}{dt}$	V	mutual	M in henry
$L = \frac{\mu_0 N^2 A}{l}$	H	long solenoid	$N = 500, l = 0.25, A = 4 \text{ cm}^2 : 0.50 \text{ mH}$
$U = \frac{1}{2} LI^2$	J	magnetic energy	$u = B^2 / 2\mu_0$
$M = \mu_0 n_1 n_2 \pi r_1^2 l$	H	coaxial solenoids	

N1

C06-N01

Ch 6

Easy

Numerical

BOARD-STYLE PRACTICE

2 marks

A coil of 200 turns and area 0.05 m^2 has its plane perpendicular to a field that rises uniformly from 0.2 T to 0.8 T in 0.1 s . Find the induced emf.

GIVEN $N = 200, A = 0.05 \text{ m}^2, \Delta B = 0.6 \text{ T}, \Delta t = 0.1 \text{ s}, \theta = 0^\circ$

FIND ε

FORMULA $|\varepsilon| = N \frac{A \Delta B}{\Delta t}$

WORKING $\varepsilon = 200 \times \frac{0.05 \times 0.6}{0.1} = 60 \text{ V}.$

Answer: 60 V

▲ Flux per turn is BA ; multiply by N .

N2

C06-N02

Ch 6

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

A rod of length 0.5 m moves at 4 m s^{-1} on rails in a perpendicular field of 0.3 T . The circuit resistance is 0.2Ω . Find the emf, current, the force needed to keep v constant and the power supplied.

GIVEN $l = 0.5 \text{ m}, v = 4 \text{ m/s}, B = 0.3 \text{ T}, R = 0.2 \Omega$

FIND ε, I, F, P

FORMULA $\varepsilon = Blv, I = \varepsilon/R, F = BIl, P = Fv$

WORKING $\varepsilon = 0.3 \times 0.5 \times 4 = 0.6 \text{ V}; I = 0.6/0.2 = 3 \text{ A}; F = 0.3 \times 3 \times 0.5 = 0.45 \text{ N}; P = Fv = 0.45 \times 4 = 1.8 \text{ W}$
(check: $I^2 R = 9 \times 0.2 = 1.8 \text{ W}$).

Answer: 0.6 V; 3 A; 0.45 N; 1.8 W

▲ Mechanical power equals electrical heating power.

N3

C06-N03

Ch 6

Medium

Numerical

BOARD-STYLE PRACTICE

2 marks

A current of 4 A in a primary coil produces a flux of $2 \times 10^{-3} \text{ Wb}$ through a secondary coil. Find M . If the primary current changes at 10 A s^{-1} , find the emf induced.

GIVEN $I_1 = 4 \text{ A}, \Phi_2 = 2 \times 10^{-3} \text{ Wb}, dI_1/dt = 10 \text{ A/s}$

FIND M, ε_2

FORMULA $M = \Phi_2/I_1, \varepsilon_2 = M dI_1/dt$

WORKING $M = \frac{2 \times 10^{-3}}{4} = 5 \times 10^{-4} \text{ H}; \varepsilon_2 = 5 \times 10^{-4} \times 10 = 5 \times 10^{-3} \text{ V}.$

Answer: $M = 0.5 \text{ mH}; \varepsilon_2 = 5 \text{ mV}$

▲ Magnitude only; sign by Lenz.

6.6 More numericals and subjective practice

N4 C06-N04 Ch 6 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A solenoid has 500 turns, length 25 cm and cross-section 4 cm^2 (air core). Find L and the energy stored when it carries 2 A.

GIVEN $N = 500, l = 0.25 \text{ m}, A = 4 \times 10^{-4} \text{ m}^2, I = 2 \text{ A}$

FIND L, U

FORMULA $L = \frac{\mu_0 N^2 A}{l}, U = \frac{1}{2} LI^2$

WORKING $L = \frac{4\pi \times 10^{-7} \times 500^2 \times 4 \times 10^{-4}}{0.25} = 5.03 \times 10^{-4} \text{ H}; U = \frac{1}{2} \times 5.03 \times 10^{-4} \times 2^2 = 1.0 \times 10^{-3} \text{ J}.$

Answer: $L \approx 0.50 \text{ mH}; U \approx 1.0 \text{ mJ}$

▲ Convert cm^2 to m^2 .

N5 C06-N05 Ch 6 Medium Numerical BOARD-STYLE PRACTICE 2 marks

A 10-turn coil of resistance 10Ω sees the flux through each turn fall from 0.05 Wb to 0.01 Wb . Find the charge that flows through it.

GIVEN $N = 10, R = 10 \Omega, \Delta\Phi = 0.04 \text{ Wb}$

FIND Q

FORMULA $Q = I\Delta t = \frac{N\Delta\Phi}{R}$

WORKING $Q = \frac{10 \times 0.04}{10} = 0.04 \text{ C}.$

Answer: 0.04 C

▲ Independent of how fast the flux changes.

N6 C06-N06 Ch 6 Easy Numerical BOARD-STYLE PRACTICE 2 marks

A circular loop of radius 10 cm lies perpendicular to a field increasing at 0.02 T s^{-1} . Find the induced emf.

GIVEN $r = 0.1 \text{ m}, dB/dt = 0.02 \text{ T/s}$

FIND ε

FORMULA $\varepsilon = A \frac{dB}{dt}$

WORKING $A = \pi (0.1)^2 = 0.0314 \text{ m}^2; \varepsilon = 0.0314 \times 0.02 = 6.3 \times 10^{-4} \text{ V}.$

Answer: 0.63 mV

▲ Direction: opposes the increase (anticlockwise for $\mathbf{B}$ into page).

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C06-S01 Ch 6 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. State Lenz's law. How does it illustrate conservation of energy?

Model answer: The induced current flows so as to oppose the change in flux that caused it. If it aided the change, flux and current would grow without external work — creating energy from nothing. So mechanical work is needed and is converted to electrical energy.

C06-S02 Ch 6 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. Why is a laminated core used in a transformer?

Model answer: Thin insulated sheets break up the paths of eddy currents, increasing the resistance and reducing $I^2 R$ heat loss, so the core stays cool and efficient.

C06-S03 Ch 6 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S3. A magnet is dropped through a long copper tube and through a plastic tube of equal size. Compare the fall times and explain.

Model answer: It takes longer in the copper tube: eddy currents in the copper oppose the motion (Lenz's law), so the magnet reaches a low terminal speed. Plastic does not conduct, so no opposition and the fall is nearly free.

C06-CB01 Ch 6 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. A rod of length 0.4 m moves at 5 m/s on rails in a 0.2 T field into the page; $R = 0.1 \Omega$. (i) Find ε . (ii) Find I . (iii) Force needed. (iv) State the direction of the induced current.

Model answer: (i) $\varepsilon = 0.2 \times 0.4 \times 5 = 0.4 \text{ V}$. (ii) $I = 0.4/0.1 = 4 \text{ A}$. (iii) $F = BIl = 0.2 \times 4 \times 0.4 = 0.32 \text{ N}$. (iv) Flux into the page increases, so I is anticlockwise (producing a field out of the page).

6.7 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C06-M01 Ch 6 Easy Conceptual BOARD-STYLE PRACTICE

1. The SI unit of magnetic flux is

- (a) tesla (b) weber
(c) henry (d) gauss

Ans: (b) $1 \text{ Wb} = 1 \text{ T m}^2$.

C06-M02 Ch 6 Medium Numerical

BOARD-STYLE PRACTICE

2. A flux of $5 \times 10^{-3} \text{ Wb}$ through a single turn falls to zero in 0.01 s. The induced emf is

- (a) 0.5 V (b) 5 V
(c) 50 V (d) 0.05 V

Ans: (a) $\varepsilon = \Delta\Phi/\Delta t = 0.5 \text{ V}$.

C06-M03 Ch 6 Medium Conceptual

BOARD-STYLE PRACTICE

3. The induced emf in a coil does **not** depend on

- (a) rate of change of flux (b) number of turns
(c) resistance of the coil (d) area of the coil

Ans: (c) Resistance affects current, not emf.

C06-M04 Ch 6 Medium Numerical

BOARD-STYLE PRACTICE

4. A 2 m rod moves at 3 m/s perpendicular to a 0.5 T field. The emf between its ends is

- (a) 1 V (b) 3 V
(c) 0.5 V (d) 6 V

Ans: (b) $\mathcal{E} = Blv = 0.5 \times 2 \times 3 = 3 \text{ V}$.

C06-M05 Ch 6 Medium Application

BOARD-STYLE PRACTICE

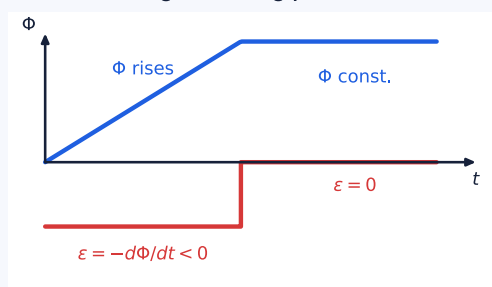
5. Eddy currents are reduced in a transformer core by

- (a) using a solid core (b) laminating it
(c) raising the frequency (d) using steel

Ans: (b) Lamination increases the resistance of the eddy path.

C06-M06 Ch 6 Medium Graph BOARD-STYLE PRACTICE

6. The graph shows flux against time rising linearly then constant. During the rising part the emf is



- (a) zero (b) constant, non-zero
(c) increasing (d) decreasing

Ans: (b) Constant slope gives constant emf.

C06-M07 Ch 6 Medium Numerical

BOARD-STYLE PRACTICE

7. The inductance of a coil is 2 H. The emf induced when the current changes at 3 A/s is

- (a) 1.5 V (b) 6 V
(c) 0.67 V (d) 5 V

Ans: (b) $\mathcal{E} = L dI/dt = 6 \text{ V}$.

C06-M08 Ch 6 Hard Application BOARD-STYLE PRACTICE

8. A square loop is pulled out of a magnetic field at constant speed. The power required is proportional to

- (a) v (b) v^2
(c) $1/v$ (d) v^0

Ans: (b) $P = B^2 l^2 v^2 / R$.

C06-AR01 Ch 6 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. **A:** A falling magnet in a copper tube experiences a retarding force.**R:** Induced currents oppose the cause producing them.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Lenz's law.

C06-AR02 Ch 6 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. **A:** Self-induction is called the inertia of electric current.**R:** The back emf opposes any change of current.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Opposition to change is like inertia.

★ EXAMINER-STYLE REMINDERS

- Always write the sign/direction using Lenz's law, not only the magnitude.
- State assumptions: uniform $\mathbf{B}$, $\mathbf{v} \perp \mathbf{B}$.
- Quote the unit of L and M : henry.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are **BOARD-STYLE PRACTICE**; none is claimed to be a past board question.

R Chapter 6 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Magnet approaching a coil with Lenz direction
2. Rod on rails with $\mathbf{B}$, $\mathbf{v}$, I , F
3. Two coaxial coils
4. $\Phi-t$ and $\varepsilon-t$ graphs
5. Eddy currents in a plate

★ 7-DAY REVISION PLAN

Day 1	Flux, Faraday
Day 2	Lenz's law
Day 3	Motional emf
Day 4	Mutual inductance
Day 5	Self-inductance
Day 6	N1–N6
Day 7	MCQs; redraw diagrams

Chapter 6 at a glance — Electromagnetic Induction

Flux

$$\Phi = BA \cos \theta.$$

Faraday

$$\varepsilon = -N d\Phi/dt.$$

Lenz

opposes the *change*; energy conservation.

Motional

$$\varepsilon = Blv; P = B^2 l^2 v^2 / R.$$

Inductance

$$L = \mu_0 N^2 A / l; U = \frac{1}{2} LI^2.$$

Eddy

braking, furnace; laminate to reduce.

Formula map

$$-N \frac{d\Phi}{dt}$$

emf

$$Blv$$

motional

$$\frac{B^2 l^2 v}{R}$$

force

$$\frac{N \Delta \Phi}{R}$$

charge

$$-M \frac{dI}{dt}$$

mutual

$$\frac{\mu_0 N^2 A}{l}$$

$$\frac{1}{2} LI^2$$

energy

$$\frac{B^2}{2\mu_0}$$

energy density

$$\frac{\mu_0 n_1 n_2 \pi r^2 l}{M}$$

! COMMON MISTAKES

- Dropping N in ε .
- Wrong direction of induced current.
- Using the angle with the plane for flux.
- Thinking resistance affects the emf.
- Forgetting that no flux change means no emf.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can state Faraday and Lenz
- ☐ I can derive $\varepsilon = Blv$
- ☐ I can derive L of a solenoid
- ☐ I know M of coaxial solenoids
- ☐ I can explain eddy currents
- ☐ I solved N1–N6
- ☐ I did the MCQs

✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

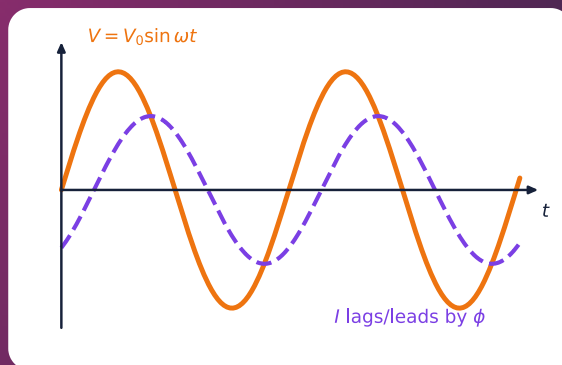
1. Unit of L ? | **Ans:** Henry
2. emf if flux is constant? | **Ans:** Zero
3. Lenz's law and which conservation? | **Ans:** Energy
4. Energy density in a field? | **Ans:** $B^2/2\mu_0$
5. Reduce eddy currents? | **Ans:** Laminate

UNIT IV · ELECTROMAGNETIC INDUCTION AND ALTERNATING CURRENTS

7

Alternating Current

Almost all grid electricity is alternating because transformers can step it up and down. Every ac circuit is solved with one picture: a phasor diagram.



Chapter roadmap

ac, rms

R, L, C in ac

Series LCR

Power

Generator & transformer

✓ LEARNING OUTCOMES

- Describe alternating voltage and current; define peak and rms values and derive $I_{rms} = I_0/\sqrt{2}$
- Derive the response of a resistor, inductor and capacitor to an ac source; define reactance
- Analyse a series LCR circuit with phasors: impedance, phase, resonance and Q
- Find average power, power factor and wattless current
- Explain LC oscillations, the ac generator and the transformer

I PREREQUISITES TO REVISE

- Faraday's law and Lenz's law (Chapter 6)
- Sine and cosine, vectors as phasors
- Energy in L and C
- Series and parallel resistors

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: alternating current; peak and rms value of alternating current/voltage; reactance and impedance; LC oscillations (qualitative treatment only); LCR series circuit; resonance; power in ac circuits and wattless current; ac generator; transformer.

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (rms, phase, resonance); 2-mark (reactance, power factor); 3-mark numericals (impedance, resonance, transformer); 5-mark derivation of series LCR response or transformer; case-based on a series circuit. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

At a glance

F REACTANCES AND IMPEDANCE

$$X_L = \omega L \text{ (rises with } f\text{); } X_C = \frac{1}{\omega C} \text{ (falls with } f\text{);}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}; \text{ resonance at } \omega_0 = 1/\sqrt{LC}.$$

★ MEMORY AIDS

CIVIL: in a **C** capacitor **I** leads **V**; in an **L** inductor **V** leads **I**. Power $P = V_{rms} I_{rms} \cos \phi$.

7.1 Alternating voltage, rms value and phasors

1 ALTERNATING VOLTAGE AND CURRENT

An ac source gives $v = V_0 \sin \omega t$ with $\omega = 2\pi f$; the current through a load is $i = I_0 \sin (\omega t \pm \phi)$. India's mains: $f = 50$ Hz, $V_{rms} = 220$ V, $V_0 = 311$ V. The average over a cycle is zero, so the useful measure is the **rms** (the dc equivalent giving the same heating):

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707 I_0, \quad V_{rms} = \frac{V_0}{\sqrt{2}}$$

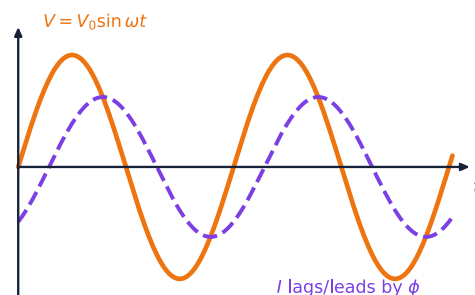


Fig. 7.1 Voltage and current sinusoids with a phase difference.

rms value of an alternating current

2 MARKS

Principle The heating effect depends on i^2 ; I_{rms} is the steady current giving the same mean heat.

Assumptions $i = I_0 \sin \omega t$ flowing in a resistor R .

- ① Mean heat power: $\langle i^2 \rangle R$, and $\langle \sin^2 \omega t \rangle = \frac{1}{2}$ over a cycle, so $\langle i^2 \rangle = \frac{1}{2} I_0^2$.
- ② Equate to $I_{rms}^2 R$: $I_{rms}^2 = \frac{1}{2} I_0^2$.

FINAL RESULT

$$I_{rms} = \frac{I_0}{\sqrt{2}} \quad \left(\text{similarly } V_{rms} = \frac{V_0}{\sqrt{2}} \right)$$

Limits: valid for a sinusoid. A dc meter reads zero; ac meters are calibrated in rms.

How to write it in the exam: Show the $\langle \sin^2 \rangle = \frac{1}{2}$ step (or integrate over 0 to T).

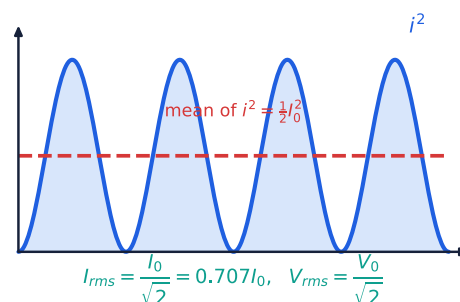


Fig. 7.2 The mean of i^2 over a cycle is $I_0^2/2$.

! COMMON MISTAKES

- Using peak values in $P = VI \cos \phi$ (use rms).
- Believing an ac ammeter reads the average value (zero) — it reads rms.
- Mixing ω and f in X_L and X_C .

7.2 ac through R, L and C

1 PURE R, PURE L, PURE C

Phasors are rotating vectors whose projection on the vertical axis gives the instantaneous value. With source $v = V_0 \sin \omega t$:

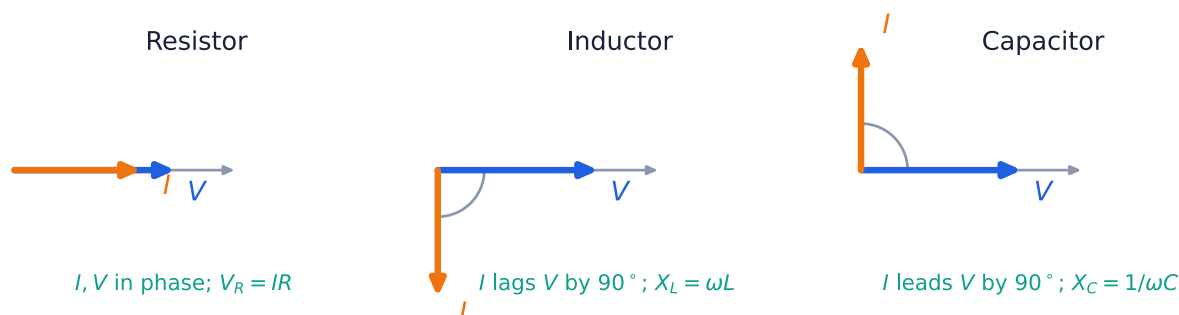


Fig. 7.3 Phasor diagrams for R, L and C.

Element	Relation	Phase of I w.r.t. V	Opposition	Dependence on f
Resistor	$V = IR$	in phase	R	none
Inductor	$V_L = IX_L$	lags by 90°	$X_L = \omega L$	$\propto f$ (passes low f)
Capacitor	$V_C = IX_C$	leads by 90°	$X_C = 1/\omega C$	$\propto 1/f$ (blocks dc)

ac through an inductor

3 MARKS

Principle Back emf $-L di/dt$; Kirchhoff's loop rule for the source and the inductor.

Assumptions Pure inductor (no resistance), $v = V_0 \sin \omega t$.

① Loop rule: $v - L \frac{di}{dt} = 0 \Rightarrow \frac{di}{dt} = \frac{V_0}{L} \sin \omega t$.

② Integrate: $i = -\frac{V_0}{\omega L} \cos \omega t = \frac{V_0}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right)$.

③ So $I_0 = V_0/X_L$ with $X_L = \omega L$, and i lags v by $\pi/2$. Average power over a cycle is zero.

FINAL RESULT

$$I_0 = \frac{V_0}{\omega L}, \quad X_L = \omega L, \quad \text{lag} = 90^\circ$$

Limits: for a capacitor: $q = Cv \Rightarrow i = C dv/dt = \omega C V_0 \cos \omega t$, so $X_C = 1/\omega C$ and i leads by 90° .

How to write it in the exam: State the loop equation and the phase result.

7.3 Series LCR circuit and resonance

Impedance and phase of a series LCR circuit

5 MARKS

Principle The same current flows through R, L, C; add the voltage phasors vectorially.

Assumptions $v = V_0 \sin \omega t$; $i = I_0 \sin (\omega t - \phi)$.

- 1 Voltage phasors: $V_R = IR$ (along I), $V_L = IX_L$ (90° ahead), $V_C = IX_C$ (90° behind).
- 2 Net reactive voltage $V_L - V_C$ is perpendicular to V_R :

$$V^2 = V_R^2 + (V_L - V_C)^2 = I^2 [R^2 + (X_L - X_C)^2]$$
- 3 Hence $Z = \frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$ and

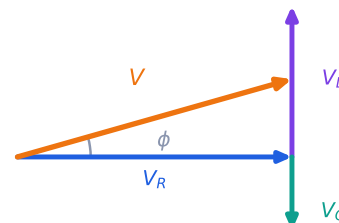
$$\tan \phi = \frac{X_L - X_C}{R}$$

FINAL RESULT

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}, \quad \tan \phi = \frac{\omega L - 1/\omega C}{R}$$

Limits: $X_L > X_C$: inductive (current lags); $X_C > X_L$: capacitive (current leads); $X_L = X_C$: resistive.

How to write it in the exam: Draw the phasor triangle with V_R , $V_L - V_C$ and V .



$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\tan \phi = \frac{X_L - X_C}{R}$$

Fig. 7.4 Phasor diagram (here $X_L > X_C$).

Resonance

3 MARKS

Principle $I_0 = V_0/Z$ is maximum when Z is minimum, i.e. $X_L = X_C$.

Assumptions Series LCR with fixed V_0 ; source frequency varied.

- 1 Set $\omega L = \frac{1}{\omega C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$, $f_0 = \frac{1}{2\pi\sqrt{LC}}$.
- 2 At resonance $Z = R$, $\phi = 0$, $I_{max} = V_0/R$.
- 3 Quality factor $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} = \frac{1}{R} \sqrt{\frac{L}{C}}$; bandwidth
 $\Delta\omega = \frac{R}{L}$, so $Q = \omega_0/\Delta\omega$.

FINAL RESULT

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad I_{max} = \frac{V_0}{R}, \quad Q = \frac{\omega_0 L}{R}$$

Limits: smaller R gives a sharper peak (larger Q): better tuning of radio receivers.

How to write it in the exam: Quote $Z = R$ and $\phi = 0$ at resonance.

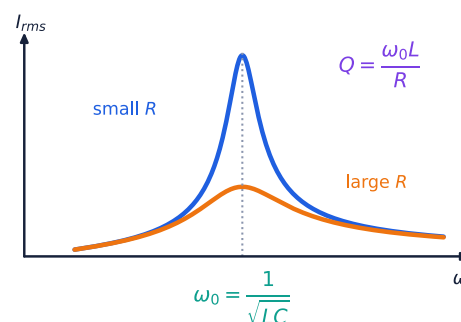


Fig. 7.5 Resonance curves for two values of R .

7.4 Power, LC oscillations, generator and transformer

F POWER IN AN AC CIRCUIT

$$P_{avg} = V_{rms} I_{rms} \cos \phi = I_{rms}^2 R, \quad \cos \phi = \frac{R}{Z}$$

$\cos \phi$ = power factor. For pure R: 1; pure L or C: 0 (**wattless current** — the current flows but no net power is used). At resonance $\cos \phi = 1$.

Average power in an ac circuit

3 MARKS

Principle $p = vi$ averaged over a cycle.

Assumptions $v = V_0 \sin \omega t$, $i = I_0 \sin (\omega t + \phi)$.

- $p = V_0 I_0 \sin \omega t \sin (\omega t + \phi)$, and
 $\sin A \sin B = \frac{1}{2} [\cos (A - B) - \cos (A + B)]$
 gives $p = \frac{1}{2} V_0 I_0 \cos \phi - \frac{1}{2} V_0 I_0 \cos (2\omega t + \phi)$.
- The average of $\cos (2\omega t + \phi)$ over a cycle is zero, so $P = \frac{1}{2} V_0 I_0 \cos \phi = V_{rms} I_{rms} \cos \phi$.

FINAL RESULT

$$P = V_{rms} I_{rms} \cos \phi$$

Limits: $\phi = 90^\circ$ gives $P = 0$.

How to write it in the exam: Show the product-to-sum step.

~ LC OSCILLATIONS (QUALITATIVE)

A charged capacitor discharging through an inductor: energy shuttles between $U_E = q^2/2C$ and $U_B = \frac{1}{2}LI^2$ with natural frequency $\omega_0 = 1/\sqrt{LC}$, the total staying constant (ideal, no resistance) — the electrical analogue of a spring-mass oscillator. With R present the oscillations are damped.

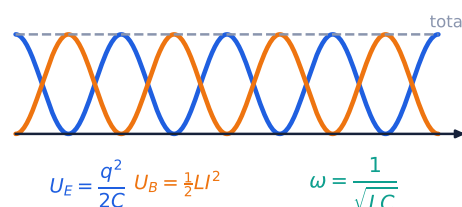


Fig. 7.6 Energy exchange in an ideal LC circuit.

I TRANSFORMER

Two coils on a laminated soft-iron core. Alternating flux links both: for an ideal transformer $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$. Step-up: $N_s > N_p$ (current decreases); step-down: $N_s < N_p$. Efficiency $\eta = \frac{P_{out}}{P_{in}}$. Losses: copper loss ($I^2 R$), eddy currents (laminates), hysteresis (soft iron), flux leakage.

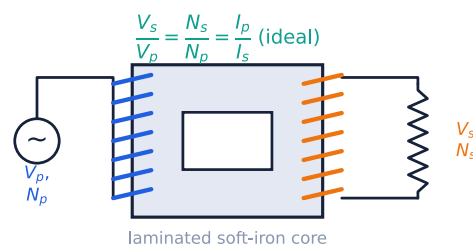


Fig. 7.7 Step-down transformer.

I AC GENERATOR

A coil of N turns, area A rotating at ω in a uniform field B : flux $\Phi = NBA \cos \omega t$, so $\varepsilon = -d\Phi/dt = NBA\omega \sin \omega t$, i.e. $\varepsilon_0 = NBA\omega$. Slip rings and brushes deliver the output.

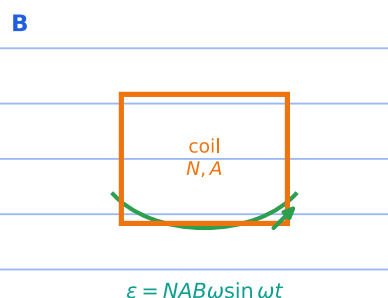


Fig. 7.8 Rotating coil in a uniform field.

7.5 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$I_{rms} = I_0/\sqrt{2}$	A	sinusoid	$V_0 = 311 \text{ V}$: 220 V
$X_L = \omega L = 2\pi fL$	Ω	f in Hz	0.1 H, 50 Hz: 31.4 Ω
$X_C = \frac{1}{2\pi fC}$	Ω	blocks dc	100 μF , 50 Hz: 31.8 Ω
$Z = \sqrt{R^2 + (X_L - X_C)^2}$	Ω	series LCR	
$\tan \phi = \frac{X_L - X_C}{R}$	—	$\phi > 0$: inductive	
$f_0 = \frac{1}{2\pi\sqrt{LC}}$	Hz	resonance	2 mH, 0.5 μF : 5.0 kHz
$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$	—	series	sharpness of resonance
$P = V_{rms}I_{rms} \cos \phi$	W	$\cos \phi = R/Z$	
$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$	—	ideal transformer	

N1 C07-N01 Ch 7 Easy Numerical BOARD-STYLE PRACTICE 2 marks

A voltage is $v = 311 \sin (100\pi t)$ V. Find the rms voltage, frequency and period.

GIVEN $V_0 = 311 \text{ V}$, $\omega = 100\pi \text{ rad/s}$

FIND V_{rms} , f , T

FORMULA $V_{rms} = V_0/\sqrt{2}$, $f = \omega/2\pi$

WORKING $V_{rms} = 311/1.414 = 220 \text{ V}$; $f = 100\pi/2\pi = 50 \text{ Hz}$; $T = 1/f = 0.02 \text{ s}$.

Answer: 220 V; 50 Hz; 20 ms

▲ Read ω directly from the argument of the sine.

N2 C07-N02 Ch 7 Medium Numerical BOARD-STYLE PRACTICE 5 marks

A series circuit has $R = 30 \Omega$, $L = 0.2 \text{ H}$ and $C = 100 \mu\text{F}$ across a 220 V, 50 Hz supply. Find X_L , X_C , Z , the current, the phase angle and the power factor.

GIVEN $R = 30 \Omega$, $L = 0.2 \text{ H}$, $C = 10^{-4} \text{ F}$, $V_{rms} = 220 \text{ V}$, $f = 50 \text{ Hz}$

FIND X_L , X_C , Z , I , ϕ , $\cos \phi$

FORMULA $X_L = 2\pi fL$, $X_C = 1/2\pi fC$, $Z = \sqrt{R^2 + (X_L - X_C)^2}$

WORKING $X_L = 2\pi \times 50 \times 0.2 = 62.8 \Omega$; $X_C = \frac{1}{2\pi \times 50 \times 10^{-4}} = 31.8 \Omega$; $Z = \sqrt{30^2 + 31.0^2} = 43.1 \Omega$;
 $I = 220/43.1 = 5.1 \text{ A}$; $\tan \phi = 31.0/30 \Rightarrow \phi = 45.9^\circ$ (current lags); $\cos \phi = 30/43.1 = 0.70$.

Answer: 62.8 Ω ; 31.8 Ω ; 43.1 Ω ; 5.1 A; 45.9°; 0.70

▲ Inductive circuit: $X_L > X_C$.

N3 C07-N03 Ch 7 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A series LCR circuit has $L = 2 \text{ mH}$, $C = 0.5 \mu\text{F}$, $R = 10 \Omega$. Find the resonant frequency and Q.

GIVEN $L = 2 \times 10^{-3} \text{ H}$, $C = 5 \times 10^{-7} \text{ F}$, $R = 10 \Omega$

FIND f_0 , Q

FORMULA $f_0 = \frac{1}{2\pi\sqrt{LC}}$, $Q = \frac{1}{R}\sqrt{L/C}$

WORKING $LC = 10^{-9}$, $\sqrt{LC} = 3.16 \times 10^{-5} \text{ s}$; $f_0 = \frac{1}{2\pi \times 3.16 \times 10^{-5}} = 5.03 \text{ kHz}$. $Q = \frac{1}{10}\sqrt{\frac{2 \times 10^{-3}}{5 \times 10^{-7}}} = \frac{63.2}{10} = 6.3$.

Answer: $f_0 \approx 5.0 \text{ kHz}$; $Q \approx 6.3$

▲ Q has no unit.

7.6 More numericals and subjective practice

N4 C07-N04 Ch 7 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A transformer has 1000 primary and 250 secondary turns, with 220 V on the primary. The secondary delivers 2 A and the efficiency is 90%. Find V_s and the primary current.

GIVEN $N_p = 1000, N_s = 250, V_p = 220 \text{ V}, I_s = 2 \text{ A}, \eta = 0.9$

FIND V_s, I_p

FORMULA $V_s = V_p N_s / N_p; \eta = \frac{V_s I_s}{V_p I_p}$

WORKING $V_s = 220 \times \frac{250}{1000} = 55 \text{ V}; P_{out} = 55 \times 2 = 110 \text{ W}; P_{in} = 110/0.9 = 122.2 \text{ W}; I_p = 122.2/220 = 0.56 \text{ A}.$

Answer: $V_s = 55 \text{ V}; I_p \approx 0.56 \text{ A}$

▲ For an ideal transformer I_p would be 0.5 A.

N5 C07-N05 Ch 7 Easy Numerical BOARD-STYLE PRACTICE 2 marks

The rms voltage across a circuit is 220 V and the rms current 2 A, lagging by 60° . Find the average power and power factor.

GIVEN $V_{rms} = 220 \text{ V}, I_{rms} = 2 \text{ A}, \phi = 60^\circ$

FIND $P, \cos \phi$

FORMULA $P = V_{rms} I_{rms} \cos \phi$

WORKING $\cos 60^\circ = 0.5; P = 220 \times 2 \times 0.5 = 220 \text{ W}.$

Answer: $P = 220 \text{ W}; \text{power factor } 0.5$

▲ Apparent power is 440 VA.

N6 C07-N06 Ch 7 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A generator coil of 100 turns and area 0.2 m^2 rotates at 50 rev/s in a field of 0.05 T. Find the peak emf and the rms emf.

GIVEN $N = 100, A = 0.2 \text{ m}^2, B = 0.05 \text{ T}, f = 50 \text{ Hz}$

FIND $\varepsilon_0, \varepsilon_{rms}$

FORMULA $\varepsilon_0 = NBA\omega$

WORKING $\omega = 2\pi \times 50 = 314 \text{ rad/s}; \varepsilon_0 = 100 \times 0.05 \times 0.2 \times 314 = 314 \text{ V}; \varepsilon_{rms} = 314/\sqrt{2} = 222 \text{ V}.$

Answer: 314 V; 222 V

▲ Rotation per second means f , then $\omega = 2\pi f$.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C07-S01 Ch 7 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. Why does a capacitor block dc but allow ac?

Model answer: For dc, $\omega = 0$ so $X_C = 1/\omega C \rightarrow \infty$ (no steady current after charging). For ac the capacitor is charged and discharged every cycle, so a current flows with $X_C = 1/\omega C$.

C07-S02 Ch 7 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. What is wattless current? When does an ac circuit have zero average power although the current is large?

Model answer: The current component that does no net work (phase 90° with voltage), $I \sin \phi$. In a pure inductor or capacitor $\phi = 90^\circ$, so $P = V_{rms} I_{rms} \cos 90^\circ = 0$.

C07-S03 Ch 7 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S3. List four energy losses in a real transformer and one remedy for each.

Model answer: Copper loss — thick wire; eddy currents — laminated core; hysteresis — soft iron core; flux leakage — winding coils on top of each other / closed core.

C07-CB01 Ch 7 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. A radio tuner uses $L = 0.5 \text{ mH}$ and a variable capacitor to receive 1000 kHz. (i) Find the capacitance. (ii) What is Z at resonance if $R = 20 \Omega$? (iii) What is the phase angle at resonance? (iv) Why is a small R better?

Model answer: (i) $C = \frac{1}{(2\pi f)^2 L} = \frac{1}{(2\pi \times 10^6)^2 \times 0.5 \times 10^{-3}} = 5.07 \times 10^{-11} \text{ F} = 50.7 \text{ pF}.$ (ii) $Z = R = 20 \Omega.$ (iii) $0^\circ.$ (iv) Larger Q gives a sharper resonance and better station selection.

7.7 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C07-M01 Ch 7 Easy Conceptual

BOARD-STYLE PRACTICE

1. The rms value of an alternating current of peak 10 A is nearly

- (a) 5 A (b) 7.07 A
(c) 14.1 A (d) 10 A

Ans: (b) $10/\sqrt{2} = 7.07$ A.

C07-M02 Ch 7 Medium Numerical

BOARD-STYLE PRACTICE

2. A 0.1 H inductor at 50 Hz has a reactance of about

- (a) 5Ω (b) 15.7Ω
(c) 31.4Ω (d) 314Ω

Ans: (c) $2\pi \times 50 \times 0.1 = 31.4$.

C07-M03 Ch 7 Medium Conceptual

BOARD-STYLE PRACTICE

3. In a purely capacitive circuit the current

- (a) lags V by 90° (b) leads V by 90°
(c) is in phase (d) is zero

Ans: (b) CIVIL: I leads in C .

C07-M04 Ch 7 Medium Numerical

BOARD-STYLE PRACTICE

4. The resonant frequency of $L = 1$ H, $C = 1 \mu\text{F}$ is nearly

- (a) 159 Hz (b) 1 kHz
(c) 16 Hz (d) 1590 Hz

Ans: (a) $f_0 = 1/(2\pi\sqrt{10^{-6}}) = 159$ Hz.

C07-M05 Ch 7 Medium Application

BOARD-STYLE PRACTICE

5. At resonance in a series LCR circuit the

- (a) impedance is maximum (b) current is minimum
(c) $\phi = 0$ (d) $X_L > X_C$

Ans: (c) $Z = R, \phi = 0$.

C07-M06 Ch 7 Medium Graph

BOARD-STYLE PRACTICE

6. In the resonance curves, a sharper peak corresponds to

- (a) larger R (b) smaller R
(c) larger C only (d) dc source

Ans: (b) $Q = \omega_0 L/R$.

C07-M07 Ch 7 Medium Numerical

BOARD-STYLE PRACTICE

7. A transformer steps 220 V to 11 V. The turns ratio N_p/N_s is

- (a) $1/20$ (b) 20
(c) 10 (d) 2

Ans: (b) $220/11 = 20$.

C07-M08 Ch 7 Hard Application

BOARD-STYLE PRACTICE

8. The power factor of a circuit with $R = 3 \Omega$, $X_L = 4 \Omega$ is

- (a) 0.6 (b) 0.8
(c) 0.75 (d) 1

Ans: (a) $\cos \phi = R/Z = 3/5$.

C07-AR01 Ch 7 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: A choke coil is preferred to a resistor to control ac current.

R: A choke has very small average power consumption.

- (a) both true, R explains A; (b) both true, R does not explain A;
(c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Inductor ($\phi \approx 90^\circ$) wastes almost no power.

C07-AR02 Ch 7 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: A transformer works only on ac.

R: It needs a changing flux in the core to induce an emf in the secondary.

- (a) both true, R explains A; (b) both true, R does not explain A;
(c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Faraday's law needs $d\Phi/dt \neq 0$.

★ EXAMINER-STYLE REMINDERS

- Draw the phasor diagram for every LCR derivation.
- Use rms values in power formulas; quote $\cos \phi = R/Z$.
- State $Z = R, \phi = 0$ at resonance.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are **BOARD-STYLE PRACTICE**; none is claimed to be a past board question.

R Chapter 7 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Phasor diagrams for R, L, C
2. Series LCR phasor triangle
3. Resonance curve for two R values
4. Transformer with primary and secondary
5. AC generator coil in B

★ 7-DAY REVISION PLAN

Day 1	rms, phasors
Day 2	R, L, C
Day 3	Series LCR
Day 4	Resonance and Q
Day 5	Power
Day 6	Transformer, generator
Day 7	N1–N6; MCQs

Chapter 7 at a glance — Alternating Current

ac basics

$$I_{rms} = I_0/\sqrt{2}; \text{ mains } 50 \text{ Hz, } 220 \text{ V.}$$

R, L, C

$$\text{phase } 0, -90^\circ, +90^\circ; X_L = \omega L, \\ X_C = 1/\omega C.$$

LCR

$$Z = \sqrt{R^2 + (X_L - X_C)^2}.$$

Resonance

$$\omega_0 = 1/\sqrt{LC}; Q = \omega_0 L/R.$$

Power

$$P = V_{rms} I_{rms} \cos \phi.$$

Devices

$$\text{transformer } V_s/V_p = N_s/N_p; \\ \text{generator } NBA\omega.$$

Formula map

$$\frac{I_0}{\sqrt{2}}$$

$$\frac{\omega L}{X_L}$$

$$\frac{1}{\omega C}$$

$$\frac{\sqrt{R^2 + (X_L - X_C)^2}}{Z}$$

$$\frac{1}{2\pi\sqrt{LC}}$$

$$\frac{\omega_0 L}{R}$$

$$V_{rms} I_{rms} \cos \phi$$

$$\frac{N_s}{N_p}$$

turns ratio

$$NBA\omega$$

peak emf

! COMMON MISTAKES

- Using peak values for power.
- Mixing up which of L and C leads.
- Using f in place of ω .
- Saying Z is minimum at resonance but writing I minimum.
- Ignoring the sign of $X_L - X_C$ for the phase.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can derive $I_{rms} = I_0/\sqrt{2}$
- ☐ I can derive the response of L and C
- ☐ I can derive Z and $\tan \phi$
- ☐ I can find f_0 and Q
- ☐ I can derive $P = VI \cos \phi$
- ☐ I know the transformer ratios and losses
- ☐ I solved N1–N6

✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

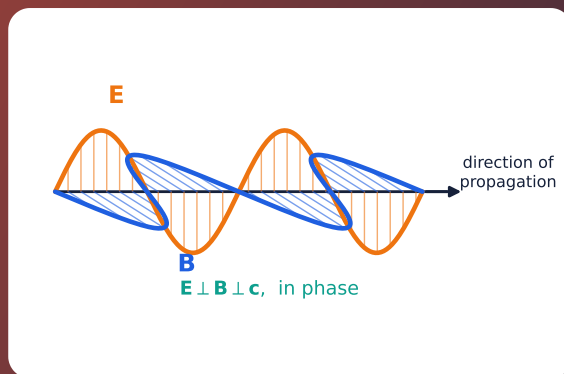
1. V_{rms} for $V_0 = 311$? | **Ans:** 220 V
2. Phase of I in an inductor? | **Ans:** Lags 90°
3. Z at resonance? | **Ans:** R
4. Power factor of a pure capacitor? | **Ans:** 0
5. Why laminate a core? | **Ans:** Reduce eddy losses

UNIT V · ELECTROMAGNETIC WAVES

8

Electromagnetic Waves

Faraday: a changing $\mathbf{B}$ makes an $\mathbf{E}$. Maxwell: a changing $\mathbf{E}$ makes a $\mathbf{B}$. Together they sustain each other in empty space — that is light.



Chapter roadmap

Displacement current

Maxwell's idea

EM waves

Spectrum

Uses

✓ LEARNING OUTCOMES

- Explain displacement current and why Ampere's law needed Maxwell's correction
- State the properties of electromagnetic waves: transverse, $E \perp B \perp \text{propagation}$, $c = E_0/B_0$, no medium needed
- Place the seven regions of the spectrum in order of wavelength and give uses
- Calculate wavelength, frequency, E_0 and B_0 for an EM wave

I PREREQUISITES TO REVISE

- Ampere's law (Chapter 4)
- Faraday's law (Chapter 6)
- Capacitor and electric flux (Chapter 2)
- Wave relation $v = f\lambda$

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: basic idea of displacement current; electromagnetic waves, their characteristics and transverse nature (qualitative ideas only); electromagnetic spectrum (radio waves, microwaves, infrared, visible, ultraviolet, X-rays, gamma rays) including elementary facts about their uses.

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (order of spectrum, $E/B = c$); 2-mark (displacement current, uses of waves); 3-mark numericals (B_0 from E_0 , wavelength); case-based on the spectrum or on a charging capacitor. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Key relations

F SPEED AND RATIO

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m s}^{-1}; \frac{E_0}{B_0} = c; c = f\lambda; \text{ in a medium } v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{n}$$

★ THE ONE-LINE STORY

Changing $\mathbf{E} \Rightarrow$ magnetic field; changing $\mathbf{B} \Rightarrow$ electric field; this self-propagating disturbance moves at c .

8.1 Displacement current and Maxwell's idea

1 WHY AMPERE'S LAW NEEDS A NEW TERM

While a capacitor charges, a conduction current I_c flows in the wires but no charge crosses the gap, yet there is a magnetic field around the gap. Maxwell proposed that a **changing electric flux** acts like a current, the **displacement current**:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}, \quad \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (I_c + I_d)$$

In a charging parallel-plate capacitor $I_d = I_c$ so the current is continuous. I_d is not a flow of charge; it is a changing field.

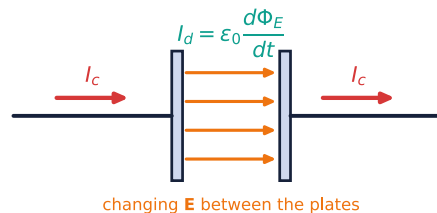


Fig. 8.1 Between the plates the displacement current carries on the conduction current.

Displacement current equals conduction current (capacitor)

2 MARKS

Principle Gauss's law gives E between the plates; differentiate the flux.

Assumptions Parallel-plate capacitor, plate area A , charge q increasing.

$$\textcircled{1} E = \frac{q}{\varepsilon_0 A} \text{ so } \Phi_E = EA = \frac{q}{\varepsilon_0}.$$

$$\textcircled{2} I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \frac{dq}{dt} = I_c.$$

FINAL RESULT

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = I_c$$

Limits: magnitude equal, so B at the gap is the same as in the wire's neighbourhood.

How to write it in the exam: Write $\Phi_E = q/\varepsilon_0$ and differentiate.

~ MAXWELL'S FOUR EQUATIONS (CONCEPTUAL SUMMARY)

(1) Gauss's law for $\mathbf{E}$: charges are sources of $\mathbf{E}$. (2) Gauss's law for $\mathbf{B}$: no magnetic monopoles. (3) Faraday's law: changing $\mathbf{B}$ induces $\mathbf{E}$. (4) Ampere–Maxwell law: current *and* changing $\mathbf{E}$ produce $\mathbf{B}$. Together they predict waves of speed $c = 1/\sqrt{\mu_0 \varepsilon_0}$, equal to the measured speed of light; Hertz produced and detected them (1887).

! COMMON MISTAKES

- Treating I_d as moving charges.
- Saying EM waves need a medium (they travel in vacuum).
- Thinking $\mathbf{E}$ and $\mathbf{B}$ are 90° out of phase (they are in phase).

8.2 Electromagnetic waves

I CHARACTERISTICS

- **Transverse:** $\mathbf{E}$ and $\mathbf{B}$ are perpendicular to each other and to the direction of travel ($\mathbf{E} \times \mathbf{B}$ points along propagation).
- **In phase,** with $E_0/B_0 = c$.
- **Speed in vacuum** $c = 1/\sqrt{\mu_0 \epsilon_0} = 3 \times 10^8 \text{ m s}^{-1}$, independent of frequency.
- No medium is needed; carry energy and momentum; not deflected by electric or magnetic fields (uncharged).
- Show reflection, refraction, interference, diffraction and polarisation.

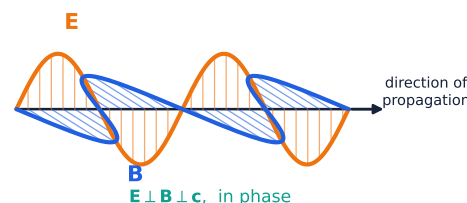


Fig. 8.2 Plane EM wave travelling to the right.

F WAVE EQUATIONS AND ENERGY

$$E = E_0 \sin(kx - \omega t), \quad B = B_0 \sin(kx - \omega t), \quad k = \frac{2\pi}{\lambda}$$

Energy density: equal electric and magnetic parts,

$$u_E = \frac{1}{2}\epsilon_0 E^2 = u_B = \frac{B^2}{2\mu_0}. \text{ Intensity } I = \frac{1}{2}\epsilon_0 c E_0^2 \text{ (extension; not required for the board).}$$

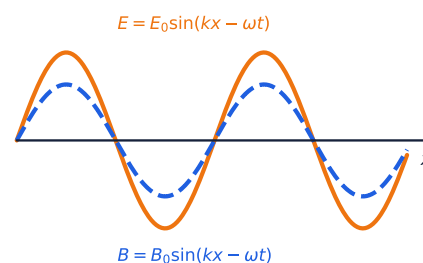


Fig. 8.3 E and B vary in phase.

✓ CHECKPOINT

A wave has $E_0 = 30 \text{ V/m}$. What is B_0 ? $B_0 = E_0/c = 1.0 \times 10^{-7} \text{ T}$. If the wave travels along $+x$ and $\mathbf{E}$ along $+y$, along which axis is $\mathbf{B}$? $+z$ (since $\mathbf{E} \times \mathbf{B} \parallel +x$).

Speed of an electromagnetic wave (idea, not a full derivation)

EXTENSION

Principle Faraday's law links E and B ; the Ampere–Maxwell law links them back.

Assumptions Plane wave along x in vacuum.

- Faraday's law for a thin strip gives $\frac{\partial E}{\partial x} = -\frac{\partial B}{\partial t}$, so $E_0 k = B_0 \omega$, i.e. $\frac{E_0}{B_0} = \frac{\omega}{k} = c$.
- The Ampere–Maxwell law gives $-\frac{\partial B}{\partial x} = \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$, so $B_0 k = \mu_0 \epsilon_0 E_0 \omega$.
- Multiplying the two results: $c^2 = \frac{\omega^2}{k^2} = \frac{1}{\mu_0 \epsilon_0}$.

FINAL RESULT

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m s}^{-1}$$

Limits: in a medium $\mu_0 \epsilon_0 \rightarrow \mu \epsilon$.

How to write it in the exam: Board only needs the result and the transverse picture (qualitative treatment).

8.3 The electromagnetic spectrum

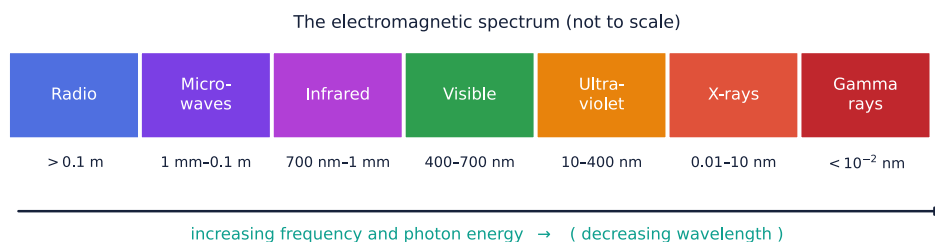


Fig. 8.4 All EM waves travel at the same speed in vacuum; they differ in frequency and wavelength.

Wave	Wavelength (approx.)	Source / production	Uses
Radio waves	$> 0.1 \text{ m}$	oscillating circuits, antennas	radio, TV, mobile communication
Microwaves	$1 \text{ mm} - 0.1 \text{ m}$	klystron, magnetron	radar, microwave oven, satellite links
Infrared	$700 \text{ nm} - 1 \text{ mm}$	hot bodies, vibrating molecules	remote controls, thermal imaging, physiotherapy, greenhouse effect
Visible	$400 - 700 \text{ nm}$	hot bodies, atomic transitions	vision, photography, optical fibres
Ultraviolet	$10 - 400 \text{ nm}$	Sun, UV lamps, arc	sterilising water, LASIK eye surgery, forged-note detection
X-rays	$0.01 - 10 \text{ nm}$	X-ray tubes, inner-shell transitions	medical imaging, airport security, crystal structure
Gamma rays	$< 0.01 \text{ nm}$	radioactive nuclei, nuclear reactions	cancer therapy, sterilisation, tracing

★ MEMORY AID

Order of increasing frequency: **Radio Micro Infra Visible Ultra X-rays Gamma** — “**Rich Men In Very Ugly X-mas Gifts**”. Visible light: **VIBGYOR** from 400 nm (violet) to 700 nm (red).

! COMMON MISTAKES

- Saying X-rays have longer wavelength than UV.
- Giving ozone as a source of UV (it absorbs it).
- Mixing the ranges of IR and microwaves.

I GREENHOUSE EFFECT

Sunlight (visible) passes through the atmosphere and warms the ground, which re-radiates infrared. Gases such as CO_2 and CH_4 absorb IR and re-emit it downward, raising the Earth's surface temperature.

I OZONE LAYER

The stratospheric ozone layer absorbs most of the Sun's harmful ultraviolet radiation. Loss of ozone increases UV reaching the ground (skin damage).

8.4 Formula bank, numericals and practice

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$	m s^{-1}	vacuum	3×10^8
$c = f\lambda$	m s^{-1}		100 MHz: 3 m
$\frac{E_0}{B_0} = c$	$\text{V m}^{-1} / \text{T}$		60 V/m: $2 \times 10^{-7} \text{ T}$
$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$	A		$= I_c$ for a capacitor
$v = \frac{c}{n}$	m s^{-1}	in a medium	$n = 1.5$: 2×10^8

N1 C08-N01 Ch 8 Easy Numerical BOARD-STYLE PRACTICE 2 marks

Calculate the speed of light from $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$ and $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$.

GIVEN μ_0, ϵ_0 as given

FIND c

FORMULA $c = 1/\sqrt{\mu_0 \epsilon_0}$

WORKING $\mu_0 \epsilon_0 = 4\pi \times 10^{-7} \times 8.85 \times 10^{-12} = 1.112 \times 10^{-17}$; $c = 1/\sqrt{1.112 \times 10^{-17}} = 3.0 \times 10^8 \text{ m s}^{-1}$.

Answer: $3.0 \times 10^8 \text{ m/s}$

▲ This agreement convinced Maxwell that light is an EM wave.

N2 C08-N02 Ch 8 Medium Numerical BOARD-STYLE PRACTICE 2 marks

The electric field of an EM wave in vacuum is $E_0 = 60 \text{ V m}^{-1}$. Find B_0 . For a 100 MHz FM wave find the wavelength.

GIVEN $E_0 = 60 \text{ V/m}$, $f = 10^8 \text{ Hz}$

FIND B_0, λ

FORMULA $B_0 = E_0/c$, $\lambda = c/f$

WORKING $B_0 = \frac{60}{3 \times 10^8} = 2 \times 10^{-7} \text{ T}$; $\lambda = \frac{3 \times 10^8}{10^8} = 3 \text{ m}$.

Answer: $2 \times 10^{-7} \text{ T}$; 3 m

▲ Use $c = 3 \times 10^8$ unless told otherwise.

N3 C08-N03 Ch 8 Medium Numerical BOARD-STYLE PRACTICE 3 marks

(a) Find the frequency of light of wavelength 600 nm. (b) Two plates of a capacitor of area 0.01 m^2 have the field between them changing at $10^{12} \text{ V m}^{-1} \text{ s}^{-1}$. Find the displacement current.

GIVEN (a) $\lambda = 6 \times 10^{-7} \text{ m}$; (b) $A = 0.01 \text{ m}^2$, $dE/dt = 10^{12}$

FIND f, I_d

FORMULA $f = c/\lambda$; $I_d = \epsilon_0 A \frac{dE}{dt}$

WORKING (a) $f = \frac{3 \times 10^8}{6 \times 10^{-7}} = 5 \times 10^{14} \text{ Hz}$. (b) $I_d = 8.85 \times 10^{-12} \times 0.01 \times 10^{12} = 0.0885 \text{ A}$.

Answer: (a) $5 \times 10^{14} \text{ Hz}$; (b) 0.089 A

▲ $\Phi_E = EA$ so $d\Phi_E/dt = A dE/dt$.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C08-S01 Ch 8 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. What is displacement current? How is it different from conduction current?

Model answer: The current due to a time-varying electric flux, $I_d = \epsilon_0 d\Phi_E/dt$, existing even in vacuum or a dielectric; no charge carriers move. Conduction current is the flow of charges.

C08-S02 Ch 8 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. Name the EM radiation used (a) in a remote control, (b) to sterilise water, (c) in a RADAR, with the reason.

Model answer: (a) Infrared — emitted by LEDs, invisible. (b) Ultraviolet — kills microbes. (c) Microwaves — short wavelength reflected by objects, directional.

C08-CB01 Ch 8 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. An EM wave of frequency $5 \times 10^{14} \text{ Hz}$ travels in glass of refractive index 1.5. (i) Find the speed. (ii) Find the wavelength in glass. (iii) What happens to the frequency? (iv) To which region does it belong?

Model answer: (i) $v = c/n = 2 \times 10^8 \text{ m/s}$. (ii) $\lambda = v/f = 2 \times 10^8 / 5 \times 10^{14} = 4 \times 10^{-7} \text{ m} = 400 \text{ nm}$. (iii) Unchanged. (iv) Visible (violet edge).

8.5 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C08-M01 Ch 8 Easy Conceptual

BOARD-STYLE PRACTICE

1. Which of the following has the shortest wavelength?

- (a) radio waves (b) microwaves
(c) X-rays (d) infrared

Ans: (c) Order: radio > micro > IR > visible > UV > X > gamma.

C08-M02 Ch 8 Medium Numerical

BOARD-STYLE PRACTICE

2. The magnetic field of an EM wave has amplitude 10^{-7} T. The electric amplitude is

- (a) 3 V/m (b) 30 V/m
(c) 300 V/m (d) 0.3 V/m

Ans: (b) $E_0 = cB_0 = 30$ V/m.

C08-M03 Ch 8 Medium Conceptual

BOARD-STYLE PRACTICE

3. Displacement current arises from

- (a) moving charges (b) a changing electric flux
(c) a changing magnetic flux (d) a steady field

Ans: (b) $I_d = \epsilon_0 d\Phi_E/dt$.

C08-M04 Ch 8 Medium Numerical

BOARD-STYLE PRACTICE

4. The wavelength of a 3 GHz microwave is

- (a) 10 cm (b) 1 m
(c) 1 cm (d) 100 m

Ans: (a) $3 \times 10^8 / 3 \times 10^9 = 0.1$ m.

C08-M05 Ch 8 Medium Application

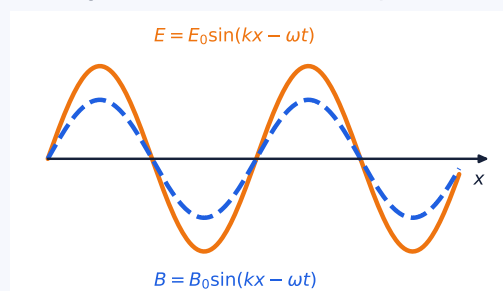
BOARD-STYLE PRACTICE

5. Ultraviolet radiation is used for

- (a) night vision (b) water sterilisation
(c) TV remote control (d) long-distance radio

Ans: (b) UV kills bacteria.

C08-M06 Ch 8 Medium Graph BOARD-STYLE PRACTICE

6. The figure shows E and B of a plane wave. They are

- (a) in phase (b) 90° out of phase
(c) 180° out of phase (d) unrelated

Ans: (a) $E \propto B$ at all points.

C08-M07 Ch 8 Medium Application

BOARD-STYLE PRACTICE

7. EM waves are NOT deflected by electric or magnetic fields because they

- (a) have no energy (b) are neutral
(c) are transverse (d) travel at c

Ans: (b) Photons carry no charge.

C08-M08 Ch 8 Hard Numerical BOARD-STYLE PRACTICE

8. In a medium of refractive index 2 the speed of an EM wave is

- (a) 3×10^8 (b) 1.5×10^8
(c) 6×10^8 (d) 2×10^8

Ans: (b) c/n .

C08-AR01 Ch 8 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: EM waves can travel through vacuum.

R: A changing electric field produces a magnetic field and vice versa.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Mutual generation sustains the wave.

C08-AR02 Ch 8 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: A microwave oven heats food quickly.

R: Microwave frequency matches the natural frequency of water molecules' rotation.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Resonant absorption by polar molecules.

★ EXAMINER-STYLE REMINDERS

- Quote $E_0/B_0 = c$ and the transverse nature in every EM-wave answer.
- For spectrum questions give *source + use*.
- State 'in vacuum' when using c .

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are **BOARD-STYLE PRACTICE**; none is claimed to be a past board question.

R Chapter 8 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Plane EM wave with $\mathbf{E}$, $\mathbf{B}$, direction
2. Charging capacitor with I_c and I_d
3. Spectrum bar with wavelengths

★ 7-DAY REVISION PLAN

Day 1	Displacement current
Day 2	Maxwell's idea
Day 3	EM wave properties
Day 4	Spectrum
Day 5	Numericals
Day 6	MCQs
Day 7	Revision

Chapter 8 at a glance — Electromagnetic Waves

Displacement current

$I_d = \epsilon_0 d\Phi_E/dt$; $I_d = I_c$ in a capacitor.

Wave

transverse; $E \perp B \perp c$; in phase;
 $E_0/B_0 = c$.

Speed

$$c = 1/\sqrt{\mu_0 \epsilon_0} = f\lambda.$$

Spectrum

R M I V U X G.

Uses

radar, remote, sterilising, imaging, therapy.

Formula map

$$\frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$f\lambda$$

wave speed

$$\frac{E_0}{B_0} = c$$

$$\epsilon_0 \frac{d\Phi_E}{dt}$$

$$\frac{c}{n}$$

in a medium

$$\frac{2\pi}{\lambda}$$

! COMMON MISTAKES

- Treating I_d as moving charge.
- Wrong order of the spectrum.
- Using 90° phase between E and B .
- Forgetting c changes in a medium.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can explain displacement current
- ☐ I can list properties of EM waves
- ☐ I can order the spectrum with uses
- ☐ I can compute B_0 and λ
- ☐ I solved N1–N3
- ☐ I did the MCQs

✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

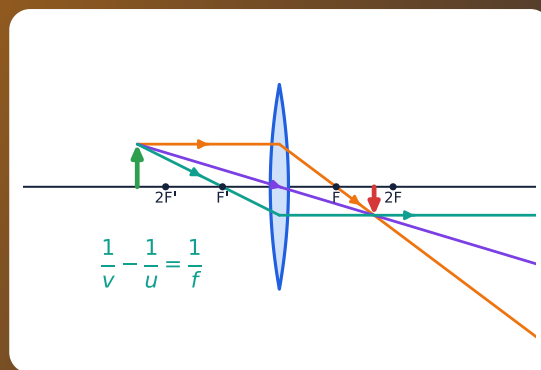
1. E_0/B_0 ? | **Ans:** c
2. Is an EM wave longitudinal? | **Ans:** No, transverse
3. Which has higher f : UV or IR? | **Ans:** UV
4. Medium needed? | **Ans:** No
5. Source of gamma rays? | **Ans:** Radioactive nuclei

UNIT VI · OPTICS

9

Ray Optics and Optical Instruments

Almost every optical device — spectacles, cameras, microscopes, telescopes, fibre-optic internet — uses just two things: refraction at a curved surface and total internal reflection.



Chapter roadmap

Mirrors

Refraction & TIR

Spherical surface
& lens

Prism

Instruments

✓ LEARNING OUTCOMES

- Apply the Cartesian sign convention to mirrors and lenses; use mirror and lens formulas and magnification
- Explain refraction, total internal reflection and optical fibres; use Snell's law and $\sin \theta_c = 1/n$
- Derive the refraction formula for a spherical surface, the lens-maker's formula and the thin-lens formula
- Define power; find the focal length and power of lenses in contact
- Derive the prism formula and describe dispersion
- Describe the compound microscope and astronomical telescope and derive their magnifying powers

I PREREQUISITES TO REVISE

- Laws of reflection and refraction
- Similar triangles and small-angle approximations
- Sign convention in coordinate geometry
- Ratio and proportion

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: reflection of light, spherical mirrors, mirror formula; refraction of light, total internal reflection and optical fibres; refraction at a spherical surface, lenses, thin-lens formula, lens-maker's formula, magnification, power of a lens, combination of thin lenses in contact; refraction of light through a prism; optical instruments: microscopes and astronomical telescopes (reflecting and refracting) and their magnifying powers.

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (TIR, power, sign convention); 2-mark (refraction formulas, critical angle); 3-mark numericals (lens, mirror, prism); 5-mark derivation of the lens-maker's formula, prism formula or telescope magnification; case-based on lenses and fibres. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Cartesian sign convention

F RULES

1) Light travels left to right. 2) All distances are measured from the pole (mirror) or optical centre (lens). 3) Distances along the direction of incident light are +, against it –. 4) Heights above the principal axis are +, below –. Concave mirror: $f < 0$; convex lens: $f > 0$.

★ MASTER FORMULAS

Mirror: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, $m = -\frac{v}{u}$. Lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $m = \frac{v}{u}$. Lens-maker: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$.
Power $P = \frac{1}{f}$ (diopetre, f in m).

9.1 Reflection at spherical mirrors

1 MIRRORS AND THE MIRROR FORMULA

For a spherical mirror of radius of curvature R : $f = \frac{R}{2}$ (paraxial rays). With the Cartesian convention $f < 0$ for a concave and $f > 0$ for a convex mirror.

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$m < 0$: real, inverted image; $m > 0$: virtual, erect image. A convex mirror always forms a virtual, erect, diminished image (used in vehicle rear-view mirrors).

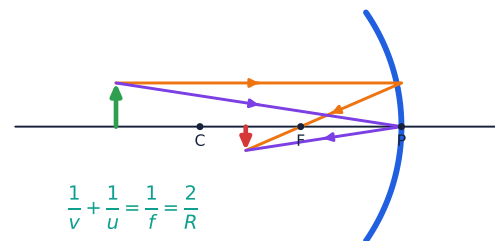


Fig. 9.1 Image formed by a concave mirror (object beyond C). All three principal rays are shown.

Mirror formula

3 MARKS

Principle Similar triangles for a ray striking the pole and the law of reflection ($\angle i = \angle r$).

Assumptions Small aperture (paraxial), concave mirror; $u < 0$, $v < 0$, $R < 0$ in the Cartesian convention.

- ① Triangles ABP and $A'B'P$ (ray reflected at the pole): $\frac{A'B'}{AB} = \frac{PB'}{PB}$, i.e. $\frac{h_i}{h_o} = -\frac{v}{u}$ (this is m).
- ② Triangles $A'B'F$ and MPF ($MP = AB$ for the parallel ray): $\frac{A'B'}{AB} = \frac{FB'}{FP} = \frac{v-f}{-f}$ in the sign convention.
- ③ Equate: $-\frac{v}{u} = \frac{f-v}{f}$ $\Rightarrow \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$.

FINAL RESULT

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Limits: paraxial rays only; $f = R/2$.

How to write it in the exam: Draw the ray diagram with $A'B'$, F , P and label with signs.

Object position (concave)	Image position	Nature	Size
At infinity	at F	real, inverted	point
Beyond C	between F and C	real, inverted	diminished
At C	at C	real, inverted	same size
Between C and F	beyond C	real, inverted	magnified
At F	at infinity	real, inverted	highly magnified
Between F and P	behind the mirror	virtual, erect	magnified

9.2 Refraction, total internal reflection and optical fibres

I SNELL'S LAW AND REFRACTIVE INDEX

$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = {}_1n_2$; $n = \frac{c}{v}$. A ray bends towards the normal on entering a denser medium. Frequency is unchanged; v and λ decrease: $\lambda' = \lambda/n$. **Apparent depth** (viewed normally): $d_{app} = d_{real}/n$. A glass slab of thickness t shifts the image by $t(1 - 1/n)$.

I TOTAL INTERNAL REFLECTION (TIR)

When light goes from a denser to a rarer medium and the angle of incidence exceeds the **critical angle** θ_c , the whole light is reflected inside:

$$\sin \theta_c = \frac{n_2}{n_1} = \frac{1}{n} \quad (\text{denser to air}), \quad \theta_c \text{ for glass} \approx 42^\circ, \text{ water} \approx 49^\circ, \text{ diamond} \approx 24^\circ$$

Conditions: (i) light travels from denser to rarer; (ii) $i > \theta_c$.

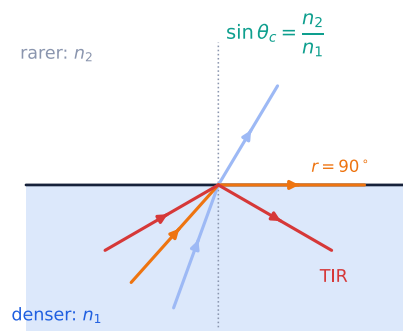


Fig. 9.2 Refraction, the critical case and total internal reflection.

~ OPTICAL FIBRES AND TIR APPLICATIONS

A glass/plastic core (high n) surrounded by cladding (lower n): rays entering within the acceptance angle strike the core–cladding boundary beyond θ_c and are guided by repeated TIR with almost no loss. Uses: communication (internet, telephone), endoscopy. Other TIR effects: sparkle of diamond, mirage, prism reflectors in binoculars.

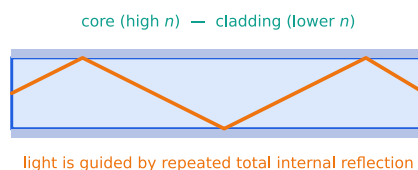


Fig. 9.3 Light guided in an optical fibre.

! COMMON MISTAKES

- Using $\sin \theta_c = n$ instead of $1/n$.
- Saying TIR can happen going from rare to dense.
- Forgetting frequency does not change on refraction.

Apparent depth (normal view)

3 MARKS

Principle Snell's law for small angles at the water–air surface.

Assumptions Object at real depth d in a medium n , viewed from air almost vertically.

- ① A ray from O meets the surface at small angle i (in n) and leaves at r (in air): $n \sin i = \sin r$, so $n i \approx r$.
- ② With horizontal distance x to the point of incidence: $i \approx x/d$ and $r \approx x/d_{app}$. Hence $d_{app} = d/n$.

FINAL RESULT

$$d_{app} = \frac{d_{real}}{n}$$

Limits: valid for near-normal viewing; for a slab of thickness t , shift $= t\left(1 - \frac{1}{n}\right)$.

How to write it in the exam: Draw the two rays and state the small-angle approximation.

9.3 Refraction at a spherical surface; lens-maker's formula

Refraction at a convex spherical surface

5 MARKS

Principle Snell's law for small angles and geometry of the ray OA meeting the surface at A .

Assumptions Paraxial rays; object O in medium n_1 , surface of radius R , image I in medium n_2 . Cartesian signs: $u < 0$, $v > 0$, $R > 0$.

① Let α, β, γ be the angles of OA , IA and CA with the axis. In $\triangle OAC$: $i = \alpha + \gamma$; in $\triangle AIC$: $\gamma = r + \beta \Rightarrow r = \gamma - \beta$.

② Snell's law for small angles:

$$n_1 i = n_2 r \Rightarrow n_1 (\alpha + \gamma) = n_2 (\gamma - \beta).$$

③ For paraxial rays $\alpha \approx \frac{h}{-u}$, $\beta \approx \frac{h}{v}$, $\gamma \approx \frac{h}{R}$. Substituting:

$$-\frac{n_1}{u} + \frac{n_1}{R} = \frac{n_2}{R} - \frac{n_2}{v}, \text{ i.e. } \frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}.$$

FINAL RESULT

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Limits: paraxial rays; signs by the Cartesian convention. For a plane surface $R \rightarrow \infty$: $v = \frac{n_2}{n_1}u$ (apparent depth).

How to write it in the exam: Draw the figure with $\alpha, \beta, \gamma, i, r$ and state the small-angle approximation.

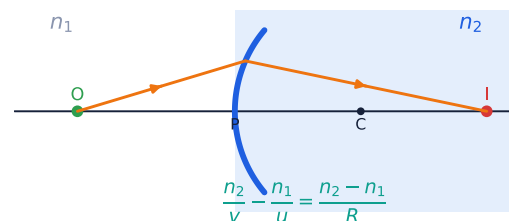


Fig. 9.4 Refraction at a spherical surface.

Lens-maker's formula and the thin-lens formula

5 MARKS

Principle Apply the surface formula twice: image of the first surface is the object for the second.

Assumptions Thin lens of refractive index n in air, surfaces of radii R_1, R_2 ; thickness negligible.

① First surface (air to glass): $\frac{n}{v_1} - \frac{1}{u} = \frac{n - 1}{R_1}$.

② Second surface (glass to air), object at v_1 : $\frac{1}{v} - \frac{n}{v_1} = \frac{1 - n}{R_2}$.

③ Add: $\frac{1}{v} - \frac{1}{u} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$. For $u \rightarrow -\infty$, $v = f$: lens-maker's formula; hence $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$.

FINAL RESULT

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Limits: in a medium of index n_m : $\frac{1}{f} = \left(\frac{n}{n_m} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ — a glass lens in water has longer f .

How to write it in the exam: Show both surface equations and the addition step.

9.4 Thin lenses: magnification, power and combinations

F THIN-LENS RESULTS

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \quad m = \frac{v}{u}, \quad P = \frac{1}{f} \text{ (D)}, \quad P = P_1 + P_2 + \dots, \quad \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

Convex lens: $f > 0$, $P > 0$ (converging); concave lens: $f < 0$, $P < 0$ (diverging). Lenses in contact: total power is the sum. Two thin lenses separated by d : $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$ (extension).

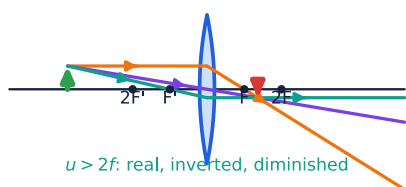


Fig. 9.5 (a) object beyond $2F$.

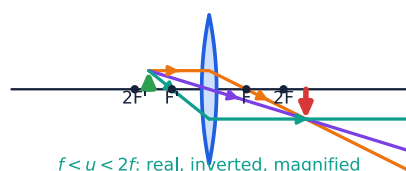


Fig. 9.6 (b) between F and $2F$.

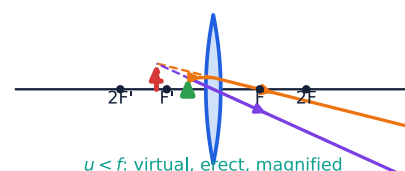


Fig. 9.7 (c) inside F : magnifier.

Focal length of two thin lenses in contact

2
MARKS

Principle The first lens forms an image that acts as the object for the second.

Assumptions Lenses f_1, f_2 in contact; thickness negligible, so the same distance from both.

① Lens 1: $\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$. Lens 2: $\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$.

② Add: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{f}$.

FINAL RESULT

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}, \quad P = P_1 + P_2$$

Limits: also magnification $m = m_1 m_2$.

How to write it in the exam: Write both lens equations and add.

★ STRATEGY FOR LENS NUMERICALS

1. Write u with a negative sign (real object on the left).
2. Substitute in $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$.
3. Positive v means a real image on the right; negative means virtual on the left.
4. Check $m = v/u$: negative means inverted.

✓ CHECKPOINT

A convex lens of $f = 10$ cm forms a real image the same size as the object. Where is the object? At $2f = 20$ cm; the image is at 20 cm, $m = -1$.

9.5 Refraction through a prism; dispersion

Prism formula

5 MARKS

Principle Geometry of the prism and Snell's law at each face; at minimum deviation the ray passes symmetrically.

Assumptions Prism of angle A , refractive index n ; ray with angle of incidence i , emergence e , refraction angles r_1, r_2 .

① Deviation:

$\delta = (i - r_1) + (e - r_2) = i + e - (r_1 + r_2)$. In the quadrilateral formed by the normals: $r_1 + r_2 = A$. So $\delta = i + e - A$.

② The graph of δ against i has a minimum δ_m when $i = e$ and $r_1 = r_2 = A/2$.

③ At minimum deviation: $i = \frac{A + \delta_m}{2}$, $r = \frac{A}{2}$, so $n = \frac{\sin i}{\sin r}$.

FINAL RESULT

$$n = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}; \quad \text{thin prism: } \delta = (n - 1) A$$

Limits: for a thin prism (A small, small i): $\delta = (n - 1) A$, independent of i .

How to write it in the exam: Draw the prism with normals; mark $i, r_1, r_2, e, A, \delta$.

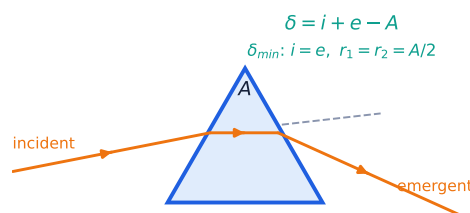


Fig. 9.8 Ray through a prism.

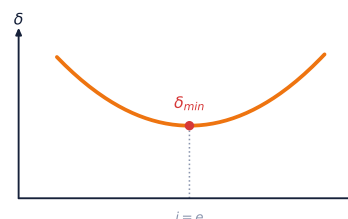


Fig. 9.9 δ against i .

DISPERSION

The refractive index depends on wavelength (Cauchy: $n \approx A + B/\lambda^2$): violet bends most, red least. White light entering a prism is split into the spectrum VIBGYOR. The rainbow is formed by dispersion and TIR inside raindrops. Angular dispersion $= \delta_v - \delta_r = (n_v - n_r) A$.

RAINBOW

Sunlight refracts into a raindrop, is dispersed, totally reflected at the back surface (primary bow: one internal reflection) and refracts out again. Violet emerges at about 40° and red at 42° to the line from the Sun: red outside, violet inside.

CHECKPOINT

A thin prism ($A = 4^\circ$, $n = 1.5$): deviation? $\delta = (n - 1) A = 2^\circ$. Which colour deviates more, red or violet? *Violet (n is larger).*

Colour	Violet	Blue	Green	Yellow	Orange	Red
Approx. n (crown glass)	1.532	1.528	1.519	1.517	1.514	1.513
Relative deviation	largest					smallest

9.6 Optical instruments

Magnifying power of a compound microscope

5 MARKS

Principle $m = m_o \times m_e$; objective forms a magnified real image I_1 that lies within the focus of the eyepiece.

Assumptions Objective f_o (short), eyepiece f_e ; final image at the near point $D = 25$ cm.

① Objective: $m_o = \frac{v_o}{u_o}$ (magnitude $\frac{v_o}{|u_o|}$; approx $\frac{L}{f_o}$ if $v_o \approx L$, tube length).

② Eyepiece as a simple magnifier at the near point:
 $m_e = 1 + \frac{D}{f_e}$ (at infinity: D/f_e).

③ $m = m_o m_e = -\frac{v_o}{|u_o|} \left(1 + \frac{D}{f_e} \right) \approx -\frac{L}{f_o} \cdot \frac{D}{f_e}$.

FINAL RESULT

$$m = \frac{v_o}{|u_o|} \left(1 + \frac{D}{f_e} \right) \text{ (near point)}, \quad m \approx \frac{LD}{f_o f_e} \text{ (at infinity)}$$

Limits: for high m : small f_o , f_e and long tube length.

How to write it in the exam: Draw the ray diagram with I_1 inside the eyepiece focus.

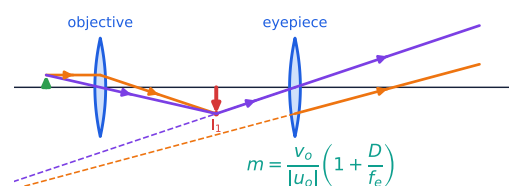


Fig. 9.10 Compound microscope (final image virtual).

Magnifying power of an astronomical (refracting) telescope

5 MARKS

Principle Angle subtended by the final image divided by the angle subtended by the object at the unaided eye.

Assumptions Objective f_o (large), eyepiece f_e ; final image at infinity (normal adjustment); tube length $L = f_o + f_e$.

① Distant object subtends α at the objective: image I_1 at its focus, height $h = f_o \alpha$.

② I_1 lies at the focus of the eyepiece, so the final image is at infinity and subtends $\beta = \frac{h}{f_e}$.

③ $m = \frac{\beta}{\alpha} = \frac{f_o}{f_e}$ (magnitude).

FINAL RESULT

$$m = -\frac{f_o}{f_e} \text{ (final image at infinity)}, \quad L = f_o + f_e$$

Limits: at the near point $m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$. Reflecting telescope: concave mirror objective — no chromatic aberration, lighter, larger aperture.

How to write it in the exam: Draw parallel rays from a distant object to the objective, I_1 at F_o .

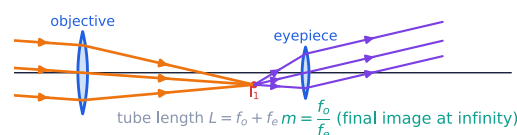


Fig. 9.11 Astronomical telescope in normal adjustment.

9.7 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	m	mirror	$f = R/2$; $m = -v/u$
$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$	—	Snell	$n = c/v$
$\sin \theta_c = \frac{1}{n}$	—	denser to air	water 48.8°
$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$	m	spherical surface	
$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	m^{-1}	thin lens in air	
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$; $m = \frac{v}{u}$	m	thin lens	
$P = \frac{1}{f}$; $P = P_1 + P_2$	D	f in metres	
$n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$	—	prism	
$m = \frac{f_o}{f_e}$	—	telescope	$L = f_o + f_e$

N1 C09-N01 Ch 9 Medium Numerical BOARD-STYLE PRACTICE 3 marks

An object is placed 20 cm in front of a concave mirror of focal length 15 cm. Find the image position and magnification.

GIVEN $u = -20 \text{ cm}$, $f = -15 \text{ cm}$

FIND v , m

FORMULA $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$, $m = -\frac{v}{u}$

WORKING $\frac{1}{v} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60} \Rightarrow v = -60 \text{ cm}$; $m = -\frac{-60}{-20} = -3$.

Answer: $v = -60 \text{ cm}$ (real, in front); $m = -3$ (inverted, $3 \times$ larger)

▲ Object between F and C gives an image beyond C .

N2 C09-N02 Ch 9 Easy Numerical BOARD-STYLE PRACTICE 2 marks

A ray enters glass ($n = 1.5$) from air at 30° . Find the angle of refraction. Find the critical angle for water ($n = 1.33$).

GIVEN $i = 30^\circ$, $n = 1.5$; $n_w = 1.33$

FIND r , θ_c

FORMULA $\sin r = \frac{\sin i}{n}$; $\sin \theta_c = \frac{1}{n}$

WORKING $\sin r = \frac{0.5}{1.5} = 0.333 \Rightarrow r = 19.5^\circ$. $\sin \theta_c = \frac{1}{1.33} = 0.752 \Rightarrow \theta_c = 48.8^\circ$.

Answer: $r = 19.5^\circ$; $\theta_c = 48.8^\circ$

▲ Bends towards the normal in the denser medium.

N3 C09-N03 Ch 9 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A convex lens of focal length 20 cm forms an image of an object placed 30 cm from it. Find v and m .

GIVEN $f = +20 \text{ cm}$, $u = -30 \text{ cm}$

FIND v , m

FORMULA $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$, $m = \frac{v}{u}$

WORKING $\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60} \Rightarrow v = 60 \text{ cm}$; $m = \frac{60}{-30} = -2$.

Answer: $v = +60 \text{ cm}$ (real); $m = -2$ (inverted, doubled)

▲ Object between F and $2F$.

9.8 More numericals and subjective practice

N4 C09-N04 Ch 9 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A biconvex lens of $n = 1.5$ has both radii 20 cm. Find f and power. If it is placed in contact with a concave lens of $f = -30$ cm, find the combination's f and power.

GIVEN $n = 1.5$, $R_1 = +20$ cm, $R_2 = -20$ cm; $f_2 = -30$ cm

FIND f , P , f_{comb} , P_{comb}

FORMULA $\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$; $P = P_1 + P_2$

WORKING $\frac{1}{f} = 0.5 \left(\frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20} \Rightarrow f = 20$ cm, $P = 5$ D. Concave: $P_2 = -\frac{100}{30} = -3.33$ D. $P = 5 - 3.33 = 1.67$ D,
 $f = \frac{100}{1.67} = 60$ cm.

Answer: 20 cm, +5 D; combination 60 cm, +1.67 D

▲ Convert f to metres before using dioptres.

N5 C09-N05 Ch 9 Medium Numerical BOARD-STYLE PRACTICE 2 marks

A prism of angle 60° has minimum deviation 30° . Find its refractive index and the angle of incidence at minimum deviation.

GIVEN $A = 60^\circ$, $\delta_m = 30^\circ$

FIND n , i

FORMULA $n = \frac{\sin \frac{A+\delta_m}{2}}{\sin \frac{A}{2}}$, $i = \frac{A+\delta_m}{2}$

WORKING $i = \frac{60+30}{2} = 45^\circ$; $n = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{0.707}{0.5} = 1.41$.

Answer: $n = \sqrt{2} \approx 1.41$; $i = 45^\circ$

▲ At minimum deviation $r = A/2$.

N6 C09-N06 Ch 9 Hard Numerical BOARD-STYLE PRACTICE 5 marks

A compound microscope has $f_o = 2$ cm, $f_e = 5$ cm. An object is 2.5 cm from the objective and the final image is formed at 25 cm. Find v_o and the magnifying power. For a telescope with $f_o = 100$ cm, $f_e = 5$ cm in normal adjustment find m and the tube length.

GIVEN $f_o = 2$, $f_e = 5$, $|u_o| = 2.5$ cm, $D = 25$ cm; telescope: $f_o = 100$, $f_e = 5$ cm

FIND v_o , m ; m_T , L

FORMULA $\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o}$; $m = \frac{v_o}{|u_o|} \left(1 + \frac{D}{f_e} \right)$

WORKING $\frac{1}{v_o} = \frac{1}{2} - \frac{1}{2.5} = \frac{1}{10} \Rightarrow v_o = 10$ cm; $m_o = 4$; $m_e = 1 + \frac{25}{5} = 6$; $m = 24$. Telescope: $m = \frac{100}{5} = 20$; $L = 105$ cm.

Answer: $v_o = 10$ cm; $m = 24$ (inverted); telescope $m = 20$, $L = 105$ cm

▲ Microscope image is inverted and virtual.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C09-S01 Ch 9 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. State two conditions for total internal reflection. Why does a diamond sparkle?

Model answer: (i) Light travels from a denser to a rarer medium; (ii) angle of incidence exceeds the critical angle. Diamond has a small θ_c (24°) so light entering is totally reflected many times before emerging, producing brilliance.

C09-S02 Ch 9 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. A convex lens made of glass ($n = 1.5$) is immersed in water ($n = 1.33$). How does its focal length change? Why?

Model answer: The relative index falls ($1.5/1.33 = 1.13$), so $(n_{\text{rel}} - 1)$ decreases from 0.5 to 0.13 and f increases (about $4 \times$).

C09-CB01 Ch 9 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. An optical fibre has a core of $n = 1.50$ and cladding of $n = 1.45$. **(i)** Find the critical angle at the core-cladding boundary. **(ii)** Why must the cladding have lower n ? **(iii)** Name the principle. **(iv)** Name one use.

Model answer: (i) $\sin \theta_c = 1.45/1.50 = 0.967 \Rightarrow \theta_c = 75.2^\circ$. (ii) Light must travel from a denser to a rarer medium for TIR. (iii) Total internal reflection. (iv) Data communication or endoscopy.

9.9 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C09-M01 Ch 9 Easy Conceptual BOARD-STYLE PRACTICE

1. A concave mirror forms a virtual, erect, magnified image when the object is

- (a) beyond C (b) at C
(c) between F and C (d) between P and F

Ans: (d) Only inside F .

C09-M02 Ch 9 Medium Numerical

BOARD-STYLE PRACTICE

2. A convex lens has $f = 10$ cm. An object at 15 cm from it gives an image at

- (a) 30 cm (b) 6 cm
(c) -30 cm (d) 25 cm

Ans: (a) $1/v = 1/10 - 1/15 = 1/30$.

C09-M03 Ch 9 Medium Conceptual

BOARD-STYLE PRACTICE

3. The critical angle for glass ($n = 1.5$) is nearly

- (a) 24° (b) 42°
(c) 49° (d) 60°

Ans: (b) $\sin \theta_c = 2/3$.

C09-M04 Ch 9 Medium Numerical

BOARD-STYLE PRACTICE

4. The power of a lens of focal length -25 cm is

- (a) +4 D (b) -4 D
(c) +0.04 D (d) -0.25 D

Ans: (b) $P = 1/(-0.25) = -4$ D.

C09-M05 Ch 9 Medium Application

BOARD-STYLE PRACTICE

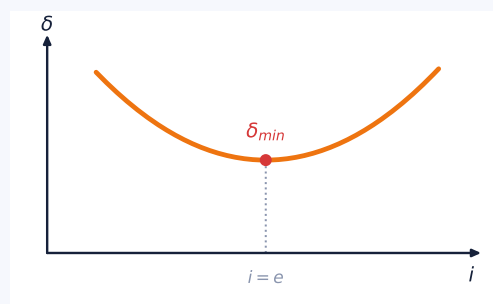
5. Two lenses of powers +3 D and -1 D are in contact. The focal length of the combination is

- (a) 0.5 m (b) 0.33 m
(c) 2 m (d) 4 m

Ans: (a) $P = 2$ D.

C09-M06 Ch 9 Medium Graph BOARD-STYLE PRACTICE

6. The graph shows deviation δ against angle of incidence i for a prism. At the minimum point



- (a) $i = e$ (b) $i > e$
(c) $i < e$ (d) $r_1 \neq r_2$

Ans: (a) Symmetric passage at minimum deviation.

C09-M07 Ch 9 Medium Numerical

BOARD-STYLE PRACTICE

7. An astronomical telescope has $f_o = 80$ cm and $f_e = 2$ cm. Its magnification in normal adjustment is

- (a) 40 (b) 160
(c) 78 (d) 82

Ans: (a) $f_o/f_e = 40$.

C09-M08 Ch 9 Hard Application BOARD-STYLE PRACTICE

8. A glass slab of thickness $t = 6$ cm and $n = 1.5$ is placed over a mark. The mark appears raised by

- (a) 2 cm (b) 3 cm
(c) 4 cm (d) 1.5 cm

Ans: (a) $t(1 - 1/n) = 6 \times 1/3 = 2$ cm.

C09-AR01 Ch 9 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: A convex mirror is used as a rear-view mirror.
R: It always forms a virtual, erect, diminished image and gives a wider field of view.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Wider field of view with an erect image.

C09-AR02 Ch 9 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: White light is dispersed by a glass prism.

R: The refractive index of glass is different for different colours.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) $n_v > n_r$, so violet deviates more.

★ EXAMINER-STYLE REMINDERS

- State the sign convention and write u as negative.
- Draw the ray diagram with arrows and labelled F , C , I .
- For prism answers mention $i = e$ at δ_m .

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are BOARD-STYLE PRACTICE; none is claimed to be a past board question.

R Chapter 9 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Concave mirror image beyond C
2. Convex lens: three object positions
3. Ray through a prism with i, r, e, δ
4. Compound microscope
5. Astronomical telescope (normal adjustment)

★ 7-DAY REVISION PLAN

Day 1	Mirrors
Day 2	Refraction and TIR
Day 3	Spherical surface
Day 4	Lens formula and power
Day 5	Prism
Day 6	Instruments
Day 7	N1–N6; MCQs

Chapter 9 at a glance — Ray Optics and Optical Instruments

Mirror

$$1/v + 1/u = 1/f; f = R/2.$$

Refraction

Snell; $\sin \theta_c = 1/n$; TIR in fibres.

Spherical surface

$$n_2/v - n_1/u = (n_2 - n_1)/R.$$

Lens

$$1/f = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right);$$

$$P = 1/f.$$

Prism

$$n = \sin \frac{A + \delta_m}{2} / \sin \frac{A}{2}.$$

Instruments

$$\text{microscope} \approx \frac{LD}{f_o f_e}; \text{telescope } f_o/f_e.$$

Formula map

$$\frac{1}{v} + \frac{1}{u}$$

mirror

$$\frac{1}{v} - \frac{1}{u}$$

lens

$$\frac{1}{\sin \theta_c}$$

$= n$

$$(n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$1/f$

$$P_1 + P_2$$

contact

$$(n - 1) A$$

thin prism

$$\frac{f_o}{f_e}$$

telescope

$$1 + \frac{D}{f_e}$$

eyepiece

$$\frac{LD}{f_o f_e}$$

microscope

! COMMON MISTAKES

- Wrong sign of u .
- Using $\sin \theta_c = n$.
- Forgetting f in metres for P .
- TIR from rare to dense.
- Mixing mirror and lens formulas.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can derive the mirror formula
- ☐ I can derive the spherical-surface and lens-maker's formulas
- ☐ I can use lens power and combinations
- ☐ I can derive the prism formula
- ☐ I can derive microscope and telescope magnifications
- ☐ I solved N1–N6
- ☐ I did the MCQs

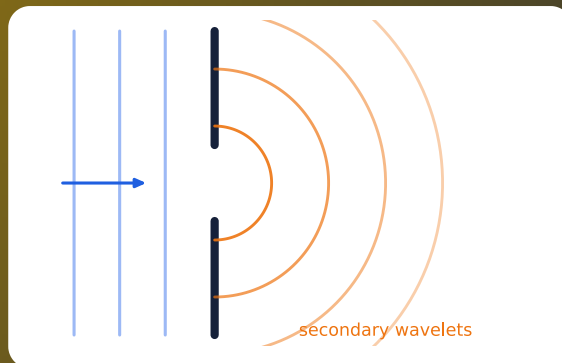
✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

1. Unit of lens power? | **Ans:** Dioptre
2. Convex lens f sign? | **Ans:** Positive
3. m of a real image? | **Ans:** Negative
4. TIR needs? | **Ans:** Denser to rarer, $i > \theta_c$
5. Telescope tube length? | **Ans:** $f_o + f_e$

UNIT VI · OPTICS

10 Wave Optics

Light that makes sharp shadows is also a wave — add two beams and you can get darkness. Interference and diffraction are the experimental proof.



Chapter roadmap

Wavefronts

Reflection & refraction

Interference

Young's experiment

Diffraction

✓ LEARNING OUTCOMES

- State Huygens' principle and use it to prove the laws of reflection and refraction
- Define coherent sources and derive the conditions and fringe width $\beta = \lambda D/d$ in Young's experiment
- Describe the single-slit diffraction pattern and the width of the central maximum
- Compare interference and diffraction patterns; solve fringe and slit numericals

1 PREREQUISITES TO REVISE

- Superposition of waves; phase and path difference
- Ray optics: Snell's law (Chapter 9)
- Small-angle approximation
- Wave speed $v = f\lambda$

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: wavefront and Huygens' principle; reflection and refraction of a plane wave at a plane surface using wavefronts; proof of the laws of reflection and refraction using Huygens' principle; interference, Young's double-slit experiment and expression for fringe width (final expression only), coherent sources and sustained interference of light; diffraction due to a single slit, width of the central maximum (qualitative treatment only).

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (conditions for interference, fringe width); 2-mark (coherent sources, differences between patterns); 3-mark numericals (fringe width, path difference, slit width); 5-mark proof of laws using Huygens' principle or of fringe width; case-based on Young's experiment. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

At a glance

F INTERFERENCE VS DIFFRACTION

Young: $\beta = \frac{\lambda D}{d}$ (equal fringes). Single slit: central width $= \frac{2\lambda D}{a}$; minima at $a \sin \theta = n\lambda$ ($n = \pm 1, \pm 2, \dots$).

★ PATH DIFFERENCE RULES

Bright: $\Delta = n\lambda$. Dark: $\Delta = \left(n + \frac{1}{2}\right)\lambda$. Phase difference $\phi = \frac{2\pi}{\lambda}\Delta$.

10.1 Wavefronts and Huygens' principle

1 WAVEFRONT AND HUYGENS' PRINCIPLE

A **wavefront** is the locus of points vibrating in phase; rays are perpendicular to it. **Huygens' principle:** every point on a wavefront is a source of secondary wavelets spreading with the speed of the wave; the new wavefront after time t is the forward envelope (common tangent) of these wavelets.

Shapes: point source $\rightarrow$ spherical; far source $\rightarrow$ plane; line source $\rightarrow$ cylindrical.

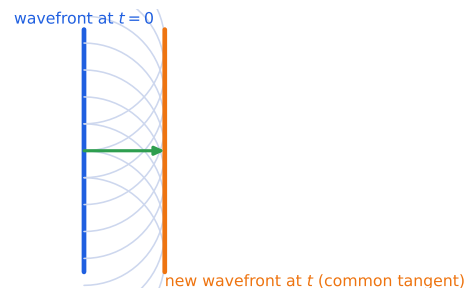


Fig. 10.1 Construction of a new wavefront.

Law of reflection using Huygens' principle

3 MARKS

Principle Wavelets from points on the incident wavefront reach the reflecting surface at different times.

Assumptions Plane wavefront AB incident at angle i on a plane mirror; B reaches the surface at C .

- ① In the time τ that B travels $BC = v\tau$, the wavelet from A travels $AE = v\tau$ in the same medium. So $AE = BC$.
- ② The reflected wavefront is the tangent CE to the circle of radius AE about A .
- ③ Triangles ABC and AEC are right-angled, share the hypotenuse AC and have $BC = AE$; hence they are congruent: $\angle BAC = \angle ECA$, i.e. $i = r$.

FINAL RESULT

$i = r$ (angle of incidence = angle of reflection)

Limits: the incident ray, reflected ray and normal lie in one plane.

How to write it in the exam: Draw AB , CE and the circle from A ; state $AE = BC$.

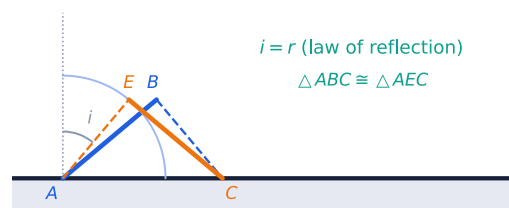


Fig. 10.2 Reflection of a plane wavefront.

! COMMON MISTAKES

- Drawing wavefronts parallel to rays.
- Saying a wavefront and a ray are the same.
- Forgetting that $AE = BC$ because both travel in the same medium for the same time.

10.2 Refraction using wavefronts

Snell's law using Huygens' principle

5 MARKS

Principle Wave speed changes at the boundary: v_1 in medium 1, v_2 in medium 2.

Assumptions Plane wavefront AB incident on a plane boundary at angle i ; refracted at r in medium 2 (denser, $v_2 < v_1$).

① While B travels $BC = v_1 \tau$ in medium 1, the wavelet from A travels $AE = v_2 \tau$ in medium 2.

② In $\triangle ABC$: $\sin i = \frac{BC}{AC}$. In $\triangle AEC$: $\sin r = \frac{AE}{AC}$.

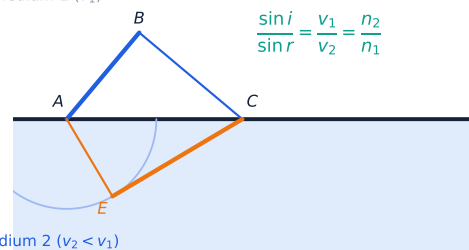
③ Divide: $\frac{\sin i}{\sin r} = \frac{BC}{AE} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$ (constant).

FINAL RESULT

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1} = {}_1n_2$$

Limits: frequency is unchanged on refraction, so $\lambda_2/\lambda_1 = v_2/v_1$. Waves bend towards the normal where they are slower.

How to write it in the exam: Draw AB , CE and mark $BC = v_1 \tau$, $AE = v_2 \tau$.

medium 1 (v_1)

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

Fig. 10.3 Refraction of a plane wavefront into a denser medium.

~ WHY LIGHT TRAVELS SLOWER IN GLASS

In a medium $v = c/n$. Experiment (Foucault, Michelson) found light slower in water, supporting the wave model (corpuscular theory predicted faster).

✓ CHECKPOINT

Light of $\lambda = 600 \text{ nm}$ enters glass ($n = 1.5$). New v and λ ? $v = 2 \times 10^8 \text{ m/s}$; $\lambda' = 400 \text{ nm}$; $f = 5 \times 10^{14} \text{ Hz}$ is unchanged.

~ POLARISATION (EXTENSION — CONFIRM WHETHER EXAMINABLE IN 2026–27)

Light is a transverse wave: its $\mathbf{E}$ oscillates perpendicular to the ray. A polaroid transmits one plane of vibration. **Malus's law:** $I = I_0 \cos^2 \theta$ for plane-polarised light through an analyser at angle θ . **Brewster's law:** reflected light is fully plane polarised when $\tan i_B = n$ (reflected and refracted rays then 90° apart). Unpolarised light through a polaroid: $I = I_0/2$.

10.3 Interference and Young's double-slit experiment

I COHERENCE AND INTERFERENCE

Coherent sources have a constant phase difference (and same frequency). Superposition gives a stable pattern: **constructive** (bright) where $\Delta = n\lambda$, **destructive** (dark) where $\Delta = (n + \frac{1}{2})\lambda$. Two independent bulbs are not coherent (phases change randomly $\sim 10^{-10}$ s) — so in Young's experiment both slits are lit from one source. Resultant intensity $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$;

$$I_{\max}/I_{\min} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2}.$$

Fringe width in Young's double-slit experiment

3–5 MARKS

Principle Path difference between the two waves at a point P on the screen.

Assumptions Slits S_1, S_2 separated by d ; screen at distance $D \gg d$; P at height y from the central point O ; $y \ll D$.

$$\textcircled{1} S_2P^2 - S_1P^2 = \left(D^2 + (y + d/2)^2 \right) - \left(D^2 + (y - d/2)^2 \right) = 2yd$$

$$\textcircled{2} \text{ Since } S_2P + S_1P \approx 2D: \text{ path difference } \Delta = S_2P - S_1P \approx \frac{yd}{D}.$$

$$\textcircled{3} \text{ Bright fringe: } \frac{yd}{D} = n\lambda \Rightarrow y_n = \frac{n\lambda D}{d}. \text{ Dark: } y_n = (n + \frac{1}{2}) \frac{\lambda D}{d}.$$

$$\textcircled{4} \text{ Fringe width = separation of consecutive bright (or dark) fringes:}$$

$$\beta = y_{n+1} - y_n = \frac{\lambda D}{d}.$$

FINAL RESULT

$$\beta = \frac{\lambda D}{d} \quad (\text{equal width, equally spaced})$$

Limits: $\beta \propto \lambda$ (red fringes are wider than blue); in a medium $\beta' = \beta/n$; the central fringe is bright for monochromatic light and white for white light with coloured fringes beyond. *The board requires only the final expression; the steps help recall.*

How to write it in the exam: Draw the geometry with S_1, S_2, P, O, D, d .

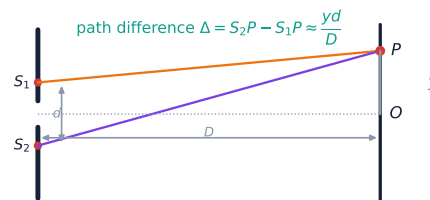


Fig. 10.4 Geometry of Young's experiment.

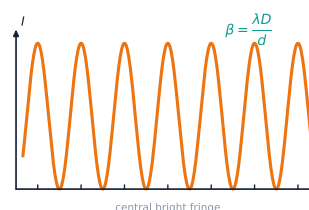


Fig. 10.5 Intensity of the fringes.

10.4 Diffraction at a single slit

I DIFFRACTION

Bending of waves around the edges of an obstacle or aperture, noticeable when the size of the opening is comparable to the wavelength. For a single slit of width a illuminated by a plane wave, the screen shows a broad bright **central maximum** flanked by weaker maxima, with minima at

$$a \sin \theta = n\lambda \quad (n = \pm 1, \pm 2, \dots)$$

Angular half-width of the central maximum $\theta \approx \frac{\lambda}{a}$; its width on a screen at distance D is $2\lambda D/a$. Narrower slit $\rightarrow$ wider pattern. Secondary maxima have much lower intensity (about 4.5%, 1.6% ... of the central peak).

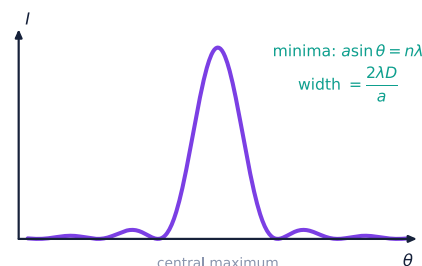


Fig. 10.6 Single-slit diffraction pattern.

Why the first minimum is at

$$a \sin \theta = \lambda \quad (\text{qualitative}) \quad 3 \text{ MARKS}$$

Principle Divide the slit into two halves; pair points separated by $a/2$.

Assumptions Plane wave, slit of width a .

- At an angle θ the path difference between the extreme rays is $a \sin \theta$. Take $a \sin \theta = \lambda$.
- Pair each point in the top half with a point $a/2$ lower: their path difference is $\lambda/2$, so their contributions cancel. All pairs cancel $\rightarrow$ zero intensity.
- Hence the first minimum: $a \sin \theta = \lambda$; similarly $a \sin \theta = n\lambda$.

FINAL RESULT

$$a \sin \theta = n\lambda \text{ (minima),} \quad \text{central width} = \frac{2\lambda D}{a}$$

Limits: qualitative only for the board; maxima lie roughly midway:
 $a \sin \theta \approx \left(n + \frac{1}{2}\right) \lambda$.

How to write it in the exam: State the pairing argument in words.

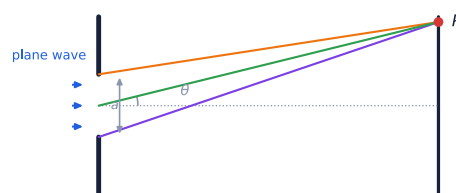


Fig. 10.7 Path difference across the slit.

Feature	Interference (Young)	Diffraction (single slit)
Origin	superposition of two coherent beams	superposition of wavelets from different parts of one slit
Fringe width	all equal, $\lambda D/d$	central is twice as wide as the others
Intensity	all bright fringes similar	falls rapidly away from the centre
Minima	zero intensity (equal amplitudes)	not quite zero
Condition	d small, $d \gg \lambda$	$a \sim \lambda$

10.5 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$\Delta = \frac{yd}{D}$	m	$y \ll D$	$y = 3 \text{ mm}; 1.5 \mu\text{m}$
bright $y_n = \frac{n\lambda D}{d}$	m		$n = 10; 12 \text{ mm}$
$\beta = \frac{\lambda D}{d}$	m	Young	600 nm, $d = 0.5 \text{ mm}, D = 1 \text{ m}; 1.2 \text{ mm}$
$\beta' = \beta/n$	m	in a medium	water: 0.90 mm
$a \sin \theta = n\lambda$	—	minima, slit	$\theta \approx \lambda/a$
width $= 2\lambda D/a$	m	central max	$a = 0.2 \text{ mm}; 5 \text{ mm}$
$\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2}$	—		$a_1 : a_2 = 3 : 1 \rightarrow 4$
$I = I_0 \cos^2 \theta$	W m^{-2}	Malus (ext.)	

N1 C10-N01 Ch 10 Easy Numerical BOARD-STYLE PRACTICE 2 marks

In Young's experiment $d = 0.5 \text{ mm}$, $D = 1 \text{ m}$ and $\lambda = 600 \text{ nm}$. Find the fringe width, and the position of the 10th bright fringe.

GIVEN $d = 5 \times 10^{-4} \text{ m}$, $D = 1 \text{ m}$, $\lambda = 6 \times 10^{-7} \text{ m}$

FIND β, y_{10}

FORMULA $\beta = \frac{\lambda D}{d}$, $y_n = n\beta$

WORKING $\beta = \frac{6 \times 10^{-7} \times 1}{5 \times 10^{-4}} = 1.2 \times 10^{-3} \text{ m}; y_{10} = 10 \times 1.2 = 12 \text{ mm}.$

Answer: $\beta = 1.2 \text{ mm}; y_{10} = 12 \text{ mm}$

▲ Fringe width is the same for bright and dark.

N2 C10-N02 Ch 10 Medium Numerical BOARD-STYLE PRACTICE 3 marks

With the same set-up, is the point 3 mm from the centre bright or dark? Find the path difference.

GIVEN $y = 3 \times 10^{-3} \text{ m}$, $d = 5 \times 10^{-4} \text{ m}$, $D = 1 \text{ m}$, $\lambda = 600 \text{ nm}$

FIND Δ , nature

FORMULA $\Delta = \frac{yd}{D}$

WORKING $\Delta = \frac{3 \times 10^{-3} \times 5 \times 10^{-4}}{1} = 1.5 \times 10^{-6} \text{ m}; \frac{\Delta}{\lambda} = \frac{1.5 \times 10^{-6}}{6 \times 10^{-7}} = 2.5 \rightarrow \text{odd multiple of } \lambda/2.$

Answer: Dark fringe ($\Delta = 2.5\lambda$)

▲ $2.5 = (2 + \frac{1}{2})$, so destructive.

N3 C10-N03 Ch 10 Medium Numerical BOARD-STYLE PRACTICE 2 marks

A single slit of width 0.2 mm is illuminated by 500 nm light; the screen is 1 m away. Find the width of the central maximum.

GIVEN $a = 2 \times 10^{-4} \text{ m}$, $\lambda = 5 \times 10^{-7} \text{ m}$, $D = 1 \text{ m}$

FIND width

FORMULA $w = \frac{2\lambda D}{a}$

WORKING $w = \frac{2 \times 5 \times 10^{-7} \times 1}{2 \times 10^{-4}} = 5 \times 10^{-3} \text{ m}.$

Answer: 5 mm

▲ Narrowing the slit widens the pattern.

10.6 More numericals and subjective practice

N4 C10-N04 Ch 10 Medium Numerical BOARD-STYLE PRACTICE 2 marks

The Young set-up of N1 is immersed in water ($n = 1.33$). Find the new fringe width and wavelength.

GIVEN $\beta = 1.2 \text{ mm}$, $\lambda = 600 \text{ nm}$, $n = 1.33$

FIND β' , λ'

FORMULA $\lambda' = \lambda/n$, $\beta' = \beta/n$

WORKING $\lambda' = \frac{600}{1.33} = 451 \text{ nm}$; $\beta' = \frac{1.2}{1.33} = 0.90 \text{ mm}$.

Answer: $\lambda' = 451 \text{ nm}$; $\beta' = 0.90 \text{ mm}$

▲ Fringes crowd closer in a medium.

N5 C10-N05 Ch 10 Medium Numerical BOARD-STYLE PRACTICE 3 marks

A plane wave in air meets a glass plate at 60° to the normal; the wave speed in glass is $2 \times 10^8 \text{ m s}^{-1}$. Find n and the angle of refraction.

GIVEN $i = 60^\circ$, $v_2 = 2 \times 10^8 \text{ m/s}$, $v_1 = 3 \times 10^8 \text{ m/s}$

FIND n , r

FORMULA $\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$

WORKING $n = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$; $\sin r = \frac{\sin 60^\circ}{1.5} = 0.577 \Rightarrow r = 35.3^\circ$.

Answer: $n = 1.5$; $r = 35.3^\circ$

▲ Same result as Snell's law.

N6 C10-N06 Ch 10 Hard Numerical BOARD-STYLE PRACTICE 2 marks

Two coherent sources have intensity ratio 9:1. Find the ratio $I_{\max}/I_{\min}$ in the interference pattern.

GIVEN $I_1:I_2 = 9:1$

FIND $I_{\max}/I_{\min}$

FORMULA $a \propto \sqrt{I}$; $\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$

WORKING $a_1:a_2 = 3:1$; $\frac{I_{\max}}{I_{\min}} = \left(\frac{3+1}{3-1} \right)^2 = 4$.

Answer: 4:1

▲ Intensity is proportional to amplitude squared.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C10-S01 Ch 10 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. Why are two independent sodium lamps not able to produce sustained interference?

Model answer: Their phase difference changes randomly in $\sim 10^{-10} \text{ s}$; they are incoherent, so the pattern shifts too fast to be seen.

C10-S02 Ch 10 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. In Young's experiment the slits are covered by red and blue filters respectively. What do you see? What happens if the filters are replaced by one red filter?

Model answer: Red and blue light are not coherent (different wavelengths), so no fringes; only uniform illumination. With a single red filter coherent red light gives the usual fringes.

C10-S03 Ch 10 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S3. The slit width in a single-slit experiment is halved. What happens to the width of the central maximum and to its intensity?

Model answer: Width doubles ($2\lambda D/a$); the intensity decreases because less light passes through.

C10-CB01 Ch 10 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. A student uses $\lambda = 500 \text{ nm}$, $d = 0.2 \text{ mm}$ and $D = 1.2 \text{ m}$ in a Young's set-up. (i) Find β . (ii) Distance between the 3rd bright fringe on one side and 3rd dark on the other side of the centre. (iii) Effect of using violet light. (iv) What happens if d is doubled?

Model answer: (i) $\beta = \frac{5 \times 10^{-7} \times 1.2}{2 \times 10^{-4}} = 3 \text{ mm}$. (ii) $y_3 = 3\beta = 9 \text{ mm}$; third dark = $2.5\beta = 7.5 \text{ mm}$; separation 16.5 mm . (iii) Violet has smaller λ , so β decreases. (iv) β halves.

10.7 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C10-M01 Ch 10 Easy Conceptual

BOARD-STYLE PRACTICE

1. Huygens' principle is used to

- (a) find the speed of light (b) construct a new wavefront
(c) measure wavelength (d) explain the photoelectric effect

Ans: (b) It constructs wavefronts.

C10-M02 Ch 10 Medium Numerical

BOARD-STYLE PRACTICE

2. In Young's experiment the fringe width is 2 mm. If D is halved and d is doubled, the new fringe width is

- (a) 4 mm (b) 1 mm
(c) 0.5 mm (d) 2 mm

Ans: (c) $\beta \propto D/d \rightarrow \frac{1}{2} \times 2 = 1$ mm.

C10-M03 Ch 10 Medium Conceptual

BOARD-STYLE PRACTICE

3. The central fringe in Young's experiment with white light is

- (a) dark (b) white
(c) red (d) violet

Ans: (b) All colours interfere constructively at $\Delta = 0$.

C10-M04 Ch 10 Medium Numerical

BOARD-STYLE PRACTICE

4. A slit of width a gives the first minimum at 30° for $\lambda = 500$ nm. Then a is

- (a) 500 nm (b) 1000 nm
(c) 250 nm (d) 2000 nm

Ans: (b) $a = \lambda / \sin 30^\circ = 1000$ nm.

C10-M05 Ch 10 Medium Application

BOARD-STYLE PRACTICE

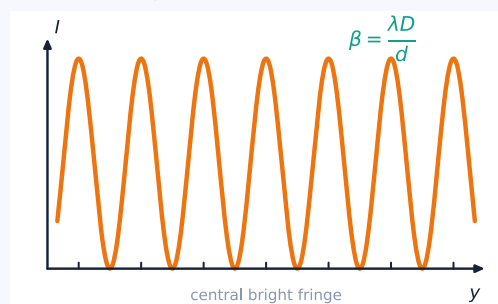
5. When light enters glass from air, its

- (a) frequency decreases (b) wavelength decreases
(c) frequency increases (d) speed increases

Ans: (b) f is unchanged; v and λ decrease.

C10-M06 Ch 10 Medium Graph BOARD-STYLE PRACTICE

6. The intensity pattern shown is for



- (a) single-slit diffraction (b) double-slit interference
(c) a plane mirror (d) a prism

Ans: (b) Equal bright fringes: Young.

C10-M07 Ch 10 Medium Numerical

BOARD-STYLE PRACTICE

7. The path difference for a dark fringe is

- (a) $n\lambda$ (b) $(n + \frac{1}{2})\lambda$
(c) $2n\lambda$ (d) $\lambda/4$

Ans: (b) Destructive interference.

C10-M08 Ch 10 Hard Application

BOARD-STYLE PRACTICE

8. The ratio of the width of the central maximum to that of a secondary maximum in single-slit diffraction is

- (a) 1:1 (b) 2:1
(c) 1:2 (d) 4:1

Ans: (b) Central is twice as wide.

C10-AR01 Ch 10 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: Two torches cannot show interference fringes.
R: They are not coherent sources.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Incoherent sources give no stable pattern.

C10-AR02 Ch 10 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: In diffraction the central maximum becomes wider if the slit is made narrower.

R: Width of the central maximum is $2\lambda D/a$.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) $w \propto 1/a$.

★ EXAMINER-STYLE REMINDERS

- Name coherence as the condition for sustained interference.
- Draw the wavefront figures with $AE = v_2\tau$ and $BC = v_1\tau$ marked.
- State clearly: β for Young and $2\lambda D/a$ for a slit.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are BOARD-STYLE PRACTICE; none is claimed to be a past board question.

R Chapter 10 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Huygens construction (plane wavefront)
2. Reflection of a plane wavefront
3. Refraction of a plane wavefront
4. Young's geometry
5. Single-slit intensity pattern

★ 7-DAY REVISION PLAN

Day 1	Wavefronts
Day 2	Reflection proof
Day 3	Refraction proof
Day 4	Young's experiment
Day 5	Diffraction
Day 6	N1–N6
Day 7	MCQs

Chapter 10 at a glance — Wave Optics

Wavefront

Huygens' construction; rays $\perp$ wavefront.

Laws

$$i = r; \sin i / \sin r = v_1 / v_2.$$

Interference

coherent sources; $\Delta = n\lambda$ bright.

Young

$$\beta = \lambda D / d.$$

Diffraction

$$a \sin \theta = n\lambda; \text{width } 2\lambda D / a.$$

Formula map

$$\frac{\lambda D}{d}$$

$$\beta$$

$$\frac{yd}{D}$$

$$\Delta$$

$$\beta$$

$$n$$

in medium

$$a \sin \theta$$

minima

$$\frac{2\lambda D}{a}$$

central width

$$\left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$$

$I_{\max} / I_{\min}$

! COMMON MISTAKES

- Using $\sin i / \sin r = v_2 / v_1$.
- Confusing fringe width with fringe position.
- Assuming fringe width changes with y .
- Forgetting central diffraction width is $2\lambda D / a$ (not $\lambda D / a$).

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can prove $i = r$ with Huygens
- ☐ I can prove Snell's law with Huygens
- ☐ I know the coherence condition
- ☐ I can derive $\beta = \lambda D / d$
- ☐ I can describe single-slit diffraction
- ☐ I solved N1–N6
- ☐ I did the MCQs

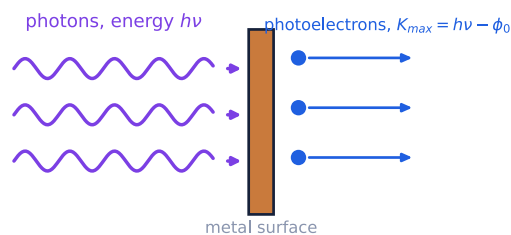
✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

1. Condition for sustained interference? | **Ans:** Coherent sources
2. What stays unchanged on refraction? | **Ans:** Frequency
3. β for violet vs red? | **Ans:** Smaller
4. Central max width? | **Ans:** $2\lambda D / a$
5. Bright fringe Δ ? | **Ans:** $n\lambda$

UNIT VII · DUAL NATURE OF RADIATION AND MATTER

11 Dual Nature of Radiation and Matter

Light behaves as a wave in interference and as a particle in photoemission; electrons, which we think of as particles, also show diffraction. Nature has a dual character.



Chapter roadmap

Photoelectric effect

Failure of wave theory

Einstein's equation

Photons

Matter waves

✓ LEARNING OUTCOMES

- Describe the photoelectric experiment and state the observations (Hertz, Lenard) that the wave theory could not explain
- State Einstein's photoelectric equation and use it: threshold frequency, work function, stopping potential, K_{max}
- Interpret the I - V and K_{max} - ν graphs; find h from a graph
- Describe photon properties and de Broglie's hypothesis; calculate de Broglie wavelengths
- State the conclusion of the Davisson–Germer experiment

1 PREREQUISITES TO REVISE

- Energy in electronvolts
- Electromagnetic waves (Chapter 8)
- Work–energy and potential difference
- Wave properties: λ , f

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: dual nature of radiation; photoelectric effect; Hertz and Lenard's observations; Einstein's photoelectric equation — particle nature of light; matter waves — wave nature of particles; de Broglie relation; Davisson–Germer experiment (only conclusion should be explained).

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (threshold, $\lambda = h/p$); 2-mark (observations, photon properties); 3-mark numericals (stopping potential, de Broglie λ); 5-mark derivation of the photoelectric equation consequences or graph questions; case-based on a photocell. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Key relations

F PHOTON AND ELECTRON

$$E = h\nu = \frac{hc}{\lambda}, p = \frac{h}{\lambda}; K_{max} = h\nu - \phi_0 = eV_0;$$

$$\lambda_{dB} = \frac{h}{p} = \frac{h}{\sqrt{2mK}}; \text{electron through } V:$$

$$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm.}$$

★ USEFUL CONSTANTS

$$h = 6.63 \times 10^{-34} \text{ J s}; hc = 1240 \text{ eV nm}; 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}; m_e = 9.1 \times 10^{-31} \text{ kg.}$$

11.1 The photoelectric effect

1 WHAT IT IS

Emission of electrons from a metal surface when light of suitable frequency falls on it. The emitted electrons are **photoelectrons**. Discovered by Hertz (1887); studied by Hertz, Hallwachs and Lenard.

Key terms: **work function** ϕ_0 — minimum energy to remove an electron; **threshold frequency** $\nu_0 = \phi_0/h$; **stopping potential** V_0 — retarding potential that just stops the fastest photoelectrons.

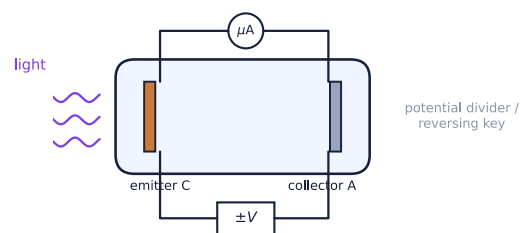


Fig. 11.1 Photoelectric apparatus with a variable (reversible) potential.

★ EXPERIMENTAL OBSERVATIONS

1. Photocurrent is proportional to the **intensity** of light (for $\nu > \nu_0$).
2. The stopping potential, hence K_{max} , depends on **frequency** and not on intensity.
3. Below the threshold frequency no emission occurs, however intense the light.
4. Emission is **instantaneous** ($\sim 10^{-9}$ s) even for faint light above ν_0 .

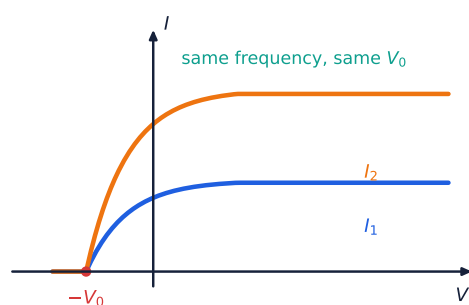


Fig. 11.2 Photocurrent against V for two intensities: larger saturation current, same V_0 .

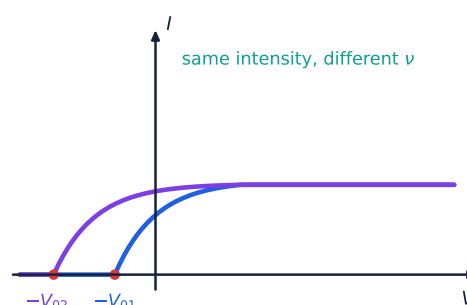


Fig. 11.3 For a higher frequency the stopping potential increases (same saturation current).

! WHY THE WAVE THEORY FAILS

- Predicts K_{max} should increase with intensity — observed to be independent.
- Predicts emission at any frequency if intense enough or given time — observed threshold.
- Predicts a time delay to accumulate energy — observed instantaneous emission.

11.2 Einstein's photoelectric equation

Einstein's photoelectric equation and its consequences

5 MARKS

Principle Light consists of photons of energy $h\nu$; one photon is absorbed by one electron (energy conservation).

Assumptions Photon energy $h\nu \geq \phi_0$; emitter at constant temperature.

- ① Energy of a photon: $E = h\nu$. An electron absorbs it whole: part ϕ_0 frees it, the remainder is kinetic energy.
- ② $K_{max} = h\nu - \phi_0 = h(\nu - \nu_0)$. The most loosely bound electrons leave with K_{max} ; others with less.
- ③ Stopping potential: $eV_0 = K_{max} \Rightarrow V_0 = \frac{h}{e}(\nu - \nu_0)$.

FINAL RESULT

$$K_{max} = h\nu - \phi_0, \quad eV_0 = h\nu - \phi_0, \quad \nu_0 = \frac{\phi_0}{h}$$

Limits: explains: threshold ($\nu < \nu_0 \Rightarrow K < 0$, no emission); K_{max} independent of intensity; instantaneous emission (one-to-one collision); current $\propto$ number of photons $\propto$ intensity.

How to write it in the exam: Write the energy balance and name each term; list the explanations.

GRAPH OF K_{max} AGAINST ν

- Straight line, not through the origin.
- Intercept on the ν -axis: threshold frequency ν_0 .
- Slope $= h$ (same for all metals); intercept on the K -axis $= -\phi_0$.
- For V_0 against ν : slope $= h/e$ — used by Millikan to find h .

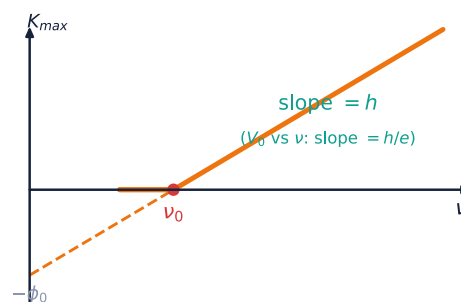


Fig. 11.4 K_{max} against frequency.

PHOTON PROPERTIES

Energy $E = h\nu = hc/\lambda$ (in eV: $E = 1240/\lambda_{nm}$); momentum $p = E/c = h/\lambda$; zero rest mass; uncharged; not deflected by fields; speed c in vacuum; intensity = photons per unit area per second $\times h\nu$. Collision with an electron conserves energy and momentum.

11.3 Matter waves and de Broglie's hypothesis

I DE BROGLIE RELATION

If light (a wave) can act like particles, matter should show wave behaviour. de Broglie proposed that a particle of momentum p has an associated wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}}$$

For a charged particle q accelerated through V : $\lambda = \frac{h}{\sqrt{2mqV}}$

Electron: $\lambda \text{ (nm)} = \frac{1.227}{\sqrt{V}}$. Matter waves are not EM waves and need no medium; their wavelength is measurable only for tiny masses (a 0.2 kg ball at 10 m/s: $\lambda \sim 10^{-34}$ m).

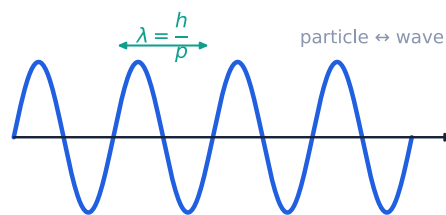


Fig. 11.5 Wave associated with a moving particle.

Wavelength of an electron accelerated through a potential V

3 MARKS

Principle Work done by the field equals kinetic energy; then $\lambda = h/p$.

Assumptions Electron starting from rest; non-relativistic.

$$\textcircled{1} eV = \frac{1}{2}m_e v^2 \Rightarrow p = m_e v = \sqrt{2m_e eV}.$$

$$\textcircled{2} \lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e eV}}.$$

$$\textcircled{3} \text{ Numerically } \lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} V}} = \frac{1.227 \times 10^{-9}}{\sqrt{V}} \text{ m}.$$

FINAL RESULT

$$\lambda = \frac{h}{\sqrt{2m_e eV}} = \frac{1.227}{\sqrt{V}} \text{ nm}$$

Limits: 100 V gives 0.123 nm, comparable to atomic spacing — hence electron diffraction by crystals.

How to write it in the exam: State $p = \sqrt{2meV}$ then substitute.

I DAVISSON-GERMER EXPERIMENT (CONCLUSION ONLY)

A beam of electrons (accelerated through 54 V) was scattered from a nickel crystal. The intensity of the scattered beam showed a strong maximum at $\theta = 50^\circ$, as expected for diffraction. The measured wavelength $\lambda = 0.165$ nm agreed with de Broglie's value $1.227/\sqrt{54} = 0.167$ nm — direct confirmation of the **wave nature of electrons**.

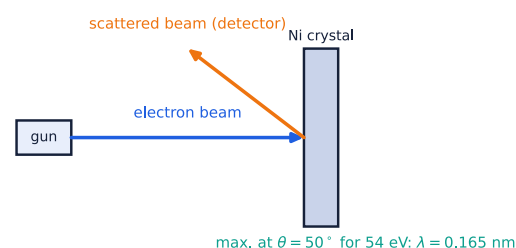


Fig. 11.6 Electron scattering from a nickel crystal.

11.4 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$E = h\nu = \frac{hc}{\lambda}$	J or eV	$hc = 1240 \text{ eV nm}$	400 nm: 3.1 eV
$p = \frac{h}{\lambda}$	kg m s ⁻¹	photon	500 nm: 1.33×10^{-27}
$K_{max} = h\nu - \phi_0$	eV		3.1 – 2.3 = 0.8 eV
$eV_0 = K_{max}$	V		$V_0 = 0.8 \text{ V}$
$\nu_0 = \phi_0/h$	Hz	threshold	2.3 eV: $5.6 \times 10^{14} \text{ Hz}$
$\lambda_{dB} = \frac{h}{mv}$	m		0.2 kg, 10 m/s: 3.3×10^{-34}
$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$	nm	electron	100 V: 0.123 nm
$\lambda = \frac{h}{\sqrt{2mK}}$	m		

N1 C11-N01 Ch 11 Easy Numerical BOARD-STYLE PRACTICE 3 marks

Light of wavelength 400 nm falls on sodium (work function 2.3 eV). Find the photon energy, K_{max} , the stopping potential and the threshold wavelength.

GIVEN $\lambda = 400 \text{ nm}$, $\phi_0 = 2.3 \text{ eV}$, $hc = 1240 \text{ eV nm}$

FIND E , K_{max} , V_0 , λ_0

FORMULA $E = \frac{hc}{\lambda}$; $K_{max} = E - \phi_0$; $\lambda_0 = \frac{hc}{\phi_0}$

WORKING $E = \frac{1240}{400} = 3.1 \text{ eV}$; $K_{max} = 3.1 - 2.3 = 0.8 \text{ eV}$; $V_0 = 0.8 \text{ V}$; $\lambda_0 = \frac{1240}{2.3} = 539 \text{ nm}$.

Answer: 3.1 eV; 0.8 eV; 0.8 V; 539 nm

▲ $\lambda < \lambda_0$ is required for emission.

N2 C11-N02 Ch 11 Medium Numerical BOARD-STYLE PRACTICE 3 marks

In a Millikan-type experiment the stopping potentials are 0.48 V at $6 \times 10^{14} \text{ Hz}$ and 1.73 V at $9 \times 10^{14} \text{ Hz}$. Find h and the work function.

GIVEN $(\nu_1, V_1) = (6 \times 10^{14}, 0.48)$, $(\nu_2, V_2) = (9 \times 10^{14}, 1.73)$

FIND h , ϕ_0

FORMULA $V_0 = \frac{h}{e}(\nu - \nu_0)$, slope = $\frac{h}{e}$

WORKING slope = $\frac{1.73 - 0.48}{3 \times 10^{14}} = 4.17 \times 10^{-15} \text{ V s}$; $h = e \times \text{slope} = 1.6 \times 10^{-19} \times 4.17 \times 10^{-15} = 6.7 \times 10^{-34} \text{ J s}$.
 $\phi_0 = h\nu_1 - eV_1 = 4.17 \times 10^{-15} \times 6 \times 10^{14} - 0.48 = 2.0 \text{ eV}$.

Answer: $h \approx 6.7 \times 10^{-34} \text{ J s}$; $\phi_0 \approx 2.0 \text{ eV}$

▲ Use two points on the straight V_0 - ν line.

N3 C11-N03 Ch 11 Medium Numerical BOARD-STYLE PRACTICE 2 marks

Find the de Broglie wavelength of an electron accelerated through 100 V.

GIVEN $V = 100 \text{ V}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$

FIND λ

FORMULA $\lambda = \frac{h}{\sqrt{2m_e eV}}$

WORKING $p = \sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 100} = 5.4 \times 10^{-24}$; $\lambda = \frac{6.63 \times 10^{-34}}{5.4 \times 10^{-24}} = 1.23 \times 10^{-10} \text{ m}$.

Answer: 0.123 nm

▲ Check: $1.227/\sqrt{100} = 0.123 \text{ nm}$.

11.5 More numericals and subjective practice

N4 C11-N04 Ch 11 Medium Numerical BOARD-STYLE PRACTICE 2 marks

Find the de Broglie wavelength of a 0.2 kg ball moving at 10 m s^{-1} , and the momentum of a 500 nm photon.

GIVEN $m = 0.2 \text{ kg}$, $v = 10 \text{ m/s}$; $\lambda = 500 \text{ nm}$

FIND λ_{dB} , p

FORMULA $\lambda = \frac{h}{mv}$; $p = \frac{h}{\lambda}$

WORKING $\lambda_{dB} = \frac{6.63 \times 10^{-34}}{0.2 \times 10} = 3.3 \times 10^{-34} \text{ m}$. $p = \frac{6.63 \times 10^{-34}}{5 \times 10^{-7}} = 1.3 \times 10^{-27} \text{ kg m s}^{-1}$.

Answer: $3.3 \times 10^{-34} \text{ m}$ (unobservable); $1.3 \times 10^{-27} \text{ kg m/s}$

▲ Macroscopic wavelengths are far too small to detect.

N5 C11-N05 Ch 11 Hard Numerical BOARD-STYLE PRACTICE 3 marks

A 1 mW beam of 400 nm light falls on a surface; 1 photon in 1000 ejects an electron. Find the photon rate and the photocurrent.

GIVEN $P = 10^{-3} \text{ W}$, $E = 3.1 \text{ eV} = 4.96 \times 10^{-19} \text{ J}$, efficiency 10^{-3}

FIND n , I

FORMULA $n = P/E$; $I = \eta ne$

WORKING $n = \frac{10^{-3}}{4.96 \times 10^{-19}} = 2.0 \times 10^{15} \text{ s}^{-1}$; electrons per s $= 2.0 \times 10^{12}$; $I = 2.0 \times 10^{12} \times 1.6 \times 10^{-19} = 3.2 \times 10^{-7} \text{ A}$.

Answer: $2.0 \times 10^{15} \text{ photons/s}$; $0.32 \mu\text{A}$

▲ Current is proportional to intensity.

N6 C11-N06 Ch 11 Medium Numerical BOARD-STYLE PRACTICE 2 marks

If the kinetic energy of a particle is quadrupled, by what factor does its de Broglie wavelength change? Also find the ratio of wavelengths of a proton and an electron with equal kinetic energy.

GIVEN $K \rightarrow 4K$; $m_p = 1836 m_e$

FIND λ'/λ ; λ_p/λ_e

FORMULA $\lambda = \frac{h}{\sqrt{2mK}}$

WORKING $\lambda \propto K^{-1/2}$; $\lambda'/\lambda = 1/\sqrt{4} = \frac{1}{2}$. For equal K : $\frac{\lambda_p}{\lambda_e} = \sqrt{\frac{m_e}{m_p}} = \frac{1}{\sqrt{1836}} = 0.023$.

Answer: halved; $\lambda_p/\lambda_e \approx 0.023$

▲ Heavier particles have shorter wavelengths.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C11-S01 Ch 11 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. What is the work function? Give one reason why an alkali metal is preferred for a photocell.

Model answer: Minimum energy needed to free an electron from the metal surface. Alkali metals have low work functions (about 2 eV), so even visible light causes emission.

C11-S02 Ch 11 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. Light of frequency ν_1 and intensity I gives stopping potential V_1 . What happens to the saturation current and the stopping potential if (a) intensity doubles, (b) frequency is raised?

Model answer: (a) Saturation current doubles; V_0 unchanged. (b) V_0 increases ($= h(\nu - \nu_0)/e$); the saturation current depends on the number of photons per second, so it is unchanged if that number is kept the same.

C11-S03 Ch 11 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S3. State the conclusion of the Davisson–Germer experiment and the relation it verified.

Model answer: Electrons undergo diffraction from a crystal, showing that they have a wave nature with wavelength $\lambda = h/p$, verifying de Broglie's hypothesis.

C11-CB01 Ch 11 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. A metal has threshold wavelength 540 nm. It is illuminated with 360 nm light. (i) Find the work function. (ii) Find K_{max} . (iii) Find V_0 . (iv) Will 600 nm light cause emission?

Model answer: (i) $\phi_0 = 1240/540 = 2.30 \text{ eV}$. (ii) $E = 1240/360 = 3.44 \text{ eV}$; $K_{max} = 3.44 - 2.30 = 1.14 \text{ eV}$. (iii) $V_0 = 1.14 \text{ V}$. (iv) No: 600 nm $>$ 540 nm (photon energy $2.07 \text{ eV} < \phi_0$).

11.6 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C11-M01 Ch 11 Easy Conceptual

BOARD-STYLE PRACTICE

1. Photoelectric emission depends on

- (a) intensity only (b) frequency of the light
(c) time of exposure (d) colour of the metal

Ans: (b) Requires $\nu > \nu_0$.

C11-M02 Ch 11 Medium Numerical

BOARD-STYLE PRACTICE

2. The energy of a photon of wavelength 620 nm is nearly

- (a) 1 eV (b) 2 eV
(c) 3 eV (d) 4 eV

Ans: (b) $1240/620 = 2$ eV.

C11-M03 Ch 11 Medium Conceptual

BOARD-STYLE PRACTICE

3. If the intensity of light is doubled (same $\nu > \nu_0$), the stopping potential

- (a) doubles (b) halves
(c) remains the same (d) becomes zero

Ans: (c) V_0 depends on ν .

C11-M04 Ch 11 Medium Numerical

BOARD-STYLE PRACTICE

4. The de Broglie wavelength of an electron accelerated through 150 V is nearly

- (a) 0.1 nm (b) 1 nm
(c) 10 nm (d) 0.01 nm

Ans: (a) $1.227/\sqrt{150} = 0.1$ nm.

C11-M05 Ch 11 Medium Application

BOARD-STYLE PRACTICE

5. The slope of the V_0 - ν graph gives

- (a) h (b) h/e
(c) ϕ_0 (d) e/h

Ans: (b) $V_0 = (h/e)(\nu - \nu_0)$.

C11-M06 Ch 11 Medium Graph

BOARD-STYLE PRACTICE

6. The K_{max} - ν graph for two metals of different work functions has

- (a) different slopes (b) the same slope
(c) zero slope (d) negative slope

Ans: (b) Slope = h .

C11-M07 Ch 11 Medium Numerical

BOARD-STYLE PRACTICE

7. The ratio of de Broglie wavelengths of two particles with the same speed and masses in ratio 1:4 is

- (a) 4:1 (b) 1:4
(c) 2:1 (d) 1:2

Ans: (a) $\lambda \propto 1/m$.

C11-M08 Ch 11 Hard Application

BOARD-STYLE PRACTICE

8. A photon and an electron have the same wavelength. Then

- (a) same energy (b) same momentum
(c) same speed (d) same kinetic energy

Ans: (b) $p = h/\lambda$ for both.

C11-AR01 Ch 11 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: Photoelectric emission does not occur below the threshold frequency.

R: A photon below ν_0 has energy less than the work function.

- (a) both true, R explains A; (b) both true, R does not explain A;
(c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Single-photon absorption.

C11-AR02 Ch 11 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: Matter waves of a moving cricket ball are not observed.

R: Its wavelength is extremely small.

- (a) both true, R explains A; (b) both true, R does not explain A;
(c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) $\lambda = h/mv$ is tiny.

★ EXAMINER-STYLE REMINDERS

- Name the photon's role: one photon, one electron.
- For graph questions state slope and intercept meaning.
- Use $hc = 1240$ eV nm to save time.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are **BOARD-STYLE PRACTICE**; none is claimed to be a past board question.

R Chapter 11 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Photoelectric circuit
2. I - V for two intensities and two frequencies
3. K_{max} - ν graph
4. Davisson–Germer set-up

★ 7-DAY REVISION PLAN

Day 1	Photoelectric observations
Day 2	Wave theory failure
Day 3	Einstein's equation
Day 4	Graphs
Day 5	Photon
Day 6	de Broglie
Day 7	N1–N6; MCQs

Chapter 11 at a glance — Dual Nature of Radiation and Matter

Photoelectric effect

$\nu > \nu_0$; instantaneous; $I \propto$ intensity.

Einstein

$$K_{max} = h\nu - \phi_0 = eV_0.$$

Graphs

V_0 vs ν : slope h/e ; K vs ν : slope h .

Photon

$$E = h\nu, p = h/\lambda.$$

de Broglie

$\lambda = h/p$; electron $1.227/\sqrt{V}$ nm; Davisson–Germer.

Formula map

$$\frac{h\nu}{\text{photon energy}}$$

$$\frac{hc}{\lambda} = 1240/\lambda \text{ eV}$$

$$\frac{h\nu - \phi_0}{K_{max}}$$

$$\frac{\phi_0}{h\nu_0}$$

$$\frac{h}{p} = \lambda_{dB}$$

$$\frac{1.227}{\sqrt{V}} \text{ electron nm}$$

! COMMON MISTAKES

- Expecting K_{max} to depend on intensity.
- Forgetting ϕ_0 in eV vs J.
- Using ν below threshold.
- Writing $\lambda = h/E$ (use momentum).
- Mixing h with h/e for slopes.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I know the 4 observations
- ☐ I can derive and use Einstein's equation
- ☐ I can read both graphs
- ☐ I know photon properties
- ☐ I can find λ_{dB}
- ☐ I solved N1–N6
- ☐ I did the MCQs

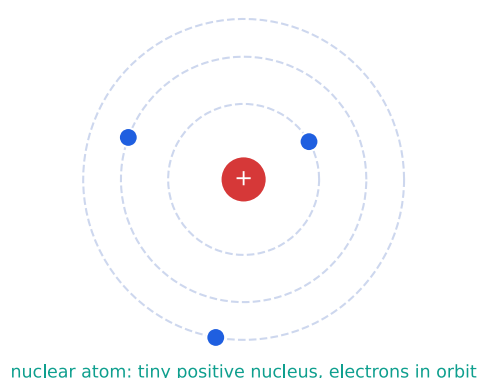
✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

1. Unit of h ? | **Ans:** J s
2. Photon rest mass? | **Ans:** Zero
3. V_0 vs ν slope? | **Ans:** h/e
4. Matter wave of ball seen? | **Ans:** No (too small)
5. Davisson–Germer proves? | **Ans:** Wave nature of electrons

UNIT VIII · ATOMS AND NUCLEI

12 Atoms

Rutherford showed that an atom is mostly empty with a tiny dense nucleus. Bohr then explained why atoms are stable and why hydrogen emits only certain colours.



Chapter roadmap

Alpha scattering

Rutherford model

Bohr postulates

Orbits & energies

Line spectra

✓ LEARNING OUTCOMES

- Describe Rutherford's α -scattering experiment and its conclusions
- State the limitations of Rutherford's model
- State Bohr's postulates and derive r_n , v_n and E_n for hydrogen
- Explain the hydrogen line spectrum and the Lyman, Balmer and Paschen series; use the Rydberg formula
- Solve numericals on orbit radii, energies and spectral lines

I PREREQUISITES TO REVISE

- Coulomb's law (Chapter 1)
- Circular motion and angular momentum
- de Broglie wavelength (Chapter 11)
- Energy in electronvolts

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: α -particle scattering experiment; Rutherford's model of the atom; Bohr model of the hydrogen atom, expression for the radius of the n th possible orbit, velocity and energy of the electron in the n th orbit; hydrogen line spectra (qualitative treatment only).

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (series names, energy levels); 2-mark (limitations of Rutherford, Bohr postulates); 3-mark numericals (wavelength of a line, r_n , E_n); 5-mark derivation of r_n and E_n or of the line spectra; case-based on energy-level diagram. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Key results

F BOHR MODEL OF HYDROGEN

$$r_n = 0.529n^2 \text{ \AA}; v_n = \frac{2.19 \times 10^6}{n} \text{ m s}^{-1};$$

$$E_n = -\frac{13.6}{n^2} \text{ eV}; mv_n r_n = \frac{nh}{2\pi}.$$

★ SPECTRAL LINES

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{ with } R = 1.097 \times 10^7 \text{ m}^{-1};$$

$$h\nu = E_i - E_f.$$

12.1 Rutherford's model

I THE α -SCATTERING EXPERIMENT (GEIGER-MARSDEN)

A beam of α particles (~ 5.5 MeV) from a radioactive source was directed at a thin gold foil ($\sim 10^{-7}$ m). A rotatable detector counted the scattered particles.

- Most α particles passed straight through.
- A small fraction was deflected through large angles.
- About 1 in 8000 (some books: 1 in 10^4) was turned back through more than 90° .

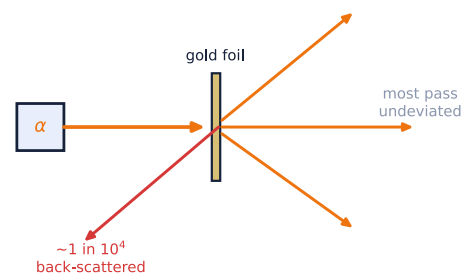


Fig. 12.1 Rutherford's experiment (schematic).

★ CONCLUSIONS

1. Most of the atom is **empty space** (undeviated passage).
2. The positive charge and almost all the mass are in a tiny **nucleus** (size $\sim 10^{-15}$ m versus atom $\sim 10^{-10}$ m).
3. Large deflection needs a strong repulsive force from a concentrated charge (Coulomb repulsion); head-on collisions return the particle.
4. Electrons orbit the nucleus like planets under the Coulomb attraction.

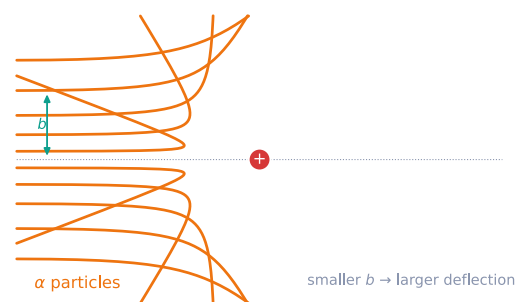


Fig. 12.2 Trajectories for different impact parameters b (Coulomb repulsion).

Distance of closest approach (head-on)

2 MARKS

Principle Energy conservation: the α particle stops momentarily when all its kinetic energy has become electrostatic potential energy.

Assumptions α particle ($q = 2e$) of kinetic energy K approaching a nucleus of charge Ze head-on.

① At closest approach r_0 all K is potential energy: $K = \frac{1}{4\pi\epsilon_0} \frac{(2e)(Ze)}{r_0}$.

② So $r_0 = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K}$.

FINAL RESULT

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K} \quad \left(\approx 4.5 \times 10^{-14} \text{ m for } 5 \text{ MeV on gold} \right)$$

Limits: gives an upper bound for the nuclear size.

How to write it in the exam: State energy conservation and the substitution.

! LIMITATIONS OF RUTHERFORD'S MODEL

- An orbiting (accelerating) electron should radiate and spiral into the nucleus in $\sim 10^{-8}$ s — atoms would collapse.
- It cannot explain why atoms emit only discrete spectral lines (it predicts a continuous spectrum).

12.2 Bohr's model of the hydrogen atom

★ BOHR'S POSTULATES

1. An electron revolves in certain **stable (stationary) circular orbits** without radiating energy.
2. The allowed orbits satisfy the **quantum condition**:
angular momentum $L = mvr = \frac{nh}{2\pi}$, $n = 1, 2, 3, \dots$
3. Radiation is emitted or absorbed when the electron jumps between orbits: $h\nu = E_i - E_f$.

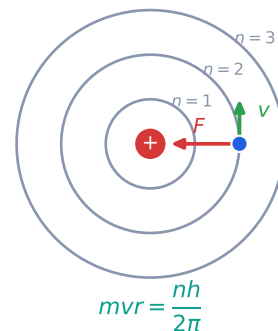


Fig. 12.3 Allowed orbits $n = 1, 2, 3$.

Radius, speed and energy of the electron in the n th orbit

5 MARKS

Principle Coulomb force provides the centripetal force; combine with the quantum condition.

Assumptions Hydrogen-like atom (charge Ze ; $Z = 1$ for H), nucleus at rest, circular orbit.

① Centripetal force: $\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2}$ (1). Quantum condition: $mvr = \frac{nh}{2\pi}$ (2).

② From (2) $v = \frac{nh}{2\pi mr}$; substitute in (1): $r_n = \frac{n^2 h^2 \epsilon_0}{\pi m Ze^2} = 0.529 \frac{n^2}{Z} \text{ \AA}$.

③ Speed: $v_n = \frac{Ze^2}{2\epsilon_0 nh} = \frac{2.19 \times 10^6 Z}{n} \text{ m s}^{-1}$.

④ Energy: $K = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r}$, $U = -\frac{Ze^2}{4\pi\epsilon_0 r}$, so $E = K + U = -\frac{Ze^2}{8\pi\epsilon_0 r_n} = -\frac{13.6 Z^2}{n^2} \text{ eV}$.

FINAL RESULT

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m Ze^2}, \quad v_n = \frac{Ze^2}{2\epsilon_0 nh}, \quad E_n = -\frac{13.6 Z^2}{n^2} \text{ eV}$$

Limits: $r_n \propto n^2$, $v_n \propto 1/n$, $E_n \propto -1/n^2$. Ground state $E_1 = -13.6 \text{ eV}$; ionisation energy 13.6 eV ; $K = -E$, $U = 2E$.

How to write it in the exam: Show (1) and (2), then each of r , v , E in turn.

~ DE BROGLIE'S EXPLANATION OF THE QUANTUM CONDITION

Treat the electron as a standing wave on the orbit: $2\pi r_n = n\lambda$ with $\lambda = h/mv$, which gives $mvr = nh/2\pi$ — the same condition as Bohr's second postulate. For $n = 1$: $\lambda = h/mv = 3.3 \times 10^{-10} \text{ m} = 2\pi r_1$.

12.3 Energy levels and the hydrogen spectrum

1 ENERGY LEVELS AND SPECTRAL SERIES

An electron in n_i falling to $n_f < n_i$ emits a photon:

$$h\nu = E_{n_i} - E_{n_f} = 13.6 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{ eV, i.e.}$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), R = 1.097 \times 10^7 \text{ m}^{-1}.$$

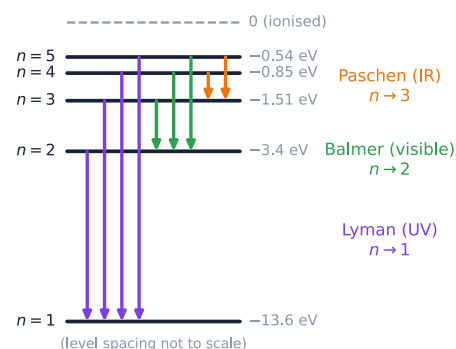


Fig. 12.4 Energy-level diagram of hydrogen showing the Lyman, Balmer and Paschen series.

Series	n_f	n_i	Region	Longest λ	Series limit
Lyman	1	2,3,...	ultraviolet	121.6 nm	91.2 nm
Balmer	2	3,4,...	visible (and near UV)	656.3 nm	364.6 nm
Paschen	3	4,5,...	infrared	1875 nm	820.4 nm
Brackett	4	5,6,...	infrared	4051 nm	1458 nm
Pfund	5	6,7,...	far infrared	7460 nm	2279 nm

Balmer series: visible lines of hydrogen



Fig. 12.5 The four visible Balmer lines of hydrogen.

1 EXCITATION AND IONISATION

Excitation energy of level n : $E_n - E_1$ (first excited state: 10.2 eV). **Ionisation energy** from level n : $-E_n$ (ground state 13.6 eV). Absorption lines are the same wavelengths as emission lines. From the level n , the number of different spectral lines possible is $\frac{n(n-1)}{2}$.

1 LIMITATIONS OF BOHR'S MODEL

- Works only for hydrogen-like (one-electron) atoms.
- Cannot explain fine structure, relative intensity of lines or the Zeeman effect.
- Treats the electron as a particle with a definite orbit; quantum mechanics replaces this.

12.4 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$r_n = 0.529 n^2 / Z \text{ Å}$	m	H: $a_0 = 0.529 \text{ Å}$	$n = 3$: 4.76 Å
$v_n = 2.19 \times 10^6 Z/n$	m s^{-1}		$n = 2$: 1.1×10^6
$E_n = -13.6 Z^2/n^2$	eV		$n = 3$: -1.51 eV
$mvr = nh/2\pi$	J s	quantum condition	
$h\nu = E_i - E_f$	J		$3 \rightarrow 2$: 1.89 eV, 656 nm
$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$	m^{-1}	$R = 1.097 \times 10^7$	Lyman limit 91.2 nm
$r_0 = \frac{2Ze^2}{4\pi\epsilon_0 K}$	m	head-on	5 MeV, Au: $4.5 \times 10^{-14} \text{ m}$
$N_{\text{lines}} = \frac{n(n-1)}{2}$	—	from level n	$n = 4$: 6 lines

N1

C12-N01

Ch 12

Easy

Numerical

BOARD-STYLE PRACTICE

3 marks

For hydrogen find the radius and energy of the electron in the $n = 3$ orbit. Find the wavelength of the photon emitted in the $3 \rightarrow 2$ transition.

GIVEN $n = 3$; $a_0 = 0.529 \text{ Å}$; $E_1 = -13.6 \text{ eV}$

FIND r_3 , E_3 , λ

FORMULA $r_n = n^2 a_0$, $E_n = -13.6/n^2$, $\lambda = hc/\Delta E$

WORKING $r_3 = 9 \times 0.529 = 4.76 \text{ Å}$; $E_3 = -\frac{13.6}{9} = -1.51 \text{ eV}$. $\Delta E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = 1.89 \text{ eV}$; $\lambda = \frac{1240}{1.89} = 656 \text{ nm}$ ($\text{H}\alpha$, red).

Answer: 4.76 Å; -1.51 eV; 656 nm

▲ Photon energy equals the level difference.

N2

C12-N02

Ch 12

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

Using $R = 1.097 \times 10^7 \text{ m}^{-1}$, find the shortest wavelength of the Lyman series and the longest and shortest wavelengths of the Balmer series.

GIVEN $R = 1.097 \times 10^7 \text{ m}^{-1}$

FIND $\lambda_{L, \min}$, $\lambda_{B, \max}$, $\lambda_{B, \min}$

FORMULA $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

WORKING Lyman limit ($n_f = 1, n_i = \infty$): $\lambda = \frac{1}{R} = 91.2 \text{ nm}$. Balmer longest ($2 \leftarrow 3$): $\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{9} \right) \Rightarrow \lambda = 656 \text{ nm}$.
Balmer limit ($2 \leftarrow \infty$): $\lambda = \frac{4}{R} = 365 \text{ nm}$.

Answer: 91.2 nm; 656 nm; 365 nm

▲ Shortest $\lambda \leftrightarrow n_i \rightarrow \infty$.

N3

C12-N03

Ch 12

Easy

Numerical

BOARD-STYLE PRACTICE

2 marks

Find the energy needed to excite hydrogen from the ground state to $n = 2$, the wavelength absorbed, and the energy needed to ionise it from $n = 2$.

GIVEN $E_1 = -13.6$, $E_2 = -3.4 \text{ eV}$

FIND ΔE , λ , E_{ion} ($n = 2$)

FORMULA $\Delta E = E_2 - E_1$; $\lambda = 1240/\Delta E$

WORKING $\Delta E = -3.4 - (-13.6) = 10.2 \text{ eV}$; $\lambda = \frac{1240}{10.2} = 121.6 \text{ nm}$ (Lyman α); ionisation from $n = 2$: 3.4 eV.

Answer: 10.2 eV; 121.6 nm; 3.4 eV

▲ Ionisation energy = $-E_n$.

12.5 More numericals and subjective practice

N4 C12-N04 Ch 12 Medium Numerical BOARD-STYLE PRACTICE 2 marks

A 5 MeV α particle approaches a gold nucleus ($Z = 79$) head-on. Find the distance of closest approach. (Use $e^2/4\pi\epsilon_0 = 1.44 \text{ MeV fm}$.)

GIVEN $K = 5 \text{ MeV}$, $Z = 79$, $q_\alpha = 2e$

FIND r_0

FORMULA $r_0 = \frac{2Ze^2}{4\pi\epsilon_0 K}$

WORKING $r_0 = \frac{2 \times 79 \times 1.44 \text{ MeV fm}}{5 \text{ MeV}} = 45.5 \text{ fm}$.

Answer: $r_0 \approx 4.6 \times 10^{-14} \text{ m}$

▲ Nuclear radius of gold is about 7 fm, so it never touches.

N5 C12-N05 Ch 12 Hard Numerical BOARD-STYLE PRACTICE 3 marks

Find the orbital period, frequency and equivalent current of the electron in the ground state of hydrogen ($r_1 = 0.529 \text{ \AA}$, $v_1 = 2.19 \times 10^6 \text{ m/s}$).

GIVEN $r_1 = 5.29 \times 10^{-11} \text{ m}$, $v_1 = 2.19 \times 10^6 \text{ m/s}$

FIND T, f, I

FORMULA $T = \frac{2\pi r}{v}$, $f = \frac{1}{T}$, $I = ef$

WORKING $T = \frac{2\pi \times 5.29 \times 10^{-11}}{2.19 \times 10^6} = 1.52 \times 10^{-16} \text{ s}$; $f = 6.6 \times 10^{15} \text{ Hz}$; $I = 1.6 \times 10^{-19} \times 6.6 \times 10^{15} = 1.05 \times 10^{-3} \text{ A}$.

Answer: $1.5 \times 10^{-16} \text{ s}$; $6.6 \times 10^{15} \text{ Hz}$; 1.05 mA

▲ Used again for the magnetic moment in Chapter 5.

N6 C12-N06 Ch 12 Medium Numerical BOARD-STYLE PRACTICE 2 marks

A hydrogen atom in the $n = 4$ state returns to the ground state. How many different spectral lines are possible, and what is the energy of the photon in the $4 \rightarrow 1$ jump?

GIVEN $n = 4$

FIND $N, \Delta E$

FORMULA $N = \frac{n(n-1)}{2}$; $\Delta E = 13.6 \left(1 - \frac{1}{16}\right)$

WORKING $N = \frac{4 \times 3}{2} = 6 \text{ lines}$ ($4 \rightarrow 3, 4 \rightarrow 2, 4 \rightarrow 1, 3 \rightarrow 2, 3 \rightarrow 1, 2 \rightarrow 1$). $\Delta E = 13.6 \times \frac{15}{16} = 12.75 \text{ eV}$.

Answer: 6 lines; 12.75 eV

▲ Count all paths, not only those to the ground state.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C12-S01 Ch 12 Easy Short answer (2 marks) BOARD-STYLE PRACTICE

S1. State two limitations of Rutherford's atomic model.

Model answer: (i) An accelerated orbiting electron should radiate and spiral into the nucleus, so the atom would be unstable. (ii) It predicts a continuous spectrum, whereas atoms show discrete line spectra.

C12-S02 Ch 12 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S2. The electron in a hydrogen atom jumps from $n = 1$ to $n = 3$. By what factor do the orbit radius, speed and energy change?

Model answer: $r \propto n^2$: increases $9 \times$. $v \propto 1/n$: decreases to $1/3$. $E_n = -13.6/n^2$: from -13.6 to -1.51 eV (increases by 12.1 eV , becoming $\frac{1}{9}$ in magnitude).

C12-S03 Ch 12 Medium Short answer (3 marks) BOARD-STYLE PRACTICE

S3. Show that the kinetic energy of the electron in the n th orbit equals the magnitude of its total energy.

Model answer: $K = \frac{Ze^2}{8\pi\epsilon_0 r}$, $U = -\frac{Ze^2}{4\pi\epsilon_0 r}$, so $E = K + U = -\frac{Ze^2}{8\pi\epsilon_0 r} = -K$; hence $|E| = K$ and $U = 2E$.

C12-CB01 Ch 12 Medium Case-based (4 marks) BOARD-STYLE PRACTICE

Case. Light from a hydrogen discharge tube is examined with a spectrometer. (i) Which transition gives the red line at 656 nm ? (ii) Which series lies in the visible region? (iii) Find the energy of the photon at 486 nm . (iv) Which transition gives it?

Model answer: (i) $n = 3 \rightarrow 2$. (ii) Balmer series. (iii) $E = 1240/486 = 2.55 \text{ eV}$. (iv) $13.6 \left(\frac{1}{4} - \frac{1}{n^2}\right) = 2.55 \Rightarrow \frac{1}{n^2} = 0.0625 \Rightarrow n = 4$: the $4 \rightarrow 2$ jump ($H\beta$).

12.6 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C12-M01 Ch 12 Easy Conceptual BOARD-STYLE PRACTICE

1. In Rutherford's experiment most α particles pass undeviated because

- (a) nuclei are heavy (b) the atom is mostly empty space
(c) electrons repel them (d) foil is thin only

Ans: (b) Nucleus is tiny.

C12-M02 Ch 12 Medium Numerical

BOARD-STYLE PRACTICE

2. The radius of the second Bohr orbit of hydrogen is

- (a) $0.529 \text{ \AA}$ (b) $1.06 \text{ \AA}$
(c) $2.12 \text{ \AA}$ (d) $4.76 \text{ \AA}$

Ans: (c) $4 \times 0.529 = 2.12 \text{ \AA}$.

C12-M03 Ch 12 Medium Conceptual

BOARD-STYLE PRACTICE

3. The ground state energy of hydrogen is -13.6 eV . The energy of the $n = 2$ level is

- (a) -6.8 eV (b) -3.4 eV
(c) -1.5 eV (d) -0.85 eV

Ans: (b) $-13.6/4$.

C12-M04 Ch 12 Medium Numerical

BOARD-STYLE PRACTICE

4. The ionisation energy from the first excited state of hydrogen is

- (a) 13.6 eV (b) 10.2 eV
(c) 3.4 eV (d) 1.5 eV

Ans: (c) $|E_2| = 3.4 \text{ eV}$.

C12-M05 Ch 12 Medium Application

BOARD-STYLE PRACTICE

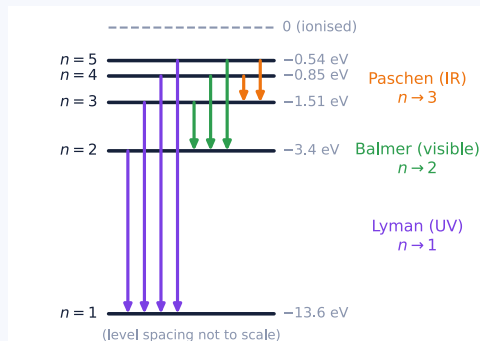
5. The series of hydrogen lines in the visible region is the

- (a) Lyman series (b) Balmer series
(c) Paschen series (d) Pfund series

Ans: (b) $n_f = 2$.

C12-M06 Ch 12 Medium Graph BOARD-STYLE PRACTICE

6. In the energy-level diagram shown, the transition emitting the longest wavelength among those drawn is



- (a) $2 \rightarrow 1$ (b) $3 \rightarrow 2$
(c) $4 \rightarrow 3$ (d) $5 \rightarrow 4$

Ans: (d) Smallest energy gap among those shown.

C12-M07 Ch 12 Medium Numerical

BOARD-STYLE PRACTICE

7. The speed of the electron in the $n = 2$ orbit of hydrogen is nearly

- (a) $2.19 \times 10^6 \text{ m/s}$ (b) $1.1 \times 10^6 \text{ m/s}$
(c) $4.4 \times 10^6 \text{ m/s}$ (d) $7.3 \times 10^5 \text{ m/s}$

Ans: (b) $v_n = 2.19 \times 10^6/n$.

C12-M08 Ch 12 Hard Application BOARD-STYLE PRACTICE

8. For a hydrogen-like ion with $Z = 3$ (Li^{2+}) the ground-state energy is

- (a) -13.6 eV (b) -40.8 eV
(c) -122.4 eV (d) -27.2 eV

Ans: (c) $-13.6 \times 9 = -122.4 \text{ eV}$.

C12-AR01 Ch 12 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: Hydrogen emits a line spectrum.

R: Electron orbits and energy levels are quantised.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Photons carry the difference of discrete levels.

C12-AR02 Ch 12 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: The kinetic energy of the electron in a Bohr orbit is positive but total energy is negative.

R: The electron is bound to the nucleus.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Potential energy ($-2K$) dominates.

★ EXAMINER-STYLE REMINDERS

- Mention both the quantum condition and Coulomb force in the Bohr derivation.
- Use $E_n = -13.6/n^2 \text{ eV}$ and $hc = 1240 \text{ eV nm}$.
- Write series name with n_f .

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are BOARD-STYLE PRACTICE; none is claimed to be a past board question.

R Chapter 12 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. α -scattering set-up
2. Rutherford trajectories
3. Bohr orbits $n = 1, 2, 3$
4. Energy-level diagram with three series

★ 7-DAY REVISION PLAN

Day 1	Rutherford
Day 2	Limitations
Day 3	Bohr postulates
Day 4	Derivation
Day 5	Spectrum
Day 6	N1–N6
Day 7	MCQs

Chapter 12 at a glance — Atoms

Rutherford

tiny dense nucleus; mostly empty atom.

Limits

radiation collapse; no line spectra.

Bohr

$$mvr = nh/2\pi; h\nu = E_i - E_f.$$

Results

$$r \propto n^2, v \propto 1/n, E = -13.6/n^2.$$

Spectrum

Lyman UV, Balmer visible, Paschen IR.

Formula map

$$0.529 n^2 \text{ \AA}$$

$$r_n$$

$$\frac{2.19 \times 10^6}{\frac{n}{v_n}}$$

$$\frac{13.6}{\frac{n^2}{E_n \text{ (eV)}}}$$

$$\frac{nh}{2\pi L}$$

$$R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$1/\lambda$$

$$\frac{n(n-1)}{2 \text{ lines}}$$

! COMMON MISTAKES

- Using $r \propto n$.
- Forgetting that E is negative.
- Mixing the series and n_f .
- Using R for non-hydrogenic atoms.
- Treating $K = E$ (it equals $-E$).

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can describe α -scattering
- ☐ I know two limits of Rutherford's model
- ☐ I can state Bohr's postulates
- ☐ I can derive r_n, v_n, E_n
- ☐ I can use the Rydberg formula
- ☐ I solved N1–N6
- ☐ I did the MCQs

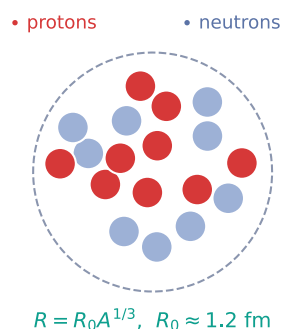
✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

1. Ground-state energy of H? | **Ans:** -13.6 eV
2. Balmer: n_f ? | **Ans:** 2
3. r_n vs n ? | **Ans:** n^2
4. Ionisation energy of H? | **Ans:** 13.6 eV
5. Why did Rutherford's atom fail? | **Ans:** Radiation collapse

UNIT VIII · ATOMS AND NUCLEI

13 Nuclei

A nucleus is 10^5 times smaller than an atom, yet holds almost all its mass — and a tiny fraction of that mass, converted to energy, powers stars and reactors.



Chapter roadmap

Composition & size

Nuclear force

Mass defect & BE

 B/A curve

Fission & fusion

✓ LEARNING OUTCOMES

- Describe the composition and size of a nucleus; use $R = R_0 A^{1/3}$ and nuclear density
- State the properties of the nuclear force
- Use $E = mc^2$ to find mass defect and binding energy; interpret the B/A curve
- Explain nuclear fission and fusion and compare their energy release
- Solve numericals on mass defect, Q -value and energy release

I PREREQUISITES TO REVISE

- Atomic structure and Rutherford's model (Chapter 12)
- Mass–energy equivalence
- Exponents and cube roots
- Energy units: eV, MeV, J

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: composition and size of the nucleus; nuclear force; mass–energy relation, mass defect; binding energy per nucleon and its variation with mass number; nuclear fission and fusion. *Radioactivity (decay law, half-life) is included in an “Extension” box below — check the official syllabus to see whether it is examinable for 2026–27.*

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (nuclear density, B/A curve); 2-mark (properties of nuclear force, definitions); 3-mark numericals (binding energy, Q -value); 5-mark long answers on the B/A curve, fission and fusion; case-based on a reactor or on the curve. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Key relations

F NUCLEAR BASICS

${}_Z^AX$: Z protons, $N = A - Z$ neutrons;
 $R = R_0 A^{1/3}$ ($R_0 = 1.2 \text{ fm}$); $\rho \approx 2.3 \times 10^{17} \text{ kg m}^{-3}$
 (same for all nuclei); $1 \text{ u} = 931.5 \text{ MeV}/c^2$.

★ ENERGY

$\Delta m = Zm_p + (A - Z)m_n - M$;
 $BE = \Delta m \times 931.5 \text{ MeV}$; $\frac{B}{A}$ peaks at 8.8 MeV near ${}^{56}\text{Fe}$; fission ($\sim 200 \text{ MeV}$), fusion ($d + t$: 17.6 MeV).

13.1 Composition, size and the nuclear force

I COMPOSITION AND SIZE

Nucleus = **protons** (+ e) + **neutrons** (neutral) = **nucleons**.
 Notation ${}_Z^AX$. **Isotopes**: same Z , different A ; **isobars**: same A ; **isotones**: same N . Atomic mass unit:
 $1\text{ u} = \frac{1}{12}m({}^{12}\text{C}) = 1.66 \times 10^{-27}\text{ kg}$.
 Radius $R = R_0 A^{1/3}$, $R_0 \approx 1.2\text{ fm}$. Volume $\propto A$, so **density is independent of A** : $\rho = \frac{3m_u}{4\pi R_0^3} \approx 2.3 \times 10^{17}\text{ kg m}^{-3}$ —
 about 10^{14} times that of water.

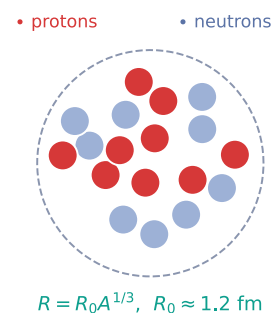


Fig. 13.1 Nucleons packed in a nucleus.

☆ PROPERTIES OF THE NUCLEAR FORCE

- The **strongest** known force (about $100 \times$ the electrostatic force at 1 fm).
- Short range**: acts up to about 2–3 fm and falls to nearly zero beyond; very strong attraction at $\sim 0.8\text{ fm}$, repulsive below.
- Charge independent**: p - p , n - n and p - n forces are the same.
- Saturated**: each nucleon interacts only with its neighbours.
- Non-central and spin dependent; not a simple inverse-square law.

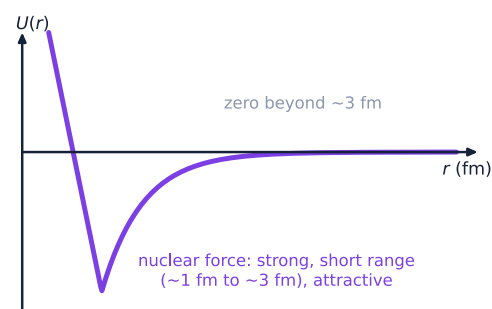


Fig. 13.2 Potential energy of two nucleons against separation.

✓ CHECKPOINT

Calculate the radius of ${}^{27}\text{Al}$. $R = 1.2 \times 27^{1/3} = 3.6\text{ fm}$. Is the density of ${}^{27}\text{Al}$ more or less than that of ${}^{238}\text{U}$?
The same — ρ does not depend on A .

Term	Meaning	Example
Isotopes	same Z , different A	${}^1\text{H}$, ${}^2\text{H}$, ${}^3\text{H}$
Isobars	same A , different Z	${}^3\text{H}$ and ${}^3\text{He}$
Isotones	same N , different Z	${}^{13}\text{C}$ and ${}^{14}\text{N}$
Nucleon number A	$Z + N$	${}^{235}\text{U}$: $A = 235$
Atomic number Z	protons (charge Ze)	${}^{235}\text{U}$: $Z = 92$, $N = 143$

13.2 Mass defect, binding energy and the B/A curve

I MASS-ENERGY AND BINDING ENERGY

Einstein: $E = mc^2$ ($1 \text{ u} \leftrightarrow 931.5 \text{ MeV}$). The mass of a nucleus is *less* than the sum of its separated nucleons: the **mass defect**

$$\Delta m = Zm_p + (A - Z)m_n - M_{\text{nucleus}} \quad (\text{usually computed with atomic masses}), \quad BE = \Delta m c^2 = \Delta m \times 931.5 \text{ MeV}$$

Binding energy = energy needed to separate a nucleus into free nucleons; **binding energy per nucleon** B/A is a measure of stability.

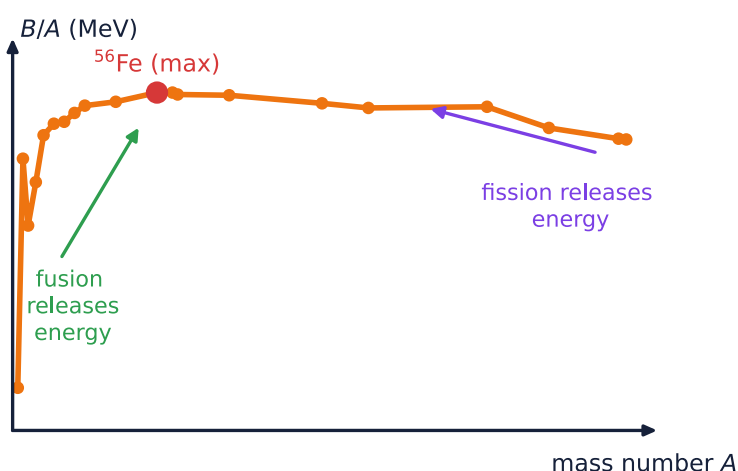


Fig. 13.3 Binding energy per nucleon against mass number (selected nuclei).

★ READING THE CURVE

- B/A rises steeply for light nuclei, is nearly constant ($\approx 8 \text{ MeV}$) for $A = 30\text{--}170$.
- Maximum $\approx 8.8 \text{ MeV}$ at ^{56}Fe : the most stable nuclei.
- Falls slowly for heavy nuclei ($\approx 7.6 \text{ MeV}$ for U).
- Fusion** of light nuclei and **fission** of heavy nuclei both move towards higher B/A , releasing energy.
- ^4He is unusually stable (high $B/A = 7.07 \text{ MeV}$).

! COMMON MISTAKES

- Forgetting to include the electron masses (use atomic masses consistently).
- Saying a high total BE means stability — compare B/A .
- Writing $BE = \Delta m \times c^2$ in u without conversion to MeV.

~ MASS-ENERGY EQUIVALENCE

- $E = mc^2$: a mass m is equivalent to an energy mc^2 .
- $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} \Rightarrow mc^2 = 1.49 \times 10^{-10} \text{ J} = 931.5 \text{ MeV}$.
- Mass of a bound system is less than the sum of its parts; the difference is the energy given out when it forms.
- In a reaction, total mass+energy is conserved, not mass alone.

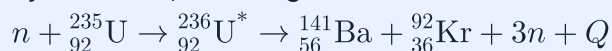
F WORKED DATA: SOME B/A VALUES

Nucleus	B/A (MeV)
^2H	1.11
^4He	7.07
^{12}C	7.68
^{56}Fe	8.79
^{208}Pb	7.87
^{235}U	7.59

13.3 Nuclear fission and fusion

I NUCLEAR FISSION

A heavy nucleus splits into two medium nuclei when struck by a neutron, releasing ~ 200 MeV and 2–3 neutrons:



Why energy is released: B/A of the products (≈ 8.5 MeV) exceeds that of U (≈ 7.6 MeV); gain $\approx 0.9 \times 235 \approx 200$ MeV. Emitted neutrons start a **chain reaction**. In a reactor: *moderator* (slows neutrons to thermal speeds: heavy water, graphite), *control rods* (Cd, B absorb neutrons; keep multiplication factor $k = 1$), *coolant*. Uncontrolled $k > 1$: atomic bomb.

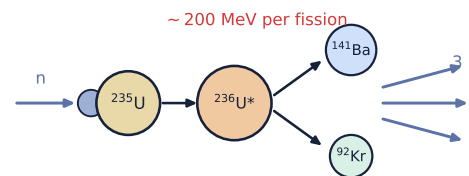
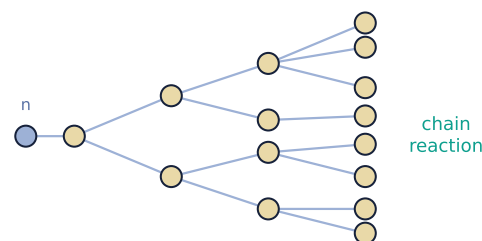


Fig. 13.4 A fission event.



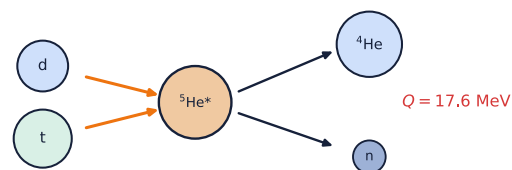
controlled in a reactor: moderator + control rods ($k = 1$)

Fig. 13.5 Chain reaction.

I NUCLEAR FUSION

Light nuclei combine to form a heavier one, releasing energy. Example (Sun/hydrogen bomb):

${}^2_1\text{H} + {}^3_1\text{H} \rightarrow {}^4_2\text{He} + n + 17.6$ MeV; in the Sun: proton–proton cycle $4 {}^1_1\text{H} \rightarrow {}^4_2\text{He} + 2e^+ + 2\nu + 26.7$ MeV. Needs $T \sim 10^7$ – 10^8 K and high density so nuclei overcome Coulomb repulsion — **thermonuclear fusion**.



needs $T \sim 10^7$ – 10^8 K to overcome Coulomb repulsion

Fig. 13.6 Deuterium–tritium fusion.

Feature	Fission	Fusion
Process	heavy nucleus splits	light nuclei join
Energy per reaction	~ 200 MeV	~ 17.6 MeV ($d + t$)
Energy per unit mass	lower (~ 0.9 MeV/nucleon)	higher (~ 3.5 MeV/nucleon)
Condition	neutron, critical mass	very high T and pressure
Waste	radioactive fission products	little long-lived waste
Use	reactors, bombs	stars, future reactors

13.4 Radioactive decay (extension)

RADIOACTIVE DECAY LAW — EXTENSION (VERIFY AGAINST THE 2026–27 SYLLABUS)

The rate of decay is proportional to the number of undecayed nuclei: $\frac{dN}{dt} = -\lambda N \Rightarrow N = N_0 e^{-\lambda t}$. **Half-life**

$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$; **mean life** $\tau = 1/\lambda$. Activity $R = \lambda N$ (becquerel). After n half-lives: $N = N_0/2^n$.

α decay: ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4\text{He}$; β^- : $n \rightarrow p + e^- + \bar{\nu}$; β^+ : $p \rightarrow n + e^+ + \nu$; γ : de-excitation photon. Penetrating power: $\gamma > \beta > \alpha$.

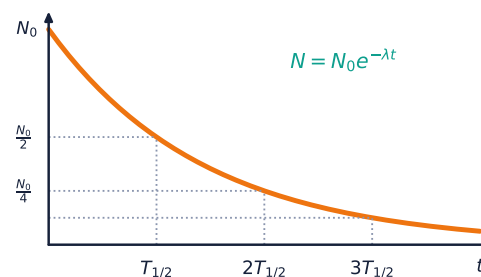


Fig. 13.7 Exponential decay.

CHECKPOINT

A sample has $T_{1/2} = 5$ days. What fraction remains after 15 days? 3 half-lives: $\frac{1}{8}$. After 20 days? $\frac{1}{16} = 6.25\%$. Decay constant? $\lambda = 0.693/5 = 0.139 \text{ day}^{-1}$.

Q-VALUE

$Q = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}} \right) c^2$. $Q > 0$: exothermic (energy released as kinetic energy of products). Conservation laws in all reactions: charge, mass number A , energy, momentum.

EXAM TIP

If masses are given as *atomic masses*, use $m({}^1\text{H}) = 1.007825 \text{ u}$ for the proton (it includes the electron) and do not add electron masses separately — they cancel in neutral atoms for a nucleus mass defect.

Radiation	Nature	Charge	Penetration	Stopped by
α	${}_2^4\text{He}$ nucleus	$+2e$	least; ionises strongly	paper, a few cm of air
β^-	fast electron	$-e$	medium	few mm of aluminium
γ	EM photon	0	greatest	thick lead / concrete

13.5 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$R = R_0 A^{1/3}$	fm	$R_0 = 1.2 \text{ fm}$	$A = 64$: 4.8 fm
$\rho = \frac{3m_u}{4\pi R_0^3}$	kg m^{-3}	indep. of A	2.3×10^{17}
$\Delta m = Zm_p + Nm_n - M$	u		${}^4\text{He}$: 0.03038 u
$BE = \Delta m \times 931.5$	MeV		28.3 MeV
$\frac{B}{A}$	MeV		${}^4\text{He}$: 7.07
$Q = \Delta m c^2$	MeV	reaction	$d + t$: 17.6
$N = N_0 e^{-\lambda t}$		extension	$T_{1/2} = 0.693/\lambda$

N1

C13-N01

Ch 13

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

Find the mass defect, binding energy and B/A of ${}^4_2\text{He}$. Atomic masses: $m({}^1_1\text{H}) = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$, $M({}^4_2\text{He}) = 4.002603 \text{ u}$.

GIVEN $Z = 2$, $N = 2$; masses as given

FIND Δm , BE , B/A

FORMULA $\Delta m = Zm_H + Nm_n - M$; $BE = \Delta m \times 931.5$

WORKING $\Delta m = 2(1.007825) + 2(1.008665) - 4.002603 = 0.030377 \text{ u}$; $BE = 0.030377 \times 931.5 = 28.3 \text{ MeV}$; $\frac{B}{A} = \frac{28.3}{4} = 7.07 \text{ MeV}$.

Answer: 0.0304 u; 28.3 MeV; 7.07 MeV/nucleon

▲ Atomic masses include electrons that cancel.

N2

C13-N02

Ch 13

Medium

Numerical

BOARD-STYLE PRACTICE

2 marks

Estimate the radius of a ${}^{64}\text{Cu}$ nucleus and the nuclear density. ($R_0 = 1.2 \text{ fm}$, $m_u = 1.66 \times 10^{-27} \text{ kg}$)

GIVEN $A = 64$, $R_0 = 1.2 \times 10^{-15} \text{ m}$

FIND R , ρ

FORMULA $R = R_0 A^{1/3}$; $\rho = \frac{Am_u}{\frac{4}{3}\pi R^3}$

WORKING $R = 1.2 \times 4 = 4.8 \text{ fm}$; $\rho = \frac{3m_u}{4\pi R_0^3} = \frac{3 \times 1.66 \times 10^{-27}}{4\pi (1.2 \times 10^{-15})^3} = 2.3 \times 10^{17} \text{ kg m}^{-3}$.

Answer: 4.8 fm; $2.3 \times 10^{17} \text{ kg/m}^3$

▲ A cancels, so density is the same for all nuclei.

N3

C13-N03

Ch 13

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

Estimate the energy released by the complete fission of 1 g of ${}^{235}\text{U}$ at 200 MeV per fission.

GIVEN $m = 1 \text{ g}$; $M = 235 \text{ g/mol}$; $E_f = 200 \text{ MeV}$

FIND E

FORMULA $N = \frac{m}{M} N_A$; $E = NE_f$

WORKING $N = \frac{1}{235} \times 6.02 \times 10^{23} = 2.56 \times 10^{21}$; $E = 2.56 \times 10^{21} \times 200 \text{ MeV} = 5.1 \times 10^{23} \text{ MeV} = 8.2 \times 10^{10} \text{ J}$ (about 23 MWh).

Answer: $8.2 \times 10^{10} \text{ J}$

▲ $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$.

13.6 More numericals and subjective practice

N4

C13-N04

Ch 13

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

Calculate the energy released in ${}_1^2\text{H} + {}_1^3\text{H} \rightarrow {}_2^4\text{He} + n$. Masses (u): $d = 2.014102$, $t = 3.016049$, $\text{He} = 4.002603$, $n = 1.008665$.

GIVEN masses as given

FIND Q

FORMULA $Q = (m_d + m_t - m_{\text{He}} - m_n) \times 931.5 \text{ MeV}$

WORKING $\Delta m = 5.030151 - 5.011268 = 0.018883 \text{ u}$; $Q = 0.018883 \times 931.5 = 17.6 \text{ MeV}$.

Answer: $Q \approx 17.6 \text{ MeV}$

▲ Positive Q : energy released.

N5

C13-N05

Ch 13

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

A reactor produces 100 MW of thermal power with 200 MeV per fission. How many fissions occur per second and how much ${}^{235}\text{U}$ is consumed per day?

GIVEN $P = 10^8 \text{ W}$, $E_f = 200 \text{ MeV} = 3.2 \times 10^{-11} \text{ J}$

FIND fission rate, mass/day

FORMULA $\text{rate} = P/E_f$; $m = \frac{\text{rate} \times t \times 235}{N_A}$

WORKING $\text{rate} = \frac{10^8}{3.2 \times 10^{-11}} = 3.1 \times 10^{18} \text{ s}^{-1}$. Per day: $3.1 \times 10^{18} \times 86400 = 2.7 \times 10^{23}$ fissions $= \frac{2.7 \times 10^{23}}{6.02 \times 10^{23}} \times 235 \text{ g} \approx 105 \text{ g}$.

Answer: 3.1×10^{18} per second; about 0.1 kg per day

▲ A coal plant needs thousands of tonnes per day for the same power.

N6

C13-N06

Ch 13

Easy

Numerical

BOARD-STYLE PRACTICE

2 marks

A radioactive sample has half-life 5 days (extension topic). Find λ and the fraction left after 15 days.

GIVEN $T_{1/2} = 5 \text{ days}$

FIND λ , N/N_0

FORMULA $\lambda = \frac{0.693}{T_{1/2}}$; $N = N_0 2^{-t/T}$

WORKING $\lambda = \frac{0.693}{5} = 0.139 \text{ day}^{-1}$; $\frac{N}{N_0} = 2^{-15/5} = \frac{1}{8}$.

Answer: 0.139 day^{-1} ; $1/8$

▲ $t/T_{1/2} = 3$.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C13-S01

Ch 13

Easy

Short answer (2 marks)

BOARD-STYLE PRACTICE

S1. Why is the nuclear density the same for all nuclei?

Model answer: Since $R = R_0 A^{1/3}$, volume $\propto A$ and mass $\approx Am_u$, so $\rho = Am_u / (\frac{4}{3}\pi R_0^3 A)$ is independent of A .

C13-S02

Ch 13

Medium

Short answer (3 marks)

BOARD-STYLE PRACTICE

S2. Explain, using the B/A curve, why energy is released in both fission and fusion.

Model answer: Fission moves a heavy nucleus ($B/A \approx 7.6 \text{ MeV}$) to medium nuclei ($\approx 8.5 \text{ MeV}$); fusion moves light nuclei (low B/A) towards the plateau. In both cases the nucleons end up more tightly bound, and the increase in total binding energy is released.

C13-S03

Ch 13

Medium

Short answer (3 marks)

BOARD-STYLE PRACTICE

S3. What is the role of a moderator and of control rods in a nuclear reactor?

Model answer: The moderator (heavy water, graphite) slows fast fission neutrons to thermal speeds so they can cause further fission of ${}^{235}\text{U}$. Control rods (Cd, B) absorb neutrons to keep the multiplication factor at 1 for a steady controlled reaction.

C13-CB01

Ch 13

Medium

Case-based (4 marks)

BOARD-STYLE PRACTICE

Case. For nuclei A, B and C the B/A values are 1.1, 8.8 and 7.6 MeV respectively. (i) Which is the most stable? (ii) Which is likely a light nucleus that can fuse? (iii) Which is likely heavy and can fission? (iv) Estimate the BE of nucleus B if $A = 56$.

Model answer: (i) B. (ii) A (e.g. deuteron). (iii) C (e.g. U). (iv) $BE = 8.8 \times 56 = 493 \text{ MeV}$.

13.7 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C13-M01 Ch 13 Easy Conceptual BOARD-STYLE PRACTICE

1. The nuclear force is

- (a) long range and charge dependent (b) short range and charge independent
(c) long range and charge independent (d) short range and charge dependent

Ans: (b) Standard properties.

C13-M02 Ch 13 Medium Numerical

BOARD-STYLE PRACTICE

2. The radius of a nucleus of mass number 8 is nearly (with $R_0 = 1.2$ fm)

- (a) 1.2 fm (b) 2.4 fm
(c) 4.8 fm (d) 9.6 fm

Ans: (b) $8^{1/3} = 2$.

C13-M03 Ch 13 Medium Conceptual

BOARD-STYLE PRACTICE

3. Which nucleus has the highest binding energy per nucleon?

- (a) ^2H (b) ^{56}Fe
(c) ^{235}U (d) ^4He

Ans: (b) Maximum near iron.

C13-M04 Ch 13 Medium Numerical

BOARD-STYLE PRACTICE

4. A mass defect of 0.01 u corresponds to about

- (a) 9.3 MeV (b) 93 MeV
(c) 0.93 MeV (d) 931 MeV

Ans: (a) 0.01×931.5 .

C13-M05 Ch 13 Medium Application

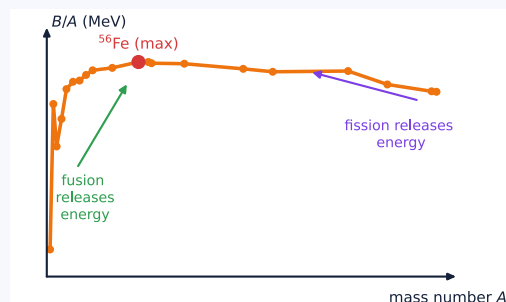
BOARD-STYLE PRACTICE

5. In a reactor, heavy water is used as a

- (a) fuel (b) coolant only
(c) moderator (d) control rod

Ans: (c) Slows neutrons.

C13-M06 Ch 13 Medium Graph BOARD-STYLE PRACTICE

6. The B/A curve shown has a maximum near

- (a) $A = 4$ (b) $A = 56$
(c) $A = 120$ (d) $A = 238$

Ans: (b) Iron peak.

C13-M07 Ch 13 Medium Numerical

BOARD-STYLE PRACTICE

7. The energy released per fission of ^{235}U is nearly

- (a) 20 MeV (b) 200 MeV
(c) 2 GeV (d) 2 MeV

Ans: (b) About 200 MeV.

C13-M08 Ch 13 Hard Application

BOARD-STYLE PRACTICE

8. After three half-lives the fraction of a radioactive sample remaining (extension) is

- (a) $1/3$ (b) $1/6$
(c) $1/8$ (d) $1/9$

Ans: (c) $(1/2)^3$.

C13-AR01 Ch 13 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: Fusion needs very high temperature.

R: Nuclei must overcome mutual Coulomb repulsion to come within nuclear-force range.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Thermonuclear reaction.

C13-AR02 Ch 13 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: Binding energy per nucleon decreases for very heavy nuclei.

R: Coulomb repulsion among protons increases with Z .

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Long-range repulsion reduces binding.

★ EXAMINER-STYLE REMINDERS

- Draw and label the B/A curve with ^{56}Fe and the fission/fusion regions.
- State units clearly: u, MeV.
- For energy calculations show Δm first.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are **BOARD-STYLE PRACTICE**; none is claimed to be a past board question.

R Chapter 13 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. B/A against A curve
2. Nuclear potential energy curve
3. Fission chain reaction
4. Reactor schematic (moderator, rods)

★ 7-DAY REVISION PLAN

Day 1	Composition, size
Day 2	Nuclear force
Day 3	BE and curve
Day 4	Fission
Day 5	Fusion
Day 6	N1–N6
Day 7	MCQs

Chapter 13 at a glance — Nuclei

Nucleus

$$p + n; R = R_0 A^{1/3}; \rho \approx 2.3 \times 10^{17}.$$

Force

strong, short range, charge independent.

Energy

$$BE = \Delta m \times 931.5 \text{ MeV}.$$

B/A curve

peak at ^{56}Fe (8.8 MeV).

Fission

~ 200 MeV; chain reaction.

Fusion

$d + t$: 17.6 MeV; $T \sim 10^7$ K.

Formula map

$$R_0 A^{1/3}$$

radius

$$\frac{3m_u}{4\pi R_0^3}$$

density

$$Zm_p + Nm_n - M$$

Δm

$$931.5 \Delta m$$

BE (MeV)

$$\frac{BE}{A}$$

stability

$$\frac{0.693}{T_{1/2}}$$

λ (ext.)

! COMMON MISTAKES

- Comparing total BE instead of B/A .
- Using proton mass without electron correction.
- Thinking nuclear force is inverse-square.
- Confusing moderator and control rod.
- Forgetting unit conversion.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I know the properties of nuclear force
- ☐ I can compute BE and B/A
- ☐ I can sketch the B/A curve
- ☐ I can explain fission and fusion
- ☐ I solved N1–N6
- ☐ I did the MCQs

✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

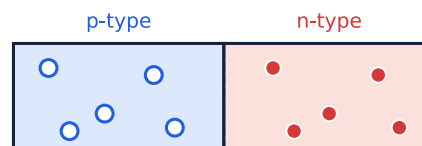
1. Unit conversion 1 u? | **Ans:** 931.5 MeV
2. Most stable nucleus? | **Ans:** ^{56}Fe
3. Nuclear density depends on A ? | **Ans:** No
4. Fission energy per event? | **Ans:** ~ 200 MeV
5. Condition for fusion? | **Ans:** Very high T

UNIT IX · ELECTRONIC DEVICES

14

Semiconductor Electronics

Every phone, laptop and LED bulb runs on one idea: a tiny junction of two doped crystals that lets current pass one way only.



p-n junction: majority carriers diffuse across the boundary

Chapter roadmap

Energy bands

Doping

p-n junction

Biasing & I-V

Rectifiers

✓ LEARNING OUTCOMES

- Distinguish conductors, insulators and semiconductors using energy bands
- Explain intrinsic and extrinsic (n-type, p-type) semiconductors
- Describe the p-n junction, depletion layer and barrier potential
- Draw and interpret the I-V characteristics in forward and reverse bias
- Explain half-wave and full-wave rectification with waveforms

I PREREQUISITES TO REVISE

- Current and Ohm's law (Chapter 3)
- Electric field and potential (Chapter 1–2)
- AC voltage, peak and rms (Chapter 7)
- Photon energy $E = hc/\lambda$ (Chapter 11)

★ OFFICIAL 2026–27 SCOPE FOR THIS CHAPTER

Prescribed: energy bands in solids (qualitative); conductors, insulators and semiconductors; intrinsic and extrinsic semiconductors; p-n junction; diode characteristics; diode as a rectifier. *Zener diode, LED, photodiode and solar cell appear in an "Extension" box — check the official syllabus for 2026–27.*

★ EXAM SNAPSHOT

Typical questions: 1-mark MCQs (band gap, majority carriers, bias); 2–3 mark theory (doping, depletion layer); 3-mark numericals; 5-mark long answers on the p-n junction, I-V curve or a rectifier circuit with waveforms. *Pattern shown is the recent board pattern; confirm with the official 2026–27 sample paper.*

Key relations

F SEMICONDUCTOR BASICS

Intrinsic: $n_e = n_h = n_i$. Doped: $n_e n_h = n_i^2$.
 Conductivity $\sigma = e(n_e \mu_e + n_h \mu_h)$. Photon from band gap: $\lambda = \frac{1240}{E_g \text{ (eV)}} \text{ nm.}$

★ DIODE AND RECTIFIER

Forward: $p \rightarrow +$, $n \rightarrow -$; silicon knee $\approx 0.7 \text{ V}$.
 Half-wave $V_{dc} = V_m/\pi$; full-wave $V_{dc} = 2V_m/\pi$,
 $f_{out} = 2f$.

14.1 Energy bands and semiconductors

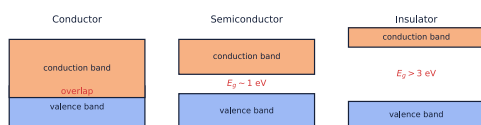


Fig. 14.1 Energy bands: conductor, semiconductor, insulator.

I ENERGY BANDS

In a crystal, atomic levels merge into bands. The filled **valence band (VB)** and empty-or-partly-filled **conduction band (CB)** are separated by the **forbidden gap** E_g .

- **Conductor:** bands overlap or CB partly filled; $E_g \approx 0$.
- **Insulator:** large gap, $E_g > 3 \text{ eV}$ (diamond 5.4 eV).
- **Semiconductor:** small gap (Si 1.1 eV, Ge 0.67 eV). Conductivity rises with temperature as electrons are excited to the CB, leaving **holes** in the VB.

I INTRINSIC SEMICONDUCTOR

Pure Si or Ge. Each thermally freed electron leaves one hole, so $n_e = n_h = n_i$. At 0 K it behaves as an insulator. Resistivity *decreases* with temperature (opposite to metals).

~ ELECTRONS AND HOLES

A hole is a vacancy that behaves as a positive carrier. Electrons move in the CB; holes move in the VB. Total current $I = I_e + I_h$.

! COMMON MISTAKES

- Saying holes are real positively charged particles — they are absences of electrons.
- Writing that semiconductor resistance increases with temperature.
- Thinking doped crystals are charged — they are electrically neutral.

✓ CHECKPOINT

Which has a smaller band gap, Si or diamond? *Si (1.1 eV)*. Why does conductivity of a semiconductor rise on heating? *More electron-hole pairs are produced.*

14.2 Extrinsic semiconductors and doping

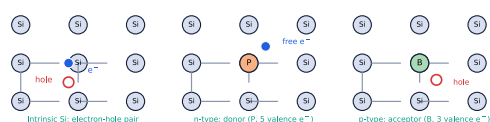


Fig. 14.2 Silicon lattice with donor (n) and acceptor (p).

I DOPING

Adding a small amount of impurity (about 1 in 10^6) gives an **extrinsic** semiconductor.

- **n-type:** pentavalent donor (P, As, Sb) in Si. Four electrons bond, the fifth is nearly free. *Majority carriers: electrons; minority: holes.*
- **p-type:** trivalent acceptor (B, Al, In). One bond is incomplete, creating a hole. *Majority: holes; minority: electrons.*

$$n_e n_h = n_i^2$$

Feature	n-type	p-type
Dopant	pentavalent (donor)	trivalent (acceptor)
Majority carrier	electrons	holes
Minority carrier	holes	electrons
Donor/acceptor level	just below CB	just above VB
Net charge	neutral	neutral

F CONDUCTIVITY

$\sigma = e(n_e \mu_e + n_h \mu_h)$, $\rho = 1/\sigma$. In doped material one term dominates.

★ EXAM TIP

Always say "majority" and "minority" carriers, and add that the crystal remains **neutral**. Both lose marks if omitted.

14.3 The p-n junction

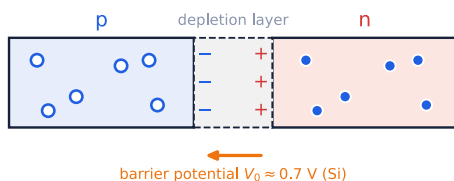


Fig. 14.3 p-n junction at equilibrium.

1 FORMATION OF THE DEPLETION LAYER

Electrons diffuse from n to p, and holes from p to n. Near the interface they recombine, leaving immobile ions: positive donor ions on the n side and negative acceptor ions on the p side. This region, free of mobile carriers, is the **depletion layer** (about $1\ \mu\text{m}$). Its internal field opposes further diffusion; the potential difference is the **barrier potential** ($\approx 0.7\ \text{V}$ for Si, $0.3\ \text{V}$ for Ge).

★ TWO CURRENTS

- **Diffusion current:** majority carriers crossing, p to n.
- **Drift current:** minority carriers swept by the field, n to p.
- At equilibrium they are equal and net current is zero.

~ DEPLETION FIELD ESTIMATE

For $V_B = 0.7\ \text{V}$ across $w = 1\ \mu\text{m}$:

$E = V_B/w = 7 \times 10^5\ \text{V m}^{-1}$ — a strong field over a very thin layer.

! COMMON MISTAKES

- Showing the depletion layer as having mobile charges.
- Calling the barrier potential an emf that can drive a bulb.
- Drawing the field from p to n.

✓ CHECKPOINT

Why does a p-n junction not conduct at equilibrium? *Diffusion and drift currents cancel.*

14.4 Biasing and I–V characteristics

1 FORWARD AND REVERSE BIAS

Forward bias (p to +, n to –): the external voltage opposes the barrier, depletion layer narrows, current rises sharply above the **knee voltage** (Si 0.7 V). **Reverse bias**: the barrier grows, the depletion layer widens, only a tiny minority-carrier current (the **reverse saturation current**, μA) flows. Beyond the **breakdown voltage** reverse current rises abruptly.

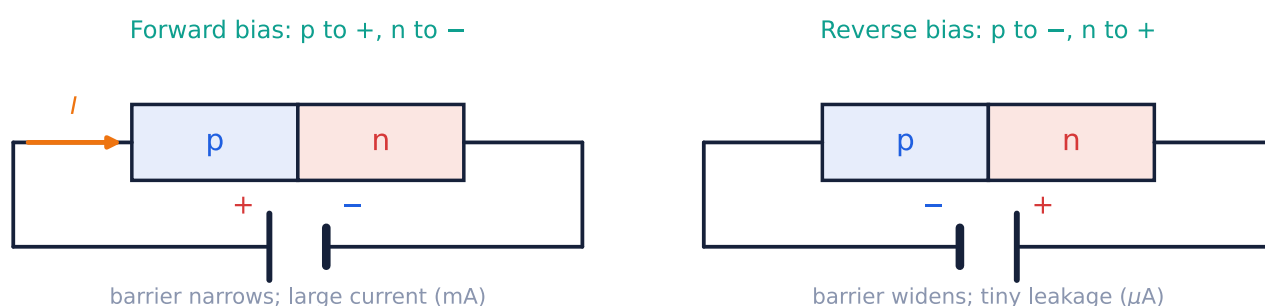


Fig. 14.4 Forward and reverse biasing of a p–n junction.

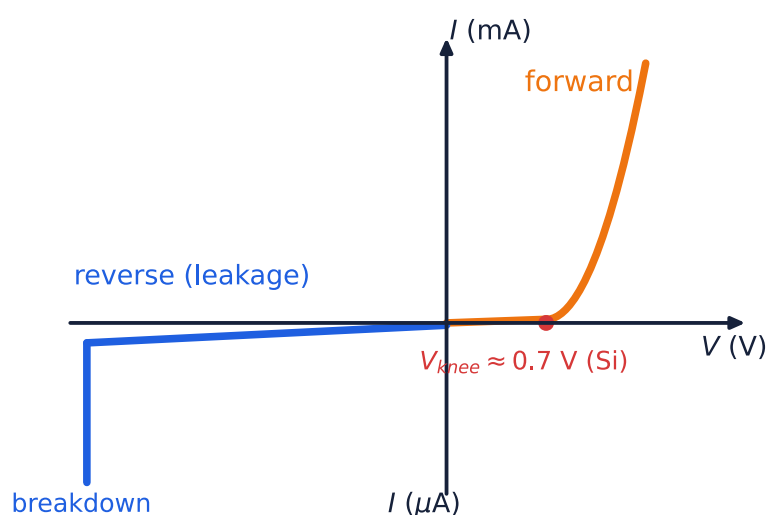


Fig. 14.5 I–V characteristic of a Si diode.

★ READING THE CURVE

- Forward: current negligible below $\approx 0.7 \text{ V}$, then rapidly increasing (non-ohmic).
- Reverse: constant small current up to breakdown.
- Forward resistance $r_f = \Delta V / \Delta I$ is small; reverse resistance is very large.
- A diode is a one-way valve for current.

★ EXAM TIP

Draw the I–V graph with **different scales** for forward (mA) and reverse (μA), mark the knee voltage and breakdown.

! COMMON MISTAKES

- Using the same scale for forward and reverse current.
- Saying a diode obeys Ohm's law.
- Forgetting that a series resistor limits forward current.

14.5 Rectifiers

I HALF-WAVE RECTIFIER

A single diode conducts only in the positive half-cycle. Output is a series of half-sine pulses with the **same frequency** as the input. $V_{dc} = V_m/\pi$, $V_{rms} = V_m/2$, ripple factor ≈ 1.21 , efficiency $\approx 40.6\%$.

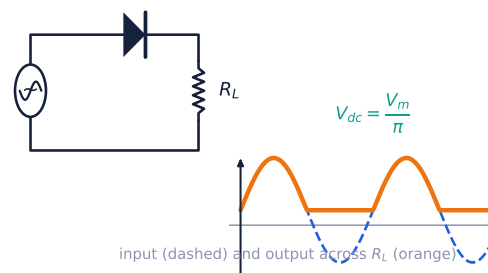


Fig. 14.6 Half-wave rectifier and output.

I FULL-WAVE RECTIFIER

Two diodes with a centre-tapped transformer (or a bridge of four) conduct on alternate half-cycles. Output frequency is **twice** the input: $f_{out} = 2f$. $V_{dc} = 2V_m/\pi$, $V_{rms} = V_m/\sqrt{2}$, ripple ≈ 0.48 , efficiency $\approx 81.2\%$. A **filter** capacitor across R_L smooths the ripple.

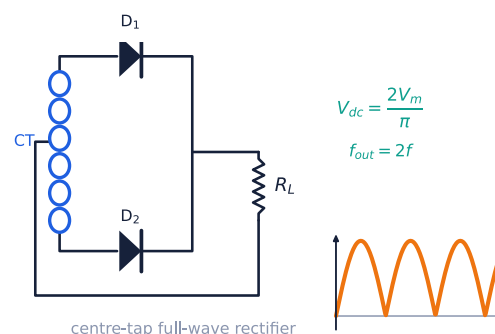


Fig. 14.7 Centre-tap full-wave rectifier.

Feature	Half-wave	Full-wave
Diodes	1	2 (CT) or 4 (bridge)
V_{dc}	V_m/π	$2V_m/\pi$
Output frequency	f	$2f$
Ripple factor	1.21	0.48
Efficiency (max)	40.6%	81.2%

14.6 Extension: special-purpose diodes

~ ZENER, LED, PHOTODIODE, SOLAR CELL — EXTENSION (VERIFY AGAINST THE 2026–27 SYLLABUS)

- **Zener diode:** heavily doped, operated in reverse breakdown; holds a nearly constant voltage V_Z across the load (voltage regulator).
- **LED:** forward-biased; electron–hole recombination emits photons of energy $\approx E_g$. Colour set by the material.
- **Photodiode:** reverse-biased; light creates electron–hole pairs and increases reverse current (light sensor).
- **Solar cell:** unbiased junction generating emf from light (photovoltaic effect).

Device	Bias	Operating principle	Use
Zener	reverse (breakdown)	constant V_Z	voltage regulator
LED	forward	recombination emits light	indicators, lighting
Photodiode	reverse	light raises current	detectors
Solar cell	none	photovoltaic emf	power generation

✓ CHECKPOINT

Which diode is operated in reverse breakdown? *Zener*. Why is an LED not made from plain Si? *Si has an indirect gap and emits mainly heat; compound semiconductors like GaAsP are used.*

★ EXAM TIP

LED colour depends on E_g : $\lambda = 1240/E_g$ nm. Larger gap, shorter wavelength.

F ENERGY OF PHOTON

$$E = hc/\lambda = \frac{1240}{\lambda \text{ (nm)}} \text{ eV.}$$

14.7 Formula bank and solved numericals

FORMULA	SYMBOLS AND SI UNITS	CONDITIONS / SIGN CONVENTION	QUICK USE
$n_e n_h = n_i^2$	m^{-6}	doped	$n_i = 1.5 \times 10^{16}$ (Si)
$\sigma = e (n_e \mu_e + n_h \mu_h)$	S m^{-1}		4.4×10^{-4} (Si)
$\lambda = 1240/E_g$	nm	E_g in eV	Si: 1127 nm
$I = \frac{V - V_k}{R}$	A	forward diode	4.3 mA
$V_{dc} = V_m/\pi$	V	half-wave	$V_m = 10$: 3.18 V
$V_{dc} = 2V_m/\pi$	V	full-wave	6.37 V

N1

C14-N01

Ch 14

Medium

Numerical

BOARD-STYLE PRACTICE

2 marks

Pure silicon has $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$. It is doped with phosphorus so that $n_e = 5 \times 10^{22} \text{ m}^{-3}$. Find n_h .

GIVEN $n_i = 1.5 \times 10^{16}, n_e = 5 \times 10^{22}$

FIND n_h

FORMULA $n_e n_h = n_i^2$

WORKING $n_h = \frac{(1.5 \times 10^{16})^2}{5 \times 10^{22}} = \frac{2.25 \times 10^{32}}{5 \times 10^{22}} = 4.5 \times 10^9 \text{ m}^{-3}$.

Answer: $4.5 \times 10^9 \text{ m}^{-3}$

▲ Hole density drops enormously on doping.

N2

C14-N02

Ch 14

Easy

Numerical

BOARD-STYLE PRACTICE

2 marks

Silicon has $E_g = 1.1 \text{ eV}$. Find the longest wavelength that can create an electron–hole pair. What wavelength does an LED with $E_g = 2.0 \text{ eV}$ emit?

GIVEN $E_{g1} = 1.1 \text{ eV}, E_{g2} = 2.0 \text{ eV}$

FIND $\lambda_{\max}, \lambda_{LED}$

FORMULA $\lambda = \frac{hc}{E_g} = \frac{1240}{E_g} \text{ nm}$

WORKING $\lambda_{\max} = \frac{1240}{1.1} = 1127 \text{ nm}$ (infrared). $\lambda_{LED} = \frac{1240}{2.0} = 620 \text{ nm}$ (orange-red).

Answer: 1127 nm; 620 nm

▲ Use E in eV.

N3

C14-N03

Ch 14

Easy

Numerical

BOARD-STYLE PRACTICE

2 marks

A silicon diode is forward biased in series with a $1 \text{ k}\Omega$ resistor and a 5 V battery. Find the current.

GIVEN $V = 5 \text{ V}, R = 1 \text{ k}\Omega, V_k = 0.7 \text{ V}$

FIND I

FORMULA $I = \frac{V - V_k}{R}$

WORKING $I = \frac{5 - 0.7}{1000} = 4.3 \times 10^{-3} \text{ A}$.

Answer: 4.3 mA

▲ The diode drops 0.7 V.

14.8 More numericals and subjective practice

N4

C14-N04

Ch 14

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

Calculate the conductivity of intrinsic Si at room temperature: $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$, $\mu_e = 0.135$, $\mu_h = 0.048 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$.

GIVEN as given

FIND σ

FORMULA $\sigma = en_i (\mu_e + \mu_h)$

WORKING $\sigma = 1.6 \times 10^{-19} \times 1.5 \times 10^{16} \times 0.183 = 4.4 \times 10^{-4} \text{ S m}^{-1}$.

Answer: $4.4 \times 10^{-4} \text{ S m}^{-1}$

▲ Compare with copper ($6 \times 10^7 \text{ S m}^{-1}$).

N5

C14-N05

Ch 14

Medium

Numerical

BOARD-STYLE PRACTICE

3 marks

A rectifier is fed with a sinusoid of peak value 10 V. Find V_{dc} and V_{rms} at the output for (a) half-wave and (b) full-wave.

GIVEN $V_m = 10 \text{ V}$

FIND V_{dc} , V_{rms}

FORMULA half: V_m/π , $V_m/2$; full: $2V_m/\pi$, $V_m/\sqrt{2}$

WORKING (a) $V_{dc} = 3.18 \text{ V}$, $V_{rms} = 5 \text{ V}$. (b) $V_{dc} = 6.37 \text{ V}$, $V_{rms} = 7.07 \text{ V}$. Efficiency (half) $= \frac{V_{dc}^2}{V_{rms}^2} = \frac{3.18^2}{5^2} = 40.6\%$.

Answer: (a) 3.18, 5 V; (b) 6.37, 7.07 V

▲ Full-wave doubles V_{dc} .

N6

C14-N06

Ch 14

Easy

Numerical

BOARD-STYLE PRACTICE

1 marks

The input to a full-wave rectifier is 50 Hz. What are the frequencies of the output for half-wave and full-wave?

GIVEN $f = 50 \text{ Hz}$

FIND f_{out}

FORMULA half: f ; full: $2f$

WORKING Half-wave: 50 Hz. Full-wave: $2 \times 50 = 100 \text{ Hz}$.

Answer: 50 Hz; 100 Hz

▲ Higher ripple frequency is easier to filter.

SUBJECTIVE PRACTICE (WITH MODEL ANSWERS)

C14-S01

Ch 14

Easy

Short answer (2 marks)

BOARD-STYLE PRACTICE

S1. What is the depletion layer? How does its width change in forward bias?

Model answer: The region near the junction depleted of mobile carriers, containing only immobile ions. In forward bias it narrows.

C14-S02

Ch 14

Medium

Short answer (3 marks)

BOARD-STYLE PRACTICE

S2. Distinguish between n-type and p-type semiconductors. Are they charged?

Model answer: n-type: pentavalent donor, majority electrons. p-type: trivalent acceptor, majority holes. Both are electrically neutral because the total positive and negative charges are equal.

C14-S03

Ch 14

Medium

Short answer (3 marks)

BOARD-STYLE PRACTICE

S3. Why does the output of a full-wave rectifier have twice the input frequency?

Model answer: Both half-cycles are conducted in the same direction through the load, so each input cycle produces two output pulses.

C14-CB01

Ch 14

Medium

Case-based (4 marks)

BOARD-STYLE PRACTICE

Case. A silicon diode is connected in series with a 330Ω resistor to a 5 V supply. (i) State the bias condition for conduction. (ii) Find the current. (iii) State the voltage across the resistor. (iv) What happens on reversing the battery?

Model answer: (i) p to +, n to -. (ii) $I = (5 - 0.7) / 330 = 13 \text{ mA}$. (iii) 4.3 V. (iv) Almost no current (leakage only); the full 5 V appears across the diode.

14.9 MCQ practice

MCQ PRACTICE — COVER THE GREEN STRIP AND ATTEMPT FIRST

C14-M01 Ch 14 Easy Conceptual

BOARD-STYLE PRACTICE

1. The majority carriers in a p-type semiconductor are

- (a) electrons (b) holes
(c) protons (d) ions

Ans: (b) Holes.

C14-M02 Ch 14 Easy Conceptual

BOARD-STYLE PRACTICE

2. Which impurity makes silicon n-type?

- (a) Boron (b) Aluminium
(c) Phosphorus (d) Indium

Ans: (c) Pentavalent donor.

C14-M03 Ch 14 Medium Numerical

BOARD-STYLE PRACTICE

3. Wavelength corresponding to a band gap of 2.48 eV is

- (a) 300 nm (b) 400 nm
(c) 500 nm (d) 600 nm

Ans: (c) $1240/2.48 = 500$ nm.

C14-M04 Ch 14 Medium Application

BOARD-STYLE PRACTICE

4. In forward bias the depletion layer

- (a) widens (b) narrows
(c) is unchanged (d) vanishes permanently

Ans: (b) Opposing bias reduces the barrier.

C14-M05 Ch 14 Medium Application

BOARD-STYLE PRACTICE

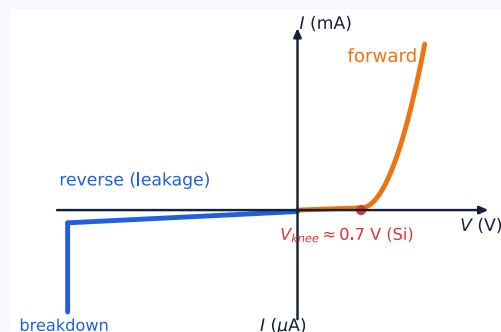
5. Input to a full-wave rectifier is 50 Hz. Output ripple frequency is

- (a) 25 Hz (b) 50 Hz
(c) 100 Hz (d) 200 Hz

Ans: (c) $2f$.

C14-M06 Ch 14 Medium Graph BOARD-STYLE PRACTICE

6. The I-V curve shown is that of



- (a) a resistor (b) a p-n junction diode
(c) a capacitor (d) an inductor

Ans: (b) One-way, non-ohmic.

C14-M07 Ch 14 Medium Numerical

BOARD-STYLE PRACTICE

7. A Si diode in series with $430\ \Omega$ and 5 V (forward) carries

- (a) 5 mA (b) 10 mA
(c) 11.6 mA (d) 0

Ans: (b) $(5 - 0.7) / 430 = 10$ mA.

C14-M08 Ch 14 Medium Conceptual

BOARD-STYLE PRACTICE

8. With increasing temperature, the resistance of a semiconductor

- (a) increases (b) decreases
(c) stays constant (d) becomes infinite

Ans: (b) More carriers.

C14-AR01 Ch 14 Medium Assertion-Reason

BOARD-STYLE PRACTICE

9. A: An n-type crystal is electrically neutral.

R: The donor ions' positive charge balances the free electrons.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Overall neutrality.

C14-AR02 Ch 14 Medium Assertion-Reason

BOARD-STYLE PRACTICE

10. A: A diode is a non-ohmic device.

R: Its V-I graph is not a straight line through the origin.

(a) both true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- (a) a (b) b
(c) c (d) d

Ans: (a) Definition of non-ohmic.

★ EXAMINER-STYLE REMINDERS

- Label p, n, depletion layer, field direction.
- Show forward and reverse regions on the I-V graph.
- Show input and output waveforms for rectifiers.

★ ABOUT PREVIOUS-YEAR QUESTIONS

All questions here are **BOARD-STYLE PRACTICE**; none is claimed to be a past board question.

R Chapter 14 revision page

★ DIAGRAMS TO PRACTISE DRAWING (EACH IS WORTH MARKS)

1. Energy bands (three types)
2. p-n junction with depletion layer
3. I–V characteristic
4. Half-wave and full-wave circuits with waveforms

★ 7-DAY REVISION PLAN

Day 1	Bands
Day 2	Doping
Day 3	Junction
Day 4	Biasing
Day 5	Rectifiers
Day 6	N1–N6
Day 7	MCQs

Chapter 14 at a glance — Semiconductor Electronics

Bands

conductor / semiconductor / insulator by E_g .

Doping

n: donor, e^- majority; p: acceptor, hole majority.

Junction

depletion layer; $V_B \approx 0.7 \text{ V}$ (Si).

Bias

forward narrows, reverse widens.

Half-wave

$$V_{dc} = V_m / \pi; f.$$

Full-wave

$$V_{dc} = 2V_m / \pi; 2f.$$

Formula map

$$\frac{n_e n_h}{= n_i^2}$$

$$en \left(\frac{\mu_e + \mu_h}{\sigma} \right)$$

$$\frac{1240}{E_g} \\ \lambda \text{ (nm)}$$

$$\frac{V - V_k}{R} \\ \text{diode current}$$

$$\frac{V_m}{\pi} \\ \text{half-wave dc}$$

$$\frac{2V_m}{\pi} \\ \text{full-wave dc}$$

! COMMON MISTAKES

- Treating holes as real positive particles.
- Forgetting that doped crystals are neutral.
- Same scale for forward and reverse current.
- Writing $f_{out} = f$ for full-wave.
- Ignoring the 0.7 V drop.

★ CHAPTER COMPLETION CHECKLIST

- ☐ I can explain energy bands
- ☐ I can explain doping
- ☐ I can describe the depletion layer
- ☐ I can draw the I–V curve
- ☐ I can draw both rectifiers
- ☐ I solved N1–N6
- ☐ I did the MCQs

✓ QUICK SELF-TEST (ANSWERS SHOWN IN GREEN)

1. Majority carrier in p-type? | **Ans:** Holes
2. Si knee voltage? | **Ans:** 0.7 V
3. Full-wave output frequency? | **Ans:** $2f$
4. Bias for conduction? | **Ans:** p to +
5. Dopant for n-type? | **Ans:** Pentavalent

F Consolidated formula sheet (1/2): Chapters 1-7

F CH 1: ELECTRIC CHARGES AND FIELDS

Coulomb	$F = \frac{kq_1q_2}{r^2}, k = 9 \times 10^9$
Field	$E = \frac{kq}{r^2}, \mathbf{F} = q\mathbf{E}$
Dipole	$p = 2qa, \tau = pE \sin \theta, U = -pE \cos \theta$
Dipole field	$E_{ax} = \frac{2kp}{r^3}, E_{eq} = \frac{kp}{r^3}$
Gauss	$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\epsilon_0}$
Line / plane	$\frac{\lambda}{2\pi\epsilon_0 r}, \frac{\sigma}{2\epsilon_0}$

F CH 2: POTENTIAL AND CAPACITANCE

Potential	$V = \frac{kq}{r}, E = -\frac{dV}{dx}$
Energy of 2 charges	$U = \frac{kq_1q_2}{r}$
Capacitance	$C = \frac{Q}{V}, C = \frac{\epsilon_0 A}{d}, \frac{K\epsilon_0 A}{d}$
Combination	series $\frac{1}{C} = \sum \frac{1}{C_i}$; parallel $C = \sum C_i$
Energy	$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$
Energy density	$u = \frac{1}{2}\epsilon_0 E^2$

F CH 3: CURRENT ELECTRICITY

Drift	$I = nAev_d, v_d = \frac{eE\tau}{m}$
Resistance	$R = \rho \frac{l}{A}, V = IR$
Temp.	$R_T = R_0 (1 + \alpha \Delta T)$
Cell	$V = E - Ir, I = \frac{E}{R+r}$
Power	$P = VI = I^2 R = \frac{V^2}{R}$
Bridge	$\frac{P}{Q} = \frac{R}{S}$; potentiometer $\frac{E_1}{E_2} = \frac{l_1}{l_2}$

F CH 4: MOVING CHARGES AND MAGNETISM

Biot-Savart	$dB = \frac{\mu_0 Idl \sin \theta}{4\pi r^2}$
Loop centre	$B = \frac{\mu_0 NI}{2R}$
Wire / solenoid	$\frac{\mu_0 I}{2\pi r}, \mu_0 nI$
Force	$\mathbf{F} = q\mathbf{v} \times \mathbf{B}, F = BIl \sin \theta$
Circular motion	$r = \frac{mv}{qB}, T = \frac{2\pi m}{qB}$
Torque / meters	$\tau = NIAB \sin \theta; S = \frac{I_g G}{I - I_g}$

F CH 5: MAGNETISM AND MATTER

Moment	$m = NIA$
Dipole fields	$B_{ax} = \frac{\mu_0 2m}{4\pi r^3}, B_{eq} = \frac{\mu_0 m}{4\pi r^3}$
Torque	$\tau = mB \sin \theta, U = -mB \cos \theta$
Materials	$B = \mu_0 (H + M), \chi = \frac{M}{H}, \mu_r = 1 + \chi$
Earth	$H = B_e \cos \delta, V = B_e \sin \delta, \tan \delta = \frac{V}{H}$

F CH 7: ALTERNATING CURRENT

Rms	$I_{rms} = \frac{I_0}{\sqrt{2}}$
Reactance	$X_L = \omega L, X_C = \frac{1}{\omega C}$
Impedance	$Z = \sqrt{R^2 + (X_L - X_C)^2}$
Phase	$\tan \phi = \frac{X_L - X_C}{R}$
Resonance	$\omega_0 = \frac{1}{\sqrt{LC}}, Q = \frac{\omega_0 L}{R}$
Power / transformer	$P = VI \cos \phi; \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$

F CH 6: ELECTROMAGNETIC INDUCTION

Flux	$\Phi = BA \cos \theta$
Faraday	$\epsilon = -N \frac{d\Phi}{dt}$
Motional	$\epsilon = Blv, \frac{1}{2}B\omega l^2$
Self	$\epsilon = -L \frac{dI}{dt}, \Phi = LI$
Solenoid	$L = \mu_0 n^2 Al$
Energy	$U = \frac{1}{2}LI^2; \text{mutual } \epsilon_2 = -M \frac{dI_1}{dt}$

F Consolidated formula sheet (2/2): Chapters 8-14

F CH 8: ELECTROMAGNETIC WAVES

Displacement current	$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$
Speed	$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}, \frac{E_0}{B_0} = c$
Energy density	$u = \frac{1}{2} \varepsilon_0 E^2 + \frac{B^2}{2\mu_0}$
Intensity	$I = \frac{1}{2} \varepsilon_0 c E_0^2$
Momentum	$p = \frac{U}{c}$

F CH 10: WAVE OPTICS

Path difference	$\Delta = d \sin \theta \approx \frac{yd}{D}$
Fringe width	$\beta = \frac{\lambda D}{d}$
Bright / dark	$y_n = \frac{n\lambda D}{d}; \left(n + \frac{1}{2}\right) \frac{\lambda D}{d}$
Single slit	min: $a \sin \theta = n\lambda$; width $\frac{2\lambda D}{a}$
Intensity	$I = 4I_0 \cos^2 \frac{\phi}{2}$

F CH 12: ATOMS

Radius	$r_n = 0.529 \frac{n^2}{Z} \text{ \AA}$
Energy	$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$
Speed	$v_n = 2.19 \times 10^6 \frac{Z}{n} \text{ m/s}$
Spectrum	$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$
Angular momentum	$L = \frac{nh}{2\pi}$
Closest approach	$r_0 = \frac{2kZe^2}{K}$

F CH 14: SEMICONDUCTOR ELECTRONICS

Carriers	$n_e n_h = n_i^2$
Conductivity	$\sigma = e (n_e \mu_e + n_h \mu_h)$
Band gap	$\lambda = \frac{1240}{E_g} \text{ nm}$
Diode	$I = \frac{V - V_k}{R}; V_k \approx 0.7 \text{ V (Si)}$
Half-wave	$V_{dc} = \frac{V_m}{\pi}, f_{out} = f$
Full-wave	$V_{dc} = \frac{2V_m}{\pi}, f_{out} = 2f$

F CH 9: RAY OPTICS

Mirror	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}, f = \frac{R}{2}, m = -\frac{v}{u}$
Refraction	$n = \frac{c}{v}, n_1 \sin i = n_2 \sin r, \sin C = \frac{1}{n}$
Spherical surface	$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$
Lens	$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$
Power	$P = \frac{1}{f} \text{ (D)}, P = P_1 + P_2$
Prism	$n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$

F CH 11: DUAL NATURE

Photon	$E = h\nu = \frac{hc}{\lambda}, p = \frac{h}{\lambda}$
Einstein	$K_{max} = h\nu - \phi_0$
Stopping	$eV_0 = K_{max}$
Threshold	$\phi_0 = h\nu_0$
de Broglie	$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$
Electron	$\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$

F CH 13: NUCLEI

Radius	$R = R_0 A^{1/3}$
Density	$\approx 2.3 \times 10^{17} \text{ kg/m}^3$
Mass defect	$\Delta m = Zm_p + Nm_n - M$
Binding energy	$BE = \Delta m \times 931.5 \text{ MeV}$
Q-value	$Q = (\Sigma m_i - \Sigma m_f) c^2$
Decay (ext.)	$N = N_0 e^{-\lambda t}, T_{1/2} = \frac{0.693}{\lambda}$

K Constants, units and dimensions

★ VALUES USED IN THIS BOOK

Quantity	Symbol	Value
Speed of light	c	3×10^8 m/s
Elementary charge	e	1.6×10^{-19} C
Planck constant	h	6.63×10^{-34} J s
Electron mass	m_e	9.1×10^{-31} kg
Proton mass	m_p	1.67×10^{-27} kg
Permittivity	ϵ_0	8.85×10^{-12} F/m
Permeability	μ_0	$4\pi \times 10^{-7}$ T m/A
Coulomb constant	k	9×10^9 N m ² /C ²
Bohr radius	a_0	0.529 Å
Rydberg constant	R	1.097×10^7 m ⁻¹
Atomic mass unit	1 u	931.5 MeV/c ² ; 1.66×10^{-27} kg
Electron volt	1 eV	1.6×10^{-19} J
hc		1240 eV nm

I SI UNITS AND DIMENSIONS

Quantity	SI unit	Dimensions
Charge	coulomb (C)	[AT]
Electric field	V m ⁻¹	[MLT ⁻³ A ⁻¹]
Potential	volt (V)	[ML ² T ⁻³ A ⁻¹]
Capacitance	farad (F)	[M ⁻¹ L ⁻² T ⁴ A ²]
Resistance	ohm (Ω)	[ML ² T ⁻³ A ⁻²]
Magnetic field	tesla (T)	[MT ⁻² A ⁻¹]
Flux	weber (Wb)	[ML ² T ⁻² A ⁻¹]
Inductance	henry (H)	[ML ² T ⁻² A ⁻²]
Power of lens	diopetre (D)	m ⁻¹
Activity (ext.)	becquerel (Bq)	s ⁻¹

★ UNIT-CHECK HABIT

Write units in every line of a numerical. A wrong unit can cost the final mark even when the number is right. Convert cm to m, μ F to F, eV to J *before* substituting.

! FREQUENT SLIPS

- Using μ_0 and ϵ_0 interchangeably.
- Mixing degrees and radians.
- Forgetting that rms $\neq$ peak.
- Not converting nm to m.

M Visual revision maps (1/2)

Ch 1 · Electric Charges and Fields

BIG IDEAS

- Charge is quantised and conserved
- Field lines never cross
- Gauss's law uses symmetry

MUST-KNOW

- $F = kq_1q_2/r^2$
- $\tau = pE \sin \theta$
- $\Phi = q/\epsilon_0$

DRAW

field lines of a dipole

TRAP

Using Gauss's law with a non-symmetric surface

Ch 2 · Electrostatic Potential and Capacitance

BIG IDEAS

- Potential is scalar
- $E = -dV/dx$
- Dielectric lowers E , raises C

MUST-KNOW

- $C = \epsilon_0 A/d$
- $U = \frac{1}{2} CV^2$
- series / parallel

DRAW

capacitor with dielectric

TRAP

Adding series capacitances directly

Ch 3 · Current Electricity

BIG IDEAS

- v_d is small
- ρ depends on material and T
- Kirchhoff = charge and energy

MUST-KNOW

- $I = nAev_d$
- $V = E - Ir$
- $P/Q = R/S$

DRAW

potentiometer circuit

TRAP

Forgetting internal resistance

Ch 4 · Moving Charges and Magnetism

BIG IDEAS

- Force on a moving charge is $\perp v$
- Ampere's law
- Torque on a loop

MUST-KNOW

- $r = mv/qB$
- $B = \mu_0 nI$
- $S = I_g G / (I - I_g)$

DRAW

cyclotron / solenoid field

TRAP

Wrong direction by the right-hand rule

Ch 5 · Magnetism and Matter

BIG IDEAS

- Field lines of a magnet are closed
- Dia/para/ferro
- Earth's field has H, V, δ

MUST-KNOW

- $\tau = mB \sin \theta$
- $\mu_r = 1 + \chi$
- $\tan \delta = V/H$

DRAW

bar magnet field lines

TRAP

Mixing χ signs

Ch 6 · Electromagnetic Induction

BIG IDEAS

- Lenz's law = energy conservation
- Motional emf
- Inductance

MUST-KNOW

- $\epsilon = -d\Phi/dt$
- $\epsilon = Blv$
- $U = \frac{1}{2} LI^2$

DRAW

rod on rails

TRAP

Forgetting the negative sign meaning

Ch 7 · Alternating Current

BIG IDEAS

- rms values
- Phasors
- Resonance

MUST-KNOW

- $Z = \sqrt{R^2 + (X_L - X_C)^2}$
- $\omega_0 = 1/\sqrt{LC}$
- $P = VI \cos \phi$

DRAW

LCR phasor diagram

TRAP

Using peak values for power

M Visual revision maps (2/2)

Ch 8 · Electromagnetic Waves

BIG IDEAS

- Displacement current
- EM waves are transverse
- Spectrum order

MUST-KNOW

- $c = 1/\sqrt{\mu_0 \epsilon_0}$
- $E_0/B_0 = c$
- $I = \frac{1}{2} \epsilon_0 c E_0^2$

DRAW

EM wave propagation

TRAP

Confusing E and B directions

Ch 9 · Ray Optics and Optical Instruments

BIG IDEAS

- Sign convention
- TIR and prisms
- Instruments

MUST-KNOW

- $1/v - 1/u = 1/f$
- $n = \sin i / \sin r$
- $m = f_o / f_e$

DRAW

ray diagrams for lens and mirror

TRAP

Wrong signs for u and f

Ch 10 · Wave Optics

BIG IDEAS

- Huygens principle
- Coherence
- Interference vs diffraction

MUST-KNOW

- $\beta = \lambda D/d$
- $a \sin \theta = n\lambda$
- fringe positions

DRAW

YDSE geometry

TRAP

Using β in the wrong medium

Ch 11 · Dual Nature of Radiation and Matter

BIG IDEAS

- Photon picture
- Threshold and stopping potential
- Matter waves

MUST-KNOW

- $K = h\nu - \phi_0$
- $\lambda = h/p$
- $eV_0 = K_{max}$

DRAW

V_0 - ν graph

TRAP

Intensity vs frequency effects

Ch 12 · Atoms

BIG IDEAS

- Rutherford scattering
- Bohr postulates
- Line spectra

MUST-KNOW

- $E_n = -13.6/n^2$
- $r_n \propto n^2$
- Rydberg formula

DRAW

energy level diagram

TRAP

Wrong series limits

Ch 13 · Nuclei

BIG IDEAS

- Constant nuclear density
- B/A curve
- Fission and fusion

MUST-KNOW

- $R_0 A^{1/3}$
- $BE = \Delta m \times 931.5$
- Q -value

DRAW

B/A curve

TRAP

Using total BE instead of B/A

Ch 14 · Semiconductor Electronics

BIG IDEAS

- Bands
- Doping
- Junction and rectifier

MUST-KNOW

- $n_e n_h = n_i^2$
- $V_{dc} = V_m / \pi$
- $f_{out} = 2f$

DRAW

I-V curve and rectifier

TRAP

Same scale for forward/reverse

Q Mixed MCQ practice (1/3)

Cover the green answer strip and attempt first. **BOARD-STYLE PRACTICE**: no question here is claimed to be a past board question.

MX-01 Medium Ch 1 · Numerical

BOARD-STYLE PRACTICE

1. The force between charges $2\ \mu\text{C}$ and $3\ \mu\text{C}$ placed 3 cm apart in air is

- (a) 6 N (b) 60 N
(c) 600 N (d) 0.6 N

Ans: (b) $9 \times 10^9 \times 6 \times 10^{-12} / 9 \times 10^{-4}$.

MX-02 Medium Ch 1 · Concept

BOARD-STYLE PRACTICE

2. A Gaussian sphere encloses charges $+2q$ and $-q$. The net flux is

- (a) q/ϵ_0 (b) $3q/\epsilon_0$
(c) $2q/\epsilon_0$ (d) zero

Ans: (a) Net enclosed charge q .

MX-03 Medium Ch 2 · Numerical

BOARD-STYLE PRACTICE

3. The potential at 0.3 m from a 3 nC point charge is

- (a) 9 V (b) 90 V
(c) 27 V (d) 270 V

Ans: (b) $kq/r = 9 \times 10^9 \times 3 \times 10^{-9} / 0.3$.

MX-04 Medium Ch 2 · Application

BOARD-STYLE PRACTICE

4. A capacitor stays connected to a battery and its plate separation is doubled. The stored energy

- (a) doubles (b) is unchanged
(c) halves (d) becomes four times

Ans: (c) C halves, V fixed, $U = \frac{1}{2}CV^2$.

MX-05 Easy Ch 3 · Numerical **BOARD-STYLE PRACTICE**

5. A 6 V battery of internal resistance $1\ \Omega$ drives a $5\ \Omega$ resistor. The current is

- (a) 0.8 A (b) 1 A
(c) 1.2 A (d) 6 A

Ans: (b) $6 / (5 + 1)$.

MX-06 Medium Ch 3 · Concept

BOARD-STYLE PRACTICE

6. The drift velocity of electrons in a copper wire carrying a few amperes is of the order of

- (a) $10^8\ \text{m/s}$ (b) $10^2\ \text{m/s}$
(c) $10^{-4}\ \text{m/s}$ (d) $10^{-12}\ \text{m/s}$

Ans: (c) About 0.1 mm/s.

MX-07 Medium Ch 3 · Numerical

BOARD-STYLE PRACTICE

7. A balanced Wheatstone bridge has $P = 2\ \Omega$, $Q = 3\ \Omega$, $R = 4\ \Omega$. The fourth arm S is

- (a) $1.5\ \Omega$ (b) $6\ \Omega$
(c) $2.7\ \Omega$ (d) $12\ \Omega$

Ans: (b) $P/Q = R/S$.

MX-08 Medium Ch 4 · Numerical

BOARD-STYLE PRACTICE

8. An electron with speed $2 \times 10^6\ \text{m/s}$ moves perpendicular to $B = 0.01\ \text{T}$. The radius of its path is

- (a) 1.1 mm (b) 11 mm
(c) 0.11 mm (d) 1.1 cm

Ans: (a)
 $mv/eB = 9.1 \times 10^{-31} \times 2 \times 10^6 / (1.6 \times 10^{-19} \times 10^{-2})$.

MX-09 Medium Ch 4 · Concept

BOARD-STYLE PRACTICE

9. The cyclotron frequency of a charged particle in a magnetic field

- (a) increases with speed (b) decreases with speed
(c) is independent of speed (d) is independent of B

Ans: (c) $f = qB/2\pi m$.

MX-10 Easy Ch 4 · Application

BOARD-STYLE PRACTICE

10. A galvanometer is converted into a voltmeter by connecting

- (a) a low resistance in parallel (b) a high resistance in series
(c) a low resistance in series (d) a high resistance in parallel

Ans: (b) Large series resistance.

Q Mixed MCQ practice (2/3)

MX-11 Easy Ch 5 · Concept **BOARD-STYLE PRACTICE**

11. Diamagnetic materials have susceptibility

- (a) small and positive (b) large and positive
(c) small and negative (d) zero

Ans: (c) They weakly oppose the field.

MX-12 Medium Ch 5 · Numerical

BOARD-STYLE PRACTICE

12. At a place $V = \sqrt{3} H$. The angle of dip is

- (a) 30° (b) 45°
(c) 60° (d) 90°

Ans: (c) $\tan \delta = \sqrt{3}$.

MX-13 Easy Ch 6 · Numerical **BOARD-STYLE PRACTICE**

13. A 1 m rod moves at 2 m/s perpendicular to $B = 0.5$ T. The emf across it is

- (a) 0.25 V (b) 1 V
(c) 2 V (d) 4 V

Ans: (b) Blv .

MX-14 Medium Ch 6 · Application

BOARD-STYLE PRACTICE

14. Eddy currents are useful in

- (a) induction furnaces (b) transformer cores
(c) electric bulbs (d) fuses

Ans: (a) Heating by induced currents.

MX-15 Medium Ch 7 · Numerical

BOARD-STYLE PRACTICE

15. A transformer has 100 primary turns on 200 V and 50 secondary turns. The secondary voltage is

- (a) 50 V (b) 100 V
(c) 200 V (d) 400 V

Ans: (b) $V_s = V_p N_s / N_p$.

MX-16 Medium Ch 7 · Concept

BOARD-STYLE PRACTICE

16. At resonance in a series LCR circuit the current is

- (a) minimum (b) maximum
(c) zero (d) infinite

Ans: (b) $Z = R$, minimum.

MX-17 Medium Ch 7 · Numerical

BOARD-STYLE PRACTICE

17. A series circuit has $R = 3 \Omega$ and net reactance 4Ω . The power factor is

- (a) 0.4 (b) 0.6
(c) 0.75 (d) 0.8

Ans: (b) $R/Z = 3/5$.

MX-18 Easy Ch 8 · Concept **BOARD-STYLE PRACTICE**

18. The electromagnetic waves used in remote controls of TVs are

- (a) microwaves (b) X-rays
(c) infrared (d) radio waves

Ans: (c) Infrared.

MX-19 Medium Ch 8 · Numerical

BOARD-STYLE PRACTICE

19. The wavelength of an EM wave of frequency 10^{15} Hz is

- (a) 3×10^{-7} m (b) 3×10^{-9} m
(c) 3×10^7 m (d) 3×10^{-5} m

Ans: (a) c/f .

MX-20 Easy Ch 9 · Numerical **BOARD-STYLE PRACTICE**

20. Light travels at 2×10^8 m/s in a medium. Its refractive index is

- (a) 0.67 (b) 1.5
(c) 2 (d) 6

Ans: (b) c/v .

Q Mixed MCQ practice (3/3)

MX-21 Easy Ch 9 · Numerical BOARD-STYLE PRACTICE

21. The power of a concave lens of focal length 25 cm is

- (a) +4 D (b) −4 D
(c) +0.25 D (d) −25 D

Ans: (b) $P = 1/f = -1/0.25$.

MX-22 Medium Ch 9 · Concept

BOARD-STYLE PRACTICE

22. Total internal reflection occurs when light goes from

- (a) denser to rarer medium at $i > C$ (b) rarer to denser medium at $i > C$
(c) denser to rarer medium at $i < C$ (d) any medium at $i = 0$

Ans: (a) Condition for TIR.

MX-23 Medium Ch 9 · Numerical

BOARD-STYLE PRACTICE

23. An astronomical telescope in normal adjustment has $f_o = 100$ cm and $f_e = 5$ cm. Its magnifying power is

- (a) 5 (b) 20
(c) 100 (d) 105

Ans: (b) f_o/f_e .

MX-24 Medium Ch 10 · Numerical

BOARD-STYLE PRACTICE

24. In Young's experiment $\lambda = 500$ nm, $D = 1$ m and $d = 1$ mm. The fringe width is

- (a) 0.25 mm (b) 0.5 mm
(c) 1 mm (d) 2 mm

Ans: (b) $\lambda D/d$.

MX-25 Medium Ch 10 · Concept

BOARD-STYLE PRACTICE

25. If the slit width in a single-slit experiment is halved, the central maximum

- (a) becomes narrower (b) becomes twice as wide
(c) is unchanged (d) disappears

Ans: (b) Width $\propto 1/a$.

MX-26 Medium Ch 11 · Numerical

BOARD-STYLE PRACTICE

26. A photon of energy 4.14 eV has wavelength nearly

- (a) 300 nm (b) 500 nm
(c) 600 nm (d) 100 nm

Ans: (a) $1240/4.14$.

MX-27 Medium Ch 11 · Concept

BOARD-STYLE PRACTICE

27. The stopping potential in photoelectric emission depends on

- (a) intensity only (b) frequency and the metal
(c) distance of the source (d) area of the metal

Ans: (b) $eV_0 = h\nu - \phi_0$.

MX-28 Medium Ch 12 · Numerical

BOARD-STYLE PRACTICE

28. The energy needed to excite hydrogen from $n = 1$ to $n = 3$ is

- (a) 1.51 eV (b) 10.2 eV
(c) 12.09 eV (d) 13.6 eV

Ans: (c) $13.6 (1 - 1/9)$.

MX-29 Medium Ch 13 · Numerical

BOARD-STYLE PRACTICE

29. The radius of a nucleus with $A = 125$ is ($R_0 = 1.2$ fm)

- (a) 3 fm (b) 6 fm
(c) 12 fm (d) 15 fm

Ans: (b) 1.2×5 .

MX-30 Medium Ch 14 · Concept

BOARD-STYLE PRACTICE

30. The depletion layer of a p-n junction contains

- (a) mobile electrons only (b) mobile holes only
(c) immobile ions (d) no charge at all

Ans: (c) Fixed donor and acceptor ions.

P Practice papers

★ THREE ORIGINAL FULL-LENGTH PAPERS

Each paper has 33 questions for 70 marks, with an answer key and model solutions (marking points in square brackets). They are **original, board-style** and **not** past papers. The section pattern is the recent pattern and is **provisional**; compare it with the official 2026–27 sample paper before relying on it.

Paper 1	Balanced: electrostatics to semiconductors
Paper 2	Emphasis on fields, EMI, optics and atoms
Paper 3	Emphasis on circuits, magnetism, AC and quantum

How to attempt a paper

1. Set a 3-hour timer and use only the data given.
2. Start with Section A, then B, C, D, E.
3. Write formula, substitution and unit for every numerical.
4. Mark yourself using the bracketed marking points.
5. Record each lost mark against its chapter on the chapter revision page.

Self-marking grid: **Paper 1** ____/70 **Paper 2** ____/70 **Paper 3** ____/70

★ PRACTICE PAPER 1 — ORIGINAL, BOARD-STYLE

Time: 3 hours

Maximum marks: 70

BOARD-STYLE PRACTICE

All questions are compulsory. Section A: 16×1 (12 MCQ + 4 assertion-reason); B: 5×2 ; C: 7×3 ; D: 2 case-based $\times 4$; E: 3×5 . This section pattern is provisional (recent pattern); confirm with the official 2026–27 sample paper. Use $c = 3 \times 10^8$ m/s, $e = 1.6 \times 10^{-19}$ C, $h = 6.63 \times 10^{-34}$ J s, $\mu_0 = 4\pi \times 10^{-7}$ SI, $\epsilon_0 = 8.85 \times 10^{-12}$ SI, $m_e = 9.1 \times 10^{-31}$ kg, $m_p = 1.67 \times 10^{-27}$ kg. No internal choices are given.

SECTION A

 $16 \times 1 = 16$

- An electric dipole has charges $\pm 2 \mu\text{C}$ separated by 1 cm. Its dipole moment is [1]
 - 2×10^{-8} C m
 - 2×10^{-4} C m
 - 2×10^{-6} C m
 - 4×10^{-8} C m
- A $10 \mu\text{F}$ capacitor is charged to 100 V. The energy stored is [1]
 - 0.5 J
 - 0.05 J
 - 5×10^{-3} J
 - 5×10^{-4} J
- A wire of resistance R is uniformly stretched to twice its length (volume constant). The new resistance is [1]
 - $R/2$
 - $2R$
 - $4R$
 - $R/4$
- A proton moving perpendicular to a uniform magnetic field describes a circle of radius r . If its speed is doubled, the radius becomes [1]
 - $r/2$
 - r
 - $2r$
 - $4r$
- Two long parallel wires carry 2 A and 3 A in the same direction, 10 cm apart. Force per metre is [1]
 - 1.2×10^{-5} N/m, attractive
 - 1.2×10^{-5} N/m, repulsive
 - 6×10^{-6} N/m, attractive
 - 2.4×10^{-5} N/m, repulsive
- The energy stored in a 0.5 H inductor carrying 2 A is [1]
 - 0.5 J
 - 1 J
 - 2 J
 - 4 J
- The resonant angular frequency of a series LCR circuit with $L = 1$ mH and $C = 10 \mu\text{F}$ is [1]
 - 10^3 rad/s
 - 10^4 rad/s
 - 10^5 rad/s
 - 10^2 rad/s
- The electric field amplitude of an EM wave in vacuum is 60 V/m. The magnetic field amplitude is [1]
 - 2×10^{-7} T
 - 2×10^{-8} T
 - 1.8×10^{10} T
 - 6×10^{-8} T
- An object is 40 cm from a convex lens of focal length 20 cm. The image distance is [1]
 - 20 cm
 - 40 cm
 - 60 cm
 - 80 cm
- In Young's experiment the screen distance is doubled, other things unchanged. The fringe width [1]
 - halves
 - remains the same
 - doubles
 - becomes four times
- Light of photon energy 6.2 eV falls on a metal of work function 5.0 eV. The stopping potential is [1]
 - 0 V
 - 1.2 V
 - 5.0 V
 - 6.2 V
- The energy of the electron in the $n = 2$ orbit of hydrogen is [1]
 - 13.6 eV
 - 6.8 eV
 - 3.4 eV
 - 1.51 eV

For Q13–Q16 choose: (a) both A and R true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

- Assertion (A):** The electric field inside a hollow charged conductor is zero. [1]
Reason (R): Excess charge resides on the outer surface, so no charge is enclosed by a Gaussian surface inside.
- Assertion (A):** A transformer does not work on dc. [1]
Reason (R): A steady current produces constant flux, so no emf is induced in the secondary.
- Assertion (A):** Nuclear density is nearly the same for all nuclei. [1]
Reason (R): The nuclear force is charge independent.
- Assertion (A):** Pure silicon behaves as an insulator at 0 K. [1]
Reason (R): At 0 K no electrons are in the conduction band.

SECTION B

 $5 \times 2 = 10$

17. State Gauss's law. Write the electric field due to an infinite plane sheet of charge density σ . [2]
18. A coil of 100 turns and area 0.02 m^2 is perpendicular to a field that falls from 0.5 T to zero in 0.1 s . Find the induced emf. [2]
19. Why is a potentiometer preferred to a voltmeter for measuring the emf of a cell? [2]
20. Define threshold frequency. Write Einstein's photoelectric equation. [2]
21. Calculate the wavelength of the first line of the Lyman series of hydrogen ($R = 1.097 \times 10^7 \text{ m}^{-1}$). [2]

SECTION C

 $7 \times 3 = 21$

22. Derive the expression for the torque on an electric dipole in a uniform electric field. [3]
23. Capacitors of 2 , 3 and $6 \mu\text{F}$ are connected in series across 12 V . Find the equivalent capacitance, the charge and the potential difference across each. [3]
24. A cell of emf 2 V and internal resistance 0.5Ω is connected to a 4.5Ω resistor. Find the current, terminal voltage and power in the resistor. [3]
25. A galvanometer of resistance 50Ω gives full-scale deflection for 2 mA . How can it be converted to an ammeter of range $0\text{--}1 \text{ A}$? [3]
26. In a series LCR circuit $R = 30 \Omega$, $X_L = 60 \Omega$ and $X_C = 20 \Omega$, connected across 100 V rms . Find Z , I_{rms} and the power consumed. [3]
27. A convex lens of focal length 15 cm forms an image of an object 10 cm from it. Find the nature, position and magnification of the image. [3]
28. An electron is accelerated from rest through 100 V . Find its de Broglie wavelength. [3]

SECTION D: case-based

 $2 \times 4 = 8$

29. **Case: Half-wave rectifier.** A student builds a half-wave rectifier with an ideal diode and a load of $1 \text{ k}\Omega$. The input is $V_m \sin \omega t$ with $V_m = 10 \text{ V}$ and $f = 50 \text{ Hz}$. [4]
- What is the frequency of the output pulses?
 - Find the peak load current.
 - Find the dc output voltage.
 - What is V_{dc} if a full-wave circuit replaces it?
30. **Case: Rod on rails.** A conducting rod of length 0.5 m moves at 4 m/s perpendicular to $B = 0.2 \text{ T}$ on rails joined by a resistor so that the circuit resistance is 2Ω . [4]
- Find the induced emf.
 - Find the current.
 - Find the force needed to keep it moving uniformly.
 - Show that the mechanical power equals the electrical power.

SECTION E

 $3 \times 5 = 15$

31. Derive the expression for the magnetic field on the axis of a circular current loop of radius a carrying current I . Hence find the field at its centre. [5]
32. Describe Young's double-slit experiment. Derive the expression for fringe width. How does it change if the experiment is done in water? [5]
33. State Bohr's postulates. Derive the radius of the n th orbit and energy of the electron in the n th orbit of hydrogen. Find the energy needed to ionise hydrogen from $n = 1$. [5]

A Paper 1: answer key and model solutions

1	2	3	4	5	6	7	8
a	b	c	c	a	b	b	a
9	10	11	12	13	14	15	16
b	c	b	c	a	a	b	a

Section A reasons: 1. $p = q \cdot 2a = 2 \times 10^{-6} \times 10^{-2}$. 2. $\frac{1}{2}CV^2 = 0.5 \times 10^{-5} \times 10^4 = 0.05 \text{ J}$. 3. $R \propto l^2$ at constant volume. 4. $r = mv/qB \propto v$. 5. $\mu_0 I_1 I_2 / 2\pi d$; same direction attract. 6. $\frac{1}{2}LI^2 = \frac{1}{2} (0.5) (4)$. 7. $1/\sqrt{LC} = 1/\sqrt{10^{-8}}$. 8. $B_0 = E_0/c$. 9. $1/v = 1/20 - 1/40$; object at $2f$. 10. $\beta = \lambda D/d$. 11. $eV_0 = h\nu - \phi$. 12. $-13.6/n^2$. 13. Gauss's law. 14. $\varepsilon = -d\Phi/dt = 0$. 15. Density is constant because $R \propto A^{1/3}$; R is true but not the reason. 16. Valence band full, CB empty.

SECTION B

model answers (2 marks)

17. Total electric flux through a closed surface $= q_{enc}/\varepsilon_0$ [1]. $E = \sigma/2\varepsilon_0$, directed normal to the sheet [1].
18. $|\varepsilon| = N \frac{\Delta\Phi}{\Delta t} = \frac{100 \times 0.02 \times 0.5}{0.1}$ [1] = 10 V [1].
19. At balance no current is drawn from the cell [1], so the potentiometer measures the true emf, whereas a voltmeter draws current and reads the terminal potential difference [1].
20. The minimum frequency below which no photoelectrons are emitted, however intense the light [1]. $K_{max} = h\nu - \phi_0$ [1].
21. $\frac{1}{\lambda} = R \left(1 - \frac{1}{4} \right)$ [1] = $0.75 \times 1.097 \times 10^7$; $\lambda = 1.215 \times 10^{-7} \text{ m} = 121.5 \text{ nm}$ [1].

SECTION C

model answers (3 marks)

22. Forces $\pm qE$ on the charges form a couple [1]. Perpendicular distance between lines of action $= 2a \sin \theta$ [1].
 $\tau = qE \cdot 2a \sin \theta = pE \sin \theta$, or $\vec{\tau} = \mathbf{p} \times \mathbf{E}$ [1].
23. $\frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$, $C = 1 \mu\text{F}$ [1]. $Q = CV = 12 \mu\text{C}$ on each [1]. $V = Q/C$: 6 V, 4 V, 2 V [1].
24. $I = \frac{2}{0.5 + 4.5} = 0.4 \text{ A}$ [1]. $V = E - Ir = 2 - 0.2 = 1.8 \text{ V}$ [1]. $P = I^2 R = 0.16 \times 4.5 = 0.72 \text{ W}$ [1].
25. Connect a shunt S in parallel [1]. $S = \frac{I_g G}{I - I_g} = \frac{0.002 \times 50}{0.998}$ [1] $\approx 0.10 \Omega$ [1].
26. $Z = \sqrt{30^2 + (60 - 20)^2} = 50 \Omega$ [1]. $I = 100/50 = 2 \text{ A}$ [1]. $P = I^2 R = 4 \times 30 = 120 \text{ W}$ [1].
27. $\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{15} - \frac{1}{10} = -\frac{1}{30}$ [1]; $v = -30 \text{ cm}$: virtual, erect, on the object side [1]. $m = v/u = 3$ [1].
28. $\lambda = \frac{h}{\sqrt{2meV}}$ [1] = $\frac{12.27}{\sqrt{V}} \text{ \AA}$ [1] = $1.227 \text{ \AA}$ ($1.23 \times 10^{-10} \text{ m}$) [1].

SECTION D

model answers (1 mark each)

29. (i) 50 Hz. (ii) $I_m = V_m/R = 10 \text{ mA}$. (iii) $V_{dc} = V_m/\pi = 3.18 \text{ V}$. (iv) $2V_m/\pi = 6.37 \text{ V}$, at 100 Hz.
30. (i) $\varepsilon = Blv = 0.2 \times 0.5 \times 4 = 0.4 \text{ V}$. (ii) $I = 0.4/2 = 0.2 \text{ A}$. (iii) $F = BIl = 0.2 \times 0.2 \times 0.5 = 0.02 \text{ N}$. (iv) $Fv = 0.02 \times 4 = 0.08 \text{ W}$; $I^2 R = 0.04 \times 2 = 0.08 \text{ W}$.

SECTION E

model answers (5 marks)

31. Take element dl at distance $r = \sqrt{a^2 + x^2}$ from axial point P [1]. Biot-Savart: $dB = \frac{\mu_0 I dl}{4\pi r^2}$ (since $dl \perp \hat{\mathbf{r}}$) [1].
 Components perpendicular to the axis cancel in pairs; $dB_x = dB \sin \alpha = dB \frac{a}{r}$ [1].
 $B = \frac{\mu_0 I a}{4\pi r^3} \int dl = \frac{\mu_0 I a \cdot 2\pi a}{4\pi r^3} = \frac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}}$ [1]. At the centre $x = 0$: $B = \frac{\mu_0 I}{2a}$ [1].
32. Coherent slits S_1, S_2 (separation d) illuminated by monochromatic light; screen at distance D [1]. Path difference at P (y from centre) $\Delta \approx \frac{yd}{D}$ [1]. Bright: $\Delta = n\lambda \Rightarrow y_n = \frac{n\lambda D}{d}$; dark: $\Delta = \left(n + \frac{1}{2}\right)\lambda$ [1]. $\beta = y_{n+1} - y_n = \frac{\lambda D}{d}$ [1]. In water $\lambda' = \lambda/\mu$, so $\beta' = \beta/\mu$: fringes become narrower [1].
33. Postulates: stable circular orbits (no radiation), $L = mvr = \frac{nh}{2\pi}$, emission/absorption $h\nu = E_i - E_f$ [1]. Centripetal:
 $\frac{mv^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2}$ with $v = \frac{nh}{2\pi mr}$ gives $r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m e^2} = 0.529 n^2 \text{ \AA}$ [2]. $E = K + U = \frac{e^2}{8\pi\varepsilon_0 r} - \frac{e^2}{4\pi\varepsilon_0 r} = -\frac{e^2}{8\pi\varepsilon_0 r_n} = -\frac{13.6}{n^2} \text{ eV}$ [1]. Ionisation energy from $n = 1$: $0 - (-13.6) = 13.6 \text{ eV}$ [1].

★ PRACTICE PAPER 2 — ORIGINAL, BOARD-STYLE

Time: 3 hours

Maximum marks: 70

BOARD-STYLE PRACTICE

All questions are compulsory. Section A: 16×1 (12 MCQ + 4 assertion-reason); B: 5×2 ; C: 7×3 ; D: 2 case-based $\times 4$; E: 3×5 . This **section pattern is provisional** (recent pattern); confirm with the official 2026–27 sample paper. Use $c = 3 \times 10^8$ m/s, $e = 1.6 \times 10^{-19}$ C, $h = 6.63 \times 10^{-34}$ J s, $\mu_0 = 4\pi \times 10^{-7}$ SI, $\epsilon_0 = 8.85 \times 10^{-12}$ SI, $m_e = 9.1 \times 10^{-31}$ kg, $m_p = 1.67 \times 10^{-27}$ kg. No internal choices are given.

SECTION A

 $16 \times 1 = 16$

1. The electric flux through a closed surface enclosing a charge of $8.85 \mu\text{C}$ is [1]
 - (a) $10^6 \text{ N m}^2 \text{ C}^{-1}$
 - (b) $10^{-6} \text{ N m}^2 \text{ C}^{-1}$
 - (c) $8.85 \times 10^{-6} \text{ N m}^2 \text{ C}^{-1}$
 - (d) zero
2. Two $4 \mu\text{F}$ capacitors in series are connected in parallel with a $2 \mu\text{F}$ capacitor. The equivalent capacitance is [1]
 - (a) $10 \mu\text{F}$
 - (b) $6 \mu\text{F}$
 - (c) $4 \mu\text{F}$
 - (d) $2.5 \mu\text{F}$
3. The current in a copper wire is doubled with everything else unchanged. The drift velocity [1]
 - (a) halves
 - (b) doubles
 - (c) remains the same
 - (d) becomes four times
4. A coil of 100 turns and radius 10 cm carries 2 A. The magnetic field at its centre is [1]
 - (a) $1.26 \times 10^{-3} \text{ T}$
 - (b) $1.26 \times 10^{-5} \text{ T}$
 - (c) $2.5 \times 10^{-3} \text{ T}$
 - (d) $6.3 \times 10^{-4} \text{ T}$
5. A bar magnet of moment 2 A m^2 lies at 30° to a field of 0.5 T. The torque is [1]
 - (a) 0.25 N m
 - (b) 0.5 N m
 - (c) 1 N m
 - (d) 0.87 N m
6. Lenz's law is a consequence of the conservation of [1]
 - (a) charge
 - (b) momentum
 - (c) energy
 - (d) mass
7. A 220 V, 50 Hz supply is connected to a pure capacitor of $\frac{100}{\pi} \mu\text{F}$. The rms current is [1]
 - (a) 0.22 A
 - (b) 2.2 A
 - (c) 22 A
 - (d) 4.4 A
8. Which of these has the shortest wavelength? [1]
 - (a) radio waves
 - (b) microwaves
 - (c) visible light
 - (d) X-rays
9. A concave mirror has $f = 10$ cm. An object is 15 cm in front. The image is [1]
 - (a) 30 cm in front, magnified, real
 - (b) 6 cm behind, virtual
 - (c) 30 cm behind, virtual
 - (d) 6 cm in front, diminished
10. A single slit of width 0.2 mm is illuminated by 600 nm light. The angular width of the central maximum is [1]
 - (a) $3 \times 10^{-3} \text{ rad}$
 - (b) $6 \times 10^{-3} \text{ rad}$
 - (c) $1.2 \times 10^{-3} \text{ rad}$
 - (d) $6 \times 10^{-4} \text{ rad}$
11. The photon energy for light of wavelength 620 nm is nearly [1]
 - (a) 1.0 eV
 - (b) 2.0 eV
 - (c) 3.1 eV
 - (d) 4.0 eV
12. A mass defect of 0.0304 u corresponds to a binding energy of about [1]
 - (a) 2.8 MeV
 - (b) 28 MeV
 - (c) 280 MeV
 - (d) 0.28 MeV

For Q13–Q16 choose: (a) both A and R true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

13. **Assertion (A):** Inserting a dielectric slab (battery disconnected) increases the capacitance. [1]
Reason (R): The dielectric reduces the field and hence the potential difference between the plates.
14. **Assertion (A):** Magnetic field lines form closed loops. [1]
Reason (R): Isolated magnetic monopoles do not exist.
15. **Assertion (A):** No photoelectrons are emitted below the threshold frequency, however intense the light. [1]
Reason (R): Each electron absorbs a single photon; if $h\nu < \phi$ it cannot escape.
16. **Assertion (A):** The hydrogen spectrum consists of discrete lines. [1]
Reason (R): Energy levels of the hydrogen atom are discrete.

SECTION B

5 × 2 = 10

17. Define drift velocity and mobility. Write the relation between them. [2]
18. A $10\ \Omega$ wire is cut into two equal parts, which are joined in parallel. Find the equivalent resistance. [2]
19. State Lenz's law. How does it illustrate conservation of energy? [2]
20. In Young's experiment $d = 0.5\text{ mm}$, $D = 1\text{ m}$ and $\lambda = 500\text{ nm}$. Find the fringe width. [2]
21. What is displacement current? Write its expression. [2]

SECTION C

7 × 3 = 21

22. Derive the expression for the energy stored in a charged capacitor. [3]
23. In a meter bridge the unknown resistance X is in the left gap and a $10\ \Omega$ resistor in the right gap. Balance is at 40 cm from the left end. Find X . State one source of error. [3]
24. Two long parallel wires carry 5 A and 10 A in the same direction, 20 cm apart. Find the force per unit length and its nature. [3]
25. A solenoid has 500 turns, length 0.5 m and cross-section 10^{-3} m^2 and carries 2 A. Find B inside and its self-inductance. [3]
26. A step-down transformer converts 2200 V to 220 V with 4400 primary turns. Find the secondary turns. If it delivers 1 kW at 90% efficiency, find the primary current. [3]
27. A prism of angle 60° has a minimum deviation of 30° . Find its refractive index and the angle of incidence at minimum deviation. [3]
28. Light of wavelength 400 nm falls on a metal of work function 2.0 eV. Find the maximum kinetic energy and stopping potential. [3]

SECTION D: case-based

2 × 4 = 8

29. **Case: Simple microscope.** A student uses a convex lens of focal length 10 cm as a magnifying glass. Take the least distance of distinct vision $D = 25\text{ cm}$. [4]
- What is the power of the lens?
 - Find the angular magnification when the image is at infinity.
 - Find the magnification when the image is at the near point.
 - Find the object distance in the latter case.
30. **Case: Hydrogen atom.** An electron in a hydrogen atom makes a transition from $n = 3$ to $n = 2$. [4]
- Find the energy of the electron in the $n = 3$ state.
 - Find the energy of the emitted photon.
 - Find its wavelength.
 - Name the spectral series.

SECTION E

3 × 5 = 15

31. Using Gauss's law derive the electric field due to a uniformly charged infinite plane sheet. State how the field varies with distance. [5]
32. Derive the expression for motional emf of a rod of length l moving with velocity v perpendicular to B . Obtain the force needed and the power, and show energy is conserved (circuit resistance R). [5]
33. Derive the relation $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ for refraction at a spherical surface (convex, paraxial rays, object in medium n_1). [5]

A Paper 2: answer key and model solutions

1	2	3	4	5	6	7	8
a	c	b	a	b	c	b	d
9	10	11	12	13	14	15	16
a	b	b	b	a	a	a	a

Section A reasons: 1. $q/\varepsilon_0 = 8.85 \times 10^{-6}/8.85 \times 10^{-12}$. 2. $2 + 2 = 4 \mu\text{F}$. 3. $I = nAev_d$. 4. $\mu_0 NI/2R$. 5. $mB \sin 30^\circ$. 6. Opposition to flux change means work must be done. 7. $X_C = 1/\omega C = 100 \Omega$. 8. EM spectrum order. 9. $1/v = 1/f - 1/u$ gives $v = -30 \text{ cm}$; $m = -2$. 10. $2\lambda/a$. 11. $1240/620$. 12. 0.0304×931.5 . 13. $C = Q/V$ with V decreased. 14. $\oint \mathbf{B} \cdot d\mathbf{A} = 0$. 15. Photon picture. 16. Transitions between discrete levels.

SECTION B

model answers (2 marks)

17. Drift velocity: average velocity of free electrons in the direction opposite to the field [1]. Mobility $\mu = v_d/E$ (drift speed per unit field), SI unit $\text{m}^2\text{V}^{-1}\text{s}^{-1}$ [1].
18. Each half is 5Ω [1]. Parallel: $5 \times 5/10 = 2.5 \Omega$ [1].
19. The induced current opposes the change in flux that produces it [1]. Work must be done against this opposition, and that work appears as electrical energy; otherwise energy would be created from nothing [1].
20. $\beta = \frac{\lambda D}{d}$ [1] $= \frac{500 \times 10^{-9} \times 1}{0.5 \times 10^{-3}} = 1 \text{ mm}$ [1].
21. Current due to a time-varying electric flux (not due to charge flow) that makes Ampere's law consistent [1]. $I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$ [1].

SECTION C

model answers (3 marks)

22. Work to add charge dq at potential q/C : $dW = \frac{q}{C} dq$ [1]. $W = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C}$ [1]. $U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV$ [1].
23. $\frac{X}{10} = \frac{l}{100-l} = \frac{40}{60}$ [1]; $X = 6.67 \Omega$ [1]. Error: end resistances / non-uniform wire / contact heating [1].
24. $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$ [1] $= \frac{2 \times 10^{-7} \times 50}{0.2} = 5 \times 10^{-5} \text{ N/m}$ [1]; attractive [1].
25. $B = \mu_0 nI = 4\pi \times 10^{-7} \times 1000 \times 2 = 2.5 \times 10^{-3} \text{ T}$ [1]. $L = \frac{\mu_0 N^2 A}{l}$ [1] $= 6.3 \times 10^{-4} \text{ H}$ [1].
26. $N_s = 4400 \times \frac{220}{2200} = 440$ [1]. Input power $= 1000/0.9 = 1111 \text{ W}$ [1]. $I_p = 1111/2200 = 0.51 \text{ A}$ [1].
27. $n = \frac{\sin \frac{A+\delta_m}{2}}{\sin \frac{A}{2}}$ [1] $= \frac{\sin 45^\circ}{\sin 30^\circ} = 1.41$ [1]. $i = \frac{A+\delta_m}{2} = 45^\circ$ [1].
28. $E = \frac{1240}{400} = 3.1 \text{ eV}$ [1]. $K_{\max} = 3.1 - 2.0 = 1.1 \text{ eV}$ [1]. $V_0 = 1.1 \text{ V}$ [1].

SECTION D

model answers (1 mark each)

29. (i) $P = 1/0.1 = 10 \text{ D}$. (ii) $m = D/f = 2.5$. (iii) $m = 1 + D/f = 3.5$. (iv) $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{25} - \frac{1}{10}$; $u = -7.1 \text{ cm}$.
30. (i) $-13.6/9 = -1.51 \text{ eV}$. (ii) $13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = 1.89 \text{ eV}$. (iii) $1240/1.89 = 656 \text{ nm}$. (iv) Balmer series (H_α line, visible red).

SECTION E

model answers (5 marks)

31. Sheet with surface charge density σ . By symmetry $\mathbf{E}$ is perpendicular to the sheet and the same magnitude on both sides [1]. Take a cylindrical Gaussian surface (area A at each end, axis perpendicular to sheet) [1]. Flux only through the two end faces: $\Phi = 2EA$ [1]. Enclosed charge σA : $2EA = \sigma A/\varepsilon_0$ [1]. $E = \frac{\sigma}{2\varepsilon_0}$, independent of distance (uniform field) [1].
32. Free charges q in the rod experience $F = qvB$; they shift to one end until $qE = qvB$ [1]. Emf $\varepsilon = Blv$ (also from $-d\Phi/dt = -Bl dx/dt$) [1]. Current $I = Blv/R$ [1]. Magnetic force on the rod $F = BIl = \frac{B^2 l^2 v}{R}$ opposes the motion; an equal external force is needed [1]. Power $P = Fv = \frac{B^2 l^2 v^2}{R} = I^2 R$; mechanical work equals Joule heating [1].
33. Ray from object O meets the surface at M , refracts to image I ; C is the centre, α, β, γ angles at O, C, I [1]. $i = \alpha + \beta$ (exterior angle of $\triangle OMC$), $\beta = r + \gamma$ so $r = \beta - \gamma$ [1]. Snell for small angles: $n_1 i = n_2 r$ [1]. $n_1 (\alpha + \beta) = n_2 (\beta - \gamma)$ with $\alpha \approx \frac{MN}{u'}$, $\beta \approx \frac{MN}{R}$, $\gamma \approx \frac{MN}{v}$ [1]. With sign convention ($u < 0$): $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ [1].

★ PRACTICE PAPER 3 — ORIGINAL, BOARD-STYLE

Time: 3 hours

Maximum marks: 70

BOARD-STYLE PRACTICE

All questions are compulsory. Section A: 16×1 (12 MCQ + 4 assertion-reason); B: 5×2 ; C: 7×3 ; D: 2 case-based $\times 4$; E: 3×5 . This section pattern is provisional (recent pattern); confirm with the official 2026–27 sample paper. Use $c = 3 \times 10^8$ m/s, $e = 1.6 \times 10^{-19}$ C, $h = 6.63 \times 10^{-34}$ J s, $\mu_0 = 4\pi \times 10^{-7}$ SI, $\epsilon_0 = 8.85 \times 10^{-12}$ SI, $m_e = 9.1 \times 10^{-31}$ kg, $m_p = 1.67 \times 10^{-27}$ kg. No internal choices are given.

SECTION A

 $16 \times 1 = 16$

1. Two point charges in air repel with force F . In a medium of dielectric constant 4 at the same separation the force is [1]
 - (a) $4F$
 - (b) $2F$
 - (c) $F/2$
 - (d) $F/4$
2. For a uniform electric field the equipotential surfaces are [1]
 - (a) concentric spheres
 - (b) planes perpendicular to the field
 - (c) planes parallel to the field
 - (d) cylinders
3. Kirchhoff's junction rule is a statement of conservation of [1]
 - (a) energy
 - (b) charge
 - (c) momentum
 - (d) angular momentum
4. The potential gradient along a potentiometer wire is 0.2 V/m. A cell is balanced at 6 m. Its emf is [1]
 - (a) 0.12 V
 - (b) 1.2 V
 - (c) 3.0 V
 - (d) 30 V
5. A charged particle enters a magnetic field perpendicular to it. Its kinetic energy [1]
 - (a) increases
 - (b) decreases
 - (c) remains constant
 - (d) becomes zero
6. The angle of dip at the magnetic equator is [1]
 - (a) 0°
 - (b) 45°
 - (c) 90°
 - (d) 180°
7. The magnetic flux through a coil is $\Phi = 5t^2 + 3t + 1$ (Wb). The emf at $t = 2$ s is [1]
 - (a) 13 V
 - (b) 23 V
 - (c) 21 V
 - (d) 11 V
8. The peak value of an alternating voltage is 311 V. Its rms value is nearly [1]
 - (a) 155 V
 - (b) 220 V
 - (c) 311 V
 - (d) 440 V
9. The speed of EM waves in vacuum is given by [1]
 - (a) $\sqrt{\mu_0 \epsilon_0}$
 - (b) $1/\sqrt{\mu_0 \epsilon_0}$
 - (c) μ_0/ϵ_0
 - (d) ϵ_0/μ_0
10. The critical angle for glass ($n = 1.5$) to air is nearly [1]
 - (a) 30°
 - (b) 42°
 - (c) 48°
 - (d) 60°
11. The path difference at the third bright fringe in Young's experiment is [1]
 - (a) λ
 - (b) 2λ
 - (c) 3λ
 - (d) $3\lambda/2$
12. The radius of a nucleus of mass number 27 is ($R_0 = 1.2$ fm) [1]
 - (a) 1.2 fm
 - (b) 3.6 fm
 - (c) 5.4 fm
 - (d) 32 fm

For Q13–Q16 choose: (a) both A and R true, R explains A; (b) both true, R does not explain A; (c) A true, R false; (d) A false, R true.

13. Assertion (A): An ammeter is connected in series in a circuit. [1]
Reason (R): An ammeter has very low resistance.
14. Assertion (A): The induced emf is zero when a coil moves with constant velocity inside a uniform magnetic field (without leaving it). [1]
Reason (R): The flux through the coil does not change.
15. Assertion (A): The photoelectric current increases when light intensity increases (frequency fixed above threshold). [1]
Reason (R): More photons per second eject more electrons per second.
16. Assertion (A): A p-type semiconductor is positively charged. [1]
Reason (R): The majority carriers in p-type are holes.

SECTION B

5 × 2 = 10

17. Define electric flux and its SI unit. What is the flux through each face of a cube with charge q at its centre? [2]
18. A bulb is rated 100 W, 220 V. Find its resistance and the current drawn. [2]
19. State the Biot–Savart law in vector form. [2]
20. Distinguish between a real and a virtual image with an example of each. [2]
21. Find the de Broglie wavelength of a ball of mass 0.1 kg moving at 10 m/s. Why is the wave nature not observed? [2]

SECTION C

7 × 3 = 21

22. Derive the force on a straight conductor of length l carrying current I in a uniform field B from the force on a moving charge. [3]
23. A 4 V cell (internal resistance $1\ \Omega$) and a 2 V cell (internal resistance $1\ \Omega$) are connected in series, with opposing polarities, to a $2\ \Omega$ resistor. Find the current and the terminal voltage of the 4 V cell. [3]
24. A galvanometer of $100\ \Omega$ gives full scale for 1 mA. How is it converted to a voltmeter of range 10 V? State the resistance of the voltmeter. [3]
25. A 2 H inductor carries 5 A which falls to zero in 0.1 s. Find the average induced emf and the energy dissipated. [3]
26. A 100 mH inductor is connected to a 220 V, 50 Hz supply. Find the reactance, the rms current, the power consumed and the phase difference. [3]
27. A convex lens of refractive index 1.5 has radii 20 cm and 20 cm. Find its focal length in air and in water ($n = 1.33$). [3]
28. Calculate the binding energy and binding energy per nucleon of ${}^2_1\text{H}$. ($m_H = 1.007825\ \text{u}$, $m_n = 1.008665\ \text{u}$, $M({}^2\text{H}) = 2.014102\ \text{u}$) [3]

SECTION D: case-based

2 × 4 = 8

29. **Case: LED.** An LED of wavelength 620 nm is operated from a 5 V supply through a series resistor. The LED drops 2.0 V when carrying 20 mA. [4]
- In which bias is the LED operated?
 - Find the photon energy emitted.
 - Find the series resistance.
 - What decides the colour of an LED?
30. **Case: Charged particle in a field.** A proton is accelerated from rest through 1 kV and enters a uniform field $B = 0.1\ \text{T}$ perpendicular to its velocity. [4]
- Find the speed of the proton.
 - Find the radius of its circular path.
 - Find the time period.
 - How much work does the magnetic field do?

SECTION E

3 × 5 = 15

31. Describe the principle of a potentiometer. Explain how it is used to compare the emfs of two cells. State two advantages over a voltmeter. [5]
32. A resistor R , inductor L and capacitor C are connected in series to $v = v_m \sin \omega t$. Using phasors, derive the impedance and the phase angle. State the resonance condition and resonant frequency. [5]
33. State the laws of photoelectric emission. Write Einstein's equation and use it to explain the laws. Sketch the graph of stopping potential against frequency and state what its slope gives. [5]

A Paper 3: answer key and model solutions

1	2	3	4	5	6	7	8
d	b	b	b	c	a	b	b
9	10	11	12	13	14	15	16
b	b	c	b	a	a	a	d

Section A reasons: 1. $F \propto 1/K$. 2. Potential changes only along **E**. 3. Charge cannot accumulate at a junction. 4. $E = kl = 0.2 \times 6$. 5. Force is perpendicular to velocity, so no work. 6. Field horizontal at the equator. 7. $d\Phi/dt = 10t + 3$. 8. $V_m/\sqrt{2}$. 9. Maxwell's result. 10. $\sin C = 1/1.5$. 11. Bright: $n\lambda$. 12. $1.2 \times 27^{1/3}$. 13. Low resistance avoids changing the current. 14. $d\Phi/dt = 0$. 15. Intensity gives number of photons. 16. A false: the crystal is neutral. R is true.

SECTION B

model answers (2 marks)

17. $\Phi = \mathbf{E} \cdot \mathbf{A}$; unit $\text{N m}^2\text{C}^{-1}$ (V m) [1]. Total flux q/ϵ_0 shared by six faces: $q/6\epsilon_0$ each [1].
18. $R = V^2/P = 48400/100 = 484 \Omega$ [1]. $I = P/V = 0.45 \text{ A}$ [1].
19. $d\mathbf{B} = \frac{\mu_0 I d\mathbf{l} \times \hat{\mathbf{r}}}{4\pi r^2}$ [1], where $d\mathbf{l}$ is the current element and $\mathbf{r}$ the position vector of the field point; direction by right-hand rule [1].
20. Real: formed by actual convergence of rays, can be taken on a screen (convex lens image of a distant object) [1]. Virtual: rays only appear to diverge from it, cannot be taken on a screen (image in a plane mirror) [1].
21. $\lambda = h/mv = 6.63 \times 10^{-34}/1 = 6.6 \times 10^{-34} \text{ m}$ [1]. It is far smaller than any size we can detect or any aperture [1].

SECTION C

model answers (3 marks)

22. Charge carriers (density n , charge q , area A) have drift speed v_d ; force on one carrier $q\mathbf{v}_d \times \mathbf{B}$ [1]. Number in length l : nAl [1]. $\mathbf{F} = nAlq\mathbf{v}_d \times \mathbf{B} = Il \times \mathbf{B}$ since $I = nqAv_d$; $F = BIl \sin \theta$ [1].
23. Net emf $= 4 - 2 = 2 \text{ V}$; total resistance $= 1 + 1 + 2 = 4 \Omega$ [1]. $I = 0.5 \text{ A}$ [1]. $V = 4 - 0.5 \times 1 = 3.5 \text{ V}$ [1].
24. Connect a high resistance R in series [1]. $R = \frac{V}{I_g} - G = \frac{10}{10^{-3}} - 100 = 9900 \Omega$ [1]. Voltmeter resistance $= 10,000 \Omega$ [1].
25. $|\epsilon| = L \frac{\Delta I}{\Delta t} = 2 \times \frac{5}{0.1} [1] = 100 \text{ V}$ [1]. $U = \frac{1}{2}LI^2 = 25 \text{ J}$ [1].
26. $X_L = 2\pi fL = 31.4 \Omega$ [1]. $I = 220/31.4 = 7.0 \text{ A}$ [1]. $P = VI \cos 90^\circ = 0$; current lags voltage by $\pi/2$ [1].
27. Air: $\frac{1}{f} = 0.5 \left(\frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20}$, $f = 20 \text{ cm}$ [1]. Water: $\frac{1}{f} = \left(\frac{1.5}{1.33} - 1 \right) \frac{2}{20} = 0.0128$ [1]; $f = 78 \text{ cm}$; lens becomes weaker [1].
28. $\Delta m = 1.007825 + 1.008665 - 2.014102 = 0.002388 \text{ u}$ [1]. $BE = 0.002388 \times 931.5 = 2.22 \text{ MeV}$ [1]. $BE/A = 1.11 \text{ MeV/nucleon}$ [1].

SECTION D

model answers (1 mark each)

29. (i) Forward bias. (ii) $1240/620 = 2.0 \text{ eV}$, so $E_g \approx 2.0 \text{ eV}$. (iii) $R = (5 - 2.0)/0.02 = 150 \Omega$. (iv) The band gap of the semiconductor ($\lambda = 1240/E_g \text{ nm}$).
30. (i) $v = \sqrt{2eV/m} = 4.4 \times 10^5 \text{ m/s}$. (ii) $r = mv/eB = 4.6 \times 10^{-2} \text{ m}$. (iii) $T = 2\pi m/eB = 6.6 \times 10^{-7} \text{ s}$. (iv) Zero; the force is always perpendicular to the velocity.

SECTION E

model answers (5 marks)

31. Principle: potential drop across a uniform wire carrying steady current is proportional to its length, $V = kl$ [1]. Circuit: driver cell, rheostat, key and long wire; cells E_1, E_2 connected through a galvanometer and jockey (positive terminals to the high-potential end) [1]. Balance lengths l_1, l_2 at zero galvanometer deflection: $E_1 = kl_1, E_2 = kl_2$ [1]. $\frac{E_1}{E_2} = \frac{l_1}{l_2}$ [1]. Advantages: draws no current at balance (measures true emf); sensitivity can be increased by a lower potential gradient [1].
32. Same current $i = i_m \sin \omega t$ in all; $V_R = iR$ in phase, $V_L = iX_L$ leads by 90° , $V_C = iX_C$ lags by 90° [1]. Phasor diagram: $V_L - V_C$ perpendicular to V_R [1]. $V_m = i_m \sqrt{R^2 + (X_L - X_C)^2}$, so $Z = \sqrt{R^2 + (X_L - X_C)^2}$ [1]. $\tan \phi = \frac{X_L - X_C}{R}$ [1]. Resonance when $X_L = X_C$: $Z = R$ (minimum), $\phi = 0$, $\omega_0 = \frac{1}{\sqrt{LC}}$, $f_0 = \frac{1}{2\pi\sqrt{LC}}$ [1].
33. Laws: photocurrent $\propto$ intensity (frequency fixed); K_{max} independent of intensity, increases with frequency; threshold frequency below which no emission; instantaneous ($\sim 10^{-9} \text{ s}$) [2]. $K_{max} = h\nu - \phi_0$ (photon gives all energy to one electron) [1]. Explains: intensity = number of photons, so current $\propto$ intensity; K_{max} depends only on ν ; $\nu < \nu_0$ gives $h\nu < \phi_0$; one-to-one photon-electron collision is instantaneous [1]. $V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$: straight line meeting the ν -axis at ν_0 ; slope $= h/e$ [1].

G Glossary and symbol index

Amplitude. Maximum displacement of an oscillating quantity.

Binding energy. Energy needed to separate a nucleus into free nucleons.

Capacitance. Charge stored per unit potential difference, $C = Q/V$.

Coherent sources. Sources with constant phase difference.

Dielectric. Insulator that reduces the field inside a capacitor.

Diffraction. Bending of waves around obstacles or apertures.

Displacement current. Current due to changing electric flux.

Drift velocity. Average velocity of charge carriers due to a field.

Eddy currents. Circulating currents induced in a conductor by changing flux.

Emf. Energy supplied per unit charge by a source.

Equipotential. Surface of constant potential.

Fermi / forbidden gap. Energy range with no allowed states between VB and CB.

Flux. Field passing through a surface, $\Phi = \mathbf{B} \cdot \mathbf{A}$.

Half-life. Time for half the nuclei to decay.

Impedance. Total opposition to ac in a circuit, Z .

Inductance. Flux linkage per unit current.

Isotopes. Nuclei with same Z but different A .

Magnetic moment. $m = NIA$ for a current loop.

Mobility. Drift speed per unit field.

Photon. Quantum of EM radiation, $E = h\nu$.

Potential. Work per unit charge to bring a charge from infinity.

Power factor. $\cos \phi$ in an ac circuit.

Reactance. Opposition of L or C to ac.

Resonance. Maximum current when $X_L = X_C$.

Retentivity. Magnetisation left after the field is removed.

Stopping potential. Reverse voltage that stops photocurrent.

Threshold frequency. Minimum frequency for photoemission.

Wavefront. Locus of points in the same phase.

Symbol index

Symbol	Meaning
ϵ_0	permittivity of free space
μ_0	permeability of free space
σ, ρ, λ	surface density, resistivity/density, wavelength/line density
ϕ, Φ	phase angle / work function, flux
τ	torque / relaxation time
χ, μ_r	susceptibility, relative permeability
δ	dip angle / deviation
ω	angular frequency
ν, f	frequency
α	temperature coefficient
Z, A, N	atomic number, mass number, neutron number
E_g	band gap
V_k	knee voltage

S Source notes

• SOURCE NOTES AND HONESTY STATEMENT

- **Syllabus.** Chapter sequence follows the standard NCERT Class 12 Physics sequence (14 chapters). The official 2026–27 syllabus and any deleted or added topics were **not** available while writing and must be checked.
- **Paper pattern and marking scheme.** The official sample question paper and marking scheme could **not** be accessed. The pattern shown (section sizes, marks) is the recent pattern and is provisional.
- **Previous-year questions.** No previous-year question has been verified, so none is reproduced or claimed. Every question is **BOARD-STYLE PRACTICE** written for this book. No year is attached to any question.
- **Content.** Concepts, derivations and worked examples are written from standard textbook physics. Numerical answers were computed in code and cross-checked for arithmetic. The book has **not** been through independent subject-expert review or classroom trials.
- **Extensions.** Polarisation (Ch 10), radioactive decay (Ch 13) and special-purpose diodes (Ch 14) are marked as extensions pending syllabus confirmation.
- **Branding.** No official board or NCERT logo is used and no endorsement is implied. All diagrams were drawn for this book.

★ AFTER THE EXAM SEASON

Found an error, a topic that should be added or removed, or a verified past-paper question you want credited? Note the question ID (for example C09-N03) and send it to the author for the next edition.